

A Short Path Quantum Algorithm for Exact Optimization

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We give a quantum algorithm to exactly solve certain problems in combinatorial optimization, including weighted MAX-2-SAT as well as problems where the objective function is a weighted sum of products of Ising variables, all terms of the same degree D ; this problem is called weighted MAX-ED-LIN2. We require that the optimal solution be unique for odd D and doubly degenerate for even D ; however, we expect that the algorithm still works without this condition and we show how to reduce to the case without this assumption at the cost of an additional overhead. While the time required is still exponential, the algorithm provably outperforms Grover's algorithm assuming a mild condition on the number of low energy states of the target Hamiltonian. The detailed analysis of the runtime depends on a tradeoff between the number of such states and algorithm speed: having fewer such states allows a greater speedup. This leads to a natural hybrid algorithm that finds either an exact or approximate solution.

I. INTRODUCTION

While quantum algorithms are useful for many problems involving linear algebra (for example, finding eigenvalues^[18], solving linear equations^[15], and performing Hadamard transforms^[11]), there are few proven speedups for combinatorial optimization problems. The most basic such speedup is Grover's algorithm^[14], which gives a quadratic speedup over a brute-force search. For a problem such as finding the ground state of an Ising model on N spins, this can lead to a speedup from a brute force time $\mathcal{O}^*(2^N)$ (where $\mathcal{O}^*(\cdot)$ is big-O notation up to polylogarithmic factors, in this case polynomials in N) to $\mathcal{O}^*(2^{N/2})$. In the black box setting, Grover's algorithm is optimal^[4], but for problems with a structure one might try to find a further speedup.

One attempt to find a speedup is the adiabatic algorithm^[10]. Here, let H_Z be a Hamiltonian which is diagonal in the computational basis, such as

$$H_Z = \sum_{i,j} J_{i,j} Z_i Z_j, \quad (1)$$

where i labels different qubits and Z_i is the Pauli Z matrix on the i -th qubit. Then, consider the Hamiltonian

$$H_s = -(1-s)X + sH_Z, \quad (2)$$

where

$$X = \sum_i X_i \quad (3)$$

and X_i is the Pauli X matrix on the i -th qubit. At $s = 0$, the ground state of this Hamiltonian can be easily prepared. At $s = 1$, the ground state is the ground state of H_Z . If the spectral gap of H_s between the ground and first excited state is only polynomially small for $s \in [0, 1]$, then one may adiabatically evolve the ground state from $s = 0$ to $s = 1$ in polynomial time.

Unfortunately, the gap may become superpolynomially small. Indeed, it was argued using ideas from Anderson localization^[2] that the gap may become as small as $N^{-\text{const.} \times N}$ so that the time required for adiabatic evolution is of order $N^{\text{const.} \times N}$ which is much slower than even classical brute force search. While the actual behavior for random instances may be more complicated than this^[20], specific examples^[33] can show this behavior.

This is perhaps not surprising. The problem of finding the ground state of H_Z is extremely hard, even if we restrict $J_{ij} \in \{-1, 0, 1\}$ for all i, j . While some speedups are known for bounded degree (here, we consider the graph with spins as vertices and an edge between vertices if the corresponding element of J is nonzero, and degree refers to the degree of this graph^{[11][12][16][31]}, if we only allow polynomial space then the fastest classical algorithms for arbitrary J take a time $\mathcal{O}^*(2^N)$. If we allow exponential space then it is possible to reduce this time to $\mathcal{O}^*(2^{\omega N/3})$, where ω is the matrix multiplication exponent^[34]. However, not only does that algorithm require exponential space, but it is not known how to give a Grover speedup of this algorithm, so that no quantum algorithm is known taking time $\mathcal{O}^*(2^{cN/2})$ for any $c < 1$. Finally, this algorithm is specific to constraint satisfaction problems where each constraint only involves a pair of variables, rather than a triple or more.

Here we present a quantum algorithm to find the ground state of H_Z which improves on Grover's algorithm in many cases. In the next subsection we give the problem definition and define which H_Z we consider.

A. Problem Definition

The problem is to find the ground state of H_Z (i.e., eigenstate of H_Z with minimum eigenvalue; eigenvalues of H_Z, H_s will often be called “energies”) assuming that the ground state energy E_0 is known. We impose the following conditions on H_Z . Let N be the number of qubits. We let H_Z be any Hamiltonian that is a weighted sum of products of Pauli Z operators, with each product containing exactly D such operators on distinct qubits for some given D . That is, the case $D = 2$ is the Ising model, the case $D = 4$ is a sum of terms $Z_i Z_j Z_k Z_l$ for i, j, k, l all distinct, and so on. We take $D = \mathcal{O}(1)$. We emphasize that all products must have the same D so that we do not allow for example $H_Z = Z_1 Z_2 + Z_1 Z_2 Z_3$; this is for a reason explained later.

Each product has a weight that is an integer. We define J_{tot} to be the sum of the absolute values of the weights. We require that $J_{tot} = \mathcal{O}(\text{poly}(N))$. That is, we fix some $\beta > 0$ and require $J_{tot} = \mathcal{O}(N^\beta)$. (If all weights are chosen from $\{-1, +1\}$ we have $J_{tot} = \mathcal{O}(N^2)$.)

We require that H_Z has a unique ground state for D odd while for D even we require that H_Z has a doubly degenerate ground state. For even D , the operator $\prod_i X_i$ commutes with the Hamiltonian, flipping all spins, so that every eigenvalue has an even degeneracy. We call this assumption the “degeneracy assumption”. We will analyze the performance of the algorithm without this assumption in future work. In an Appendix we show how to reduce to the case without this assumption at the cost of an additional time overhead.

We give two theorems [12] which describe the performance of the algorithm; the different theorems correspond to different choices of the parameters in the algorithm. Both theorems show, roughly, that at least one of two things holds: the algorithm finds the ground state in a certain expected time or H_Z has a large number of low energy states, i.e., eigenstates with eigenvalue close to E_0 .

These theorems can be applied in one of two ways. One way is to define a promise problem, in which we are promised that H_Z does *not* have such a large number of low energy states as well as promised the degeneracy assumption on H_0 . Then, the algorithm solves this promise problem. For this promise problem, while the algorithm is given E_0 , it is not necessary to know E_0 in advance since one can try all possible E_0 with only polynomial overhead.

Alternatively, we can consider H_Z without such a promise on the number of low energy states but still including the promise on the degeneracy assumption. Then, we give a hybrid algorithm which tries both running the quantum algorithm here to find the exact ground state as well as random sampling (or a Grover search) to find an approximate ground state. This hybrid algorithm will be explained after theorem [12].

The problem of minimizing H_Z is also known as MAX-ED-LIN2, as we assume that each term is exactly of degree D , rather than having degree at most D ; of course, there is an overall sign difference also, as we try to minimize H_Z rather than maximizing. Due to the uniqueness conditions on H_Z , we call this problem UNIQUE-MAX-ED-LIN2.

However, we can reduce any instance of MAX-2-LIN2 with a unique solution to UNIQUE-MAX-E2-LIN2, where MAX-2-LIN2 allows terms to have degree 1 or 2. Up to a sign difference, MAX-2-LIN2 with a unique solution means that we consider $H_Z = \sum_{i,j} J_{i,j} Z_i Z_j + \sum_i h_i Z_i$, with integer $J_{i,j}, h_i$ and $\sum_{i,j} |J_{i,j}| + \sum_i |h_i| = J_{tot}$, with H_Z required to have a unique ground state. We can find the ground state of this problem by finding one of the two ground states of a problem on $N + 1$ spins defined by $H_Z = \sum_{i,j} J_{i,j} Z_i Z_j + \sum_i h_i Z_0 Z_i$, where 0 is the added spin.

B. Main Results

We now give the main results. The parameters B, K enter into the definition of the algorithm, given later; the algorithm makes use of a family of Hamiltonians $H_s = H_Z - sB(X/N)^K$ so that B sets the strength of a term that is proportional to a power (set by K) of the transverse field. In each theorem, the statement about the number of low energy states of H_Z is first expressed as a statement about probability distributions with high entropy and low energy which then implies the statement about the number of low energy states.

We give two distinct theorems. The first theorem considers the case that $K = C \log(N)$ for $C > 0$ a fixed constant. The second theorem is more general, and allows K to be an arbitrary odd integer with $K \geq 3$; K may depend on N and the first theorem follows from the second. The first theorem is simply given as a way to explain some of the results in the particular case of a logarithmically growing K , when some of the results simplify; the second theorem gives stronger results but they are more complicated. Even in the case of a logarithmically growing K , the second theorem is stronger.

Fixed K , compared to logarithmically growing K , leads to a larger speedup but requires a stronger promise.

In an abuse of notation, when we write $K = C \log(N)$, we mean that K is taken to be the smallest odd integer larger than $C \log(N)$. We explicitly write the dependence on these constants in the equations, not hiding them in big-O notation. The quantity b is a real number, $0 < b < 1$. We take $b < 1$ else the second possibility (item 2 in the list) is trivially true and does not imply any interesting constraint on the density of states.

We let $W(E)$ denote the number of computational basis states with expectation value E for H_Z .

All entropies are measured in bits rather than nats. All logarithms are to base 2 unless otherwise specified.

Theorem 1. Assume that H_Z obeys the degeneracy assumption. Suppose that $B = -bE_0$ and $K = C \log(N)$. Then, at least one of the following holds:

1. The algorithm finds the ground state in expected time

$$\mathcal{O}^*\left(2^{N/2} \exp\left[-\frac{b}{2CD} \frac{N}{\log(N)}\right]\right).$$

2. There is some probability distribution $p(u)$ on computational basis states with entropy at least

$$S^{(comp)} \geq N \cdot (1 - \mathcal{O}(1)/C)$$

and with expected value of H_Z at most $(1-b)E_0 + \mathcal{O}(1) \cdot \frac{J_{tot}}{N^2} C^2 D^2 \log(N)^2$. Further, for any $\eta > 0$, for some

$$E \leq E_0 + (1+\eta) \left(b|E_0| + \mathcal{O}(1) \cdot \frac{J_{tot}}{N^2} C^2 D^2 \log(N)^2 \right),$$

we have

$$\log(W(E)) \geq N \cdot (1 - \mathcal{O}(1) \frac{1+\eta}{\eta} \frac{1}{C}) - \frac{1+\eta}{\eta} \mathcal{O}(\log(N)).$$

Theorem 1 implies the following corollary about a hybrid algorithm:

Corollary 1. There is an algorithm that, given H_Z and E_0 , either outputs “approximate” or “exact”. If it outputs “approximate”, it returns also a state with energy at most $E_{approx} \equiv E_0 + (1.01) \left(b|E_0| + \mathcal{O}(1) \cdot \frac{J_{tot}}{N^2} C^2 D^2 \log(N)^2 \right)$ and takes time at most $\mathcal{O}^*(2^{\mathcal{O}(1)N/C})$. If it outputs “exact”, it outputs an exact ground state of H_Z . The expected run time of the algorithm (averaged over both approximate and exact outputs, rather than conditioned on an output) is $\mathcal{O}^*\left(2^{N/2} \exp\left[-\frac{b}{2CD} \frac{N}{\log(N)}\right]\right)$.

Proof. Take $\eta = 0.01$.

Run the following algorithm. First, try repeated random sampling of states to find a state with energy at most E_{approx} . Take a total of n_{samp} samples. If any sample succeeds, terminate the algorithm, returning “approximate” and the given state. If item 2 of the theorem holds, each sample succeeds in finding such a state with probability at least $2^{-\mathcal{O}(1)N/C - \mathcal{O}(\log(N))} = 2^{-\mathcal{O}(1)N/C} / \text{poly}(N)$, so one can choose $n_{samp} = \mathcal{O}^*(2^{\mathcal{O}(1)N/C})$ such samples and succeed with probability at least $1 - 2^{-N}$.

If no sample succeeds, then run the quantum algorithm of this paper in parallel with a brute force search, until one of them finds an exact ground state, returning “exact” and the state found.

The run time bound on the approximate output holds by construction. We now show the expected run time bounds. If item 1 holds, then the algorithm of this paper succeeds in expected time $\mathcal{O}^*\left(2^{N/2} \exp\left[-\frac{b}{2CD} \frac{N}{\log(N)}\right]\right)$ so the expected run time bound holds. If item 2 holds and the repeated random sampling fails to find an approximate state then the brute force search will find the ground state in time $\mathcal{O}^*(2^N)$; since the probability that the repeated random sampling fails is $\leq 2^{-N}$, this adds a negligible amount to the expected time of the algorithm.

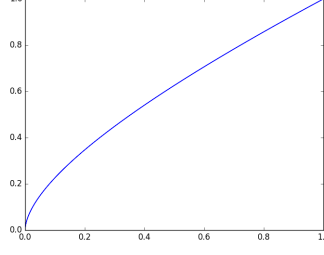
Remark: the repeated random sampling can be quadratically improved with Grover search but this only improves constants which are hidden in the big-O notation. $\square$

We now give the case of arbitrary K . The function $\tau(\cdot)$ in theorem 2 is a continuous increasing function, taking $[0, 1]$ to $[0, 1]$. It is defined in lemma 11. It is differentiable on this interval, except at zero; for small σ ,

$$\tau(\sigma) = \Theta\left(\sqrt{\frac{\sigma}{-\log(\sigma)}}\right). \quad (4)$$

In Fig. 1 we plot $\tau(\cdot)$.

Theorem 2. Assume that H_Z obeys the degeneracy assumption. Suppose that $B = -bE_0$ with $0 \leq b < 1$. Suppose that $K \geq 3$ and $K > \beta$, with K odd. Then, at least one of the following holds:

FIG. 1: A plot of $\tau(\cdot)$.

1. The algorithm finds the ground state in expected time

$$\mathcal{O}^*\left(2^{N/2} \exp\left[-\frac{b}{2DK}N\right]\right).$$

2. For some $X_0 \geq X_{\min} = N \cdot (4B)^{-1/K}$, there is some probability distribution $p(u)$ on computational basis states with entropy in bits at least

$$S^{(comp)} \geq N\tau^{-1}\left(\frac{X_0 - X_{\min}/K}{N}\right)$$

and with expected value of H_Z at most

$$E_0 + \mathcal{O}(1)\frac{J_{tot}K^2D^2}{X_{\min}^2} + B\left(\frac{X_0 + X_{\min}/K}{N}\right)^K \cdot \mathcal{O}(1).$$

Further, for some function $F(S)$ with

$$F(S) = E_0 + \mathcal{O}(1)\frac{J_{tot}K^2D^2}{X_{\min}^2} + B(\tau(S/N))^K \cdot \mathcal{O}(1), \quad (5)$$

then for some $E > E_0$ we have $\log(W(E)) \geq F^{-1}(E) - \mathcal{O}(\log(N))$.

There are two parameters b, K that can be adjusted in this algorithm, so the implications of this theorem take some unpacking. We now argue, however, that for most problems for many choices of b, K the conditions of items 2. of the theorem will not be satisfied and hence a nontrivial speedup will occur from item 1. The next paragraph will deliberately be more heuristic, and will draw on some physics intuition, since the goal is not to prove a result but rather to argue that certain conditions do not occur in most examples.

Let us consider a simple example, $K = 99, b = 1/99$. Then, either the algorithm gives a speedup over a Grover search, taking expected time $\mathcal{O}^*(2^{cN/2})$ for a c slightly less than 1, or there are a large number of computational basis states with low energy for H_Z . Let us see whether we can satisfy $\log(W(E)) \geq F^{-1}(E)$ for any $E > E_0$. For E near 0, we have $\log(W(E))$ close to N since a typical state has energy E near 0. If we had taken $b = 1$, we could then satisfy $N \geq F^{-1}(0)$. However, the function $F(S)$ is bounded at $E_0 + \mathcal{O}(1)\frac{J_{tot}K^2D^2}{X_{\min}^2} + B\mathcal{O}(1)$, and the constant hidden in the big-O notation $\mathcal{O}(1)$ is such that for $b = 1/99$ we have $F(S)$ bounded away from zero. Indeed, $F(S)$ will be upper bounded by roughly $(1 - b\mathcal{O}(1))E_0$. Let us, for clarity in explanation pretend that the constant hidden in the $\mathcal{O}(1)$ is equal to 1. Then, we could have a problem if $F^{-1}(\frac{98}{99}E_0) \approx N$. However, this means an enormous (entropy almost equal to N) number of states have energy which is $98/99$ times the ground state energy. Now let us instead ask whether the conditions of item 2. can be satisfied for X_0 close to 0. Indeed, for an example such as the Sherrington-Kirkpatrick spin glass^[17], the entropy $\log(W(E))$ has a power law dependence on $E - E_0$, with the scaling behavior well-understood^{[9][25]} from the Parisi solution^{[26][27]} giving $\log(W(E)) \sim N \cdot ((E - E_0)/|E_0|)^{2/3}$. By choosing K large, from studying the limiting behavior of c we find that we might satisfy the condition at small $E - E_0$ if $\log(W(E)) \sim N \cdot ((E - E_0)/|E_0|)^{2/99}$ up to some polylogs in energy. So, again, we would not expect to obey item 2. at small X_0 since $2/99 < 2/3$. Note also that for $K = 99$, since $B \leq \mathcal{O}(N^2)$ we have $X_{\min} = \Omega(N^{1-2/99}) = \Omega(N^{97/99})$ so that $X_{\min}$ cannot be too small.

On the other hand, a model such as the “toy model” of Ref. [33] has a unique ground state but has $\log(W(E)) \approx N/2$ for $E = E_0 + 1$. Thus, for this model, we cannot consider fixed K without satisfying item 2. of the theorem but we can still obtain a nontrivial speedup by taking $K = C \log(N)$, albeit not giving $\mathcal{O}^*(2^{cN/2})$ with $c < 1$.

Even in the case $K = C \log(N)$, theorem 2 gives tighter bounds than theorem 1. The entropy in theorem 1 is the entropy for $X_0 = X_{min}$, i.e., it is the minimum possible, while the energy given in theorem 1 is the largest possible that can occur from theorem 2 for $K = C \log(N)$ and this value of energy would not occur for $X_0 = X_{min}$. Thus, theorem 2 gives stronger results involving a functional relationship between entropy and energy.

C. Outline

The algorithm has three distinct ideas. First we use a short path evolution, where the initial state is not a ground state of the initial Hamiltonian. That is, rather than trying to follow the full evolution from a transverse field Hamiltonian that is diagonal in the X basis to H_Z which is diagonal in the Z basis, we instead follow only a “short path”, keeping a term proportional to H_Z in the Hamiltonian fixed and varying a transverse field-like term from a small value to zero. We called this term “transverse field-like” rather than a transverse field, because we add a more complicated term $-B(X/N)^K$ that is proportional to a power of the transverse field. This is the second idea and plays a key role in avoiding small spectral gaps. The third idea uses measurements to obtain adiabatic evolution with exponentially small error; similar ideas are in Ref. 28,32.

We will show that either the algorithm succeeds in giving a nontrivial speedup or a certain spectral gap becomes small. However, for such a gap to become small, there must be some state with a large expectation value for $-B(X/N)^K$ and with a small expectation value for H_Z . Roughly, the idea then is that for a large K , this imposes strong constraints on the expectation value of X . For example, if the state is an eigenstate of X , then we would have $\langle X/N \rangle = \langle (X/N)^K \rangle^{1/K}$, and so for large K , even a small expectation value of $(X/N)^K$ would give a large expectation value for X/N . Technically, the implementation is more complicated than this because there may be large fluctuations in X in the state. Ignoring these technical details for the moment, the next idea is that given a large expectation value of X , we use log-Sobolev estimates on entropy to relate the expectation value of X to the entropy in the computational basis. This is then used to show that there must be many eigenstates of H_Z with low energy.

Other technical tools used include a Brillouin-Wigner perturbation theory to compute overlaps and a method of localizing wavefunctions in X ; this method is used to deal with the problem of large fluctuations in X .

In sections II,III we give the algorithm. In section IV we explain the Brillouin-Wigner perturbation theory. We then consider two cases, depending on the spectrum of the Hamiltonian QH_sQ , where Q projects onto the states of H_Z with energy greater than E_0 . Theorem 3 in section V assumes the first case and gives bounds on the spectral gap of H_s and shows the speedup of the algorithm. Theorem 4 in section VI assumes the second case and shows the existence of probability distributions over computational basis states with high entropy and low energy. Putting these results together, theorems 1,2 follow as shown in section VII; there, we use the high entropy and low energy probability distributions to lower bound $W(E)$. In section VIII, we briefly discuss some other results and also remark on a hybrid algorithm.

Section VI will use the ground state degeneracy assumption for convenience, but we will show how to remove this assumption for the results of this section. Section V will rely more heavily on the ground state degeneracy assumption; we will remove this assumption in future work using a modified perturbation theory.

II. THE SHORT PATH ALGORITHM

The algorithm that we use is based on applying amplitude amplification 33 to another algorithm. This latter algorithm is Algorithm 1. Remark: The case $K = 1, B = cN$ corresponds to a transverse magnetic field of strength c . Also, note the direction of evolution from $s = 1$ to $s = 0$.

To analyze this algorithm for even D we need a definition. For D even, we define the “even subspace” to be the eigenspace of $\prod_i X_i$ with eigenvalue $+1$ and define the “odd subspace” to be the eigenspace of $\prod_i X_i$ with eigenvalue -1 . The Hamiltonian H_s commutes with $\prod_i X_i$ and ψ_+ is in the even subspace; the measurement algorithm will be chosen to preserve the eigenvalue of $\prod_i X_i$. When we analyze the algorithm for even D , all references for the remainder of the paper to the Hamiltonian will refer to the Hamiltonian projected into the even subspace and all vectors will be in the even subspace. We will remark on this as needed.

At $s = 0$, H_s has a unique ground state; call this state $|0\rangle$; for even D this state $|0\rangle$ is the unique ground state of H_Z in the even subspace and it is an equal amplitude superposition of two computational basis states. Let $\psi_{0,s}$ be the ground state of H_s . By Perron-Frobenius this state is unique for $s \geq 0$ for K odd; to see this, note that one can flip any given spin K times to attain a single spin flip so that the matrix H_s is irreducible in the computational basis, i.e., consider the term X_i^K for any given i . When we refer to uniqueness of a state, we mean uniqueness up to an overall phase.

Algorithm 1 Short-Path (unamplified version)

1. Prepare the wavefunction in the state $\psi_+ = |+\rangle^{\otimes N}$.
2. Use the measurement algorithm of section III to evolve under the Hamiltonian H_s from $s = 1$ to $s = 0$. where

$$H_s = H_Z - sB(X/N)^K, \quad (6)$$

where

$$X = \sum_i X_i. \quad (7)$$

where K is a positive odd integer and B is a scalar chosen later.

When we refer to a Hamiltonian H_s , it will be assumed that $s \in [0, 1]$ unless explicitly stated otherwise. We pick $B > 0$ so that H_s has all off-diagonal entries non-positive in the computational basis.

3. Measure the state in the computational basis and compute the value of H_Z after measuring. If this value is equal to E_0 then declare success and output the computational basis state.
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We will compute the squared overlap

$$P_{ov} \equiv |\langle \psi_+ | \psi_{0,1} \rangle|^2. \quad (8)$$

later. All state vectors that we write will be assumed to be normalized to have unit norm, except where explicitly stated otherwise later.

Suppose that the gap remains $\Omega(1/\text{poly}(N))$ along the path. Indeed, later we will give conditions under which the gap is $\Omega(1)$. Then, for any $P_{succ} < 1, \epsilon > 0$ we can use the measurement algorithm explained in section III to produce, with probability at least $P_{ov}P_{succ}$, a state equal to $\psi_{0,0}$ up to some error ϵ . This algorithm takes a time that is $\mathcal{O}(\text{poly}(N, (1 - P_{succ})^{-1}, \log(1/\epsilon)))$. This measurement algorithm has the advantage, compared to adiabatic evolution, that we can achieve a better scaling with the error ϵ .

Thus, the quantum algorithm described above succeeds with probability at least $P_{ov}P_{succ} - \epsilon$ in finding the ground state of H_Z . We take $\epsilon = 2^{-N}$ so that the error ϵ is negligible compared to P_{ov} computed later. We choose $P_{succ} = 1/2$. Hence, applying the method of amplitude amplification to the evolution, one obtains an algorithm which succeeds in producing the ground state of H_Z in expected time $\mathcal{O}(P_{ov}^{-1/2}P_{succ}^{-1/2}\text{poly}(N, \log(1/\epsilon)))$.

III. MEASUREMENT ALGORITHM FOR ADIABATIC EVOLUTION WITH SMALLER ERROR

We now explain how to do step 2. of algorithm I. We explain the method in a general setting and then show that in the particular case here, it can be reduced to a single measurement.

Suppose that we have a path of Hamiltonians H_s , for $s \in [0, 1]$. Suppose further that all of these Hamiltonians have a unique ground state $\psi_{0,s}$, separated from the rest of the spectrum by a gap at least $\Delta = \Omega(1/\text{poly}(N))$ and suppose that $\| \partial_s H_s \| = \mathcal{O}(\text{poly}(N))$. Suppose that for any s , we can simulate time evolution under H_s for time t up to error δ in a time that is $\mathcal{O}(\text{poly}(N, t, \log(1/\delta)))$. This holds for the Hamiltonians H_s considered above so long as $B = \mathcal{O}(\text{poly}(N))$ using any of several different algorithms in the literature which achieve this time [\[5, 7, 22, 23\]](#).

We seek a quantum algorithm that, taking a state ψ as input with overlap $P_{ov} = |\langle \psi | \psi_{0,1} \rangle|^2$, will succeed, with probability at least $P_{ov}P_{succ}$ for P_{succ} close to 1, in giving an output state which is equal to $\psi_{0,0}$ up to some error ϵ , with ϵ exponentially small. A natural way to do this is to follow adiabatic evolution of the Hamiltonians, i.e., to simulate a time-dependent Hamiltonian which slowly changes from H_1 to H_0 , taking ψ as input to the evolution and ψ as output. Unfortunately, this method has two problems, one minor and one major. We will explain these problems before giving a different approach using measurements that avoids these problems.

The minor problem is that we must then simulate time-dependent evolution of a Hamiltonian; this problem is not too serious as for example the Taylor series approach [\[7\]](#) can simulate this evolution in a time that is $\mathcal{O}(\text{poly}(N, t, \log(1/\delta)))$ even for slowly varying Hamiltonians. The major problem is that there may be diabatic transitions from the ground state $\psi_{0,s}$ to some excited state along the path. While there is some controversy about the exact error estimates in the adiabatic theorem [\[3, 29\]](#), even the best estimates give an error that is super-polynomially small but not exponentially small in the evolution time. Hence, even taking these estimates, in order to obtain an error δ that is exponentially small in N as we will need, we need a super-polynomial evolution time.

It is possible that a careful analysis of the error terms in the adiabatic algorithm would show that the required evolution time is not in fact too large. However, given some question about even simpler error estimates in the adiabatic theorem, we prefer not to use this method.

Instead we use the following algorithm [2]. Later we will show that for the specific H_s considered in this paper, it suffices to use the simpler algorithm [3] given later. We present algorithm [2] here because it also works for more general paths of Hamiltonians H_s , so long as a spectral gap is present and the adiabatic evolution can be simulated.

Algorithm 2 Measurement Algorithm

1. Let ψ be the input state.
 2. Phase estimate ψ using H_1 . If the energy estimate is larger than $E_{0,1} + \Delta/2$, then terminate the algorithm and return failure. Else continue.
 3. Adiabatically evolve ψ from H_1 to H_0 .
 4. Phase estimate ψ using H_1 . If the energy estimate is larger than $E_{0,1} + \Delta/2$, then terminate the algorithm and return failure. Else declare success and return ψ .
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One can also replace the adiabatic evolution of step 3. with a sequence of measurements as in Ref. [28]. This has the advantage that one does not need to simulate a time dependent evolution.

In algorithm [2], we use the initial phase estimation to approximately project onto $\psi_{0,1}$. Then we adiabatically evolve. Finally we repeat with another phase estimation to project onto $\psi_{0,0}$. We now consider the error in the phase estimation. We consider the phase estimation scheme of Ref. [19] though others are possible. There are two ways to quantify the error. One is the probability of error and the other is the precision. For us, it suffices to have a precision that is smaller than $\Delta/2$ so that one distinguishes the energy of the ground and first excited states. We take the probability of error in any step to be ϵ so that the total probability of error is $\mathcal{O}(\epsilon)$. If controlled time evolution can be implemented exactly, the phase estimation has an overhead that is only logarithmic in the error. For constant K , we can use any of the time evolution algorithms above or use [28] to implement the controlled time evolution with error $\mathcal{O}(\epsilon)$ for a time $t \sim \Delta^{-1}$ with a cost that is $\mathcal{O}(\text{poly}(\Delta^{-1}, N, \log(1/\epsilon)))$.

For $K = C \log(N)$, implementing the controlled time evolution is slightly more difficult. The Hamiltonian is no longer sparse when expressed in the computational basis. However, we can still use the approach of Ref. [23] or the related approach in Ref. [28] using a sequence of measurements to implement time evolution as follows. The term $-sB(X/N)^K$ can be expressed as a sum of terms, each of which is proportional to a projector onto an eigenspace of X , with eigenvalues $-N, -N+2, \dots, N$ so that there are $N+1$ such terms. We can measure each such term using polynomially many gates by applying a Hadamard to all qubits, use an adder to compute the total X , then uncomputing the addition and undoing the Hadamard. A similar approach might also work for the Taylor series approach [7]. Even without this way of decomposing $-sB(X/N)^K$, the overhead due to the non-sparsity of the Hamiltonian would be quasi-polynomial and would be negligible compared to the improvements in theorems [1, 2].

We choose the adiabatic evolution so that for input state $\psi = \psi_{0,1}$, the output state has squared overlap with $\psi_{0,0}$ equal to $1 - P_{diab}$ for some P_{diab} close to 0. From the adiabatic theorems quoted above, the adiabatic evolution time required is only polynomial in P_{diab}^{-1} . A similar estimate comes from using the repeated measurements of Ref. [28]. Hence,

Lemma 1. *For any ψ , and $P_{diab} > 0, \epsilon > 0$, the algorithm succeeds with probability at least $|\langle \psi_{0,1} | \psi \rangle|^2 (1 - P_{diab}) - \mathcal{O}(\epsilon)$. It takes a time $\mathcal{O}(\text{poly}(N, \log(1/\epsilon), P_{diab}^{-1}))$. Assuming success, the output state has squared overlap at least $1 - \epsilon$ with $\psi_{0,0}$.*

Choosing P_{diab} and ϵ small we obtain the desired P_{succ} close to 1.

In fact, for the Hamiltonian H_s from section II, we will show later that (under a spectral gap assumption and under some assumptions on B, K that we give later that)

$$|\langle \psi_{0,1} | \psi_{0,0} \rangle|^2 = \Omega(1). \quad (9)$$

Hence, for this problem, we can use a single measurement, phase estimating $H_{0,1}$, and not use any adiabatic evolution. There is no need for the final phase estimation on H_0 since this measurement in the computational basis projects into an eigenstate of H_Z whose energy can then be computed classically.

This leads to the following simple algorithm Algorithm [3] which replaces Algorithm [1].

As a technical remark, the above algorithm assumes that we know $E_{0,1}$. However, in the application that we consider, we do have good estimates on $E_{0,1}$, given later.

As a further remark, for even D many of the Hamiltonian simulation algorithms that we refer to can be chosen to preserve the even subspace exactly. Even if they do not preserve it exactly, they preserve it up to some error ϵ which is chosen to be negligible. Hence, if desired, one can do the simulation using H_s not projected into the even subspace.

Algorithm 3 Simplified Short-Path (unamplified version)

1. Let $\psi = \psi_+$ be the input state.
 2. Phase estimate ψ using Hamiltonian H_1 . If the energy estimate is greater than $E_{0,s} + \Delta/2$, then terminate the algorithm and return failure.
 3. Measure the state in the computational basis and compute the value of H_Z after measuring. If this value is equal to E_0 then declare success and output the computational basis state.
-

IV. BRILLOUIN-WIGNER PERTURBATION THEORY

We use Brillouin-Wigner perturbation theory to compute the ground state of H_s as a function of s . In the case that H_0 has a unique ground state (as we consider here), Brillouin-Wigner perturbation theory gives a particularly simple result for the ground state of H_s , given in Eq. (13). There are many references for Brillouin-Wigner perturbation theory; a useful reference is Ref. [21] which also gives some convergence results. We will however derive the needed results below.

For even D , we work in the even subspace throughout this section. We remark on the appropriate choice of basis states later.

A. Introduction

We begin with some general results on Brillouin-Wigner perturbation theory. The results in this subsection do not use any properties of the specific choice of H_s above, except that we assume that H_0 has a unique ground state (one great feature of this perturbation theory, however, is that it becomes only slightly more complicated when the ground state is degenerate, while Rayleigh-Schrodinger perturbation theory becomes much more complicated). We consider a Hamiltonian $H_s = H_0 + sV$ in this subsection with H_0, V arbitrary, later taking $V = -B(X/N)^K$.

Let $|0\rangle$ be the ground state of H_0 . Let $Q = 1 - |0\rangle\langle 0|$. Let $\phi_{0,s}$ be the ground state of H_s . To define the Brillouin-Wigner perturbation theory, it is convenient to normalize the states differently, rather than normalizing them to have unit norm. We will use the normalization that

$$\langle \phi_{0,s} | 0 \rangle = 1. \quad (10)$$

so that

$$\psi_{0,s} = \frac{\phi_{0,s}}{|\phi_{0,s}|}. \quad (11)$$

Let $E_{0,s}$ denote the energy of $\psi_{0,s}$ for Hamiltonian H_s . Let

$$G_0(\omega) = (Q(\omega - H_0)Q)^{-1}, \quad (12)$$

where ω is a scalar and where the inverse is computed in the subspace which is the range of Q and let $(1 - Q)G_0 = G_0(1 - Q) = 0$. That is, $G_0(\omega)$ is a Moore-Penrose pseudo-inverse of $Q(\omega - H_0)Q$, so that $G_0(\omega)(\omega - H_0) = (\omega - H_0)G_0(\omega) = Q$.

Then Brillouin-Wigner perturbation theory gives

$$\phi_{0,s} = |0\rangle + \sum_{k \geq 1} \left(s G_0(E_{0,s}) V \right)^k |0\rangle. \quad (13)$$

Note that by definition $G_0(\omega)V = G_0(\omega)QV$.

Note that $E_{0,s}$ appears in the power series of $\phi_{0,s}$. Thus, in applications of this perturbation theory to compute eigenvectors or eigenvalues, it is necessary to self-consistently compute $E_{0,s}$, using

$$E_{0,s} = E_0 + s \sum_{k \geq 0} \langle 0 | V \left(s G_0(E_{0,s}) V \right)^k | 0 \rangle \quad (14)$$

We will give bounds on $E_{0,s}$ later.

The correctness of Eq. (13) as a formal power series in s can be readily verified by computing $(H - E_{0,s})\phi_{0,s}$. In subsequent subsections, we will give conditions for convergence of this power series and we will compute $|\phi_{0,s}|$. Eq. (10) is immediate from Eq. (13).

B. Overlap

Now consider the specific choices of $H_0 = H_Z$ and $V = -B(X/N)^K$ using Eq. (6). We will compute the overlap $\langle \psi_+ | \phi_{0,1} \rangle$, assuming convergence of the series (13) and assuming a bound on $E_0 - E_{0,1}$. Note that $E_{0,1} < E_0$.

For odd D , let us use $|u\rangle, |v\rangle, \dots$ to denote basis states in the computational basis. For even D , we use $|u\rangle$ to denote the equal amplitude superposition of a pair of basis states in the computational basis, with the two basis states related by flipping all the spins. Each u labels a bit string of length N ; let $\bar{u}$ denote the bit string with all bits flipped. Then, $|u\rangle$ and $|\bar{u}\rangle$ denote the same basis state for even D .

Let $E_u, E_v, \dots$ denote the corresponding eigenvalues for these states for Hamiltonian H_0 . Then,

$$\begin{aligned} \langle \psi_+ | \phi_{0,1} \rangle &= \langle \psi_+ | 0 \rangle \\ &+ B \sum_{u \neq 0} \langle \psi_+ | u \rangle \frac{\langle u | (X/N)^K | 0 \rangle}{E_u - E_{0,1}} \\ &+ B^2 \sum_{u \neq 0} \sum_{v \neq 0} \langle \psi_+ | u \rangle \frac{\langle u | (X/N)^K | v \rangle}{E_u - E_{0,1}} \frac{\langle v | (X/N)^K | 0 \rangle}{E_v - E_{0,1}} \\ &+ \dots \end{aligned} \tag{15}$$

For any u , we have $\langle \psi_+ | u \rangle = 2^{-N/2}$.

Before continuing, we need two technical lemmas. First, the following lemma which estimates $\langle 0 | (X/N)^L | 0 \rangle$.

Lemma 2. For $0 < L < N$ even, $\langle 0 | (X/N)^L | 0 \rangle \leq L!!/N^L \leq (L/N)^{L/2}$ where $(L-1)!! = (L-1) \cdot (L-3) \cdot \dots$. For L odd, $\langle 0 | (X/N)^L | 0 \rangle = 0$.

For L, L' both even with $0 < L < L'$ we have $\langle 0 | (X/N)^L | 0 \rangle > \langle 0 | (X/N)^{L'} | 0 \rangle$, hence for $L > N/2$ we have $\langle 0 | (X/N)^L | 0 \rangle \leq 2^{-N/4}$.

Proof. We have $\langle 0 | (X/N)^L | 0 \rangle = N^{-L} \sum_{i_1, \dots, i_L} \langle 0 | X_{i_1} \dots X_{i_L} | 0 \rangle$.

First consider odd D . The expectation value vanishes unless for all j , there are an even number of a such that $i_a = j$. In that case, the expectation value is equal to 1. Thus, there must be some $a > 1$ such that $i_a = i_1$. There are N possible choices of i_1 and $L-1$ possible choices of $a > 1$. Hence, $\langle 0 | (X/N)^L | 0 \rangle \leq ((L-1)/N) \langle (X/N)^{L-2} | 0 \rangle$. So, $\langle 0 | (X/N)^L | 0 \rangle \leq L!!/N^L \leq (L/N)^{L/2}$, where $L!! = (L-1)(L-3) \dots$.

For even D , for $L \geq N$, there are additional terms in the expectation where for all j , there are an odd number of a such that $i_a = j$. However, since we have chosen $L < N$, such terms do not occur.

The monotonic decrease with increasing L is immediate when working in the X basis since $(X/N)^L > (X/N)^{L'}$. $\square$

Second we need the following inequality

Lemma 3. Let $x_1, \dots, x_k$ be positive random variables. The variables need not be independent of each other. Then $\mathbb{E}[\prod_{i=1}^k x_i^{-1}] \geq \prod_{i=1}^k (\mathbb{E}[x_i])^{-1}$.

Proof. We have $\mathbb{E}[\prod_{i=1}^k x_i^{-1}] = \mathbb{E}[\exp(-\sum_{i=1}^k \ln(x_i))] \geq \exp(-\mathbb{E}[\sum_{i=1}^k \ln(x_i)])$, where the inequality is by convexity of the exponential. However, $-\mathbb{E}[\ln(x_i)] \geq -\ln(\mathbb{E}[x_i])$, again by convexity. So, $\mathbb{E}[\prod_{i=1}^k x_i^{-1}] \geq \exp(-\sum_{i=1}^k \ln(\mathbb{E}[x_i])) = \prod_{i=1}^k (\mathbb{E}[x_i])^{-1}$. $\square$

We now show that

Lemma 4. Assume that $E_{0,1} \geq E_0 - 1$ and assume that series (15) is convergent. Let $B = \mathcal{O}(\text{poly}(N))$. Then,

$$\langle \psi_+ | \phi_{0,1} \rangle \geq 2^{-N/2} \left(\exp \left[\frac{BN}{(2DK + \mathcal{O}(1/N^3))|E_0|} \right] - o(1) \right) \frac{1}{\text{poly}(N)}. \tag{16}$$

Proof. Note that all terms in Eq. (15) are non-negative. We re-express the series in terms of a random walk on the basis states $|u\rangle$ as follows. The random walk starts in state $|0\rangle$ at time 0. If the random walk is in some state $|u_t\rangle$ at time t , then the state of the random walk at time $t+1$ is given by repeating K times the process of picking a random spin and flipping that spin. Note that we can flip the same spin more than once in a single step of the random walk (indeed, it may be flipped up to K times) although this is unlikely for $K \ll \sqrt{N}$. That is, each step of the random walk we consider here is K steps of a random walk on the Boolean hypercube.

For $t > 0$ we say that the random walk “returns at time t ” if the state of the random walk at time t is $|0\rangle$. For $t > 0$ we say that the random walk “returns by time t ” if the random walk returns at some time s with $0 < s \leq t$.

Let $P_{nr}(t)$ denote the probability that the random walk does not return by time t . Let $\mathbb{E}_{nr,t}$ denote an expectation value conditioned on the random walk not returning by time t . Then, we have

$$\langle \psi_+ | \phi_{0,1} \rangle = 2^{-N/2} \sum_{t=0}^{\infty} B^t \mathbb{E}_{nr,t} \left[\prod_{m=1}^t \frac{1}{E_{u_m} - E_{0,1}} \right] P_{nr,t}. \quad (17)$$

where the random walk has a sequence of states $u_1, \dots, u_t$.

We can estimate $P_{nr,t}$ from lemma 2. We have

$$P_{nr,t} \leq \sum_{0 < s \leq t} \langle 0 | (X/N)^{Ks} | 0 \rangle,$$

where $\langle 0 | (X/N)^{Ks} | 0 \rangle$ is the probability that it returns at time s . By lemma 2 for $t = \mathcal{O}(\text{poly}(N))$ and $K \geq 3$, we have $P_{nr,t} = \mathcal{O}(1/N^3)$.

Eq. (17) requires computing the expectation value of $\prod_{m=1}^t \frac{1}{E_{u_m} - E_{0,1}}$. Applying lemma 3 to this expectation value we have

$$\mathbb{E}_{nr,t} \left[\prod_{m=1}^t \frac{1}{E_{u_m} - E_{0,1}} \right] \geq \prod_{m=1}^t \frac{1}{\mathbb{E}_{nr,t}[E_{u_m} - E_{0,1}]}. \quad (18)$$

Suppose the state of the random walk at time t is given and has some energy E_{u_t} . Then, if we pick a single spin at random and flip it, the expectation value of the energy of the resulting state is equal to

$$\left(1 - 2\frac{D}{N}\right) E_{u_t}.$$

To see this, consider any term in H_0 which is a product of D spins; the probability that we flip one of these spins, changing the sign of this term, is D/N . Remark: this is the point at which we use that all terms in H_Z have the same degree D ; otherwise, the dependence of the average energy on m may be more complicated.

Repeating mK times, we find that the expectation value of the energy at time $m+1$, for given u_m , is equal to

$$\begin{aligned} \mathbb{E}[E_{u_m}] &= \left(1 - 2\frac{D}{N}\right)^{mK} E_0 \\ &\leq \left(1 - 2\frac{DmK}{N}\right) E_0. \end{aligned} \quad (19)$$

Since $E_{u_m} \geq E_0$ for all u_m , we have for $t = \mathcal{O}(\text{poly}(N))$,

$$\mathbb{E}[E_{u_m}] \geq (1 - P_{nr,t}) E_0 + P_{nr,t} \mathbb{E}_{nr,t}[E_{u_m}]. \quad (20)$$

So, for $t = \mathcal{O}(\text{poly}(N))$ where $P_{nr,t} = \mathcal{O}(1/N^3)$ we have

$$\mathbb{E}_{nr,t}[E_{u_m}] \leq \left(1 - 2\frac{DmK}{N} - \mathcal{O}(1/N^3)\right) E_0. \quad (21)$$

So, by Eqs. (18,21) we have

$$\begin{aligned} &\mathbb{E}_{nr,t} \left[\prod_{m=1}^t \frac{1}{E_{u_m} - E_{0,1}} \right] \\ &\geq \frac{1}{1 + (\frac{2DK}{N} + \mathcal{O}(1/N^3))|E_0|} \cdot \frac{1}{1 + (\frac{4DK}{N} + \mathcal{O}(1/N^3))|E_0|} \cdots \frac{1}{1 + (\frac{2tDK}{N} + \mathcal{O}(1/N^3))|E_0|} \\ &\geq \left(\frac{1}{(\frac{2DK}{N} + \mathcal{O}(1/N^3))|E_0|} \right)^t \cdot \frac{1}{t!} \cdot \frac{1}{1 + (\frac{2DK}{N} + \mathcal{O}(1/N^3))|E_0|} \cdot \frac{1}{1 + (\frac{2DK}{N} + \mathcal{O}(1/N^3))|E_0|} \cdots \frac{1}{1 + (\frac{2DK}{N} + \mathcal{O}(1/N^3))|E_0|} \\ &\geq \left(\frac{1}{(\frac{2DK}{N} + \mathcal{O}(1/N^3))|E_0|} \right)^t \cdot \frac{1}{t!} \cdot \exp\left(-\frac{\log(t) + 1}{1 + (\frac{2DK}{N} + \mathcal{O}(1/N^3))|E_0|} \right). \end{aligned} \quad (22)$$

We may assume that $|E_0| = \Omega(N)$ as otherwise the ground state is degenerate. So, for t only polynomially large, the quantity $\exp\left(-\frac{\log(t)+1}{1+\frac{1}{(\frac{2DK}{N}+\mathcal{O}(1/N^3))|E_0|}}\right)$ is only polynomially small; indeed for $|E_0| = \omega(N \log(N))$, this quantity is $\Omega(1)$. So, the sum in Eq. (15) obeys

$$\begin{aligned} \langle \psi_+ | \phi_{0,1} \rangle &\geq 2^{-N/2} \sum_{t=0}^{\mathcal{O}(\text{poly}(N))} B^t P_{nr,t} \left(\frac{N}{(2DK + \mathcal{O}(1/N^3))|E_0|} \right)^t \frac{1}{t! \text{poly}(N)} \\ &= 2^{-N/2} \left(\exp \left[\frac{BN}{(2DK + \mathcal{O}(1/N^3))|E_0|} \right] - o(1) \right) \frac{1}{\text{poly}(N)}. \end{aligned} \quad (23)$$

Here, we have used the fact that the power series expansion of $\exp(\alpha)$ is given by $\sum_{t \geq 0} \alpha^t / t!$ and the fact that $\sum_{t \geq s} \alpha^t / t!$ is negligible for $s \gg \alpha$. Indeed, $\sum_{t \geq s} \alpha^t / t!$ is exponentially small in α for fixed ratio s/α with $s/\alpha > e$. Hence, we can choose the polynomial in the limits on the first line of Eq. (23) to be large compared to $\frac{BN}{(2DK + \mathcal{O}(1/N^3))|E_0|}$ and then the sum of remaining terms (i.e., the terms in the series expansion of the exponential which are *not* included in the first line of Eq. (23)) is $o(1)$. $\square$

V. CONVERGENCE PROPERTIES, ENERGY SHIFT, AND NORM

In this subsection, we consider convergence of the series (15), we bound the shift on energy $E_0 - E_{0,1}$ and we bound the norm $|\phi_{0,1}|$, and we consider the gap of H_s . Most of the results will be based on considering the spectrum of QH_sQ . Again we take $H_0 = H_Z$ and $V = -B(X/N)^K$.

We first show the following three results. Eq. (27) below implies that Eq. (9) holds.

Lemma 5. 1. The series (15) always converges, assuming that the value of $E_{0,1}$ in the series indeed is equal to the ground state energy of H_1 .

2. Consider the Hamiltonian QH_sQ . Let $E_{0,s}^Q$ be the smallest eigenvalue of this Hamiltonian in the subspace spanned by the range of Q . Then, $\partial_s E_{0,s}^Q \leq 0$.

3. Finally, assume that $E_{0,1}^Q \geq E_0 + 1/2$. Then,

$$E_{0,1} \geq E_0 - B \langle 0 | (X/N)^K | 0 \rangle - 2B^2 \langle 0 | (X/N)^{2K} | 0 \rangle. \quad (24)$$

and

$$\begin{aligned} |\phi_{0,1}|^2 &\leq 1 + 4B^2 \left| Q(X/N)^K | 0 \rangle \right|^2 \\ &\leq 1 + 4B^2 \langle 0 | (X/N)^{2K} | 0 \rangle. \end{aligned} \quad (25)$$

For K odd with $\langle 0 | B^2 (X/N)^{2K} | 0 \rangle \leq 1/2$, we have that

$$E_{0,1} \geq E_0 - 1, \quad (26)$$

and

$$|\phi_{0,1}| \leq 2. \quad (27)$$

Proof. Defining

$$G_s(\omega) = \left(Q(\omega - H_s)Q \right)^{-1}, \quad (28)$$

where again the inverse is a Moore-Penrose inverse, the power series (13) is a series expansion of

$$\phi_{0,s} = |0\rangle + sG_s(E_{0,s})V|0\rangle \quad (29)$$

in powers of s . We now consider the singularities of $G_s(\omega)$ considered as a function of s . Then, the radius of convergence of the series (13) is equal to the distance from the origin $s = 0$ to the closest singularity (as a function of s) of $G_s(\omega)$ in the complex plane.

First, we will show that the singularities are simple poles at values of s where QH_sQ has an eigenvector in the range of Q with eigenvalue equal to ω . Let $g = (Q(\omega - H_0)Q)^{-1/2}$ and let $v = QVQ$, where the exponent $-1/2$ means a square-root of the Moore-Penrose inverse. For ω smaller than all eigenvalues of H_0 (which holds for our problem since $E_{0,1} < E_0$), this inverse is well-defined and the square-root can be taken pure imaginary so that g is anti-Hermitian. We choose the sign of the square-root arbitrarily. Then,

$$G_s(\omega) = g(1 - sgvg)^{-1}g. \quad (30)$$

To see Eq. (30), we have $Q(\omega - H_0 - sV)Qg(1 - sgvg)^{-1}g = Q(g^{-1} - sVg)(1 - sgvg)^{-1}g = Qg^{-1}(1 - sgVg)(1 - sgvg)^{-1}g = Q$, where all inverses are Moore-Penrose inverses. The matrix gvg is Hermitian and so can be diagonalized. Thus, the singularities of $G_s(\omega)$ are simple poles at values of s equal to eigenvalues of $(gvg)^{-1}$. Note that while elsewhere in the paper we assume that $s \in [0, 1]$, here we allow s to be an arbitrary complex number. Let ζ be an eigenvector of gvg with eigenvalue s^{-1} , so that $(1 - sgvg)\zeta = 0$. Hence, $0 = g(g^{-1} - sVg)\zeta = g(g^{-2} - sV)g\zeta = g(\omega - H_s)g\zeta$ so that $g\zeta$ is an eigenvector of QH_sQ with eigenvalue ω , as claimed.

The above results about singularities hold for arbitrary H_0, V, ω . Now to show convergence, we consider $H_0 = H_Z, V = -B(X/N)^K, \omega = E_{0,1} \leq E_0$. Suppose some matrix element of $G_s(E_{0,1})$ in the computational basis has radius of convergence r . Every matrix element of $G_s(E_{0,1})$ in the computational basis has a series expansion with all coefficients the same sign. So, by Pringsheim's theorem, there must be a singularity for $s = r$, i.e., there must be a singularity on the positive real axis. So, the series for $G_1(E_{0,1})$ converges unless $E_{0,t}^Q = E_{0,1}$ for some $t \in [0, 1]$.

Since all off-diagonal terms of QH_sQ and QVQ are negative, we have $\partial_s E_{0,s}^Q \leq 0$ by Perron-Frobenius, thus proving item 2. Hence, the series for $s = 1$ is convergent if $E_{0,1}^Q > E_{0,1}$. However, $E_{0,1}^Q > E_{0,1}$ also by Perron-Frobenius, thus proving item 1.

With this definition of G_s , we have

$$E_{0,s} = E_0 + s\langle 0|V|0\rangle + s^2\langle 0|VG_s(E_{0,s})V|0\rangle. \quad (31)$$

Let $H_0 = H_Z, V = -B(X/N)^K$. Assume that $E_{0,1}^Q \geq E_0 + 1/2$. Since $E_{0,1} \leq E_0$, $G_1(E_{0,1})$ has operator norm bounded by 2. So,

$$E_{0,1} \geq E_0 - B\langle 0|(X/N)^K|0\rangle - 2B^2\langle 0|(X/N)^{2K}|0\rangle. \quad (32)$$

Also, if $G_1(E_{0,1})$ has operator norm bounded by 2, we have

$$\begin{aligned} |\phi_{0,1}|^2 &\leq 1 + 4B^2\left|Q(X/N)^K|0\rangle\right|^2 \\ &\leq 1 + 4B^2\langle 0|(X/N)^{2K}|0\rangle. \end{aligned} \quad (33)$$

This proves Eqs. (24,25). Eqs. (26,27) are immediate. $\square$

We now consider the gap of H_s :

Lemma 6. *The Hamiltonian H_s has a gap between ground and first excited states that is greater than or equal to $E_{0,s}^Q - E_0$. Since $\partial_s E_{0,s}^Q \leq 0$, the gap of H_s is greater than or equal to $E_{0,1}^Q - E_0$.*

Proof. This is a special case of a general result. Consider a Hamiltonian

$$H = E_0|0\rangle\langle 0| + \sum_{a \neq 0} E_a|a\rangle\langle a| + \sum_{a \neq 0} v_a(|a\rangle\langle 0| + \text{h.c.}), \quad (34)$$

where v_a is an arbitrary vector. We will show that for any E_a, v_a, E_0 , the spectral gap of this Hamiltonian is greater than or equal to $\min_{a \neq 0}(E_a - E_0)$. Then, to apply this result to the spectral gap of H_s , we take the states $|a\rangle$ in Eq. (34) to be eigenstates of QH_sQ and take v_a to be matrix elements of sV between $|0\rangle$ and those eigenstates.

Define the Green's function $G(\omega) = (\omega - H)^{-1}$. We have

$$\langle 0|G|0\rangle = \left(\omega - E_0 - \Sigma(\omega)\right)^{-1}, \quad (35)$$

where

$$\Sigma(\omega) = \sum_{a \neq 0} |v_a|^2 (\omega - E_a)^{-1}. \quad (36)$$

For $\omega < \min_{a \neq 0} E_a$, we have $\Sigma(\omega) < 0$. Hence, $\langle 0|G|0 \rangle$ does not have any poles in the interval $E_0 < \omega < \min_{a \neq 0} E_a$. Hence, if H has an eigenvalue in this interval, then the corresponding eigenvector has vanishing amplitude on $|0\rangle$; however, any such eigenvector has eigenvalue equal to E_a for some a , so no such eigenvector exists.

Thus, all eigenvalues of H , for any v are contained in $(-\infty, E_0] \cup [\min_a(E_a), \infty)$. If $v_a = 0$ for all a , there is exactly one eigenvalue in the interval $(-\infty, E_0]$ and so this cannot change as v_a changes. $\square$

Hence it follows that:

Theorem 3. *Consider the Hamiltonian QH_sQ . Let $E_{0,s}^Q$ be the smallest eigenvalue of this Hamiltonian in the subspace spanned by the range of Q . Assume that $E_{0,1}^Q \geq E_0 + 1/2$ and assume that $\langle 0|B^2(X/N)^{2K}|0 \rangle \leq 1/2$. Then*

1. *The Hamiltonian H_s has gap at least $1/2$ between the ground and first excited state.*
2. *We have*

$$\langle \psi_+ | \psi_{0,1} \rangle \geq \frac{1}{2} 2^{-N/2} \left(\exp \left[\frac{BN}{(2DK + \mathcal{O}(1/N^3))|E_0|} \right] - o(1) \right) \frac{1}{\text{poly}(N)}. \quad (37)$$

Remark: for $B = -bE_0$, $J_{\text{tot}} = \mathcal{O}(N^\beta)$, $K \geq \beta$, the condition $\langle 0|B^2(X/N)^{2K}|0 \rangle \leq 1/2$ holds for all sufficiently large N .

VI. GAP ASSUMPTION AND ENTROPY

In this section we assume that $E_{0,1}^Q < E_0 + 1/2$ and prove some consequences of that. In subsection [VIA](#) we show how to construct states with large expectation value for X . In subsection [VIB](#), we show entropic consequences of this using a log-Sobolev inequality. In subsection [VIC](#), we put these results together.

For technical convenience later, it is easier to work with an eigenvector of H_1 . So we show:

Lemma 7. *Assume $E_{0,1}^Q < E_0 + 1/2$. Then, there is an eigenvector Ψ of H_1 with eigenvalue at most $E_0 + 1/2$ such that $\langle \Psi | B(X/N)^K | \Psi \rangle \geq 1/4$.*

Proof. H_1 has at least two eigenvalues with energy at most $E_0 + 1/2$. This is because the ground state of QH_1Q in the subspace spanned by Q and the state $|0\rangle$ span a two dimensional space. Further, the average of energy over the corresponding eigenvectors is at most $E_0 + 1/4$ since the ground state energy of H_1 is $\leq E_0$. On the other hand, the average of H_Z over these eigenvectors is at least $E_0 + 1/2$. Hence, the average of $B(X/N)^K$ over these eigenvectors is at least $1/4$. So, at least one such Ψ exists. $\square$

Remark: the above lemma is the only place in this section in which the degeneracy of H_Z is used. We used the degeneracy to show that the average of H_Z over these eigenvectors is at least $E_0 + 1/2$. However, it is not hard to remove the degeneracy assumption here, at the cost of slightly worse constants. We sketch this in the discussion.

A. Localizing in X

In this section, we show how given an eigenvector with a large expectation value for $B(X/N)^K$ we can construct a wavefunction with a large expectation value for X and whose expectation value for H_Z is only slightly changed. This will be necessary to apply log-Sobolev bounds in the next section.

We begin with a lemma that considers states of a Hamiltonian describing a single particle hopping in one-dimension:

Lemma 8. *Let h be a real Hermitian tridiagonal matrix. Label rows and columns of h by an integer x , so that $h_{x,y} = 0$ if $|x - y| > 1$. Let ψ be a real eigenvector of h with eigenvalue E . Let $\psi(x)$ denote the x -th entry of ψ . Let h_{od} denote the off-diagonal part of h , i.e, h_{od} has the same entries as h off the main diagonal and is zero on the main diagonal. Then,*

1. *For any integer $\ell > 0$, for any y , there is a state ξ with $|\xi| = 1$ such that*

$$\langle \xi | h | \xi \rangle \leq E + \mathcal{O}(1/\ell^2) \|h_{od}\|, \quad (38)$$

and such that $\xi(x)$ is non-vanishing only for $x < y + \ell$ or $\xi(x)$ is non-vanishing for $x > y - \ell$.

2. For small $\epsilon > 0$, the state ψ_ϵ defined by $\psi_\epsilon(x) = \exp(\epsilon x)\psi(x)$ obeys

$$\frac{\langle \psi_\epsilon | h | \psi_\epsilon \rangle}{|\psi_\epsilon|^2} \leq E + \mathcal{O}(\epsilon^2) \|h_{od}\|. \quad (39)$$

3. If the state ψ_ϵ defined above has

$$\sum_{x \geq z} \psi_\epsilon(x)^2 \geq \frac{1}{2} \sum_x \psi_\epsilon(x)^2 \quad (40)$$

for some z , then there is a state ξ with $|\xi| = 1$ such that

$$\langle \xi | h | \xi \rangle \leq E + \mathcal{O}(1/\ell^2 + \epsilon^2) \|h_{od}\| \quad (41)$$

and such that either $\xi(x)$ is non-vanishing only for $y - \ell < x < y + \ell$ for some $y \geq z$.

Proof. Assume without loss of generality that $E = 0$ (otherwise add a scalar to h so that $E = 0$). Assume without loss of generality that $\psi(x)$ is real. Let $f(x)$ be any function. Let $\hat{f}$ be the diagonal matrix with entries given by $f(\cdot)$. We compute

$$\begin{aligned} & \langle \hat{f}\psi | h | \hat{f}\psi \rangle \quad (42) \\ &= \sum_x \left(f(x)^2 h_{x,x} \psi(x)^2 + f(x)f(x+1)h_{x,x+1}\psi(x)\psi(x+1) + f(x)f(x-1)h_{x,x-1}\psi(x)\psi(x-1) \right) \\ &= \sum_x f(x)^2 \psi(x) \left(h_{x,x}\psi(x) + h_{x,x-1}\psi(x-1) + h_{x,x+1}\psi(x+1) \right) \\ &\quad + \sum_x \left[f(x)(f(x+1) - f(x))h_{x,x+1}\psi(x)\psi(x+1) + f(x)(f(x-1) - f(x))h_{x,x-1}\psi(x)\psi(x-1) \right] \\ &= \sum_x \left[f(x)(f(x+1) - f(x))h_{x,x+1}\psi(x)\psi(x+1) + f(x)(f(x-1) - f(x))h_{x,x-1}\psi(x)\psi(x-1) \right] \\ &= \sum_x \left[f(x)(f(x+1) - f(x))h_{x,x+1}\psi(x)\psi(x+1) + f(x+1)(f(x) - f(x+1))h_{x+1,x}\psi(x+1)\psi(x) \right] \\ &= - \sum_x (f(x) - f(x+1))^2 h_{x,x+1}\psi(x)\psi(x+1), \end{aligned}$$

where the third equality follows from the fact that $h_{x,x}\psi(x) + h_{x,x-1}\psi(x-1) + h_{x,x+1}\psi(x+1) = 0$ since ψ is an eigenvector with eigenvalue 0 and the fourth equality follows by shifting the variable in the summation by 1.

To prove the first claim, let $\rho^< = \sum_{y-\ell < x \leq y} \psi(x)^2$ and let $\rho^> = \sum_{y \leq x < y+\ell} \psi(x)^2$. Assume $\rho^< > \rho^>$. Choosing $f(x) = \ell$ for $x \leq y$ and $f(x) = \ell - (x - y)$ for $y < x < y + \ell$ and $f(x) = 0$ for $x \geq y + \ell$, we find that $\langle \hat{f}\psi | h | \hat{f}\psi \rangle \leq \|h_{od}\| \rho^>$ and $|\hat{f}\psi|^2 \geq \ell^2 \rho^<$, so that for $\xi = \hat{f}\psi / |\hat{f}\psi|$, Eq. (38) is satisfied and $\xi(x)$ is nonvanishing only for $x < y + \ell$. If instead $\rho^< \leq \rho^>$, choose instead $f(x) = \ell$ for $x \geq y$ and $f(x) = \ell - |x - y|$ for $y - \ell < x < y$ and $f(x) = 0$ for $x \leq y - \ell$ and for $\xi = \hat{f}\psi / |\hat{f}\psi|$, Eq. (38) is satisfied and $\xi(x)$ is nonvanishing only for $x > y - \ell$.

To prove the second claim, choose $f(x) = \exp(\epsilon x)$, so that we have $\langle \hat{f}\psi | h | \hat{f}\psi \rangle = - \sum_x \psi(x)\psi(x+1)h_{x,x+1}\mathcal{O}(\epsilon^2)\exp(2\epsilon x) \leq \mathcal{O}(\epsilon^2)\|h_{od}\| \cdot |\hat{f}\psi|^2$.

To prove the third claim, define $\rho_y = \sum_{|x-y| \leq \ell/2} \exp(2\epsilon x)\psi(x)^2$. If the assumption (40) holds, then there must be some $y > z$ such that $\rho_y \geq (1/2)\rho_{y-\ell}$; to see this, suppose no such y exists, then $\rho_{z+\ell/2} < (1/2)\rho(z - \ell/2)$ and $\rho(z + 3\ell/2) < (1/4)\rho(z + \ell/2)$ and so on, so $\sum_{x \geq z} \psi_\epsilon(x)^2 < \sum_{x < z} \psi_\epsilon(x)^2$.

So, there must be some $y > z$ such that $\rho_y \geq (1/2)\rho_{y-\ell}$ and such that $\rho_y \geq (1/2)\rho_{y+\ell}$. To see this, find the largest $y > z$ such that $\rho_y \geq (1/2)\rho_{y-\ell}$; by the above paragraph at least one such y must exist. This y must have $\rho_y \geq (1/2)\rho_{y+\ell}$ as otherwise it would not be the largest.

Choose $f(x) = \exp(\epsilon x)\ell$ for $|y - x| \leq \ell$, choose $f(x) = \exp(\epsilon x)(2\ell - |y - x|)$ for $\ell \leq |y - x| \leq 2\ell$ and choose $f(x) = 0$ otherwise. $\square$

The above lemma applies to a one-dimensional Hamiltonian. However, we can apply it to an eigenvector Ψ of the Hamiltonian H_1 as follows. Let P_x project onto the eigenspace of X with eigenvalues in the interval $[xD, xD + D)$. That is, defining $q(y) = \lfloor y/D \rfloor$, then P_x projects onto the eigenspace of X with eigenvalues y such that $q(y) = x$.

Then, the Hamiltonian H_1 , projected into the space spanned by $P_x\Psi/|P_x\Psi|$ obeys the conditions of lemma 8 with $\|h_{od}\| \leq \|H_Z\|$ and with $\psi(x) = \sqrt{\langle P_x\Psi|\Psi\rangle}$.

We now show how to attain a wavefunction with a large expectation value for X . We do this in two slightly different ways depending on how large K is. The lemma will involve the constant $C > 0$; the constants hidden inside the big-O notation do *not* depend on C .

Lemma 9. Assume that $E_{0,1}^Q < E_0 + 1/2$.

1. Assume that $K = C \log(N)$. Let $B = -bE_0$ with $b \leq 1$. Then, there is a state Ξ , with $|\Xi| = 1$ such that

$$\langle \Xi | X/N | \Xi \rangle \geq 1 - \mathcal{O}(1)/C \quad (43)$$

and such that

$$\langle \Xi | H_Z | \Xi \rangle \leq E_0 + 1/2 + \mathcal{O}(1) \cdot \frac{J_{tot}}{N^2} C^2 D^2 \log(N)^2 + B. \quad (44)$$

2. Consider arbitrary K . Let $B = -bE_0$ with $b \leq 1$. Let $X_{min} = N \cdot (4B)^{-1/K}$. (Remark: for $X < X_{min}$ we have $B(X/N)^K \leq 1/4$.)

Then, there is a state Ξ with $|\Xi| = 1$ such that Ξ is supported on an eigenspace of X with eigenvalues in some interval $[X_0 - X_{min}/K, X_0 + X_{min}/K]$ for $X_0 \geq X_{min}$ and such that

$$\langle \Xi | H_Z - B(X/N)^K | \Xi \rangle \leq E_0 + 1/2 + \mathcal{O}(1) \frac{J_{tot} K^2 D^2}{X_{min}^2}, \quad (45)$$

so that

$$\begin{aligned} \langle \Xi | H_Z | \Xi \rangle &\leq E_0 + 1/2 + \mathcal{O}(1) \frac{J_{tot} K^2 D^2}{X_{min}^2} + B \left(\frac{X_0 + X_{min}/K}{N} \right)^K \\ &\leq E_0 + 1/2 + \mathcal{O}(1) \frac{J_{tot} K^2 D^2}{X_{min}^2} + eB(X_0/N)^K. \end{aligned} \quad (46)$$

Proof. The proof of both cases is the same. By lemma 7, there is an eigenstate Ψ of H_1 with $\langle \Psi | H_1 | \Psi \rangle \leq E_0 + 1/2$ and $\langle \Psi | B(X/N)^K | \Psi \rangle \geq 1/4$. Throughout, the state ψ will be an eigenvector of h constructed from Ψ as explained above by projecting into the space spanned by $P_x\Psi/|P_x\Psi|$.

Let $X_{min} = N \cdot (4B)^{-1/K}$. For $X < X_{min}$ we have $B(X/N)^K < 1/4$.

Recall that K is chosen odd. So $\exp(K(X - X_{min})/X_{min}) \geq 4B(X/N)^K$, and so

$$\exp(KX/X_{min}) \geq 4 \exp(K) B(X/N)^K. \quad (47)$$

Hence, since $\langle \Psi | B(X/N)^K | \Psi \rangle \geq 1/4$, we have

$$\langle \Psi | \exp(KX/X_{min}) | \Psi \rangle \geq \exp(K). \quad (48)$$

Recall that $q(y) = \lfloor y/D \rfloor$. Construct the state ψ_ϵ of lemma 8 using $\epsilon = KD/2X_{min}$. Then,

$$|\psi_\epsilon|^2 \geq \langle \Psi | \exp(KDq(X)/X_{min}) | \Psi \rangle \geq \exp(-2\epsilon) \langle \Psi | \exp(KX/X_{min}) | \Psi \rangle \geq \exp(-2\epsilon) \exp(K),$$

using Eq. (48). At the same time, $\sum_{x < q(X_{min} - \ln(2)X_{min}/K)} |\psi_\epsilon(x)|^2 \leq (1/2) \exp(K - 2\epsilon)$. So, the state ψ_ϵ obeys Eq. (40) for $z = q(X_{min} - \ln(2)X_{min}/K)$. Then construct state ξ using item 3 of lemma 8 choosing $\ell = \text{const.} \times \epsilon^{-1}$, using $\|h_{od}\| \leq J_{tot}$.

From state ξ , construct state $\Xi = \sum_x \xi(x) P_x\Psi/|P_x\Psi|$. We have chosen $\ell = \text{const.} \times \epsilon^{-1}$; by choosing this constant smaller than $(1 - \ln(2))$, Ξ will be supported on the given eigenspace of X . This shows Eq. (45). Eq. (44) follows because in this case $X_{min} = N \cdot (1 - \mathcal{O}(1)/C)$. $\square$

B. Log-Sobolev Inequality

Given a quantum state ψ , let $S^{(comp)}(\psi)$ be the entropy of the probability distribution of measurement outcomes when measuring the state in the computational basis. That is, if $\psi = \sum_u \psi(u)|u\rangle$, with $\sum_u |\psi(u)|^2 = 1$, then

$$S^{(comp)}(\psi) = - \sum_u |\psi(u)|^2 \log(|\psi(u)|^2). \quad (49)$$

Here we measure entropy using bits, i.e., taking logs to base 2, rather than nats. This will lead to some various $\ln(2)$ differences between our definitions and definitions in the log-Sobolev literature.

In this subsection, we relate the entropy $S^{(comp)}(\psi)$ to $\langle \psi | X | \psi \rangle$. It is clear that if $\langle \psi | X | \psi \rangle = N$, then $\psi = \psi_+$ up to an overall phase, and so $S^{(comp)}(\psi) = N$. Roughly, we will show that if $\langle \psi | X | \psi \rangle$ is extensive (i.e., equal to N times some nonzero constant), then $S^{(comp)}(\psi)$ is also extensive.

One such result uses the log-Sobolev inequality. This lemma [10](#) is not tight: it only gives a nontrivial bound on $S^{(comp)}(\psi)$ if $\langle \psi | X | \psi \rangle > (1 - \ln(2))N$. In lemma [11](#) we give a tight bound and give a precise statement of that rough extensivity claim above.

Lemma 10. *Let $\psi(u)$ be real. Then,*

$$S^{(comp)}(\psi) \geq \left(1 - \frac{1}{\ln(2)}\right)N + \frac{1}{\ln(2)} \langle \psi | X | \psi \rangle. \quad (50)$$

Proof. This follows from the log-Sobolev inequality [13,24](#). Define for any function $f(u)$,

$$\text{Ent}(f) = \mathbb{E}[f \log(\frac{f}{\mathbb{E}[f]})], \quad (51)$$

where the expectation is taken for a random choice of u in the domain of f . We have $\mathbb{E}[\psi^2] = 2^{-N} \sum_u \psi(u)^2 = 2^{-N}$. Here we abuse notation to use ψ to represent both a quantum state and a function $\psi(u)$. So,

$$\begin{aligned} \text{Ent}(\psi^2) &= 2^{-N} \sum_u \psi(u)^2 \left(N + \log(\psi(u)^2) \right) \\ &= 2^{-N} \left(N - S^{(comp)}(\psi) \right). \end{aligned} \quad (52)$$

The log-Sobolev inequality states that

$$\text{Ent}(\psi^2) \leq 2^{-N} \frac{N - \langle \psi | X | \psi \rangle}{\ln(2)}. \quad (53)$$

Hence Eq. [50](#) follows.

Remark: the log-Sobolev literature usually expresses the inequality in terms of gradients of a function on the Boolean hypercube; however, since this gradient is defined in terms of bit flips and X induces a single bit flip, after a little algebra one may see that Eq. [53](#) is equivalent to the results in the literature. $\square$

A tighter bound on $S^{(comp)}$ follows from the log-Sobolev inequality of Ref. [30](#). We have

Lemma 11. *Let $S(x) = -x \log(x) - (1-x) \log(1-x)$ be the binary entropy function (we use S rather than H to avoid confusion with the Hamiltonian H). Let*

$$\tau(\sigma) = 2 \sqrt{S^{-1}(\sigma) \left(1 - S^{-1}(\sigma)\right)}. \quad (54)$$

(The inverse of S may be chosen arbitrarily so long as the same inverse is chosen in both locations.) Then,

$$\tau\left(\frac{S^{(comp)}(\psi)}{N}\right) \geq \frac{\langle \psi | X | \psi \rangle}{N}. \quad (55)$$

The function $\tau(\cdot)$ is a continuous increasing function, taking $[0, 1]$ to $[0, 1]$. For small σ ,

$$\tau(\sigma) = \Theta\left(\sqrt{\frac{\sigma}{-\log(\sigma)}}\right), \quad (56)$$

and for small $\langle \psi | X | \psi \rangle / N$,

$$\frac{S^{(comp)}(\psi)}{N} = \Theta((X/N)^2 \log(N/X)). \quad (57)$$

Proof. This follows from theorem 1.2 of Ref. [30]. That theorem is expressed in terms of $\text{Ent}(\psi^2)$, up to multiplicative factors of $\ln(2)$ since natural logs are used in that paper. Use Eq. [52] to re-express this in terms of $S^{(comp)}(\psi)$ and then Eq. [55] follows after some algebraic manipulations. $\square$

The function $\tau(\cdot)$ has a simple interpretation. Consider the single qubit state

$$\sqrt{S^{-1}(\sigma)}|0\rangle + \sqrt{1 - S^{-1}(\sigma)}|1\rangle.$$

A product state of N qubits, all in this single qubit state, has entropy $N\sigma$ and expectation value of X equal to $2N\sqrt{S^{-1}(\sigma)(1 - S^{-1}(\sigma))}$. The result is that this product state minimizes entropy for the given expectation value of X .

Remark: in fact, while we have only considered the case of pure states, it is possible to show a generalization of Eq. [55] to mixed states. We do not need this generalization here, but it is interesting to note that it exists. Let ρ be a density matrix. Let $S^{(comp)}(\rho)$ be the entropy of the mixed state obtained by measuring ρ in the computational basis. Then, we have

$$\tau\left(\frac{S^{(comp)}(\rho)}{N}\right) \geq \frac{\text{tr}(X\rho)}{N}. \quad (58)$$

This can be proven similarly to the proof of the pure state case: one establishes it for a single qubit, and then one uses convexity of c and conditional entropy to show it for an arbitrary number of qubits.

C. Number of Low Energy Eigenstates

Theorem 4. Assume that $E_{0,1}^Q < E_0 + 1/2$.

1. Assume that $K = C \log(N)$. Let $B = -bE_0$ with $b \leq 1$. Let $C = \Theta(1)$. Then, there is some probability distribution $p(u)$ on computational basis states with entropy at least

$$S^{(comp)} \geq N \cdot (1 - \mathcal{O}(1)/C)$$

and with expected value of H_Z at most $(1 - b)E_0 + \mathcal{O}(1) \cdot \frac{J_{tot}}{N^2} C^2 D^2 \log(N)^2$.

2. Consider arbitrary K . Let $B = -bE_0$ with $b \leq 1$. Then, for some $X_0 \geq X_{min} = N \cdot (4B)^{-1/K}$, there is some probability distribution $p(u)$ on computational basis states with entropy at least

$$S^{(comp)} \geq N\tau^{-1}((X_0 - X_{min}/K)/N)$$

and with expected value of H_Z at most

$$E_0 + \mathcal{O}(1) \frac{J_{tot} K^2 D^2}{X_{min}^2} + B \left(\frac{X_0 + X_{min}/K}{N} \right)^K \cdot \mathcal{O}(1),$$

where the function $\tau(\cdot)$ is defined in lemma [11]. If $X_0 \ll N$, then

$$2^{N\tau^{-1}(X_0/N)} = 2^{\Omega((X_0/N)^2 \log(N/X_0))N}$$

Proof. This follows from lemmas [9][10][11]. We have dropped the additive $+1/2$ from lemma [9] as it is smaller than other terms hidden in the big-O notation. $\square$

VII. PROOF OF THEOREMS 1, 2

We now prove the main theorem. Either $E_{0,1}^Q \geq E_0 + 1/2$, in which case theorem 3 applies or $E_{0,1}^Q < E_0 + 1/2$ in which case theorem 4 applies. As noted, for $B = -bE_0$, for $K \geq 3$ and $K > \beta$, the condition $B(X/N)^K \leq 1/2$ in theorem 3 holds for all sufficiently large N .

Theorem 4 shows the existence of a probability distribution with large entropy and small expectation value for H_Z . To turn this into a statement about $W(E)$ as in theorems 1, 2 we need the following lemma. Remark: the factor of $\mathcal{O}(\log(N))$ in the statement of the lemma can be interpreted, for physicists, as an entropy difference arising when passing from a canonical to a microcanonical ensemble, and the need for two energies E_1, E_2 can be interpreted as a Maxwell construction.

Lemma 12. *Let $p(u)$ be a probability distribution over basis states $|u\rangle$ such that*

$$\sum_u p(u) E_u = \bar{E}. \quad (59)$$

Then, there are two energies, E_1, E_2 with $E_1 \leq \bar{E} \leq E_2$ such that for some probability $P \in [0, 1]$ we have $PE_1 + (1 - P)E_2 = \bar{E}$ and $P \log(W(E_1)) + (1 - P) \log(W(E_2)) \geq S - \mathcal{O}(\log(N))$, where $W(E)$ is the number of computational basis states with expectation value E for H_Z .

Proof. Let

$$P(E) = \sum_{u: E_u = E} p(u). \quad (60)$$

If $P(E) > 0$, then for u such that $E_u = E$, let $p(u|E) = p(u)/P(E)$ and let

$$S(E) = - \sum_{u: E_u = E} p(u|E) \log(p(u|E)). \quad (61)$$

Then,

$$S = - \sum_E P(E) \log(P(E)) + \sum_E P(E) S(E). \quad (62)$$

Since there are only $\mathcal{O}(J_{tot})$ possible values of E for which $P(E) \neq 0$, we have $-\sum_E P(E) \log(P(E)) = \mathcal{O}(\log(N))$. Hence, $\sum_E P(E) S(E) \geq S - \mathcal{O}(\log(N))$. Also, $\sum_E P(E) E = \bar{E}$. We have $S(E) \leq \log(W(E))$. Hence, $\sum_E P(E) \log(W(E)) \geq S - \mathcal{O}(\log(N))$.

Now maximize $\sum_E P(E) \log(W(E))$ subject to the linear constraints $\sum_E P(E) E = \bar{E}$ and $\sum_E P(E) = 1$, with $0 \leq P(E) \leq 1$ for all E . Introducing Lagrange multipliers λ_1, λ_2 corresponding to these constraints, one finds that for all E we have either $P(E) = 0$ or $P(E) = 1$ (and hence only one choice of E has $P(E) > 0$) or $\log(W(E)) = \lambda_1 E + \lambda_2$. Let T be the set of E such that $\log(W(E)) = \lambda_1 E + \lambda_2$. Since this constraint is linear, for any choice of $P(E)$ such that $P(E) = 0$ for $E \notin T$ and $\sum_E P(E) = 1$ and $\sum_E P(E) E = \bar{E}$ we have the same $\sum_E P(E) \log(W(E))$. Hence, to maximize $\sum_E P(E) \log(W(E))$ it suffices to consider the case that $P(E)$ is nonvanishing for at most 2 choices of E . Let these choices be E_1, E_2 with $E_1 < E_2$ and $E_1 \leq \bar{E} \leq E_2$. $\square$

Remark: In the lemma above we used the fact that H_Z has integer eigenvalues to bound the entropy $-\sum_E P(E) \log(P(E))$. For more general H_Z , where J_{ij} are chosen more generally, we could bin energies into polynomially many bins and obtain a similar result.

Using this lemma, now consider the first case of theorem 4, where $K = C \log(N)$. Theorem 4 shows the existence of a probability distribution $p(u)$ on computational basis states with entropy at least $S = S^{(comp)} \geq N \cdot (1 - \mathcal{O}(1)/C)$ and with expected value of H_Z at most $\bar{E} = (1 - b)E_0 + 1/2 + \mathcal{O}(1) \cdot \frac{J_{tot}}{N^2} C^2 D^2 \log(N)^2$. By lemma 12, there are two energies, E_1, E_2 with $E_1 \leq \bar{E} \leq E_2$ such that for some probability $P \in [0, 1]$ we have $PE_1 + (1 - P)E_2 = \bar{E}$ and $P \log(W(E_1)) + (1 - P) \log(W(E_2)) \geq S - \mathcal{O}(\log(N))$. If $\log(W(E_1)) \geq S - \mathcal{O}(\log(N))$, the conclusion follows. Otherwise, we must have $\log(W(E_2)) \geq S - \mathcal{O}(\log(N))$. If $E_2 - E_0 \geq (1 + \eta)(\bar{E} - E_0)$, then $P_2 \leq 1/(1 + \eta)$ since $E_1 - E_0 \geq 0$. Since $\log(W(E_2)) \leq N$, we have $P_1 \log(W(E_1)) + P_2 N \geq S - \mathcal{O}(\log(N))$ so $\log(W(E_1)) \geq N \cdot (1 - \mathcal{O}(1) \frac{1+\eta}{\eta} \frac{1}{C}) - \frac{1+\eta}{\eta} \mathcal{O}(\log(N))$.

This completes the proof of theorem 1.

Now consider the second case of theorem 4, where K is a constant. The theorem shows that for some $X_0 \geq X_{min} = N \cdot (4B)^{-1/K}$, there is some probability distribution $p(u)$ on computational basis states with entropy at

least $S = S^{(comp)} \geq N\tau^{-1}((X_0 - X_{min}/K)/N)$ and with expected value of H_Z at most $\bar{E} = E_0 + \mathcal{O}(1)\frac{J_{tot}K^2D^2}{X_{min}^2} + B(\frac{X_0+X_{min}/K}{N})^K \cdot \mathcal{O}(1)$. Shifting X_0 by X_{min}/K , we find that for some $X_0 \geq (1 - 1/K)N \cdot (4B)^{-1/K}$, there is some probability distribution $p(u)$ on computational basis states with entropy at least $S = S^{(comp)} \geq N\tau^{-1}(X_0/N)$ and with expected value of H_Z at most $\bar{E} = E_0 + \mathcal{O}(1)\frac{J_{tot}K^2D^2}{X_{min}^2} + B(\frac{X_0+2X_{min}/K}{N})^K \cdot \mathcal{O}(1)$. However, for the given range of X_0 , we have $(\frac{X_0+2X_{min}/K}{N})^K \leq (X_0/N)^K \mathcal{O}(1)$, so $\bar{E} \leq E_0 + \mathcal{O}(1)\frac{J_{tot}K^2D^2}{X_{min}^2} + B(X_0/N)^K \cdot \mathcal{O}(1)$. Now removing the restriction on the range of X_0 allowing $0 \leq X_0 \leq N$ and defining $m_x = X_0/N$ we find that for some $m_x \in [0, 1]$ there is a probability distribution with entropy $S \geq N\tau^{-1}(m_x)$ and with expected value of H_Z at most

$$\begin{aligned} & E_0 + \mathcal{O}(1)\frac{J_{tot}K^2D^2}{X_{min}^2} + B(m_x)^K \cdot \mathcal{O}(1) \\ & \leq E_0 + \mathcal{O}(1)\frac{J_{tot}K^2D^2}{X_{min}^2} + B(\tau(S/N))^K \cdot \mathcal{O}(1) \\ & \equiv F(S). \end{aligned} \tag{63}$$

By lemma 12, there are two energies, E_1, E_2 with $E_1 \leq \bar{E} \leq E_2$ such that for some probability $P \in [0, 1]$ we have $PE_1 + (1 - P)E_2 = \bar{E} \leq F(S)$ and $P \log(W(E_1)) + (1 - P) \log(W(E_2)) = S - \mathcal{O}(\log(N))$. The function $F(S)$ is a convex function, so either $E_1 \leq F(\log(W(E_1)) + \mathcal{O}(\log(N)))$ or $E_2 \leq F(\log(W(E_2)) + \mathcal{O}(\log(N)))$. This completes the proof of theorem 2.

VIII. DISCUSSION

We have given a quantum algorithm for exact optimization. This algorithm leads to a nontrivial speedup, assuming a bound on the number of computational basis states with low energy for H_Z . This naturally leads to a hybrid algorithm: if the bound is obeyed, we run the exact algorithm here while if the bound is not obeyed, one can find a low energy state (i.e., an approximate solution of the optimization problem) by repeated random sampling or by Grover search.

We have only considered problems where all terms in H_Z have the same degree. Eq. (19) is the point at which we used this assumption. For problems such as MAX-3-SAT, the energy dependence may be more complicated so it is not clear how well this algorithm works there. It would be interesting to study this using numerical simulation. Also we emphasize that the mere presence of a large number of low energy states of H_Z is not, on its own, sufficient to make the algorithm fail; rather, one needs to have a low energy eigenstate of QH_1Q , which requires some structure to how those states are arranged on the hypercube. That is, if a large number of low energy states are distributed in some fairly random way on the hypercube it is possible that one might not have a small $E_{0,1}^Q$. Numerical simulation may also be useful here.

Lemma 7 is the only place in section VI where we used the assumption on the degeneracy of H_Z . However, it is not hard to remove this assumption here. Assume $E_{0,1}^Q < E_0 + 1/2$. Then consider the Hamiltonian $H_Z - (5/2)B(X/N)^K$. This must have an eigenvalue $\leq E_0 - 1/4$. Hence there must be some eigenvector Ψ of $H_Z - (5/2)B(X/N)^K$ with $\langle \Psi | B(X/N)^K | \Psi \rangle \geq 1/4$. Then one could use this Ψ in the rest of section VI, the cost is that in some cases factors of B in this section get replaced by $(5/2)B$, in particular in theorem 4. One can equivalently understand this as follows: one instead assumes that $E_{0,2/5}^Q < E_0 + 1/2$. If this holds, then $E_{0,1}^Q < E_0 - 1/4$. If this assumption fails, then section 3 implies a nontrivial speedup for the algorithm using Hamiltonian $H_Z - (2/5)B(X/N)^K$. In fact, there is nothing special about our assumption that $E_{0,1}^Q < E_0 + 1/2$. The analysis of section V would work under the assumption that $E_{0,1}^Q \geq E_0 + c$ for any constant $c > 0$. Further, there is nothing special about our assumption in the lemma that $\langle \Psi | B(X/N)^K | \Psi \rangle \geq 1/4$; the analysis above would go through up to some changes in constants if we assumed that $\langle \Psi | B(X/N)^K | \Psi \rangle \geq c'$ for any $c' > 0$. So, it would suffice to consider $H_Z - c''B(X/N)^K$ for $c'' = (1 + c')/(1 - c)$ which can be made arbitrarily close to 1. In future work, we will address the degeneracy assumption in section V.

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Appendix A: Reduction to Unique Ground State

We now corollaries how to reduce the problem without the degeneracy assumption to the problem with the degeneracy assumption, at an additional time cost.

For given H_Z , consider a more general family of Hamiltonians, of the form $H_Z + \sum_i h_i Z_i$, where the h_i are chosen from $\{-1, 0, +1\}$. We will define a sequence of such Hamiltonians, $H^{(0)}, H^{(1)}, \dots, H^{(m)}$. We will set $H^{(0)} = H_Z$ so that for $H^{(0)}$ all $h_i = 0$. Then, each $H^{(a+1)}$ is constructed from $H^{(a)}$ by picking some i such that $h_i = 0$ in the Hamiltonian $H^{(a)}$ and setting h_i either equal to $+1$ or -1 in $H^{(a+1)}$, while leaving all other terms in the Hamiltonian unchanged.

Before continuing we need the following lemma:

Lemma 13. *Consider any arbitrary set S with $S \subseteq \{-1, +1\}^N$ and $|S| > 1$. Write elements of S as vectors $v = (v_1, \dots, v_N)$ with $v_i \in \{-1, +1\}$. Then, there exists some $i \in \{1, \dots, N\}$ and some $\sigma \in \{-1, +1\}$ such that for*

$$T = S \cap \{v | v_i = \sigma\}$$

we have $1 \leq |T| \leq |S|/2$. In words, the set T is the set of vectors in S such that the i -th entry of the vector is equal to σ .

Proof. For each i , let $n_i = |S \cap \{v | v_i = +1\}|$. If for some i we have $0 < n_i < |S|$, then either $1 \leq n_i \leq |S|/2$ in which case we can pick $T = S \cap \{v | v_i = +1\}$ or $|S|/2 \leq n_i \leq |S| - 1$ in which case we can pick $T = S \cap \{v | v_i = -1\}$. On the other hand, if for all i we have $n_i \in \{0, |S|\}$, then $|S| \leq 1$ which contradicts the assumptions of the lemma (proof: without loss of generality assume that $n_i = |S|$ for all i ; then, only the all $+1$ vector can be in S). $\square$

Let $H^{(a)}$ have $n_{gs}(a)$ different ground states. Write $n_{gs} = n_{gs}(0)$. We now show

Lemma 14. *For some $m \leq \log(n_{gs})$, we can choose a sequence $H^{(0)}, \dots, H^{(m)}$ such that $1 \leq n_{gs}(a+1) \leq n_{gs}(a)/2$ for all $0 \leq i \leq a-1$ with $n_{gs}(m) = 1$.*

Proof. The proof is inductive. Let $S^{(a)}$ be the set of ground states of $H^{(a)}$. We will construct each $H^{(a)}$ such that if it has $h_i \neq 0$ for any i , then all ground states of $H^{(a)}$ have $v_i = -h_i$. Then, apply lemma 13 with $S = S^{(a)}$. Find the i, σ so that for $T = S \cap \{v | v_i = \sigma\}$ we have $1 \leq |T| \leq |S|/2$. By the inductive assumption $h_i = 0$ in Hamiltonian $H^{(a)}$ as if $h_i \neq 0$ then $T = \emptyset$ or $T = S^{(a)}$. Then set $H^{(a+1)} = H^{(a)} - \sigma Z_i$. Since the added term $-\sigma Z_i$ is equal to its minimal possible value of -1 on some non-empty subset of the ground states of $H^{(a)}$, those states are the ground states of $H^{(a+1)}$ and the inductive assumption holds. Continue in this fashion, until for some a we have $n_{gs}(a) = 1$. $\square$

Hence, if H_Z has n_{gs} ground state, then there exists some Hamiltonian $H_Z + \sum_i h_i Z_i$ with the following properties. First, it has a unique ground state which is also a ground state of H_Z . Second, the number of nonzero h_i is at most $\log(n_{gs})$. That is, writing $\vec{h}$ as a vector with entries h_i , the vector $\vec{h}$ has $\log(n_{gs})$ nonzero entries.

There are

$$2^1 \binom{N}{1} + 2^2 \binom{N}{2} + \dots + 2^{n_{gs}} \binom{N}{\log(n_{gs})} \leq (2N)^{\log(n_{gs})}$$

possible choices of $\vec{h}$.

The Hamiltonians of the form $H_Z + \sum_i h_i Z_i$ do not have all terms of the same degree, as some terms have degree D and some have degree 1. However, for any such Hamiltonian $H_Z + \sum_i h_i Z_i$ with a unique ground state we can define a Hamiltonian H' on $N + D + 1$ qubits which obeys the degeneracy assumption (i.e., has a unique ground state for odd D or a doubly degenerate state for even D) so that the ground state of $H_Z + \sum_i h_i Z_i$ can be trivially obtained from a ground state of H' . To do this, add $D + 1$ qubits, labelled $N + 1, \dots, N + D + 2$. Write J equal to the sum of all possible D -th order monomials in Pauli Z operators on those added qubits, with a coefficient -1 in front of each monomial. For example, for $D = 3$, we have $J = -Z_{N+1}Z_{N+2}Z_{N+3} - Z_{N+1}Z_{N+2}Z_{N+4} - Z_{N+1}Z_{N+3}Z_{N+4} - Z_{N+2}Z_{N+3}Z_{N+4}$. Consider the Hamiltonian

$$H_Z + J + \sum_{i \leq N} h_i Z_i Z_{N+1} Z_{N+2} \dots Z_{N+D-1}.$$

The term J is minimized for even D , by all Z_i being the same for $i > N$, while for odd D is it minimized by all Z_i equaling $+1$ for $i > N$. The terms $H_Z + \sum_{i \leq N} h_i Z_i Z_{N+1} Z_{N+2} \dots Z_{N+D-1}$ can be minimized by choosing all $Z_i = +1$ and choosing the unique ground state of $H_Z + \sum_{i \leq N} h_i Z_i$ found above. Thus $H_Z + J + \sum_{i \leq N} h_i Z_i Z_{N+1} Z_{N+2} \dots Z_{N+D-1}$ obeys the degeneracy assumption and finding its ground state directly gives a ground state of H_Z .

Thus, we can reduce to the unique ground state problem at the cost of trying $(2N)^{\log(n_{gs})}$ different choices of $\vec{h}$. This number is small if n_{gs} is small, while if n_{gs} is large, one can find the ground state more rapidly by Grover search. Considering the algorithm with logarithmically growing K , note that if $n_{gs} \geq 2^{rN/\log(N)^2}$ for some scalar r then one can find the ground state using Grover search in expected time $\mathcal{O}^*(2^{N/2 - rN/\log(N)^2/2})$, while if $n_{gs} \leq 2^{rN/\log(N)^2}$ then there are only $(2N)^{rN/\log(N)^2} = 2^{rN/\log(N) + \mathcal{O}(N/\log(N)^2)}$ choices of $\vec{h}$. For $r < b/(2CD)$, theorem [1](#) shows a speedup greater than $2^{rN/\log(N)}$ and so one still has a $2^{\text{const.} \times N/\log(N)}$ speedup in this case, so in worst case one has a $2^{\text{const.} \times N/\log(N)^2}$ speedup.

Mind the gap: Achieving a super-Grover quantum speedup by jumping to the end

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Abstract

We present a quantum algorithm that has rigorous runtime guarantees for several families of binary optimization problems, including Quadratic Unconstrained Binary Optimization (QUBO), Ising spin glasses (p -spin model), and k -local constraint satisfaction problems (k -CSP). We show that either (a) the algorithm finds the optimal solution in time $O^*(2^{(0.5-c)n})$ for an n -independent constant c , a 2^{cn} advantage over Grover's algorithm; or (b) there are sufficiently many low-cost solutions such that classical random guessing produces a $(1 - \eta)$ approximation to the optimal cost value in sub-exponential time for arbitrarily small choice of η . Additionally, we show that for a large fraction of random instances from the k -spin model, and for any sufficiently close-to-regular, fully satisfiable (or slightly frustrated) k -CSP formula, statement (a) is the case. The algorithm and its analysis are largely inspired by Hastings' short-path algorithm [*Quantum* **2** (2018) 78].

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1 Overview

1.1 Motivation and the search for super-Grover speedups

It is hoped that quantum computers will outperform classical computers at solving combinatorial optimization problems, both in theory and, eventually, in practice. A primary motivation for this expectation is the existence of Grover’s quantum algorithm [1] for unstructured search: since classical algorithms for hard optimization problems often resort to some form of brute-force search, Grover’s algorithm potentially offers a quadratic speedup without exploiting any particular structure in the problem.

However, the hope to leverage Grover’s algorithm for a practical advantage over state-of-the-art classical algorithms faces two pitfalls. First, it is rare that an exhaustive search through all possible solutions is the best classical algorithm for a specific combinatorial optimization problem. For example, the most well-known constraint satisfaction problem, 3-SAT¹, admits a $2^{0.39n}$ -time classical algorithm for instances with n binary variables [2, 3], a nearly cubic speedup over a simple exhaustive search. These algorithms are still exponential time, and they typically retain elements of exhaustive search. However, they exploit structure to reduce the search space and to search more efficiently. Second, practical implementation of quantum algorithms with asymptotic quadratic speedup on actual quantum devices will suffer constant-factor slowdowns when compared to state-of-the-art classical hardware due to slower clock speeds, error-correction overheads, and general lack of parallelizability. When these factors are considered, realistic assessments of resources needed for quantum advantage using a quadratic speedup are pessimistic, suggesting that the breakeven point where quantum overtakes classical would occur only after many days, or in some cases, many years of runtime [4–6].

The outlook for practical advantage dramatically improves as the power of the polynomial speedup becomes greater than quadratic. Ref. [6] found that an algorithm with quartic speedup (i.e. the quantum runtime scales as $T^{1/4}$ where T is the classical runtime) would offer a much more viable path to actual quantum advantage. This motivates the question of whether the quadratic Grover speedup can be surpassed for combinatorial optimization.

In some cases, a super-quadratic speedup over exhaustive search can be realized by combining the quantum techniques of amplitude amplification [7] or quantum walk search [8–10]—both of which can be understood as generalizing Grover’s algorithm—with classical techniques for exploiting problem structure such as backtracking [11–14], branch-and-bound [15], nested search [16], dynamic programming [17], and Markov chain Monte Carlo methods [18–20]. Establishing that these classical techniques can be employed while retaining the quadratic quantum speedup is typically non-trivial; however, these ingredients alone can only *restore* the quadratic speedup compared to classical state-of-the-art methods, not surpass it, since the fundamentally quantum part of the algorithm generates only a Grover-like speedup.

Progress toward proving a genuine *super-Grover* speedup for combinatorial optimization is limited. The quantum adiabatic algorithm (QAA) [21, 22] is one example of an algorithm that could conceivably generate super-Grover speedups for some problems. However, its runtime is notoriously hard to study due to dependence on the minimal spectral gap of certain non-commuting Hamiltonians. Additionally, some prior work has argued that the QAA can in some cases exhibit runtime that scales super-exponentially—even worse than exhaustive search—when applied to combinatorial optimization problems [23, 24], although for random instances it is expected to

¹An instance of the 3-SAT problem is the question of whether a Boolean formula in conjunctive normal form (CNF)—i.e. where each constraint is the conjunction of at most three of the n binary variables—has a solution that satisfies all constraints. Elsewhere in the paper, we refer to this problem as 3-CNF-SAT.

scale merely exponentially [25–27]. Many other works (see, e.g., Ref. [28]), often with a focus on near-term algorithms, aim only to solve combinatorial optimization problems *approximately* rather than exactly (and typically in only polynomial time, rather than exponential time); these algorithms have the potential to be practically useful, but rigorous guarantees of quantum advantage are difficult to establish.

In this work, we focus on *exact* binary optimization problems where the goal is to find the assignment z^* that minimizes some efficiently computable, classical cost function $H(z)$, with $z \in \{+1, -1\}^n$. We desire not only the best solution z^* , but also a high degree of confidence that no better solution exists; thus, we generally expect that any algorithm that solves this problem, quantum or classical, requires exponential runtime. For exponential-time algorithms, we follow the convention of writing $O^*(2^{an})$ to mean that there is an upper bound on the runtime of the form $h(n)2^{an}$ that holds for sufficiently large n , where h is a polynomial function. Exhaustive enumeration runs in (classical) time $O^*(2^n)$. Meanwhile, Grover’s algorithm runs in (quantum) time $O^*(2^{0.5n})$. We seek a “super-Grover speedup,” that is, a quantum algorithm that runs in time $O^*(2^{(0.5-c)n})$ for some n -independent constant c , where the additional advantage over Grover’s algorithm is generated by a fundamentally quantum mechanism.

Obtaining speedups over Grover’s algorithm was also the goal of a series of works by Hastings [29–31] on an algorithm called the *short-path algorithm*. Whereas Grover’s algorithm begins in the equal superposition state $|\oplus\rangle \equiv 2^{-n/2} \sum_j |j\rangle$ and measures the classical function $H(z)$ whilst performing amplitude amplification to boost the probability of obtaining an optimal-cost measurement outcome, the short-path algorithm instead uses quantum phase estimation to measure $H + V$, where in this context H denotes the diagonal operator corresponding to the cost function and V is an off-diagonal non-commuting perturbation term. Here again, amplitude amplification is used to boost the probability of ending in the ground state of $H + V$. After the ground state is prepared, a computational basis measurement is performed to find the optimal solution. It was shown that for certain choice of V and under certain assumptions on the spectral density of H at low energies (i.e. low cost values), a super-Grover speedup could be accomplished. However, no concrete scenario was given where the spectral density assumption was proved to hold. In follow-up work [30], Hastings showed that for a specific family of cost functions with 2-local Ising-like terms, the spectral density assumption could be dropped, but in this case the super-Grover speedup constant c was not n -independent, decaying like $1/\log(n)$ or faster (depending on the amount of frustration in the cost function).

1.2 A simple quantum algorithm for exact combinatorial optimization

Our main contribution is a simple algorithm inspired by Hastings’ short-path algorithm [29] for which we can prove a super-Grover speedup for a few specific families of cost functions, including Ising spin glasses and k -local constraint satisfaction problems. The algorithm can be run on other families of cost functions and may very well have a super-Grover speedup more generally, but we present rigorous guarantees only for these specific cases.

The algorithm can be understood as an implementation of the QAA for combinatorial optimization problems with two crucial modifications, as we now explain. For simplicity, suppose here that z^* is *unique*, i.e. $H(z^*) < H(z)$ for all $z \neq z^*$, and suppose that the optimal value $E^* = H(z^*)$ is known ahead of time (these assumptions will be dropped later). By convention, we assume that H is offset such that the average cost of a uniformly random input is zero, and thus $E^* < 0$. Let $X = \sum_{i=1}^n X_i$ be the transverse-field operator, where X_i denotes the Pauli- X operator on qubit i .

Then, the QAA evolves by the Hamiltonian

$$H_b = -\frac{X}{n} + b \frac{H}{|E^*|} \quad (\text{for the QAA}) \quad (1)$$

while the parameter b is slowly tuned from $b = 0$, where the ground state of H_b is $|+\rangle \equiv |+\rangle^{\otimes n}$, to $b = \infty$, where the ground state of H_b is $|z^*\rangle$. The first modification is that $\frac{H}{|E^*|}$ is replaced by $g(\frac{H}{|E^*|})$ for a certain piecewise linear function $g : [-1, \infty) \rightarrow [-1, 0]$:

$$H_b = -\frac{X}{n} + b g\left(\frac{H}{|E^*|}\right) \quad (\text{for our algorithm}) \quad (2)$$

This modification allows us to control the spectral properties of the cost function to enable proof of our claims. The second modification is, rather than evolve continuously through values of b for which the spectral gap of H_b is small and unknown, the algorithm simply *jumps*, first from $b = 0$ to an n -independent value of $b > 0$ where the gap is guaranteed to be large, and then from that value of b all the way to the end of the algorithm ($b = \infty$). These jumps are accomplished with quantum phase estimation along with amplitude amplification to boost the success probability of projecting onto the ground state of H_b . The first jump is small in the sense that the success probability is nearly one (and little or no amplitude amplification is required). The second jump is large in the sense that the success probability is exponentially small. We will be able to show that (for some specific families of cost functions) the success probability of the second jump is larger than $2^{-(1-2c)n}$, and hence, after amplitude amplification, the runtime of the algorithm is less than $O^*(2^{(0.5-c)n})$, where c is a constant independent of n .

Both modifications described above, as well as the analysis of the algorithm, are inspired by (and in some cases closely follow) Hastings' short-path algorithm [29]. However, the algorithms are not identical, and in some sense they are dual to each other: where our algorithm makes a small jump and then a large jump, the short-path algorithm instead makes a large jump and then a small jump. Moreover, where our algorithm makes the modification $H/|E^*| \rightarrow g(H/|E^*|)$ for a piecewise linear function g (see Eqs. (1) and (2)), the short-path algorithm makes the modification $-X/n \rightarrow g(-X/n)$ for the function $g(x) = x^K$, with K an odd integer.

By taking our approach, we can prove more concrete results than was possible to show for the original short-path algorithm. This is true for subtle technical reasons. Firstly, as in Ref. [29], our proof utilizes the log-Sobolev inequality, which relates the expectation value of the X/n operator for a given state $|\phi\rangle$ to the entropy of the distribution over measurement outcomes that arises when $|\phi\rangle$ is measured in the computational basis. This step goes through more cleanly for us because the X/n operator appears directly in our Hamiltonian (see Eq. (2)), whereas $(X/n)^K$ appeared in Hastings' Hamiltonian, requiring some work to argue that the expectation of $(X/n)^K$ could be approximated by the K th power of the expectation of X/n . Secondly, by switching from Hastings' strategy of a large-then-small jump to our strategy of a small-then-large jump, (roughly speaking) Hastings' spectral density assumption switches from a statement about the number of states at low energy to a statement about the number of states at high energy, which is much easier to show using straightforward tail bounds. This switch does come at a cost: we lose Hastings' perturbation theory argument that proves the exponential advantage over Grover, assuming the spectral density assumption. We are able to replace this part of the argument in our case with a proof that utilizes an approximate ground state projector.

1.3 Overview of provable statements

Consider the following families of optimization problems over inputs in the set $\{+1, -1\}^n$.

- **MAX-Ek-LIN2:** $H(z) = p(z_1, \dots, z_n)$, where p is a polynomial consisting only of monomials of degree k .
- **Quadratic Unconstrained Binary Optimization (QUBO):** $H(z) = p(z_1, \dots, z_n)$, where p is a polynomial consisting only of monomials of degree-1 or degree-2.
- **MAX- k -CSP with limited frustration:** $H(z) = \sum_{j=1}^m \mathcal{C}_j$ where each $\mathcal{C}_j$ is a k -constraint, defined by the criteria that (i) it is a function of at most k of the n bits of the input z , and (ii) it takes the value -1 on $s_j \in [1, 2^k - 1]$ (“satisfying”) assignments to those bits and value $s_j/(2^k - s_j)$ on the other $2^k - s_j$ (“unsatisfying”) assignments, such that the average across all 2^k assignments is zero. Let E^* be the optimal value of H . The magnitude of the super-Grover speedup will depend on $|E^*|/m$ and is maximal when $|E^*|/m = 1$, i.e. when the instance is frustration free (all constraints are simultaneously satisfiable). The speedup guarantee also depends on the parameter $D = nk^{-2}m^{-2} \sum_{j=1}^n d_j^2$, where d_j is the number of constraints in which the bit z_j participates.² The parameter D measures, in a sense, how far from regular the interaction hypergraph of the CSP instance is. That is, D achieves its minimum of 1 when each bit participates in exactly mk/n constraints.

We also study a random ensemble of MAX-Ek-LIN2 instances known as the “ p -spin” model [32], which here we refer to as the “ k -spin” model so that the symbol k consistently represents the locality of the terms of H throughout the paper. An instance of the k -spin model is given by

$$H(z_1, \dots, z_n) = \sqrt{\frac{k!}{n^{k-1}}} \sum_{1 \leq i_1 < \dots < i_k \leq n} J_{i_1, \dots, i_k} z_{i_1} z_{i_2} \dots z_{i_k}, \quad (3)$$

where the weights $J_{i_1, \dots, i_k}$ are each chosen independently at random from a standard Gaussian distribution with mean 0 and variance 1. The k -spin model is used in physics to model spin glasses. When $k = 2$ it is identical to the Sherrington-Kirkpatrick (SK) model [33].

Before we state our main results, we define our usage of big- O , big- Ω notation, which appears in various places throughout the paper. Let h be a function of n , k , and $\frac{m}{|E^*|}$. Then a quantity is said to be $O(h(\cdot))$ (resp. $\Omega(h(\cdot))$) if it is upper bounded (resp. lower bounded) by some constant times $h(\cdot)$. We typically think of k as a constant as n grows, but we include the k -dependence in our expressions (rather than absorbing it into big- O) to communicate how the size of the speedup depends on k .

Our main results are the following:

- For each instance of MAX-Ek-LIN2 and QUBO, either there is a quantum algorithm with super-Grover speedup, or there is a classical algorithm that can achieve an arbitrarily good approximation ratio in sub-exponential time. More precisely, for any $\eta \in [0, 1]$ and any $\gamma \in [0, 1]$, either (a) the quantum algorithm has runtime $2^{0.5(1-\Omega(\gamma\eta/k))n}$, or (b) the classical algorithm that repeatedly samples assignments uniformly at random produces a bit string y with cost $H(y) \leq (1 - \eta)E^*$ within time $O^*(2^{\gamma n})$ (with high probability). Thus, if there is no n -independent choice of η, γ for which (a) holds, then for arbitrarily small η , the runtime

²In the [first version](#) of this paper and in the version published in the [STOC’23 proceedings](#), the proof of Theorem 7 attempted to argue that solving MAX- k -CSP for arbitrary instances can essentially be reduced to solving instances with bounded value of $D \leq O(1)$. This argument allowed us to state our main results for MAX- k -CSP without mentioning D . We thank François Le Gall for pointing out an error in this argument. This version fixes the error by reporting the D -dependence of the algorithm when it is applied to MAX- k -CSP. The results for MAX-Ek-LIN2 and QUBO are not affected.

of the classical algorithm is better than $O^*(2^{\gamma n})$ for arbitrarily small γ (i.e. sub-exponential time).

- For the k -spin model, there is a choice of $\eta = \Omega(1)$ and $\gamma = \Omega(1/k^2)$ such that case (a) is satisfied in the previous bullet for all but at most a $e^{-\Omega(n/k)}$ fraction of instances.
- For instances of MAX- k -CSP where $|E^*|/m$ is independent of n and irregularity parameter $D = O(1)$, the quantum algorithm has a super-Grover speedup. More precisely, there exists a constant $c = \Omega\left(\frac{|E^*|^3}{k^3 2^{3k} m^3 D}\right)$ such that MAX- k -CSP admits a quantum algorithm with runtime $O^*(2^{(0.5-c)n})$. The algorithm can thus solve the question of whether or not a k -CSP instance is fully satisfiable in $O^*(2^{(0.5-\Omega(1/k^3 2^{3k} D))n})$ time. When $|E^*|/m$ is independent of n but D is growing with n , we may instead make a statement similar to the first bullet—either a super-Grover speedup is available (i.e., c is lower bounded by an n -independent quantity), or else a classical algorithm can find an arbitrarily good approximate solution in sub-exponential time.

These results are stated in more formal language later in Sec. 3.4. Note that for random MAX- k -CNF-SAT, the random ensemble of MAX- k -CSP instances where each clause is the conjunction of k distinct randomly chosen variables (or their negations), we have that $D = O(1)$ for typical instances, and meanwhile the amount of frustration increases with the number of clauses. If $m \leq \alpha_c n$ for some critical value α_c (which grows like 2^k), then most instances are fully satisfiable [34], which means that $|E^*|/m = 1$. If $m = \alpha n$ for $\alpha > \alpha_c$, then most instances are not fully satisfiable (“frustrated”), but for typical instances the ratio $|E^*|/m$ is larger than some n -independent number [35]. Thus, the super-Grover speedup persists despite the frustration, although the constant c indicating the size of the 2^{cn} advantage over Grover is reduced compared to the frustration-free case. If $m/n \rightarrow \infty$ as $n \rightarrow \infty$, then $|E^*|/m \rightarrow 0$ for typical instances in that limit (in particular, it is expected to decay as $\sqrt{n/m}$) [36], and our method fails to prove a super-Grover speedup.

1.4 Comparison to classical algorithms and significance

In Tab. 1, we present several problems for which we can make concrete statements about the runtime of our algorithm, and we compare with the corresponding best known classical runtime. While our algorithm has a super-Grover speedup for the 3-CNF-SAT problem (i.e. the question of whether there is a fully satisfying assignment for a 3-CSP instance where each clause is in conjunctive normal form) on instances where $D = O(1)$, it does not have a super-quadratic speedup—in fact, it has no speedup at all—owing to the fact that there are also classical algorithms with a significant speedup over exhaustive enumeration. At large k , our algorithm does give a speedup for k -CNF-SAT on instances when $D = O(1)$,³ but the speedup is sub-quadratic. However, it is important to note that these classical algorithms for k -CNF-SAT are the product of decades of incremental improvements: for $k = 3$, Monien and Speckenmeyer found a $O^*(2^{0.70n})$ -time algorithm in 1985 [37] and the coefficient was subsequently reduced to 0.57, 0.45, 0.42, and 0.39 in Refs. [38], [39], [40], and [2], respectively, with further infinitesimal improvements in Refs. [41] and [3]. For the SK model, our algorithm also fails to deliver a speedup due to the existence of a classical branch-and-bound algorithm [15] which gives a significant provable advantage over exhaustive search. It is worth noting that the classical algorithms for k -CNF-SAT in Ref. [3] and for the SK model in Ref. [15] each admit a quadratic quantum speedup by applying amplitude amplification or quantum

³Note that the classical algorithm being compared against works for any D .

walk search techniques. The resulting quantum algorithm is the best-known quantum algorithm in both cases.

Problem	Our quantum algo	Best classical algo
3-CNF-SAT	$0.5 - (1.7 \times 10^{-6})D^{-1}$	0.39 [3]
k -CNF-SAT	$0.5 - \Omega(2^{-3k}k^{-3}D^{-1})$	$1 - \Omega(k^{-1})$ [3]
SK model	$0.5 - (2.7 \times 10^{-5})$	0.45 [15]
k -spin	$0.5 - \Omega(k^{-3})$	1

Table 1: Summary of concrete combinatorial optimization problems where we can prove an upper bound on the runtime of our algorithm, in comparison to the best known classical algorithm for the same problem. The number displayed is the coefficient of n in the exponential, i.e. if the algorithm runs in time $O^*(2^{an})$ then a appears in the table. The problem k -CNF-SAT refers to the question of whether or not a Boolean formula in conjunctive normal form with k -local constraints is fully satisfiable. The k -spin model refers to the random ensemble of MAX-E k -LIN2 instances where every k -local term appears with a random Gaussian weight, defined in Eq. (3). The Sherrington-Kirkpatrick (SK) model corresponds to the k -spin model with $k = 2$. The irregularity parameter D takes a value $D = 1$ when the interaction hypergraph is regular, and $D \leq 3$ holds for typical randomly chosen k -CNF-SAT instances.

For the k -spin model with $k \geq 3$, the branch-and-bound technique of Ref. [15] does not obviously generalize, and we do not know of a classical algorithm that has been proved to run in time $O^*(2^{(1-c)n})$ for an n -independent value c . In contrast, our algorithm runs in time $O^*(2^{(0.5-\Omega(1/k^3))n})$, a potential super-quadratic speedup. However, we believe that it is plausible that there does exist a $O^*(2^{(1-c)n})$ -time classical algorithm for k -spin: a potential candidate is classical Metropolis sampling at high temperature. Evidence that this algorithm would be effective comes from prior work on the *spherical* k -spin model, a continuous variable analogue of the “Ising” k -spin model we study here. In particular, Ref. [42] showed that for the spherical k -spin model, the Langevin dynamics are rapidly mixing when the temperature is above some n -independent threshold, allowing efficient classical sampling from the Gibbs distribution. Since the k -spin model is normalized such that typical instances have an extensive optimal cost value $|E^*| = \Omega(n)$, the Boltzmann factor $e^{-\beta E^*}$ for the optimal assignment z^* at constant inverse temperature β is exponentially large in n . This fact suggests that there exists an n -independent choice of c for which only $2^{(1-c)n}$ Gibbs samples need to be drawn to find z^* with high probability. It would be interesting to extend these results from the spherical to the Ising k -spin model and formally verify that it leads to a speedup over exhaustive enumeration.

In all cases, it is apparent that the provable advantage of our algorithm over $O^*(2^{0.5n})$ is very small. Our goal has been to prove that there exists *some* constant improvement over Grover’s algorithm, and we have not dedicated much effort to optimizing the proofs to maximize the constant. We are certain that the size of the provable speedup could be improved, and furthermore in Sec. 5, we give numerical evidence that the speedup over Grover is much more substantial than the proofs imply.

We emphasize that a key reason this speedup is interesting despite its small numerical size is that the speedup mechanism has no immediate classical analogue with comparable runtime guarantees upon which our algorithm simply applies a general technique like amplitude amplification. We connect the speedup mechanism to an observation about 1-norm vs. 2-norm localization, a feature shared by Hastings’ short-path algorithm [30]. Namely, it is possible for a wavefunction to be localized on a single basis state when using the 2-norm, yet de-localized across many basis states

when using the 1-norm, a situation with no classical analogue. See Sec. 4.5 for a more detailed discussion. With further innovations, future algorithms may be able to leverage this phenomenon for more substantial speedups.

2 Algorithm

2.1 Enacting jumps from one ground state to another

The algorithm we present in this work is conceptually simple; it consists essentially of just two steps, each of which is a jump from the ground state of one Hamiltonian to the ground state of another Hamiltonian. The time it takes to perform each jump is related to the overlap of the two ground states and the spectral gap of the two Hamiltonians. This is the same primitive step that was used in Hastings’ short-path algorithm [29], as well as various other quantum algorithms before it (e.g. [43]).

Proposition 1 (jump from $K_1 \rightarrow K_2$, simplified). *Given two n -qubit Hamiltonians K_1 and K_2 , let $|\psi_1\rangle$ be the (unique) ground state of K_1 and Π_2 be the projector onto the (possibly degenerate) ground space of K_2 . Let Δ_1 and Δ_2 denote the spectral gap above the ground space for K_1 and K_2 . Then there is a unitary U for which $U|\psi_1\rangle \propto \Pi_2|\psi_1\rangle$ that is enacted up to error δ by a quantum circuit consisting of $[\min(\Delta_1, \Delta_2)]^{-1} \|\Pi_2|\psi_1\rangle\|^{-1} \text{poly}(n, \log(\delta^{-1}))$ gates, where $\|\cdot\|$ denotes the standard Euclidean norm for a vector.*

Additionally, if K_1 or K_2 is a classical Hamiltonian (i.e. diagonal in either the computational basis or the Hadamard basis, where diagonal entries can be efficiently classically computed), then the number of gates does not depend on the corresponding gap parameter Δ_1 or Δ_2 , respectively.

To actually construct the unitary, we need to have knowledge of a lower bound on $\|\Pi_2|\psi_1\rangle\|$, upper bounds on the ground state energy of K_1 and K_2 , and lower bounds on the excited energy of K_1 and K_2 . A more complete version of the proposition that considers these factors appears as Prop. 21 in App. B, along with its proof. There we describe the unitary U , which is constructed through two steps. First, it uses phase estimation to produce unitary operators R_1 and R_2 that reflect about the state $|\psi_1\rangle$ and about the ground space of K_2 , respectively. The gate cost of approximating R_j to error δ is $O(\Delta_j^{-1} \log(\delta^{-1}))$ calls to a so-called “block-encoding” of the Hamiltonian K_j , which typically requires just $\text{poly}(n)$ gates. The exception to this statement is the case where K_j is a classical Hamiltonian. In this case, the terms of K_j commute and the energy can be measured exactly, allowing the reflection operator to be implemented exactly in $\text{poly}(n)$ gates regardless of how small Δ_j is. Second, the unitary U performs fixed-point amplitude amplification [44] to produce the state $\frac{\Pi_2|\psi_1\rangle}{\|\Pi_2|\psi_1\rangle\|}$ using the reflection operators R_1 and R_2 $O(\|\Pi_2|\psi_1\rangle\|^{-1} \log(\delta^{-1}))$ times each.

2.2 Specification of algorithm

Now we specify the main algorithm. The inputs to the algorithm are as follows:

Inputs:

- (a) A classical cost function H on n -bit binary assignments $z \in \{+1, -1\}^n$. This may be specified, for example, by giving the coefficients of $H(z)$ when it is expanded as a polynomial in $z_1, \dots, z_n$, where $z_i \in \{+1, -1\}$ denotes the i th bit of z . By convention we offset H so that it has no constant term, i.e. $\sum_z H(z) = 0$. In any case, we assume that for any z , $H(z)$ can be evaluated classically in $\text{poly}(n)$ time.

- (b) The value $E^* = \min_z H(z)$.
- (c) A value for η satisfying $0 \leq \eta < 1$.
- (d) A value for b satisfying $0 \leq b < 1$.

Note that here we assume the optimal value E^* of the cost function is known ahead of time, which may not always be the case. In App. C, we discuss how this assumption can be dropped at the expense of only polynomial overheads.

We define the piecewise-linear function $g_\eta : [-1, \infty) \rightarrow [-1, 0]$, and provide a plot for convenience:

$$g_\eta(x) = \min \left(0, \frac{x + 1 - \eta}{\eta} \right) \quad \text{---} \quad \begin{array}{c} \text{Plot of } g_\eta(x) \text{ for } \eta = 0.5 \\ \text{The function is zero for } x \geq 0. \text{ For } x < 0, \text{ it is a line with slope } -1, \text{ passing through } (-1, -1). \text{ A red dot marks the point } (-1, -1). \end{array} \quad (4)$$

We also define the Hamiltonian H_b , parameterized by $b > 0$, as

$$\begin{aligned} H_b &= -\frac{X}{n} + b g_\eta \left(\frac{H}{|E^*|} \right) \\ |\psi_b\rangle &= \text{ground state of } H_b \\ E_b &= \text{ground state energy of } H_b, \end{aligned} \quad (5)$$

where here (in a slight abuse of notation) H denotes the diagonal $2^n \times 2^n$ Hermitian operator for which $\langle z | H | z \rangle = H(z)$ and X denotes the transverse field $\sum_{i=1}^n X_i$, with X_i the Pauli- X operator on qubit i . Let $|\psi_b\rangle$ denote the ground state of H_b and E_b the ground-state energy. For illustration purposes, the spectrum of H_b for an example $n = 20$, $\eta = 0.5$ instance is shown in Fig. 1. Note that H_b is a *stoquastic* Hamiltonian, that is, when written in the computational basis, all of the off-diagonal entries of H_b are non-positive. As a consequence of being stoquastic, the ground state $|\psi_b\rangle$ can be taken to have non-negative real entries in the computational basis [45], and we assume this convention throughout.

Let Π^* denote the projector onto the (potentially degenerate) groundspace of H (spanned by computational basis states). The algorithm begins by preparing the initial state $|+\rangle \equiv |+\rangle^{\otimes n}$, the ground state of the Hamiltonian $-X/n$ (which is diagonal in the Hadamard basis). Next, the algorithm prepares $|\psi_b\rangle$, the (unique) ground state of H_b by performing a jump from Hamiltonian $-X/n$ to Hamiltonian H_b , using the unitary described in Prop. 1.⁴ Finally, the algorithm prepares $\frac{\Pi^* |\psi_b\rangle}{\|\Pi^* |\psi_b\rangle\|}$ using a second jump, from Hamiltonian H_b to the (classical) Hamiltonian $H/|E^*|$. The state $\Pi^* |\psi_b\rangle$ is a superposition of optimal solutions to the cost function H ; one of these solutions can be retrieved by measurement in the computational basis. Pseudocode for the algorithm appears in Algorithm 1, where a “jump” from Hamiltonian $K_1 \rightarrow K_2$ refers to the procedure in Prop. 1.

⁴One might be concerned about how the appearance of the operator $b g_\eta(H/|E^*|)$ might affect the construction of this unitary. Indeed, even if H can be decomposed as a sum of $\text{poly}(n)$ Pauli strings, $g_\eta(H/|E^*|)$ will not have this property. Nevertheless, in App. B, we show how a “block-encoding” of $g_\eta(H/|E^*|)$, which is diagonal, can be constructed by explicitly computing $H(z)$ into an ancilla register, computing $\theta_z = \arcsin(g_\eta(H(z)/|E^*|))$ with classical arithmetic in superposition, and subsequently performing a rotation by angle θ_z . This only works because H is a classical operator.

Algorithm 1: Pseudocode for main algorithm

Input: H, E^*, η, b , which together define H_b in Eq. (5)

Output: an optimal assignment z^* for H

- 1 Prepare $|+\rangle \equiv |+\rangle^{\otimes n} \equiv \sum_{i=1}^{2^n} |i\rangle$, the ground state of $-\frac{X}{n}$
 - 2 Prepare $|\psi_b\rangle$ up to exponentially small error with jump $-\frac{X}{n} \rightarrow H_b$
 - 3 Prepare $\frac{\Pi^* |\psi_b\rangle}{\|\Pi^* |\psi_b\rangle\|}$ up to exponentially small error with jump $H_b \rightarrow \frac{H}{|E^*|}$
 - 4 Measure in the computational basis to produce $|z^*\rangle$
-

2.3 Condition for success and overall runtime

Note that implementing the jumps in steps 2 and 3 of the algorithm requires an upper bound on E_b , a lower bound on the excited energy of H_b , and lower bounds on the quantities $|\langle +|\psi_b\rangle|$ and $\|\Pi^* |\psi_b\rangle\|$, where here $|+\rangle \equiv |+\rangle^{\otimes n}$ (see Prop. 21 in App. B for a formal presentation of jump implementation). The bounds on $|\langle +|\psi_b\rangle|$ and $\|\Pi^* |\psi_b\rangle\|$ can be “guessed” in the sense that one can try one value, see if the algorithm succeeds (it is easy to check if the output y of the algorithm satisfies $H(y) = E^*$), and if not, repeat with a guess that is smaller by a fixed constant factor. This procedure contributes at most $\text{poly}(n)$ overhead compared to if these quantities were known ahead of time. On the other hand, bounds on the eigenenergies of H_b must be shown separately, and are not guaranteed to hold for every possible cost function H and choice of parameters b, η . Accordingly, we define a condition under which success can be proved (and later we prove that the condition holds in specific cases).

Condition 1 (large-excited-energy condition). *We say the Hamiltonian H_b , as defined in Eq. (5), satisfies the large-excited-energy condition if the ground-state of H_b is non-degenerate and all excited states have energy greater than $-1 + 1/n$.*

Observe that when $b = 0$, the Hamiltonian H_b is equal to $-X/n$ and the large-excited-energy condition is satisfied since all excited states have energy at least $-1 + 2/n$. In the situations for which we prove rigorous bounds, the large-excited-energy condition will continue to hold as b is increased, up until some n -independent threshold. For example, in Fig. 1, we plot the numerically computed eigenvalues of H_b as a function of b for an $n = 20$ instance drawn from the 3-spin ensemble, with $\eta = 0.5$. For this instance, the large-excited-energy condition persists past $b = 0.8$.

Theorem 1 (runtime). *Fix a cost function H and parameters η and b , which defines H_b through Eq. (5). If H_b has the large excited-energy condition (Cond. 1), then with probability at least $1 - \exp(-\Omega(n))$, the algorithm outputs an optimal solution z^* of H and runs in time at most*

$$\text{poly}(n) \left(|\langle +|\psi_b\rangle|^{-1} + \|\Pi^* |\psi_b\rangle\|^{-1} \right), \quad (6)$$

where $|\psi_b\rangle$ is the ground state of H_b and Π^* is the projector onto the ground space of H .

Proof. Note that since $g_\eta(H/|E^*|)$ is a negative semidefinite operator, we have $\langle +|H_b|+\rangle \leq -1$ and thus, by the variational principle, the ground state energy E_b of H_b satisfies $E_b \leq -1$. The large-excited-energy condition then implies that $\Delta = 1/n$ is a lower bound on the spectral gap of H_b . We now refer to the steps described in the pseudocode of Algorithm 1. The only non-trivial steps are steps 2 and 3. Step 2 performs the jump $-X/n \rightarrow H_b$. Note that $-X/n$ is a classical Hamiltonian as it is diagonal in the Hadamard basis with efficiently computable entries. Thus, by Prop. 1, we may choose $\delta = e^{-\Omega(n)}$ and assert that step 2 prepares $|\psi_b\rangle$ up to error δ (in standard

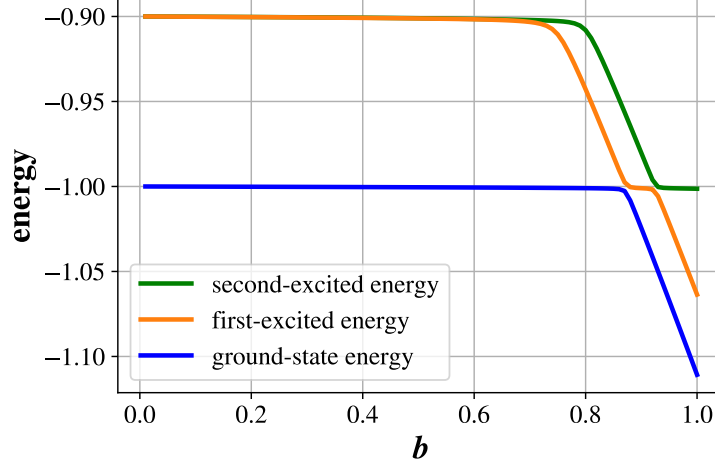


Figure 1: Plot of the lowest three eigenvalues of H_b as a function of b , for an $n = 20$ instance randomly chosen from the 3-spin ensemble, with $\eta = 0.5$. Eigenvalues were computed numerically using exact diagonalization. The two key features are that the spectral gap remains large and the ground state energy barely shifts from -1 until a relatively large value of b , namely $b \approx 0.85$.

Euclidean vector norm) and runs in time $\text{poly}(n)\Delta^{-1}|\langle +|\psi_b\rangle|^{-1}$. Similarly, step 3 performs the jump $H_b \rightarrow H/|E^*|$. Noting that $H/|E^*|$ is also a classical Hamiltonian, and that Π^* is its ground-space projector, Prop. 1 implies that it prepares $\Pi^*|\psi_b\rangle/\|\Pi^*|\psi_b\rangle\|$ up to error at most 2δ and runs in time $\text{poly}(n)\Delta^{-1}\|\Pi^*|\psi_b\rangle\|^{-1}$. This is true even if the spectral gap of $H/|E^*|$ is exponentially small. A subsequent computational basis measurement produces an optimal solution z^* with probability at least $1 - 2\delta = 1 - \exp(-\Omega(n))$. As Δ^{-1} is $\text{poly}(n)$, the theorem statement follows. $\square$

Theorem 1 shows that the large-excited-energy condition is sufficient for algorithmic success and a bound on its runtime. However, the large-excited-energy condition is not a necessary condition: a similar statement would follow for a relaxed version of the condition, where the excited energy is at least $-1 + 1/\text{poly}(n)$. Additionally, the algorithm could still succeed even if the excited energy falls beneath -1 , as long as there is a sizable gap between the ground and excited energy and a good approximation to ground/excited energy. We focus on the large-excited-energy condition because we will be able to prove that it holds under certain circumstances.

3 Proving speedup over Grover

The runtime statement in Theorem 1 implies that a super-Grover speedup can be shown given sufficient control of the spectrum of H_b (and in particular the first-excited energy) in combination with an upper bound on the quantity $|\langle +|\psi_b\rangle|^{-1} + \|\Pi^*|\psi_b\rangle\|^{-1}$, which is essentially equivalent to a lower bound on the overlaps $\langle +|\psi_b\rangle$ and $\|\Pi^*|\psi_b\rangle\|$. These tasks are accomplished separately, and in this section, we illustrate all the technical steps involved while deferring many of the mathematical proofs to the Appendices. In Sec. 3.1 we define additional conditions needed to organize our technical results. In Sec. 3.2, we show how, when these conditions are met, the runtime enjoys a super-Grover speedup. In Sec. 3.3, we show how a tail bound on the spectral density of the cost function implies that all of the conditions are met. We also discuss when such a tail bound is guaranteed to hold, and what can be said in the case there is no such tail bound.

3.1 Additional conditions and properties

In addition to the large-excited-energy condition, we define the small-ground-energy-shift condition and the α -subdepolarizing property, which will be needed to bound the runtime of the algorithm.

Condition 2 (small-ground-energy-shift condition). *We say that the Hamiltonian H_b , as defined in Eq. (5), satisfies the small-ground-energy-shift condition if the ground-state energy E_b of H_b satisfies $-1 - 1/n^3 \leq E_b \leq -1$.*

To get a qualitative sense of the idea behind Cond. 2, observe the remarkable flatness of the ground state energy of the 3-spin instance depicted in Fig. 1: E_b stays very close to -1 until relatively large values of b .

We also define a property of cost functions that we call the α -subdepolarizing property, which is important for establishing a lower bound on the overlap that determines the algorithm's runtime. To define α -subdepolarizing, we introduce the notation $y \sim x$ to denote that bit string $y \in \{+1, -1\}^n$ is generated from x by flipping a single bit chosen uniformly at random. First we define α -depolarizing before generalizing to α -subdepolarizing.

Definition 1 (α -depolarizing). *We say a cost function H is α -depolarizing if for every bit string x*

$$\mathbb{E}_{y \sim x} H(y) = (1 - \alpha)H(x) \quad (7)$$

We call the property α -depolarizing because it states that flipping a single bit at random brings the energy toward zero by exactly a fixed factor $1 - \alpha$, on average. We can immediately note that all MAX-Ek-LIN2 instances are α -depolarizing, due to the fact that every term has the same degree.

Proposition 2. *Any MAX-Ek-LIN2 instance has the α -depolarizing property with $\alpha = 2k/n$.*

Proof. If one of the n bits is flipped at random, then the sign of any degree- k monomial will flip with probability k/n . Thus the expectation value of the monomial is brought toward zero by a factor $1 - 2k/n$, and the monomial is $2k/n$ -depolarizing. Moreover, the sum of cost functions all of which possess the α -depolarizing property is also α -depolarizing, by linearity of Eq. (7). This extends the property to all MAX-Ek-LIN2 instances. $\square$

Although Hastings [29] did not use the same terminology, it was precisely the α -depolarizing property that led to an upper bound on the runtime of the short-path algorithm that suggested the possibility of super-Grover speedup. We now define the α -subdepolarizing property.

Definition 2 (α -subdepolarizing). *Consider a pair (H, g) , where H is a cost function with optimal value $E^* < 0$ and $g : [-1, \infty) \rightarrow [-1, 0]$ is a monotonic non-decreasing, concave function that is twice-differentiable at every point where it is nonzero. Let $f(x) := -g(-x)$, so that f is monotonically non-decreasing and convex. We say that (H, g) is α -subdepolarizing if for any set of constants $0 < c_1, c_2, \dots, c_T < 1$,*

$$\mathbb{E}_{y \sim x} \prod_{t=1}^T f\left(\frac{c_t H(y)}{E^*}\right) \geq \prod_{t=1}^T f\left(\frac{c_t (1 - \alpha) H(x)}{E^*}\right). \quad (8)$$

The definition appears complex, but it attempts to capture the same idea as α -depolarizing, with minor relaxations that allow for MAX- k -CSP cost functions to be included. First, note that if H is α -depolarizing, then (H, g) is α -subdepolarizing for any function g satisfying the criteria

in Def. 2, which includes g_η from Eq. (4) for any η (see Prop. 13 in App. A.2). Second, note that any MAX- k -CSP instance has the property for the function g_η for any η , as stated in the following proposition, which is proved in App. A.2.

Proposition 3. *Suppose H is a MAX- k -CSP instance with m terms and optimal value E^* . Then, for any η , (H, g_η) is α -subdepolarizing with*

$$\alpha = \frac{m}{|E^*|} \frac{k2^k}{(1-\eta)n} \quad (9)$$

In particular, if H is frustration free, i.e. fully satisfiable, then $|E^| = m$ and $\alpha = k2^k/(1-\eta)n$.*

3.2 Bounding the runtime with an approximate ground-state projector

Per Theorem 1, the runtime of the algorithm (assuming the large-excited-energy condition) depends on the quantity $|\langle +|\psi_b\rangle|^{-1} + \|\Pi^*|\psi_b\rangle\|^{-1}$. We wish to upper bound this quantity. In particular, we want to show that it is at most $2^{(0.5-c)n}$ for some constant c , implying a super-Grover speedup.

Let z^* be any optimal bit string and note that $\|\Pi^*|\psi_b\rangle\| \geq \langle z^*|\psi_b\rangle$ (recall we take the convention that all of the entries of $|\psi_b\rangle$ are positive in the computational basis). Then we have

$$\begin{aligned} \langle +|\psi_b\rangle^{-1} + \|\Pi^*|\psi_b\rangle\|^{-1} &\leq \langle +|\psi_b\rangle^{-1} + \langle \psi_b|z^*\rangle^{-1} \leq 2(\langle +|\psi_b\rangle \langle \psi_b|z^*\rangle)^{-1} \\ &= 2\langle +|\Pi_b|z^*\rangle^{-1} \end{aligned} \quad (10)$$

where $\Pi_b = |\psi_b\rangle \langle \psi_b|$ is the ground-state projector for the Hamiltonian H_b . We will replace Π_b by an *approximate ground state projector*, a tool that has been used successfully in the completely different context of proving area laws for ground states of many-body Hamiltonians [46–48]. As in the context of area laws, our approximate ground state projector will be a degree- ℓ polynomial in H_b ; however, where they used Chebyshev polynomials, we need only examine the simpler polynomial

$$P_\ell := \left(\frac{H_b}{E_b} \right)^\ell. \quad (11)$$

The operator P_ℓ approximates Π_b since $|\psi_b\rangle$ is an eigenstate with eigenvalue 1, and, assuming ℓ is sufficiently large, the other eigenvalues of P_ℓ will be close to zero. We show that $\ell = \Omega(n^2)$ is sufficiently large, assuming the large-excited-energy condition.⁵

Lemma 2. *If H_b satisfies the large-excited-energy condition (Cond. 1), then for any z and any $L \geq (\mu + 1.5 \ln(2))n^2$, the following equation holds either for $\ell = L$ or $\ell = L + 1$ (or both):*

$$\langle +|\psi_b\rangle \langle \psi_b|z\rangle \geq \langle +|P_\ell|z\rangle - 2^{-n/2}e^{-\mu n}. \quad (12)$$

Proof. Consider the operator $-X/n$, which is the first term of H_b , as defined in Eq. (5). Its maximum eigenvalue is 1, associated with eigenvector $|- \rangle \equiv 2^{-n/2}(|0\rangle - |1\rangle)^{\otimes n}$, and its second-largest eigenvalue is $1 - 2/n$. Meanwhile, the second term $bg_\eta(H/|E^*|)$ is a negative semidefinite operator. Denoting the largest eigenvalue of H_b by E'_b , and the associated eigenvector by $|\psi'_b\rangle$, we can say that $E'_b \leq 1$. Additionally, we can assert that all other eigenvalues of H_b are smaller than

⁵One could consider using Chebyshev-like polynomials for P_ℓ , which could reduce the requirement of the degree of P_ℓ to $\ell = \Omega(n^{3/2})$. However, this would provide very limited benefit to the proof, and it would bring its own additional complications; for example, in lower bounding $\langle +|P_\ell|z\rangle$, it is helpful that all the terms in the polynomial in H_b/E_b (there is only one term) have a positive coefficient.

$1 - 2/n$. To see this, suppose for contradiction that there were two eigenvectors $|p\rangle$ and $|q\rangle$ of H_b with eigenvalue greater than $1 - 2/n$. Then the state $|p\rangle - \frac{\langle -|p\rangle}{\langle -|q\rangle} |q\rangle$ is orthogonal to $|-\rangle$ and has average energy larger than $1 - 2/n$. However, this is impossible, since the average value of the $-X/n$ term can be at most $1 - 2/n$ (as the state is orthogonal to $|-\rangle$) and the average energy of the $bg_\eta(H/|E^*|)$ term can be at most 0.

The operator P_ℓ has the same eigenvectors as H_b , and for each eigenvalue λ of H_b , $(\lambda/E_b)^\ell$ is an eigenvalue of P_ℓ . Thus, in the limit of $\ell \rightarrow \infty$, P_ℓ approaches the projector $|\psi_b\rangle\langle\psi_b|$. By assumption, $\ell \geq L \geq \nu n^2$ with $\nu = \mu + 1.5 \ln(2)$. Hence, by the large-excited-energy condition, all 2^n eigenvalues of P_ℓ have magnitude at most $(1 - 1/n)^{\nu n^2} \leq e^{-\nu n} \leq 2^{-3n/2} e^{-\mu n}$, except for the eigenvalue associated with $|\psi_b\rangle$, which is 1, and (if $E'_b > 1 - 1/n$) the eigenvalue associated with $|\psi'_b\rangle$ which is $(E'_b/E_b)^\ell$. Denote these $2^n - 2$ eigenvalues by λ_i and associated eigenvectors by $|\lambda_i\rangle$ for $i = 1, \dots, 2^n - 2$. Thus, the quantity $\langle + | P_\ell | z \rangle$ is equal to

$$\langle + | \psi_b \rangle \langle \psi_b | z \rangle + \left(\frac{E'_b}{E_b} \right)^\ell \langle + | \psi'_b \rangle \langle \psi'_b | z \rangle + \sum_{i=1}^{2^n-2} \lambda_i \langle + | \lambda_i \rangle \langle \lambda_i | z \rangle$$

which is upper bounded by

$$\langle + | \psi_b \rangle \langle \psi_b | z \rangle + \left(\frac{E'_b}{E_b} \right)^\ell \langle + | \psi'_b \rangle \langle \psi'_b | z \rangle + (2^n - 2) 2^{-1.5n} e^{-\mu n} \quad (13)$$

In the case where $E'_b > 1 - 1/n > 0$, we have $E'_b/E_b < 0$ (since $E_b < 0$) and hence $(E'_b/E_b)^\ell \langle + | \psi'_b \rangle \langle \psi'_b | z \rangle$ is non-positive for exactly one of the choices $\ell = L$ or $\ell = L + 1$. For this choice of ℓ , we have

$$\langle + | \psi_b \rangle \langle \psi_b | z \rangle \geq \langle + | P_\ell | z \rangle - 2^{-n/2} e^{-\mu n}. \quad (14)$$

In the case where $E'_b \leq 1 - 1/n$, the second term of Eq. (13) can be combined with the third term to arrive at the same result in Eq. (14). $\square$

Lemma 2, together with Eq. (10), reduces the task of upper bounding the runtime of the algorithm to lower bounding the quantity $\langle + | P_\ell | z^* \rangle$ for $\ell = \Omega(n^2)$. We produce a lower bound by expanding P_ℓ as a sum of 2^ℓ terms using its definition in Eq. (11) and the definition of H_b in Eq. (5). Assuming the small-ground-energy-shift condition (Cond. 2) and that $\ell < O(n^3)$, we can say that the magnitude of the denominator $|E_b|^\ell$ is at most a constant, as $(1 + O(1/n^3))^{O(n^3)} = O(1)$. Each of these 2^ℓ terms contributes a positive amount to the sum; we lower bound the sum by lower bounding each individual term under the assumption that (H, g_η) has the α -subdepolarizing property (Def. 2). The result of this calculation is captured in Lemma 3, which is proved in App. A.3.

Lemma 3. *Given positive parameters $\eta < 1$, $b < 1$, $\alpha < (1 - b)/2$, and integer ℓ , suppose that (H, g_η) has the α -subdepolarizing property (Def. 2), that $3/\alpha^2 \leq \ell < n^3$, and that H_b satisfies the small-ground-energy shift condition (Cond. 2). Define the function $F : [0, 1] \rightarrow [0, 1]$ as follows (plot of F included for convenience).*

$$F(x) = 1 - x + x \ln(x) . \quad (15)$$

Let $z \in \{+1, -1\}^n$ be any assignment for which $\mathcal{E} := H(z)/|E^*|$ satisfies $\mathcal{E} \leq -(1 - \eta)$. Then

$$\langle + | P_\ell | z \rangle \geq 2^{-n/2} \exp \left(\frac{b}{\alpha} \frac{|\mathcal{E}|}{\eta} F \left(\frac{1 - \eta}{|\mathcal{E}|} \right) \right) (e^{-1} - 2e^{-2}), \quad (16)$$

noting that $e^{-1} - 2e^{-2} \approx 0.0972 = \Omega(1)$.

These lemmas together imply that the following conditions are sufficient for a super-Grover speedup: (i) the cost function has the α -subdepolarizing property for $\alpha = O(1/n)$, as is the case for MAX-Ek-LIN2 and MAX- k -CSP with limited frustration, and (ii) b and η are constants independent of n chosen such that H_b satisfies the large-excited-energy and small-ground-energy-shift conditions. Generally speaking, the larger b and η are, the larger the speedup that can be shown. This is formally captured in the following Theorem.

Theorem 4 (Super-Grover speedup). *Let H be a cost function on n variables with $n \geq 4$. Fix parameters η , b and a . Suppose that H has the α -subdepolarizing property (Def. 2) with $\alpha = a/n$ and $1 \leq a < n(1 - b)/2$. Suppose further that H_b satisfies the large-excited-energy condition (Cond. 1) and the small-ground-energy-shift condition (Cond. 2). Then, Algorithm 1 successfully produces an optimal solution with probability $1 - e^{-\Omega(n)}$ while running in time bounded above by*

$$\text{poly}(n) 2^{(0.5-c)n} \quad (17)$$

where

$$c = \frac{bF(1 - \eta)}{a\eta \ln(2)} \geq \frac{b\eta}{2a \ln(2)}. \quad (18)$$

Proof. Theorem 1, together with Eq. (10) and Lemma 2 imply that for $L > (\mu + 1.5)n^2$, the runtime is upper bounded by

$$\text{poly}(n) (\langle + | P_\ell | z^* \rangle - 2^{-n/2} e^{-\mu n})^{-1} \quad (19)$$

for either $\ell = L$ or $\ell = L + 1$, where $|z^*\rangle$ is any optimal solution to H . We choose $L = 3.5n^2$ (i.e. $\mu = 2$) and note that, by the assumptions of the lemma, $3/\alpha^2 \leq \ell < n^3$ holds for both $\ell = L$ and $\ell = L + 1$. Thus all the conditions of Lemma 3 hold. As $|\mathcal{E}| = 1$ in this case, we have that the runtime is upper bounded by

$$\text{poly}(n) 2^{n/2} (e^{n \frac{bF(1-\eta)}{na}} (e^{-1} - 2e^{-2}) - e^{-2n})^{-1}. \quad (20)$$

The e^{-2n} term will be smaller than the first term by a constant factor whenever $n \geq 4$. We can thus absorb it, as well as the $e^{-1} - 2e^{-2}$ factor, into the $\text{poly}(n)$ expression. This proves the theorem. Note that $F(1 - \eta)/\eta \geq \eta/2$ holds for $0 \leq \eta \leq 1$. $\square$

Our results actually say something stronger: every bit string z lying in a deep cost valley, that is, those for which $H(z) \leq (1 - \eta)E^*$, will have overlap with $|\psi_b\rangle$ that is $2^{c'n}$ larger than $2^{-n/2}$ for some n -independent constant c' . Thus, if slightly suboptimal solutions are acceptable, the probability that our algorithm will find one of these bit strings upon measurement of $|\psi_b\rangle$ is also boosted by an amount $2^{c'n}$ compared to measurement of $|+\rangle$. This is irrelevant for our particular task of finding the exactly optimal solution, but could be relevant in other situations.

3.3 Tail bound on spectral density implies conditions are met

For a family of cost functions with the α -subdepolarizing property with $\alpha = O(1/n)$, Theorem 4 reduces the task of showing super-Grover speedup to the task of proving that the large-excited-energy and small-ground-energy-shift conditions are met. In general, we do not show that the conditions are always satisfied. Rather, we show that they are satisfied whenever there is a tail bound on the cumulative number of assignments to H beneath a certain cost (i.e. integral of spectral density of H), as follows.

Lemma 5. *Let H be a cost function over assignments $\{+1, -1\}^n$, and let $C(E)$ denote the number of assignments to H with cost value beneath E , i.e.*

$$C(E) := |\{z : H(z) \leq E\}|. \quad (21)$$

Suppose that for a certain choice of $\eta < 1$

$$C((1 - \eta)E^*) \leq 2^{(1-\gamma)n}. \quad (22)$$

Suppose further that $\gamma \geq (1 + 4\log_2(n))/n$ (which holds for any n -independent γ and sufficiently large n). Then the large-excited energy condition (Cond. 1) and the small-ground-energy-shift condition (Cond. 2) are met for the Hamiltonian H_b (as defined in Eq. (5)) whenever

$$b \leq \frac{\ln(2)\gamma}{2 + \ln(2)} \approx 0.257\gamma \quad (23)$$

Lemma 5 is proved in App. A.1.4; the proof utilizes the log-Sobolev inequality and tools from statistical mechanics.

Proof sketch of Lemma 5. The basic idea is as follows. For the large-excited-energy condition to be violated, there must exist a state $|\phi\rangle$ orthogonal to $|\mathbf{+}\rangle$ for which $\langle\phi|H_b|\phi\rangle \leq -1 + 1/n$. Thus, referring to the definition of H_b in Eq. (5), there must be a number $\mathcal{U}$ satisfying $-1 \leq \mathcal{U} \leq 0$ such that the following two relations hold simultaneously:

$$\mathcal{U} = \langle\phi|g_\eta(H/|E^*|)|\phi\rangle \quad (24)$$

$$\langle\phi|X/n|\phi\rangle \geq 1 - 1/n + b\mathcal{U}. \quad (25)$$

Let $p(z) = |\langle z|\phi\rangle|^2$ denote the probability of obtaining z when measuring $|\phi\rangle$ in the computational basis. Viewing p as a probability distribution over assignments, Eq. (24) says that the average cost for the function $g_\eta(H/|E^*|)$ of a sample from p is equal to $\mathcal{U}$. Meanwhile, the log-Sobolev inequality [29, 49] allows us to turn the lower bound on $\langle\phi|X/n|\phi\rangle$ in Eq. (25) into a lower bound on S , the (log base-2) entropy of p , which reads:

$$S \geq n(1 - \ln(2)^{-1}(-b\mathcal{U} + 1/n)). \quad (26)$$

This turns the question into a statistical mechanics problem: is there a distribution p with entropy greater than $n(1 - O(|\mathcal{U}|))$, yet average cost of $g_\eta(H/|E^*|)$ equal to $\mathcal{U}$? Here, the tail bound constrains what is possible: since g_η zeroes out the cost of any assignment z for which $H(z) > (1 - \eta)E^*$, there are at most $2^{(1-\gamma)n}$ assignments z that have negative cost for $g_\eta(H/|E^*|)$. In order to have a distribution with average cost $\mathcal{U} < 0$, a large portion of the probability must be concentrated on this subset of $2^{(1-\gamma)n}$ assignments, constraining the entropy. Formally, for this step we use an elementary fact from statistical mechanics: the distribution that maximizes entropy

for a fixed average energy (average cost) is a Gibbs distribution, where assignments with cost E are allocated probability proportional to $e^{-\beta E}$ for some value of β , which physically corresponds to the inverse temperature. Applying these tools, we are able to show that whenever $b \leq 0.257\gamma$, it is not possible for Eq. (24) and Eq. (25) to simultaneously be true, implying that the large-excited-energy condition holds.

Separately, the small-ground-energy-shift condition follows from a perturbation-theory argument. Viewing $bg_\eta(H/|E^*|)$ as a perturbation to $-X/n$ in the expression for H_b in Eq. (5), the magnitude of the first-order shift in energy is $|\langle + | bg_\eta(H/|E^*|) | + \rangle|$, which is smaller than $b2^{-\gamma n}$ (an exponentially small number), simply due to the tail bound, and the fact that only $2^{(1-\gamma)n}$ entries of $g_\eta(H/|E^*|)$ have non-zero cost. The full proof also bounds higher-order contributions. $\square$

To get a sense of the interplay between the tail bound and the magnitude of the super-Grover speedup, suppose $H(z)$ comes from a family of cost functions with a unique optimal solution and for which exactly half of the eigenvalues are negative, so that $C(E^*) = 1$ and $C(0) = 2^{n-1}$. Thus, asymptotically speaking, the tail bound is satisfied with $\gamma = 1$ at $\eta = 0$, and with $\gamma = 0$ at $\eta = 1$. However, from Theorem 4, we see that the magnitude of the super-Grover speedup is zero if either $\eta = 0$ or if $\gamma = 0$ (since $\gamma = 0$ implies $b = 0$, by Lemma 5), so in both of these cases, these observations are not sufficient to show a super-Grover speedup. To show a super-Grover speedup, we need a non-trivial tail bound for non-zero η, γ to be true.

What happens if the tail bound condition is not satisfied for any choice of η, γ ? Then, regardless how small we make η , there must be *many* assignments for which $H(z) \leq (1 - \eta)E^*$. There are so many assignments that a classical algorithm could produce an assignment achieving a $1 - \eta$ approximation ratio by simple repetition in sub-exponential time for any constant value of η . Thus, instances with the α -subdepolarizing property can be partitioned into a set where our quantum algorithm has a super-Grover speedup, and a set that are unusually classically easy in a precise sense.

When can the tail bound be shown? For spin-glass-like cost functions consisting of random local terms, we often expect the spectral density to be roughly Gaussian, with the minimum cost assignment among all 2^n assignments lying $\Omega(\sqrt{n})$ standard deviations beneath 0. In this case, an exponential tail bound can be shown for any $\eta > 0$. As η increases, the corresponding bound γ decreases.

Proposition 4. *In each of the following situations, a tail bound of the form in Eq. (22) holds for any η and some choice of γ that depends on η but is independent of n .*

1. *If the cost function H is a MAX- k -CSP instance, a tail bound holds with*

$$\gamma = \left(\frac{|E^*|}{m} \right)^2 \frac{(1 - \eta)^2}{\ln(2) 2^{2k} k^2 D} \quad (27)$$

where D is a measure of how irregular the constraint hypergraph is (e.g., $D = 1$ if all n variables participate in exactly km/n constraints); specifically $D = nk^{-2}m^{-2} \sum_{j=1}^n d_j^2$, with d_j the number of constraints in which variable j participates. If the clauses $\mathcal{C}_j$ are each independently chosen to act on a set of k uniformly random bits, then $D \leq 3$ holds for most instances.

2. *If the cost function H is randomly chosen from the k -spin ensemble of MAX- Ek -LIN2 instances, then for any η a tail bound holds with probability at least $1 - 2^{-\gamma n+1}$ over choice of instance, with*

$$\gamma = \frac{(1 - \eta)^2}{32\pi \ln(2) k^2} \quad (28)$$

Proof. Item 1 is shown in Prop. 17 and item 2 is shown in Prop. 20, which appear in App. A.4. $\square$

3.4 Formal statement of main results

Theorem 6. *Let H be an instance of Quadratic Unconstrained Binary Optimization (QUBO) with $n \geq 12$ or MAX-Ek-LIN2 with $n \geq 6k$. Let $\gamma_0 = (1 + 4\log_2(n))/n$ (and note that $\gamma_0 \rightarrow 0$ as $n \rightarrow \infty$). For every $\gamma \in [\gamma_0, 1]$ and every $\eta \in [0, 1]$, one of the following must be true.*

- (a) *There is a quantum algorithm running in time $O^*(2^{(0.5-c)n})$ that produces an optimal solution to H with probability at least $1 - e^{-\Omega(n)}$, where*

$$c = \frac{1}{4(2 + \ln(2))} \frac{\gamma\eta}{k} \approx 0.0928 \frac{\gamma\eta}{k} \quad (29)$$

with $k = 2$ in the case of QUBO.

- (b) *There is a classical algorithm which repeatedly samples assignments uniformly at random that outputs a solution z for which $H(z) \leq E^*(1 - \eta)$ in expected time $O^*(2^{\gamma n})$.*

Thus, if (b) is the case for arbitrarily small η and arbitrarily small γ , we can say that the classical algorithm can achieve an arbitrarily good approximation to the optimal cost value in sub-exponential time.

Proof. Rather than analyze QUBO directly, we employ a well-known reduction from QUBO to MAX-E2-LIN2: given a QUBO instance H on n variables, construct a MAX-E2-LIN2 instance H' on $n + 1$ variables by introducing a binary variable z_0 and multiplying it with all degree-1 terms of H . Since all terms of H' have even degree, there is a $\mathbb{Z}_2$ symmetry where the value of $H'(z)$ is unchanged under a flip of all bits of z ; thus, the spectrum of H' is the same as the spectrum of H , with each energy appearing twice. The ground state of H can be determined by computing a ground state z of H' and flipping the bits $z_1, \dots, z_n$ if $z_0 = -1$. Since H and H' have the same spectrum, one will obey a tail bound of the form of Eq. (22) if and only if the other obeys the same tail bound. Thus all subsequent statements made about MAX-Ek-LIN2 will also apply to QUBO with the substitution $k = 2$.

Note that by Prop. 2, any MAX-Ek-LIN2 instance H is α -depolarizing (Def. 1) with $\alpha = 2k/n$, and thus by Prop. 13, for every η the pair (H, g_η) is α -subdepolarizing (Def. 2) with the same value of α . For a certain choice of η and γ , either H satisfies a tail bound of the form in Eq. (22) (case (a)), or it does not (case (b)).

In case (a), Lemma 5 implies that H_b satisfies the large-excited-energy and small-ground-energy-shift conditions when we choose $b = \gamma \ln(2)/(2 + \ln(2))$. Note that if $\gamma < 1$ then $b < 0.3$. So long as $n \geq 6k$, the condition $1 \leq (\alpha n) \leq n(1 - b)/2$ will hold, and we may apply Theorem 4 to prove the theorem statement.

In case (b), the fact that there are more than $2^{(1-\gamma)n}$ assignments with cost at most $(1 - \eta)E^*$ among 2^n total assignments implies that the average number of samples the classical algorithm must make before finding one of these assignments is at most $2^{\gamma n}$. Assuming the algorithm terminates as soon as it finds one such assignment, and noting that each sample can be drawn in $\text{poly}(n)$ time, the statement follows. $\square$

Theorem 7. *There is a quantum algorithm which, for any instance of MAX-k-CSP, produces an optimal solution with probability at least $1 - e^{-\Omega(n)}$ while running in time $O^*(2^{(0.5-c)n})$, where*

$$c = 0.0289 \left(\frac{|E^*|}{m} \right)^3 \frac{1}{2^{3k} k^3 D} \quad (30)$$

with m denoting the number of clauses, E^* denoting the optimal value of the instance, and D denoting the irregularity parameter defined by $D = nk^{-2}m^{-2} \sum_{j=1}^n d_j^2$, where d_j is the number of constraints that involve variable j . ($D \leq 3$ for most random MAX- k -CSP instances.) In particular, the question of whether or not the instance is fully satisfiable can be answered with probability $1 - e^{-\Omega(n)}$ in time $O^*(2^{(0.5-c)n})$ setting $|E^*|/m = 1$ in the above expression for c . If additionally $k = 3$ and $D = 3$, we find that $c = 5.87 \times 10^{-7}$.

Proof. We may assume E^* is known. If it is not known, we may simply loop through all possible values that E^* could possibly take, which incurs only polynomial overhead by the following argument. Note that, in the general setting, there are at most $m \leq n^k$ clauses and for each clause $\mathcal{C}_t$, the value of $\mathcal{C}_t$ is -1 when satisfied and $s_t/(2^k - s_t)$ when not satisfied, where $1 \leq s_t \leq 2^k - 1$ denotes the number of satisfying assignments to $\mathcal{C}_t$. Thus, for every t and every assignment, the value of $\mathcal{C}_t$ is an integral multiple of $[(2^k - 1)!]^{-1}$, and the number of distinct values E^* can possibly take is upper bounded by $(2^k - 1)!m = \text{poly}(n)$.

We fix a value of η to be specified later. By Prop. 3, the pair (H, g_η) has the α -subdepolarizing property with

$$\alpha = \frac{m}{|E^*|} \frac{k2^k}{(1 - \eta)n}. \quad (31)$$

Furthermore, by Prop. 4, H obeys a tail bound of the form of Eq. (22) with

$$\gamma = \left(\frac{|E^*|}{m} \right)^2 \frac{(1 - \eta)^2}{\ln(2)2^{2k}k^2D}, \quad (32)$$

where $D = nk^{-2}m^{-2} \sum_{j=1}^n d_j^2$, with d_j the number of constraints involving binary variable j . By Lemma 5, this implies that the large-excited-energy and small-ground-energy-shift conditions are satisfied when we choose $b = \gamma \ln(2)/(2 + \ln(2))$. By Theorem 4, and substituting our choices for b and γ above, the runtime is $O^*(2^{(0.5-c)n})$ with

$$c = \left[\frac{1}{\ln(2)(2 + \ln(2))} \frac{(1 - \eta)^3 F(1 - \eta)}{\eta} \right] \left(\frac{|E^*|}{m} \right)^3 \frac{1}{D2^{3k}k^3}. \quad (33)$$

The expression in brackets achieves its maximum of 0.0289 at $\eta = 0.189$, leading to the quoted statement.

Now consider the question of whether a certain k -CSP instance is fully satisfiable, which is equivalent to whether $E^* = -m$, or not. To solve this problem, it suffices to give an algorithm that can find a satisfying assignment, promised that one exists: we may run the algorithm assuming that $|E^*| = m$; if the instance is satisfiable then the output y will satisfy $H(y) = -m$ with high probability, whereas if the instance is not satisfiable, the output y will satisfy $H(y) > -m$ with probability 1. $\square$

Theorem 8. *There is a quantum algorithm which, for at least a fraction $1 - e^{-\Omega(n/k)}$ instances drawn from the k -spin ensemble, produces an optimal solution with probability at least $1 - e^{-\Omega(n)}$ while running in time $O^*(2^{(0.5-c)n})$, where*

$$c = (2.24 \times 10^{-4}) \frac{1}{k^3} \quad (34)$$

Proof. By Prop. 2, any MAX-E k -LIN2 instance H is α -depolarizing (Def. 1) with $\alpha = 2k/n$, and thus by Prop. 13, for every η the pair (H, g_η) is α -subdepolarizing (Def. 2) with the same value of

α . Furthermore, by Prop. 4, H obeys a tail bound of the form of Eq. (22) with

$$\gamma = \frac{(1 - \eta)^2}{32\pi \ln(2)k^2}. \quad (35)$$

By Lemma 5, this implies that the large-excited-energy and small-ground-energy-shift conditions are satisfied when we choose $b = \gamma \ln(2)/(2 + \ln(2))$. By Theorem 4, and substituting our choices for b and γ above, the runtime is $O^*(2^{(0.5-c)n})$ with

$$c = \left\lceil \frac{1}{64 \ln(2) \pi (2 + \ln(2))} \frac{(1 - \eta)^2 F(1 - \eta)}{\eta} \right\rceil \frac{1}{k^3}. \quad (36)$$

The expression in brackets achieves its maximum of 2.24×10^{-4} at $\eta = 0.405$, proving the theorem. $\square$

4 Physical intuition for algorithm and speedup

This section gives an intuition for why the algorithm works and why it is able to give a speedup over Grover's algorithm.

4.1 A rugged cost landscape

A local structure can be established by placing the 2^n assignments to the cost function H on the corners of a hypercube in n -dimensions. Flipping a bit in the assignment corresponds to traversing an edge of the hypercube from one corner to an adjacent corner. The Hamming distance between two assignments is defined as the minimum number of edges that must be traversed to get from one to the other. An assignment is a local minimum if its cost is lower than all adjacent assignments on the hypercube. Random instances of cost functions like MAX- Ek -LIN2 and MAX- k -SAT typically give rise to a rugged landscape, where there are an exponential number of local minima and an exponential number of deep cost valleys (i.e. achieving a constant fraction of the optimal cost) separated from each other by Hamming distance at least order n . This property limits the effectiveness of any algorithm that aims to find the global minimum by local search, an intuitive claim that has also been formalized in works such as Ref. [50].

By convention, the cost function is offset such that the average cost across all assignments is 0; thus the optimal cost E^* is negative. Dividing H by $|E^*|$ normalizes the cost function such that its deepest valley has cost -1 . Applying g_η to $H/|E^*|$ smooths out the cost function, setting nearly all assignments to zero cost, while preserving the local structure of the deepest valleys. One way to visualize it is that g_η has the effect of flooding the entire landscape with water, such that the water level lies at a factor $1 - \eta$ times the maximum valley depth. All dry land is set to zero cost; meanwhile, the local structure of the cost landscape underwater is preserved up to a stretching factor of $1/\eta$ such that the range of the cost function $g_\eta(H/|E^*|)$ remains $[-1, 0]$. The reason for introducing this function g_η is elaborated upon in Sec. 4.4.

4.2 Quantum phase transition from extended to localized phase

With η fixed, H_b is defined in Eq. (5) as a sum of the transverse field $-X/n$ and the diagonal operator $g_\eta(H/|E^*|)$. We may view b as a free parameter that drives one or more quantum phase transitions. When $b = 0$, we have $H_b = -\frac{X}{n}$ and $|\psi_b\rangle = |+\rangle$. The state $|+\rangle$ might be called an

extended state because it has substantial amplitude, in fact *equal* amplitude, on all 2^n computational basis states.

As we increase b , the diagonal term $b g_\eta(H_b/|E^*|)$ begins to play a role. The spectrum of this term lies in the range $[-b, 0]$, whereas the first term $-\frac{X}{n}$ has range $[-1, 1]$. Thus, when b is small, the second term may be viewed as a perturbation of the first, and we expect $|\psi_b\rangle$ to retain high overlap with $|+\rangle$ and continue to be an extended state with substantial overlap across all 2^n computational basis states. However, the distribution of amplitude is no longer uniform; assignments that lie within the valleys of $g_\eta(H_b/|E^*|)$ will collect more amplitude than assignments not lying within a valley. Note that, as previously mentioned, since H_b is a stoquastic Hamiltonian, we may take $|\psi_b\rangle$ to have positive amplitudes in the computational basis [45].

On the other end, in the limit $b \rightarrow \infty$, the first term vanishes relative to the second. The ground state $|\psi_b\rangle$ becomes *localized* onto a single computational basis state $|z^*\rangle$, associated with the optimal assignment to the cost function (or if there are multiple optimal values, then there will be a degenerate ground state in this limit). For large but finite b , the state $|\psi_b\rangle$ will have some support everywhere, but it will still be localized in the sense that most of the amplitude will be on $|z^*\rangle$ with amplitude on other computational basis states exponentially small in the Hamming distance from $|z^*\rangle$.

These observations suggest that increasing b drives a quantum phase transition from an extended phase to a localized phase at some critical value of b . This is a second-order phase transition, analogous to the thermal phase transition in a spin glass. When does the critical point occur? We know that it must occur at some value $b < 1$. This observation follows from the fact that, at $b = 1$, both terms $-X/n$ and $b g_\eta(H/|E^*|)$ have equal operator norm, and $|+\rangle$ and $|z^*\rangle$ have similar expectation values for H_b despite the former being extended and the latter being localized. Our formal proofs, in particular Lemma 5, showed that under the tail-bound assumption, the ground and first-excited states of H_b remain well separated up until an n -independent value of b , which puts a lower bound on where the critical point for this quantum phase transition occurs.

Note that there can also be first-order phase transitions within the localized phase, where increasing b can cause the wavefunction $|\psi_b\rangle$ to change from being localized within one valley of the landscape to being localized in another far-away valley. In the context of the quantum adiabatic algorithm, these first-order phase transitions have been identified as the truly problematic part of the protocol [23, 26, 51], as the size of the minimum spectral gap is hard to study and exponentially small in n . Our algorithm avoids the complication of first-order phase transitions within the localized phase by *jumping straight from the extended phase to the end of the algorithm*: as long as b is chosen at some point smaller than the critical value, H_b will have a large spectral gap, and $|\psi_b\rangle$ will be extended. The speedup leverages the fact that while $|\psi_b\rangle$ has substantial amplitude on all 2^n computational basis states, it has a factor 2^{cn} larger amplitude on the computational basis states lying within the deepest valleys of the landscape. Thus, by measuring $|\psi_b\rangle$ in the computational basis (and performing amplitude amplification), we can find the optimal assignment with a 2^{cn} advantage over Grover’s algorithm.

4.3 The α -depolarizing property and the super-Grover speedup

As Grover’s algorithm is optimal in a black-box model, any algorithm achieving a super-Grover speedup must exploit structure in the cost function. In Hastings’ work on the short-path algorithm [29], the cost function studied was MAX-Ek-LIN2, in part because it has the following nice property: for any assignment z with cost value $H(z)$, the average cost of the n neighbors of z on the hypercube is exactly $(1-\alpha)H(z)$, with $\alpha = 2k/n$; in other words, traversing a random edge brings the expected cost value toward zero by a small and consistent factor. Hastings used this property in combination

with a perturbation theory analysis to show that a certain overlap that determines the runtime of the short-path algorithm is $2^{\Omega(b/\alpha)}$ larger than what occurs in Grover's algorithm, under certain assumptions. This advantage is exponentially large in n as long as the assumptions hold for constant b . Proving that these additional assumptions hold was the weaker part of Hastings' analysis.

We call this property the α -depolarizing property (Def. 1), and then we define a generalized version: the α -subdepolarizing property (Def. 2). The idea is that cost functions consisting of k -local terms may not exactly satisfy the α -depolarizing property, but there is still a sense in which a single random bit flip will not cause the energy to change much on average. Indeed, we are able to show that MAX- k -SAT instances that are not highly frustrated (together with the function g_η) have the α -subdepolarizing property where $\alpha = O(k2^k/n)$ (see Prop. 3). While the overlap we want to compute, $\langle \psi_b | z^* \rangle$, is different than the one Hastings' needed to compute in Ref. [29], we show that the α -subdepolarizing property, along with the small-ground-energy-shift condition (Cond. 2), implies a $2^{\Omega(b/\alpha)} = 2^{cn}$ advantage over Grover's algorithm, with $c = \Omega(b/(k2^k))$. Thus, as long as the small-ground-energy shift is met for an n -independent choice of b , the super-Grover speedup quantity c will be independent of n . Where Hastings' main tool was perturbation theory, we use an approximate ground-state projector and a combinatorial analysis (see Sec. 3.2).

4.4 The role of η and establishing a lower bound on the critical point

Since the magnitude of the advantage in the exponent over Grover's algorithm is cn for a value c proportional to b , we need to establish that we can take b to be a constant independent from n and still be in the extended phase if we want to have a super-Grover speedup. It is not obvious that this would be the case, and previous studies of phase transitions in the adiabatic algorithm suggested the possibility that the critical point could occur at a value of b that shrinks with n [23, 24], although this is not expected to be the typical behavior [25].

By applying g_η to the cost function $H/|E^*|$, we can provably avoid this possibility and show that the extended phase persists up to an n -independent choice of b . The idea is again to view $b g_\eta(H/|E^*|)$ as a perturbation to $-X/n$. Assuming a tail bound of the form of Eq. (22), $g_\eta(H(z)/|E^*|) = 0$ for all but an exponentially small fraction of assignments z . As a result, the ground state does not *see* the perturbation in the sense that $\langle + | g_\eta(H(z)/|E^*|) | + \rangle$ is exponentially small. The smaller we make η , the weaker the perturbation appears, and the larger the value of b we can choose and still be sure the ground state will be extended. However, the magnitude of the speedup is roughly proportional to η , so we do not want to make η too small. The formal proof of this ingredient utilizes the log-Sobolev inequality and tools from statistical mechanics (see Sec. 3.3).

4.5 Mechanism for speedup: localization in 1-norm vs. 2-norm

In quantum mechanics, the Born rule dictates that probabilities of measurement outcomes are the square of amplitudes of the wavefunction. In other words, if we want to measure the probability that the wavefunction lies within a certain subset of the 2^n vertices of the hypercube, we should use the 2-norm. This contrasts with standard probability theory, where the 1-norm would be used.

In describing the mechanism for speedup for the short-path algorithm [30], Hastings pointed to precisely this fact. In the short-path algorithm, a state $|\phi\rangle$ was prepared that was simultaneously localized in the 2-norm and de-localized in the 1-norm, in the following sense. Define

$$w_1(z) = \frac{|\langle z | \phi \rangle|}{\sum_z |\langle z | \phi \rangle|} \quad w_2(z) = \frac{|\langle z | \phi \rangle|^2}{\sum_z |\langle z | \phi \rangle|^2}, \quad (37)$$

both of which are probability distributions over inputs z . For the state $|\phi\rangle$ prepared by the algorithm, the distribution w_2 had nearly all its mass concentrated on the optimal solution z^* , while the distribution w_1 had an exponentially small 2^{-cn} fraction of its mass concentrated on z^* . In fact, if we break the hypercube up into subsets of equal Hamming distance from z^* , the subset with the most w_1 mass was order- n bit flips away from z^* . This 1-norm delocalization is what gave rise to the super-Grover speedup during the large jump made by the algorithm, while the 2-norm localization allowed the small jump to succeed with high probability.

Our algorithm exploits a similar effect to produce a super-Grover speedup, although it does so in the Hadamard basis. That is, the same comments for $|\phi\rangle$ above apply to the state $|\psi_b\rangle$ if we redefine

$$w_1(u) = \frac{|\langle u|\psi_b\rangle|}{\sum_u |\langle u|\psi_b\rangle|}, \quad w_2(u) = \frac{|\langle u|\psi_b\rangle|^2}{\sum_u |\langle u|\psi_b\rangle|^2}, \quad (38)$$

where the states $|u\rangle$ are the 2^n n -fold tensor products of $|+\rangle$ and $|-\rangle$. We have already discussed how the state $|\psi_b\rangle$ is extended and has high overlap with $|+\rangle$: thus, it is 2-norm localized in the Hadamard basis. Now, without loss of generality, we assume the optimal solution is the computational basis state $|0^n\rangle = 2^{-n/2}(|+\rangle + |-\rangle)^{\otimes n}$. Then, the overlap that determines the runtime of our algorithm is given by

$$\langle 0^n|\psi_b\rangle = 2^{-n/2} \sum_{u \in \{+, -\}^n} \langle u|\psi_b\rangle. \quad (39)$$

The statement that $\langle 0^n|\psi_b\rangle \geq 2^{-(0.5-c)n}$ is thus equivalent to the statement $\sum_u \langle u|\psi_b\rangle \geq 2^{cn}$. This implies the denominator of the definition of w_1 in Eq. (38) is at least 2^{cn} , but meanwhile $\langle +|\psi_b\rangle \approx 1$, meaning $w_1(+)\leq 2^{-cn}$. In other words, an exponentially small amount of 1-norm weight lies on the $|+\rangle$ basis state.

Crucially, this dichotomy between the distribution of weight in the 1-norm and the 2-norm is only possible in quantum mechanics; it is not obvious how one would exploit this phenomenon with a classical algorithm. As a result, this mechanism has the potential to deliver super-Grover speedups, and, potentially, super-quadratic speedups compared to best-known classical algorithms.

4.6 Performance of analogous classical Markov chain methods

While a classical algorithm cannot exactly replicate the same 1-norm vs. 2-norm dichotomy that our quantum algorithm exploits, it is valuable to scrutinize whether a classical algorithm might nonetheless emulate our quantum algorithm. We identify two categories of classical algorithms based on sampling Markov processes that might, in a certain sense, be regarded as classical analogues of our algorithm. However, in both cases, they lack rigorous runtime guarantees similar to those we have shown about our algorithm.

The first candidate analogue is the classical algorithm mentioned in Sec. 1.4 that repeatedly samples the high-temperature classical Gibbs distribution of the cost function (or some transformation applied to the cost function), a feat which can be accomplished via classical Metropolis sampling or simulated annealing. Showing that this classical algorithm runs in time $O^*(2^{(1-c)n})$ would require proving that the associated Markov chain remains rapidly mixing up to some sufficiently large inverse temperature β . As rapid-mixing proofs often leverage the log-Sobolev inequality and its relatives (e.g., [42, 52, 53]), perhaps it is unsurprising that the log-Sobolev inequality is also helpful for us in showing lower bounds on the spectral gap of the Hamiltonian H_b . However, despite a heuristic connection between the inverse temperature β in the Metropolis sampling algorithm and the inverse transverse-field strength parameter b in our algorithm, there is no direct

relationship between the runtime of the two algorithms. Indeed, a clearer quantum analogue of simulated annealing is given by applying the “quantum simulated annealing” algorithm proposed in Ref. [18] (see also Refs. [5, 19, 54–56]), which, when combined with amplitude amplification as in Ref. [20], gives a direct quadratic (but no larger) quantum speedup of the classical algorithm described above.

The second candidate analogue of our quantum algorithm is formed by applying a Quantum Monte Carlo (QMC) approach to the Hamiltonian H_b (see, e.g., [57–61]). A reason to think that this might be successful is that our Hamiltonian H_b is stoquastic and thus it does not have the sign problem [45]. Indeed, the core step of our algorithm is the application of quantum phase estimation to the stoquastic Hamiltonian H_b , a situation that was shown in Ref. [58] to admit a classical QMC simulation whenever there exists a “guiding state.” In the case of H_b , there is no obvious state that satisfies the criteria of Ref. [58]: the state $|+\rangle$ falls short since it differs from $|\psi_b\rangle$ by an exponentially large factor on some of the amplitudes. More generally, showing that the QMC algorithm can sample (up to inverse exponential precision) the same distribution as computational basis measurements on the ground state of H_b requires proving that a certain Markov chain representing the imaginary time evolution $e^{-\beta H_b}$ remains rapidly mixing up to imaginary time β of order- n^2 (this is necessary to guarantee that the second largest eigenvalue of $e^{-\beta H_b}$ is exponentially smaller than the largest eigenvalue). For comparison, Ref. [61] showed rapid mixing for transverse-field Ising models up to imaginary time of order $(J(\Delta + 2))^{-1}$, where J is the maximum interaction strength and Δ is the interaction-degree. This encompasses the SK model (i.e. Eq. (3) with $k = 2$), where, after normalization by $|E^*|$ (order- n), we have $J^{-1} = O(n\sqrt{n})$ and $\Delta^{-1} = 1/n$, so this fails to prove a relevant statement for our situation. However, it remains an interesting question to examine the performance of QMC as a classical simulation of our algorithm. Toward that end, it is interesting to note that previous work (see fourth counterexample of Ref. [62], as well as Refs. [59, 63]) has demonstrated examples where the classical QMC algorithm struggles to simulate its quantum counterpart due in part to a discrepancy between wavefunction localization in the 1-norm versus the 2-norm, the exact effect exploited by our quantum algorithm (albeit in a different basis).

5 Numerical results

5.1 Example instance

We present an example instance to confirm that the theoretical picture of the algorithm holds in practice. Recall that in Fig. 1, we presented the lowest three energy levels of H_b as a function of b for an $n = 20$ instance chosen from the 3-spin ensemble, with $\eta = 0.5$. In Fig. 2, we show the overlap profile for the same instance, that is $|\langle +|\psi_b\rangle|$ and $|\langle z^*|\psi_b\rangle|$ as a function of b .

Our theoretical picture is largely confirmed. The ground state energy stays remarkably unshifted from -1 and the spectral gap remains open until b is quite large, around $b = 0.85$ for this instance. Meanwhile, the overlap $|\langle z^*|\psi_b\rangle|$ which determines the runtime of the algorithm grows exponentially with b , even as the overlap $|\langle +|\psi_b\rangle|$ remains close to 1 (the wavefunction remains localized in the 2-norm even as it becomes increasingly de-localized in the 1-norm, see Sec. 4.5). This confirms that for this instance the first step of the algorithm will succeed with little need for amplification even when we take b as large as 0.8, and that the second step has success probability significantly better than Grover’s algorithm.

This instance also illustrates how there might be considerable room for improvement in the theoretical analysis. Firstly, in Fig. 2, the quantity $|\langle z^*|\psi_b\rangle|$ appears to grow *super*-exponentially

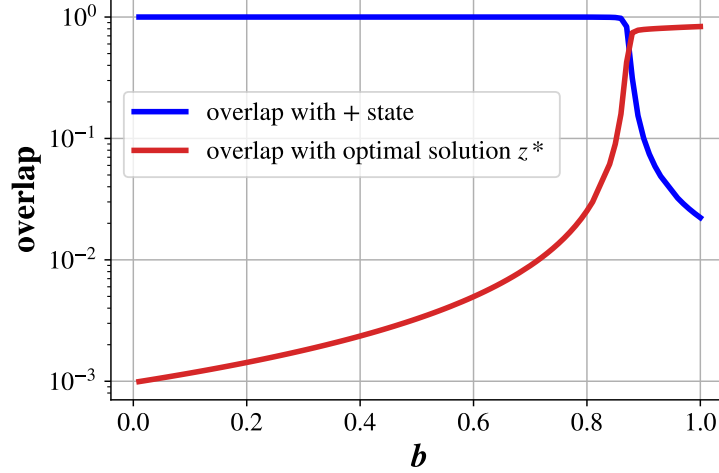


Figure 2: Plot of the relevant overlap values for the same $n = 20$, $\eta = 0.5$ 3-spin instance from Fig. 1. Overlaps are determined by exact diagonalization of H_b . The overlap $|\langle + | \psi_b \rangle|$ (blue) determines the runtime of step 2 of Algorithm 1, and the overlap $|\langle z^* | \psi_b \rangle|$ (red) determines the runtime of step 3.

with b , even as $|\langle + | \psi_b \rangle|$ remains close to 1, whereas the theoretical lower bound from Lemma 2 predicts only exponential growth. Secondly, the numerics indicate that the algorithm will work well for this $\eta = 0.5$ instance all the way up to $b = 0.8$, whereas the theoretical analysis only guarantees success up to a much smaller value of b . Looking back at Eq. (35) and the text beneath, for $k = 3$ and $\eta = 0.5$, the large-excited-energy condition (Cond. 1) is shown to hold only for $b \leq 1.02 \times 10^{-4}$ (and only for *most* random instances in the ensemble). From the theory perspective, the important conclusion is that this value is independent of n , but these numerics suggest that there is a considerable gap between the theoretical bounds and the empirical reality.

5.2 Estimation of actual size of super-Grover speedup

Our theoretical analysis shows a super-Grover speedup but with a very small lower bound on the size of the speedup. We illustrate numerically that the actual super-Grover speedup might be much more substantial. By performing exact diagonalization on 30 random $\eta = 0.5$, $b = 0.7$ instances from the 3-spin ensemble at each value $n = 17, 18, \dots, 23$, we determine the growth of the advantage with n . We chose $b = 0.7$ to give some breathing room between the maximum value at which the large-excited-energy condition (Cond. 1) failed (roughly $b = 0.8$) for the instance depicted in Fig. 1. Indeed, we found that for all of the instances we diagonalized, $b = 0.7$ fell comfortably below the phase transition point and the large-excited-energy condition was satisfied. Thus, by Theorem 1, the algorithm would succeed with runtime dependent on $|\langle z^* | \psi_{0.7} \rangle|^{-1}$. In Fig. 3, we plot the quantity $|\langle z^* | \psi_{0.7} \rangle|^{-1}$ for each instance, as well as the median among all 30 instances at each value of n . Fitting the medians to a line on a log scale, we find that the best fit is $|\langle z^* | \psi_{0.7} \rangle|^{-1} \approx 0.28 \times 2^{0.427n}$; the 95% confidence interval on the 0.427 value is $[0.415, 0.439]$. While the quality of the numerical fit is encouraging, we caution that robust conclusions are difficult to assert based on an exponential fit to just 7 data points that span the small range from roughly 40 to roughly 300, less than even one order of magnitude.

A scaling of $O^*(2^{0.43n})$ would represent a material improvement over Grover, if not a practically transformative one. A larger speedup, closer to quartic compared with exhaustive enumeration (corresponding to a quantum algorithm with $2^{0.25n}$ runtime), would be needed to make the prospect of actual quantum advantage more likely [6]. However, as we have not made much effort at optimizing the parameters b and η , we are optimistic that future examination will reveal additional speedup over Grover than what we report here.

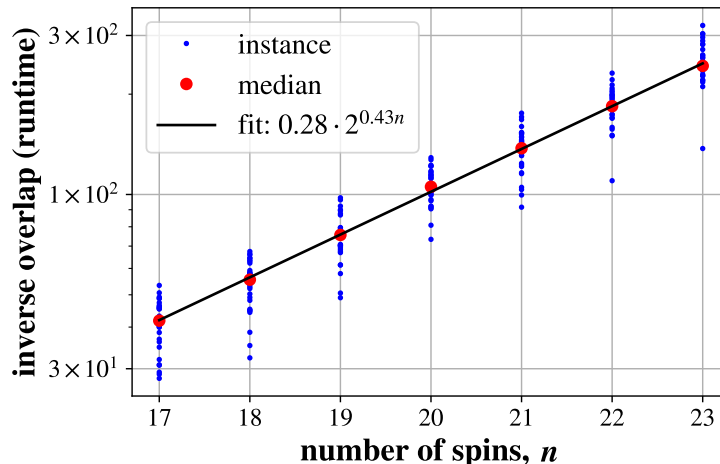


Figure 3: Plot of $|\langle z^* | \psi_b \rangle|^{-1}$ for several randomly chosen instances from the 3-spin model at $b = 0.7$ and $\eta = 0.5$ at values of n ranging from $n = 17$ to $n = 23$. For each value of n , the median value is also plotted. A fit of the medians to an exponential yields the fit $|\langle z^* | \psi_b \rangle|^{-1} = 0.276 \cdot 2^{0.427n}$.

6 Conclusion

We present a quantum algorithm that has a rigorous proof of a speedup compared to Grover’s algorithm in certain cases. On the one hand, as it stands, the algorithm is not likely to lead to a practical advantage. For one, the size of the speedup in our proof is very small, although we believe that further optimization of parameters and extensions of the theory would lead to improvements. Additionally, while numerical experiments suggest that the actual speedup is non-negligible, it still appears to fall short of a cubic or quartic speedup (in comparison to exhaustive enumeration), and thus, after accounting for other overheads, it is unlikely to lead to practical benefit without further improvements [6]. Finally, in most cases where our algorithm has rigorous guarantees, the speedup over the *best* classical algorithm is less than quadratic, owing to the existence of clever classical algorithms that exploit structure to beat exhaustive enumeration.

On the other hand, the mechanism by which our algorithm generates a speedup has no obvious classical analogue. This contrasts with many other speedups over exhaustive enumeration, which are essentially a Grover speedup combined with a classical technique. The inherent quantumness of our speedup positions our algorithm to at least have a fighting chance at achieving a super-quadratic speedup, even if it cannot yet do so when compared to mature classical algorithms that have been improved over the course of decades. Toward that end, an important open question is whether our algorithm and the speedup mechanism behind it can be combined with any of the classical techniques for exploiting problem structure, such as backtracking [11] and branch-and-bound [15].

Acknowledgments

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A Deferred proofs

A.1 Tail bound implies large-excited-energy and small-ground-energy-shift conditions

A.1.1 Short-path condition definition and consequences

First, we define a condition we call the “short-path condition” since it is analagous to the main condition used by Hastings in the analysis of the short-path algorithm Ref. [29].

Condition 3 (Short-path condition). *Let H_b be a Hamiltonian as defined in Eq. (5). Let $\bar{\Pi} = I - |\!+\rangle\langle\!+|$ to be a projector onto the excited space of $-X/n$ and let $\bar{E}_b$ denote the ground state energy of the Hamiltonian*

$$\bar{\Pi}H_b\bar{\Pi}. \quad (40)$$

Then H_b is said to obey the short-path condition if

$$\text{Short-path condition: } \bar{E}_b \geq -1 + \frac{1}{n}. \quad (41)$$

Proposition 5. *If H_b satisfies the short-path condition (Cond. 3), then H_b satisfies the large-excited-energy condition (Cond. 1).*

Proof. Suppose for contradiction that the large-excited-energy condition were false. Then there are at least two eigenvectors of H_b with eigenvalue beneath $-1 + 1/n$. Denote two such eigenvectors by $|\psi_b\rangle$ and $|\eta_b\rangle$. Furthermore, since $\langle\!+|H_b|\!+\rangle \leq -1$, we may assume without loss of generality that at least one of these eigenvectors has nonzero overlap with $|\!+\rangle$. We define the normalized nonzero state

$$|Y\rangle = \frac{\langle\!+|\eta_b\rangle|\psi_b\rangle - \langle\!+|\psi_b\rangle|\eta_b\rangle}{|\langle\!+|\eta_b\rangle|^2 + |\langle\!+|\psi_b\rangle|^2} \quad (42)$$

such that it is orthogonal to $|\!+\rangle$ and thus lies in the support of $\bar{\Pi}$, and meanwhile has expectation value of $\langle Y|\bar{\Pi}H_b\bar{\Pi}|Y\rangle$ that is less than $-1 + 1/n$. By the variational principle, the ground state of $\bar{\Pi}H_b\bar{\Pi}$ (restricted to the support of $\bar{\Pi}$) must have smaller expectation value than $|Y\rangle$, which contradicts the short-path condition (Cond. 3). $\square$

Proposition 6. *Suppose that H_b satisfies the short-path condition (Cond. 3) and H obeys the cumulative states tail bound*

$$C((1 - \eta)E^*) \leq 2^{(1-\gamma)n} \quad (43)$$

where $C(E) = |\{z : H(z) \leq E\}|$. Then

$$E_b \geq -1 - 2^{-\gamma n}(b + b^2n) \quad (44)$$

Thus, if γ, b are independent of n , the small-ground-energy-shift condition (Cond. 2) is satisfied for sufficiently large n .

Proof. Let

$$W_b = (\bar{\Pi}(E_b - H_b)\bar{\Pi})^{-1} \quad (45)$$

where the inverse is the Moore-Penrose pseudo-inverse. In the following, let G_η denote the operator $g_\eta(H/|E^*|)$. We claim that

$$E_b = -1 - b \langle + | G_\eta | + \rangle - b^2 \langle + | G_\eta W_b G_\eta | + \rangle . \quad (46)$$

This is justified as follows. First, define the (non-normalized) state $|U\rangle$ as follows:

$$|U\rangle := |+\rangle + \frac{\bar{\Pi}|\psi_b\rangle}{\langle +|\psi_b\rangle} \implies \langle +|\psi_b\rangle |U\rangle = (|+\rangle\langle +| + \bar{\Pi})|\psi_b\rangle = |\psi_b\rangle \quad (47)$$

which by noting the resolution of the identity $I = |+\rangle\langle +| + \bar{\Pi}$, is seen to be proportional to $|\psi_b\rangle$. Since by definition $|\psi_b\rangle$ has eigenvalue E_b under H_b , we can make the following assertions.

$$H_b |U\rangle = E_b |U\rangle \quad (48)$$

$$H_b \left(|+\rangle + \frac{\bar{\Pi}|\psi_b\rangle}{\langle +|\psi_b\rangle} \right) = E_b \left(|+\rangle + \frac{\bar{\Pi}|\psi_b\rangle}{\langle +|\psi_b\rangle} \right) \quad (49)$$

$$(E_b - H_b) |+\rangle = (H_b - E_b) \bar{\Pi} \frac{|\psi_b\rangle}{\langle +|\psi_b\rangle} \quad (50)$$

$$\bar{\Pi}(E_b - H_b) |+\rangle = \bar{\Pi}(H_b - E_b) \bar{\Pi} \frac{|\psi_b\rangle}{\langle +|\psi_b\rangle} \quad (51)$$

$$\bar{\Pi}(E_b + (X/n) - bG_\eta) |+\rangle = (\bar{\Pi}(H_b - E_b)\bar{\Pi}) \bar{\Pi} \frac{|\psi_b\rangle}{\langle +|\psi_b\rangle} \quad (52)$$

$$-\bar{\Pi}(bG_\eta) |+\rangle = (\bar{\Pi}(H_b - E_b)\bar{\Pi}) \bar{\Pi} \frac{|\psi_b\rangle}{\langle +|\psi_b\rangle} \quad (53)$$

$$-bW_b G_\eta |+\rangle = \frac{\bar{\Pi}|\psi_b\rangle}{\langle +|\psi_b\rangle} , \quad (54)$$

so, referring back to the definition of $|U\rangle$, we have

$$|U\rangle = |+\rangle - bW_b G_\eta |+\rangle . \quad (55)$$

Note also that since $H_b |U\rangle = E_b |U\rangle$, we have

$$E_b = \frac{\langle + | H_b | U \rangle}{\langle + | U \rangle} = \langle + | H_b | U \rangle \quad (56)$$

$$= \langle + | \left(-X/n + bG_\eta \right) \left(|+\rangle - bW_b G_\eta |+\rangle \right) \quad (57)$$

$$= -1 - b \langle + | G_\eta | + \rangle - b^2 \langle + | G_\eta W_b G_\eta | + \rangle . \quad (58)$$

Now, by the short-path assumption, the minimum eigenvalue of $\bar{\Pi}H_b\bar{\Pi}$ is greater than $-1 + 1/n$. Since $E_b \leq -1$, we can assert that the minimum eigenvalue of $\bar{\Pi}(H_b - E_b)\bar{\Pi}$ is greater than $1/n$, and thus W_b has operator norm upper bounded by n . Thus,

$$E_b \geq -1 + b \langle + | G_\eta | + \rangle - b^2 n \langle + | G_\eta^2 | + \rangle . \quad (59)$$

Recall that G_η is a diagonal operator, but by assumption, most of its 2^n diagonal entries are zero. There are at most $2^{(1-\gamma)n}$ strings z for which $G_\eta |z\rangle \neq 0$, and $\langle z| G_\eta |z\rangle \geq -1$ for all z . Since $|+\rangle = 2^{-n/2} \sum_z |z\rangle$, this implies

$$0 \leq -\langle +| G_\eta |+\rangle \leq 2^{-\gamma n} \quad (60)$$

$$0 \leq \langle +| G_\eta^2 |+\rangle \leq 2^{-\gamma n}, \quad (61)$$

which together with Eq. (59) gives the proposition. $\square$

Proposition 7 (log-Sobolev inequality). *Given a state $|\phi\rangle$, let $p : \{+1, -1\}^n \rightarrow [0, 1]$ denote the probability distribution over measurement outcomes when $|\phi\rangle$ is measured in the computational basis, that is*

$$p(z) = |\langle z|\phi\rangle|^2. \quad (62)$$

and let $S = -\sum_z p(z) \log_2(p(z))$ be the base-2 Shannon entropy of p . Then

$$S/n \geq \tau^{-1}(\langle \phi| X/n | \phi \rangle) \quad (63)$$

where

$$\tau^{-1}(y) = H_2\left(\frac{1}{2} - \frac{1}{2}\sqrt{1-y^2}\right) \quad (64)$$

with H_2 denoting the binary entropy function $H_2(q) = -q \log_2(q) - (1-q) \log_2(1-q)$. Moreover, $\tau^{-1}(y) \geq 1 + (y-1)/\ln(2)$, and thus

$$S/n \geq 1 - \ln(2)^{-1}(1 - \langle \phi| X/n | \phi \rangle). \quad (65)$$

Proof. We import the log-Sobolev inequality directly from prior literature. The bound we use originates in Ref. [49], and we state it identically as it appeared in Hastings' short-path analysis [29, Lemma 11]:

$$\tau(S/n) \geq \langle \phi| X/n | \phi \rangle \quad (66)$$

where $\tau(x) := 2\sqrt{H_2^{-1}(x)(1-H_2^{-1}(x))}$ can be verified to be inverse of the τ^{-1} function defined above. This implies Eq. (63). To show the relation $\tau^{-1}(y) \geq 1 - \ln(2)^{-1}(1-y)$, we first show that τ^{-1} is convex by proving that its second derivative is non-negative. Define $q(y) := (1 - \sqrt{1-y^2})/2$ and note that $q(1-q) = y^2/4$.

$$\frac{d^2 \tau^{-1}(y)}{dy^2} = \frac{d^2 H_2(q(y))}{dy^2} = \frac{d^2 H_2(q)}{dq^2} \left(\frac{dq}{dy}\right)^2 + \frac{dH_2(q)}{dq} \frac{d^2 q}{dy^2} \quad (67)$$

$$= -\frac{1}{\ln(2)q(1-q)} \left(\frac{y}{2\sqrt{1-y^2}}\right)^2 + \frac{1}{\ln(2)} \ln\left(\frac{1-q}{q}\right) \left(\frac{1}{2(1-y^2)^{3/2}}\right) \quad (68)$$

$$= -\frac{1}{\ln(2)} \frac{1}{1-y^2} + \frac{1}{\ln(2)} \ln\left(\frac{1+\sqrt{1-y^2}}{1-\sqrt{1-y^2}}\right) \left(\frac{1}{2(1-y^2)^{3/2}}\right) \quad (69)$$

$$\geq 0 \quad (70)$$

where the last line follows using the inequality $\ln((1+x)/(1-x)) \geq 2x$ when $0 \leq x < 1$. Since the second derivative is non-negative on $y \in [0, 1]$ and $\tau^{-1}(1) = 1$, the function satisfies $\tau^{-1}(y) \geq 1 + (y-1)a$, where a is the derivative of the function at $y = 1$. In our case, $a = (\ln(2))^{-1}$. This yields Eq. (65), which was also stated explicitly in Hastings [29, Lemma 10]. $\square$

Proposition 8. *If the short-path condition (Cond. 3) is false for H_b , then there exists a probability distribution over assignments to the cost function $p : \{+1, -1\}^n \rightarrow [0, 1]$ and a number $\mathcal{U}$ for which*

$$\frac{S}{n} \geq 1 + \frac{b}{\ln(2)}\mathcal{U} - \frac{1}{n \ln(2)} \quad (71)$$

$$\text{and} \quad -\frac{1}{bn} \geq \mathcal{U} \geq -1 \quad (72)$$

where

$$\mathcal{U} = \sum_z p(z) g_\eta \left(\frac{H(z)}{|E^*|} \right) \quad \text{is the average energy of } p \text{ for } g_\eta \left(\frac{H}{|E^*|} \right) \quad (73)$$

$$S = \sum_z -p(z) \log_2 p(z) \quad \text{is the entropy of } p \quad (74)$$

Proof. Define short-hand $G_\eta = g_\eta(H/|E^*|)$ so that $H_b = -X/n + b G_\eta$. If the short-path condition is false, then the ground state $|\bar{\psi}_b\rangle$ of $\bar{\Pi} H_b \bar{\Pi}$ is orthogonal to $|+\rangle$ and has energy $\bar{E}_b$ satisfying $\bar{E}_b < -1 + 1/n$. We choose the distribution p to be the one induced by measuring $|\bar{\psi}_b\rangle$ in the computational basis, i.e. $p(z) = |\langle z | \bar{\psi}_b \rangle|^2$. This implies $\mathcal{U} = \langle \bar{\psi}_b | G_\eta | \bar{\psi}_b \rangle$ which is bounded below by -1 . We can then say

$$\bar{E}_b = \langle \bar{\psi}_b | \bar{H}_b | \bar{\psi}_b \rangle = -\langle \bar{\psi}_b | (X/n) | \bar{\psi}_b \rangle + b \langle \bar{\psi}_b | G_\eta | \bar{\psi}_b \rangle = -\langle \bar{\psi}_b | (X/n) | \bar{\psi}_b \rangle + b\mathcal{U} \quad (75)$$

Thus, we have

$$\langle \bar{\psi}_b | (X/n) | \bar{\psi}_b \rangle > 1 - 1/n + b\mathcal{U}. \quad (76)$$

Since $|\bar{\psi}_b\rangle$ has zero overlap with $|+\rangle$, and the second largest eigenvalue of $-X/n$ is $1 - 2/n$, the left-hand-side is bounded above by $1 - 2/n$ and we find that $-1 \leq \mathcal{U} \leq -1/(bn)$. Moreover, using Eq. (65) from Prop. 7 to relate S to the left-hand side, we have

$$\frac{S}{n} > 1 + \frac{b\mathcal{U}}{\ln(2)} + \frac{1}{n \ln(2)} \quad (77)$$

proving the proposition. $\square$

A.1.2 Tools from statistical mechanics

The main technical problem solved in this section is the following question.

Suppose we have a cost function over 2^n assignments and the only information we know about it is that (i) the minimum cost value is -1 , (ii) the maximum cost value is 0 , and (iii) the number of assignments with nonzero cost value is at most $2^{(1-\gamma)n}$, where $0 < \gamma < 1$ is a known number. If p is a probability distribution over assignments with average cost equal to $\mathcal{U}$, with $-1 < \mathcal{U} < 0$, what is an upper bound on the entropy of p ?

One difficult aspect of this question is that we know that there are at most $2^{(1-\gamma)n}$ assignments with negative cost value, but we have no additional information about how these costs are distributed on the interval $[-1, 0)$, and we want an upper bound on the entropy that will hold regardless of the distribution. To answer this question, we first review some elementary machinery from statistical mechanics culminating in Prop. 10, which gives a simple condition that allows us to say that one stat mech system has a larger entropy than another. This proposition is then used in the proof of Prop. 11, where it is shown that the entropy-per-site S/n is upper bounded by $1 + \gamma\mathcal{U} + O(1/n)$. Together with Prop. 8, this implies that a tail bound implies the short-path condition as long as b is sufficiently small.

Discrete systems Let us recall basic stat mech definitions. A discrete system with Ω total states is defined by its energy levels $\{E_j\}_{j=0}^{\Omega-1}$ (which we assume are listed in non-decreasing order). A probability distribution over the Ω states is given by a function $p : \{0, 1, 2, \dots, \Omega-1\} \rightarrow [0, 1]$ that assigns each state a positive real number, such that $\sum_{j=0}^{\Omega-1} p(j) = 1$. The average energy of the distribution is

$$U = \sum_{j=0}^{\Omega-1} p(j) E_j \quad (78)$$

and the entropy is

$$S = - \sum_{j=0}^{\Omega-1} p(j) \log_2(p(j)). \quad (79)$$

Note that we define the entropy with logarithm base-2 to maintain consistency with the statement of the log-Sobolev inequality in Prop. 7, rather than following the standard stat mech convention of using natural logarithms. Standard results of statistical mechanics dictate that for any fixed choice of $\mathcal{U}$ satisfying $E_0 < \mathcal{U} < E_{\Omega-1}$, there exists a distribution p with average energy $U = \mathcal{U}$, and among all such distributions, the one that maximizes S is uniquely given by $p(j) = e^{-\beta E_j} / Z(\beta)$ for some real, finite value of β , with $Z(\beta)$ the *partition function*, given by

$$Z(\beta) = \sum_{j=0}^{\Omega-1} e^{-\beta E_j}. \quad (80)$$

This is the *Gibbs distribution* or *Boltzmann distribution* and, physically, β represents the inverse temperature (which can be negative). For the remainder of this section, we always assume systems are in their Gibbs distribution for a certain inverse temperature β . We may then view U and S as a function of β . Note the following standard identities

$$U(\beta) = - \frac{1}{Z(\beta)} \frac{dZ(\beta)}{d\beta} \quad (81)$$

$$S(\beta) = \ln(2)^{-1} [\ln(Z(\beta)) + \beta U(\beta)] \quad (82)$$

Cumulative state function We can define a cumulative state function $C : (-\infty, \infty) \rightarrow [0, \Omega]$ by

$$C(E) := |\{j : E_j \leq E\}| = \sum_{j=0}^{\Omega-1} \mathbb{1}(E_j \leq E), \quad (83)$$

where $\mathbb{1}$ denotes the indicator function. Note that $C(E)$ has a discontinuity at each value of E that appears in the list of energies $\{E_j\}_{j=0}^{\Omega-1}$, where it steps up by an integer equal to the multiplicity of E in the sequence. The cumulative state function $C(E)$ contains all of the information about a stat mech system, and we may rewrite the partition function and internal energy directly in terms of C , as shown in the following proposition.

Proposition 9. For a system with Ω states, we can rewrite $Z(\beta)$ and $U(\beta)$ in terms of the cumulative state functions as follows:

$$Z(\beta) = \begin{cases} \beta \int_{-\infty}^{\infty} dE C(E) e^{-\beta E} & \text{if } \beta > 0 \\ C(\infty) & \text{if } \beta = 0 \\ \beta \int_{-\infty}^{\infty} dE (C(E) - C(\infty)) e^{-\beta E} & \text{if } \beta < 0 \end{cases} \quad (84)$$

$$U(\beta) = \begin{cases} Z(\beta)^{-1} \int_{-\infty}^{\infty} dE C(E) (\beta E - 1) e^{-\beta E} & \text{if } \beta > 0 \\ C(\infty)^{-1} \left(- \int_{-\infty}^0 dE C(E) + \int_0^{\infty} dE (C(\infty) - C(E)) \right) & \text{if } \beta = 0 \\ Z(\beta)^{-1} \int_{-\infty}^{\infty} dE (C(E) - C(\infty)) (\beta E - 1) e^{-\beta E} & \text{if } \beta < 0 \end{cases} \quad (85)$$

Proof. When $\beta = 0$, we see from the definition of Z in Eq. (80) that $Z(0) = \Omega = C(\infty)$. Meanwhile, from the definition of U in Eq. (78),

$$U(0) = \frac{1}{Z(0)} \sum_{j=0}^{\Omega-1} E_j = \frac{1}{C(\infty)} \sum_{j=0}^{\Omega-1} \left(- \int_{\min(E_j, 0)}^0 dE + \int_0^{\max(0, E_j)} dE \right) \quad (86)$$

$$= \frac{1}{C(\infty)} \sum_{j=0}^{\Omega-1} \left(- \int_{-\infty}^0 \mathbb{1}(E_j \leq E) dE + \int_0^{\infty} (1 - \mathbb{1}(E_j \leq E)) dE \right) \quad (87)$$

$$= \frac{1}{C(\infty)} \left(- \int_{-\infty}^0 \sum_{j=0}^{\Omega-1} \mathbb{1}(E_j \leq E) dE + \int_0^{\infty} \sum_{j=0}^{\Omega-1} (1 - \mathbb{1}(E_j \leq E)) dE \right) \quad (88)$$

$$= \frac{1}{C(\infty)} \left(- \int_{-\infty}^0 dE C(E) + \int_0^{\infty} dE (C(\infty) - C(E)) \right) \quad (89)$$

When $\beta > 0$, we can rewrite

$$Z(\beta) = \sum_{j=0}^{\infty} e^{-\beta E_j} = \beta \sum_{j=0}^{\Omega-1} \int_{E_j}^{\infty} dE e^{-\beta E} = \beta \sum_{j=0}^{\Omega-1} \int_{-\infty}^{\infty} dE \mathbb{1}(E_j \leq E) e^{-\beta E} \quad (90)$$

$$= \beta \int_{-\infty}^{\infty} dE \sum_{j=0}^{\Omega-1} \mathbb{1}(E_j \leq E) e^{-\beta E} = \beta \int_{-\infty}^{\infty} dE C(E) e^{-\beta E}. \quad (91)$$

and

$$U(\beta) = \sum_{j=0}^{\infty} E_j e^{-\beta E_j} = \sum_{j=0}^{\Omega-1} \int_{E_j}^{\infty} dE (\beta E_j - 1) e^{-\beta E} = \sum_{j=0}^{\Omega-1} \int_{-\infty}^{\infty} dE \mathbb{1}(E_j \leq E) (\beta E_j - 1) e^{-\beta E} \quad (92)$$

$$= \int_{-\infty}^{\infty} dE \sum_{j=0}^{\Omega-1} \mathbb{1}(E_j \leq E) (\beta E_j - 1) e^{-\beta E} = \int_{-\infty}^{\infty} dE C(E) (\beta E - 1) e^{-\beta E}. \quad (93)$$

The formulas for $\beta < 0$ can be derived directly from the formulas for $\beta > 0$. Define a new stat mech system, denoted with a prime on each symbol, by negating all of the energies, i.e. $E'_j = -E_j$. We have $C'(E) = C(\infty) - C(E)$. Moreover, the primed system at inverse temperature β' is equivalent to the original system at inverse temperature β when $\beta = -\beta'$. Thus, negative β corresponds to positive β' and the same formulas apply with the substitution $C(E) \rightarrow C'(E) = C(\infty) - C(E)$. $\square$

So far, we have restricted to $C(E)$ that come from a system with a discrete set of energy levels $\{E_j\}$. We now generalize slightly and consider a wider class of $C(E)$ which need not always correspond to a system with discrete energy levels.

Definition 3 (finite-stepping). *We say that a cumulative state function $C(E)$ has the finite-stepping property if it (i) is monotonically non-decreasing, (ii) has a finite number of discontinuities, and (iii) takes a constant value in between consecutive discontinuities.*

Thus, when $C(E)$ comes from a discrete system of Ω energy levels, it will have the finite-stepping property and additionally each step size will be an integer; however, other functions for which the step size is non-integer also have the finite-stepping property.

The equations for $Z(\beta)$ and $U(\beta)$ in Prop. 9 are well-defined for any function $C(E)$ with the finite-stepping property. We will also extend the definition of $S(\beta)$ to all functions $C(E)$ with the finite-stepping property by continuing to impose the identity in Eq. (82).

Comparing the entropy of different systems We show a condition under which one can assert that one stat mech system has larger entropy than another for the same average energy.

Proposition 10. *Consider two statistical mechanical systems defined by their cumulative state functions $C_1(E)$ and $C_2(E)$, and assume that both have the finite-stepping property. Let $U_1(\beta)$ and $U_2(\beta)$ denote their average energy functions. Given a fixed choice of $\mathcal{U} \in \mathbb{R}$, let β_1 and β_2 be the solutions to $U_1(\beta_1) = \mathcal{U}$ and $U_2(\beta_2) = \mathcal{U}$, which we assume exist. If for every real value of E , it holds that $C_1(E) \geq C_2(E)$ and $C_1(\infty) - C_1(E) \geq C_2(\infty) - C_2(E)$, then*

$$S_1(\beta_1) \geq S_2(\beta_2), \quad (94)$$

where $S_1(\beta_1)$ and $S_2(\beta_2)$ denote the entropy of each system at fixed internal energy $\mathcal{U}$.

Proof. For $j \in \{1, 2\}$, let A_j (B_j) be the minimum (maximum) argument of C_j where it has a discontinuity. Note that $U_j(\beta)$ is a monotonically decreasing and continuous function of β and that

$$\lim_{\beta \rightarrow \infty} U_j(\beta) = A_j \quad (95)$$

$$\lim_{\beta \rightarrow -\infty} U_j(\beta) = B_j. \quad (96)$$

Thus, since we have assumed $U_j(\beta_j) = \mathcal{U}$ has a solution, it must be the case that $A_j \leq \mathcal{U} \leq B_j$ for $j \in \{1, 2\}$. The limits above correspond to where all the probability mass is concentrated on the lowest and highest energy levels, respectively, and the statement can be proved by evaluation of Eqs. (84) and (85) in the $\beta \rightarrow \pm\infty$ limits.

We now consider a one-parameter family of systems that interpolates between system 1 and system 2, with cumulative states function

$$C(E, t) = (1 - t)C_1(E) + tC_2(E) \quad (97)$$

Let $U(\beta, t)$ denote its average energy function and let $S(\beta, t)$ denote its entropy function. Since $C_1(E)$ and $C_2(E)$ have the finite-stepping property, it is easy to verify that $C(E, t)$ also has the finite-stepping property for any $t \in [0, 1]$. Moreover, if $A(t)$ ($B(t)$) is the minimum (maximum) argument of $C(\cdot, t)$ for which it has a discontinuity, we can say that $A(t) \leq \min_j A_j$ and $B(t) \geq \max_j B_j$. Thus $A(t) \leq \mathcal{U} \leq B(t)$ and for every t , there will exist a value $\beta^*(t)$ for which $U(\beta^*(t), t) = \mathcal{U}$. That is, as the cumulative states function changes with t , β^* will also change with t such that the average energy remains fixed at $\mathcal{U}$.

Starting from Eq. (82), we now compute the total derivative of S with respect to t , for fixed average energy:

$$\ln(2) \frac{dS(\beta^*(t), t)}{dt} = \frac{1}{Z(\beta^*(t), t)} \frac{dZ(\beta^*(t), t)}{dt} + \mathcal{U} \frac{d\beta^*(t)}{dt} \quad (98)$$

$$= \frac{1}{Z(\beta^*(t), t)} \frac{\partial Z(\beta^*(t), t)}{\partial t} + \left(\frac{1}{Z(\beta^*(t), t)} \frac{\partial Z(\beta^*(t), t)}{\partial \beta} + \mathcal{U} \right) \frac{d\beta^*(t)}{dt}. \quad (99)$$

From Eq. (81), we see that the quantity in parentheses vanishes, and we have

$$\frac{dS(\beta^*(t), t)}{dt} = \frac{1}{\ln(2)Z(\beta^*(t), t)} \frac{\partial Z(\beta^*(t), t)}{\partial t} \quad (100)$$

Using Eq. (84) and assuming $\beta > 0$, this yields

$$\frac{dS(\beta^*(t), t)}{dt} = \frac{\beta}{\ln(2)Z(\beta^*(t), t)} \int_{-\infty}^{\infty} dE \frac{\partial C(E, t)}{\partial t} e^{-\beta^*(t)E} \quad (101)$$

$$= \frac{\beta}{\ln(2)Z(\beta^*(t), t)} \int_{-\infty}^{\infty} dE (C_2(E) - C_1(E)) e^{-\beta E} \quad (102)$$

$$\leq 0 \quad (103)$$

since, by assumption, $C_1(E) \geq C_2(E)$ for all E . Assuming $\beta < 0$, it yields

$$\frac{dS(\beta^*(t), t)}{dt} = \frac{\beta}{\ln(2)Z(\beta^*(t), t)} \int_{-\infty}^{\infty} dE \left(\frac{\partial C(E, t)}{\partial t} - \frac{\partial C(\infty, t)}{\partial t} \right) e^{-\beta^*(t)E} \quad (104)$$

$$= \frac{\beta}{\ln(2)Z(\beta^*(t), t)} \int_{-\infty}^{\infty} dE [(C_1(\infty) - C_1(E)) - (C_2(\infty) - C_2(E))] e^{-\beta E} \quad (105)$$

$$\leq 0 \quad (106)$$

since, by assumption, $C_1(\infty) - C_1(E) \geq C_2(\infty) - C_2(E)$ (recall that $\beta < 0$).

Since the derivative of S with t is always non-positive, we have $S(\beta^*(1), 1) \leq S(\beta^*(0), 0)$, which is equivalent to $S_2(\beta_2) \leq S_1(\beta_1)$. $\square$

A.1.3 Tail bound implies short-path condition

Proposition 11. *Let H be a cost function on n bits with optimal value E^* and let $C(E)$ be its cumulative state function*

$$C(E) = |\{z : H(z) \leq E\}|. \quad (107)$$

Fix a value of η and b , which defines the Hamiltonian H_b in Eq. (5). If $C(E)$ obeys a tail bound $C(E^(1 - \eta)) \leq 2^{(1-\gamma)n}$, and*

$$b \leq \frac{\ln(2)\gamma}{2 + \ln(2)} \quad (108)$$

then H_b satisfies the short-path condition (Cond. 3).

Proof. The function $C(E)$ is the cumulative state function for the cost function H . Define $C_\eta(v)$ to be the cumulative state function for the cost function $g_\eta(H/|E^*|)$. Recalling that $g_\eta(x) = \min(0, (x + 1 - \eta)/\eta)$, we can say that

$$C_\eta(v) = \begin{cases} 0 & \text{if } v < -1 \\ C(E^*(1 - \eta v - \eta)) \leq 2^{(1-\gamma)n} & \text{if } -1 \leq v < 0 \\ 2^n & \text{if } v \geq 0, \end{cases} \quad (109)$$

and note that, because it is associated with a discrete system, $C_\eta(v)$ has the finite-stepping property defined in Def. 3. Now consider an alternate system which has $2^{(1-\gamma)n}$ states at energy -1 , and 2^n states at energy 0 . Denote quantities related to this system with an overline, and define its cumulative states function by

$$\overline{C}(v) = \begin{cases} 0 & \text{if } v < -1 \\ 2^{(1-\gamma)n} & \text{if } -1 \leq v < 0 \\ 2^{(1-\gamma)n} + 2^n & \text{if } v \geq 0. \end{cases} \quad (110)$$

The function $\overline{C}$ is constant except at two discontinuities where it steps up by a finite amount; thus it also has the finite-stepping property. Inspecting the above two expressions, we see that, due to the assumption $C(E^*(1-\eta)) \leq 2^{(1-\gamma)n}$, we have $C_\eta(v) \leq \overline{C}(v)$. We also see that, by construction, $\overline{C}(\infty) - \overline{C}(v) \geq 2^n$ whenever $v < 0$, and thus $C_\eta(\infty) - C_\eta(v) \leq \overline{C}(\infty) - \overline{C}(v)$ for all v . Hence, the requirements of Prop. 10 are satisfied.

Let $S(\mathcal{U})$ denote the maximum entropy of the system described by cumulative state function $C_\eta(\cdot)$ for fixed expected cost value (average energy) of $g_\eta(H/|E^*|)$ equal to $\mathcal{U}$. Let $\overline{S}(\mathcal{U})$ denote the same for the cumulative state function $\overline{C}(\cdot)$. Prop. 10 implies that

$$S(\mathcal{U}) \leq \overline{S}(\mathcal{U}) \quad (111)$$

for all $\mathcal{U}$ in which there exists a distribution for both systems with average energy $\mathcal{U}$, i.e. when $-1 \leq \mathcal{U} \leq 0$.

The entropy of the $\overline{C}$ system can be analyzed exactly. We compute the partition function

$$\overline{Z}(\beta) = \beta \int_{-1}^{\infty} dv \overline{C}(v) e^{-\beta v} = 2^{(1-\gamma)n} e^\beta + 2^n \quad (112)$$

which, via Eq. (81), determines the average energy

$$\overline{U}(\beta) = -\frac{1}{\overline{Z}(\beta)} \frac{d\overline{Z}(\beta)}{d\beta} = -\frac{2^{(1-\gamma)n} e^\beta}{2^{(1-\gamma)n} e^\beta + 2^n}. \quad (113)$$

For a given $\mathcal{U}$ satisfying $-1 \leq \mathcal{U} \leq 0$, we choose $\overline{\beta}$ such that $\overline{U}(\overline{\beta}) = \mathcal{U}$, giving the relation

$$e^{\overline{\beta}} = -\frac{2^{\gamma n} \mathcal{U}}{1 + \mathcal{U}} \quad (114)$$

$$\implies \overline{\beta} = \gamma n \ln(2) + \ln\left(\frac{-\mathcal{U}}{1 + \mathcal{U}}\right). \quad (115)$$

Plugging this value of $\beta = \overline{\beta}$ in to Eq. (112), as well as the definition of entropy in Eq. (82), we find:

$$\overline{Z} = \frac{2^n}{1 + \mathcal{U}} \quad (116)$$

$$\overline{S}(\mathcal{U}) = \ln(2)^{-1} [\ln(\overline{Z}) + \overline{\beta} \mathcal{U}] \quad (117)$$

$$= n - \log_2(1 + \mathcal{U}) + \gamma n \mathcal{U} - (-\mathcal{U}) \log_2(-\mathcal{U}) + (-\mathcal{U}) \log_2(1 + \mathcal{U}) \quad (118)$$

$$= n(1 + \gamma \mathcal{U}) + H_2(-\mathcal{U}) \quad (119)$$

where $H_2(q) = -q \log_2(q) - (1-q) \log_2(1-q)$ is the binary entropy. Note that $H_2(q) \leq 1$ for all q .

By assumption, we have b satisfies $b \leq \frac{\ln(2)\gamma}{2+\ln(2)}$. Thus, we can say

$$\frac{S(\mathcal{U})}{n} \leq \frac{\bar{S}(\mathcal{U})}{n} = 1 + \gamma\mathcal{U} + \frac{H_2(-\mathcal{U})}{n} \leq 1 + \gamma\mathcal{U} + \frac{1}{n} \leq 1 + \frac{(2+\ln(2))b}{\ln(2)}\mathcal{U} + \frac{1}{n} \quad (120)$$

$$= \left[1 + \frac{b}{\ln(2)}\mathcal{U} - \frac{1}{n\ln(2)} \right] + [(b\mathcal{U} + 1/n)(1 + \ln(2))^{-1}] \quad (121)$$

Thus, whenever $-1 \leq \mathcal{U} \leq -\frac{1}{bn}$, the final term is negative and $S(\mathcal{U})/n \leq 1 + \ln(2)^{-1}b\mathcal{U} - (n\ln(2))^{-1}$. By Prop. 8, this implies that the short-path condition must hold. $\square$

A.1.4 Proof of Lemma 5

Proof of Lemma 5. We assume that a tail bound of the form $C(E^*(1-\eta)) \leq 2^{(1-\gamma)n}$ holds and that $b \leq \ln(2)\gamma/(2+\ln(2))$. By Prop. 11, this means that H_b obeys the short-path condition. By Prop. 5, the short-path condition implies the large-excited-energy condition (Cond. 1). By Prop. 6, the short-path condition (Cond. 3) combined with the tail bound implies that $E_b \geq -1 - 2^{-\gamma n}(b + b^2n)$. Noting $b < 1$ and the assumption that $2^{\gamma n} \geq 2n^4 \geq n^3(b + b^2n)$, we have that $E_b \geq -1 - 1/n^3$, giving the small-ground-energy-shift condition (Cond. 2) and proving the lemma. $\square$

A.2 α -(sub)depolarizing

Proof of Proposition 3. We wish to show (H, g_η) is subdepolarizing. Let $f(x) = -g_\eta(-x)$, and note that f is monotonically non-decreasing and convex. Note also that it is twice differentiable at every place where it is nonzero. The cost function H can be written as $H = \sum_{j=1}^m \mathcal{C}_j$ for m different constraints $\mathcal{C}_j$ each depending on at most k variables. Recall that each constraint takes value -1 for some integral number s_j of the 2^k settings of these k variables, and it takes value $s_j/(2^k - s_j)$ on the other $2^k - s_j$ settings, with $0 \leq s_j < 2^k$. Given an assignment x , let $\mathcal{S} \subset [M]$ be the subset of constraints that x satisfies (i.e., $j \in \mathcal{S}$ if $\mathcal{C}_j(x) = -1$). Suppose that y is generated by flipping one bit of x at random. For each $j \in \mathcal{S}$, there is a $(n-k)/n$ chance that the flipped bit is not involved in constraint j and $\mathcal{C}_j(y) = \mathcal{C}_j(x) = -1$. Meanwhile, there is a k/n chance that the flipped bit is involved, and the clause could become unsatisfied, but even in this case, we can at least say that $\mathcal{C}_j(y) \leq 2^k - 1$, as this is an upper bound for the maximum value the constraint can take on any input (achieved when $s_j = 2^k - 1$). Thus, for any $j \in \mathcal{S}$,

$$\mathbb{E}_{y \sim x} \mathcal{C}_j(y) \leq \frac{n-k}{n}(-1) + \frac{k}{n}(2^k - 1) = -1 + \frac{k2^k}{n} = \mathcal{C}_j(x) + \frac{k2^k}{n} \quad (122)$$

For each $j \notin \mathcal{S}$, the value of $\mathcal{C}_j(x)$ is already the maximum value of $\mathcal{C}_j$, so in this case, we can assert that $\mathbb{E}_{y \sim x} \mathcal{C}_j(y) \leq \mathcal{C}_j(x)$. Overall, this gives

$$\mathbb{E}_{y \sim x} H(y) \leq H(x) + |\mathcal{S}| \frac{k2^k}{n} \leq H(x) + m \frac{k2^k}{n}. \quad (123)$$

Let $c_1, \dots, c_T$ be non-negative constants less than 1. We can then say the following, where the first step uses Jensen's inequality and the fact that $\prod_{t=1}^T f(c_t x)$ is a convex function from Proposition 12, while the second step uses the monotonicity of f along with Eq. (123) (recalling that $E^* < 0$).

$$\mathbb{E}_{y \sim x} \prod_{t=1}^T f\left(\frac{c_t H(y)}{E^*}\right) \geq \prod_{t=1}^T f\left(c_t \mathbb{E}_{y \sim x} \frac{H(y)}{E^*}\right) \geq \prod_{t=1}^T f\left(\frac{c_t H(x)}{E^*} + \frac{c_t m k 2^k}{E^* n}\right) \quad (124)$$

$$= \prod_{t=1}^T f\left(\frac{c_t H(x)}{E^*} \left(1 + \frac{m k 2^k}{H(x) n}\right)\right) \quad (125)$$

If $H(x) \geq E^*(1 - \eta)/c_t$ for at least one value of t , then $0 = \prod_{t=1}^T f(c_t H(x)/E^*) = \prod_{t=1}^T f(c_t(1 - \alpha)H(x)/E^*)$, and $\mathbb{E}_{y \sim x} \prod_{t=1}^T f(c_t H(y)/E^*) \geq \prod_{t=1}^T f(c_t(1 - \alpha)H(x)/E^*)$ is true. On the other hand, if $H(x) \leq E^*(1 - \eta)/c_t$ for all t , then we can certainly say that $H(x) \leq E^*(1 - \eta)$ and hence $1/H(x) \geq -1/|E^*|(1 - \eta)$ and by substitution in Eq. (125) and the fact that f is monotonic, we have

$$\mathbb{E}_{y \sim x} \prod_{t=1}^T f\left(\frac{c_t H(y)}{E^*}\right) \geq \prod_{t=1}^T f\left(\frac{c_t H(x)}{E^*} \left(1 - \frac{m}{|E^*|} \frac{1}{(1 - \eta)} \frac{k 2^k}{n}\right)\right) \quad (126)$$

which proves the lemma. $\square$

Proposition 12. *Suppose $f : (-\infty, 1] \rightarrow [0, 1]$ is a monotonically non-decreasing, convex function. Suppose that f is twice differentiable for every x in which $f(x) \neq 0$. Then for any non-negative constants $c_1, \dots, c_m$, the function $h(x) = \prod_{t=1}^m f(c_t x)$ is also a convex function.*

Proof. Since f is non-negative and monotonically non-decreasing, if $h(y) = 0$, then $h(x) = 0$ for all $x < y$ as well. If $h(x) \neq 0$, then h is twice-differentiable at x , since it is the product of twice-differentiable functions. To show that h is convex, it suffices to show that the second derivative of h is non-negative for all points x in which $h(x)$ is non-zero.

$$h''(x) = \sum_t c_t^2 f''(c_t x) \frac{h(x)}{f(c_t x)} + \sum_{t_1} \sum_{t_2 \neq t_1} c_{t_1} c_{t_2} f'(c_{t_1} x) f'(c_{t_2} x) \frac{h(x)}{f(c_{t_1} x) f(c_{t_2} x)} \quad (127)$$

We observe that $h''(x)$ is always non-negative, since $c_t, h(x), f(c_t x), f''(c_t x), f'(c_t x) \geq 0$ for all t and all x . $\square$

Proposition 13. *If H has the α -depolarizing property (Definition 1) and $g : [-1, \infty) \rightarrow [-1, 0]$ is a monotonic non-decreasing, concave function that is twice-differentiable at every point where it is nonzero, then (H, g) has the α -subdepolarizing property.*

Proof. This follows from application of Jensen's inequality and Prop. 12. Let $f(x) = -g(-x)$ and note that f satisfies the constraints of Prop. 12. The proposition implies that for any non-negative constants $c_1, \dots, c_T$

$$\mathbb{E}_{y \sim x} \prod_{t=1}^T f(c_t H(y)/E^*) \geq \prod_{t=1}^T f\left(c_t \mathbb{E}_{y \sim x} H(y)/E^*\right) = \prod_{t=1}^T f(c_t(1 - \alpha)H(x)/E^*) \quad (128)$$

where the second equality results from invoking the α -depolarizing property. This proves the proposition. $\square$

A.3 Overlap lower bound with approximate ground state projector

Proof of Lemma 3. First, let $\mathcal{E} = H(z)/|E^*|$ and recall the assumption $H(z) \leq (1 - \eta)E^*$. Thus $\mathcal{E} < 0$. Next, define A , B , and f by the following equations.

$$A = X/n \quad (129)$$

$$B = -b g_\eta(H/|E^*|) = b f(H/E^*) \quad (130)$$

such that, referring to the definition of H_b in Eq. (5), $H_b = -A - B$ and thus

$$P_\ell = \frac{(A + B)^\ell}{|E_b|^\ell}. \quad (131)$$

When the small-ground-energy-shift condition (Cond. 2) holds, the denominator of the above expression is no larger than $(1 + 1/n^3)^\ell \leq e$, where the inequality holds since $\ell < n^3$ by assumption. If we expand the numerator, we get a sum over 2^ℓ strings, where each string is a length- ℓ sequence of A and B operators (note that A and B do not commute). We will evaluate this sum by computing $\langle + | \sigma | z \rangle$ where σ is a string of A and B operators. To get a feel for this, we start simple and compute

$$\langle + | A | z \rangle = 2^{-n/2} \quad (132)$$

$$\langle + | B | z \rangle = b 2^{-n/2} f(|\mathcal{E}|). \quad (133)$$

Here we have utilized the eigenvalue equations $\langle + | A = \langle + |$ and $B | z \rangle = b f(H(z)/E^*) | z \rangle$, and the inner product $\langle + | z \rangle = 2^{-n/2}$.

The first non-trivial computation is $\langle + | BA | z \rangle$. We evaluate this by noting that applying A to $| z \rangle$ yields a uniform superposition over the n computational basis states that differ from z by one bit flip. As in the definition of α -subdepolarizing (Definition 2), we use the notation $y \sim z$ to denote that y is generated from z by choosing one of these n neighboring assignments at random. We have

$$\langle + | BA | z \rangle = b \mathbb{E}_{y \sim z} \langle + | f\left(\frac{H}{E^*}\right) | y \rangle = b 2^{-n/2} \mathbb{E}_{y \sim z} f\left(\frac{H(y)}{E^*}\right) \quad (134)$$

$$\geq b 2^{-n/2} f\left(\frac{H(z)}{E^*}(1 - \alpha)\right) = b f(|\mathcal{E}|(1 - \alpha)) 2^{-n/2} \quad (135)$$

where the inequality uses the α -subdepolarizing property. Similar logic yields the relation

$$\langle + | BA^k | z \rangle \geq b f(|\mathcal{E}|(1 - \alpha)^k) 2^{-n/2} \quad (136)$$

for any k , by applying the α -subdepolarizing property k times.

We can also consider

$$\langle + | BABA | z \rangle = b^2 2^{-n/2} \mathbb{E}_{y \sim z} \left[f\left(\frac{H(y)}{E^*}\right) \mathbb{E}_{w \sim y} f\left(\frac{H(w)}{E^*}\right) \right] \quad (137)$$

$$\geq b^2 2^{-n/2} \mathbb{E}_{y \sim z} \left[f\left(\frac{H(y)}{E^*}\right) f\left(\frac{H(y)(1 - \alpha)}{E^*}\right) \right] \quad (138)$$

$$\geq b^2 2^{-n/2} f\left(\frac{H(z)(1 - \alpha)}{E^*}\right) f\left(\frac{H(z)(1 - \alpha)^2}{E^*}\right) \quad (139)$$

$$= b^2 2^{-n/2} f(|\mathcal{E}|(1 - \alpha)) f(|\mathcal{E}|(1 - \alpha)^2) \quad (140)$$

where the second-to-last line again uses the α -subdepolarizing property.

In general we may write any string of A s and B s as

$$\dots AB^{x_3} AB^{x_2} AB^{x_1} AB^{x_0} \quad (141)$$

and thus associate every string σ with a sequence of non-negative integers $\sigma = (x_0, x_1, \dots)$. Let this sequence be infinitely long by padding it with an infinite number of zeros (note that no two distinct strings σ and σ' both of length ℓ can map to the same sequence of integers). By generalizing the calculations above, we can show the following proposition.

Proposition 14. *If (H, g_η) is α -subdepolarizing and $B = -b g_\eta(H/|E^*|) =: b f(H/E^*)$, then*

$$2^{n/2} \langle + | \dots AB^{x_3} AB^{x_2} AB^{x_1} AB^{x_0} | z \rangle \geq b \sum_{j=0}^{\infty} x_j \prod_{j=0}^{\infty} f(|\mathcal{E}|(1-\alpha)^j)^{x_j} =: w(\sigma) \quad (142)$$

where the right-hand side of the above equation defines $w(\sigma)$, with $\sigma = (x_0, x_1, \dots)$.

The proof of Prop. 14 appears after the conclusion of this proof. Note that if $\sigma = \sigma_1 + \sigma_2$ (with entry-wise addition at each location in the sequence), then $w(\sigma) = w(\sigma_1)w(\sigma_2)$. We can now compute a sum over $w(\sigma)$ for all σ .

Proposition 15. *Let Γ be the set of all sequences σ that have non-negative entries. If $|\mathcal{E}| \geq 1 - \eta$, then*

$$\sum_{\sigma \in \Gamma} w(\sigma) \geq \exp \left(\frac{b|\mathcal{E}|}{\eta\alpha} F \left(\frac{1-\eta}{|\mathcal{E}|} \right) \right) \quad (143)$$

where the function $F(x) := 1 - x + x \ln(x)$, as defined previously in Eq. (15).

The proof of Prop. 15 appears after the conclusion of this proof. The sum computed in Prop. 15 includes all strings, but we only want to sum over strings that appear in the expansion of $(A+B)^\ell$ for a particular choice of ℓ . Let the set of sequences that appear in this expansion be denoted Λ_ℓ . We have

$$\sum_{\sigma \in \Lambda_\ell} w(\sigma) = \sum_{\sigma \in \Gamma} w(\sigma) - \sum_{\sigma \notin \Lambda_\ell} w(\sigma). \quad (144)$$

The following proposition upper bounds the final term.

Proposition 16.

$$\sum_{\sigma \notin \Lambda_\ell} w(\sigma) \leq b|\mathcal{E}|(1-\alpha)^{\ell+1} \left(\frac{e^{2/\alpha}}{1-\alpha-b} + \alpha^{-1} \sum_{\sigma \in \Gamma} w(\sigma) \right). \quad (145)$$

The proof of Prop. 16 appears after the conclusion of this proof. Using Prop. 16, Prop. 15 and Eq. (144) we find that

$$\sum_{\sigma \in \Lambda_\ell} w(\sigma) \geq \sum_{\sigma \in \Gamma} w(\sigma) (1 - b|\mathcal{E}|\alpha^{-1}(1-\alpha)^{\ell+1}) - b|\mathcal{E}|(1-\alpha)^{\ell+1}(1-\alpha-b)^{-1}e^{2/\alpha} \quad (146)$$

$$\geq \exp \left(\frac{b|\mathcal{E}|}{\alpha\eta} F \left(\frac{1-\eta}{|\mathcal{E}|} \right) \right) \left(1 - b|\mathcal{E}| \frac{(1-\alpha)^{\ell+1}}{\alpha} \right) - \exp \left(\frac{2}{\alpha} \right) \frac{b|\mathcal{E}|(1-\alpha)^{\ell+1}}{1-\alpha-b} \quad (147)$$

Now, using the assumption $\ell > 3/\alpha^2$, we note that $(1-\alpha)^\ell \leq e^{-3/\alpha}$, and using the assumption $\alpha < (1-b)/2$, we have $1-\alpha-b > \alpha$. Further noting that

$$1 \leq \exp \left(\frac{b|\mathcal{E}|}{\alpha\eta} F \left(\frac{1-\eta}{|\mathcal{E}|} \right) \right) \quad (148)$$

allows us to rewrite Eq. (147) as

$$\sum_{\sigma \in \Lambda_\ell} w(\sigma) \geq \exp \left(\frac{b|\mathcal{E}|}{\alpha\eta} F \left(\frac{1-\eta}{|\mathcal{E}|} \right) \right) \left(1 - b|\mathcal{E}|(1-\alpha)e^{-3/\alpha}/\alpha - b|\mathcal{E}|(1-\alpha)e^{-1/\alpha}/\alpha \right). \quad (149)$$

Finally noting that $e^{-1/\alpha}/\alpha \leq 1/e$ while $b, |\mathcal{E}|, (1-\alpha) < 1$, the final term in parentheses is greater than $1 - 2e$. Recalling the observation that the denominator in Eq. (131) is bounded above by e , the lemma statement follows. $\square$

Proof of Prop. 14. Let t be the largest index for which $x_t \neq 0$. Following the examples described above the proof of Lemma 3, we can say that

$$2^{n/2} \langle + | \sigma | z \rangle = b^{\sum_{j=0}^t x_j} \mathbb{E}_{z_1 \sim z} \mathbb{E}_{z_2 \sim z_1} \cdots \mathbb{E}_{z_t \sim z_{t-1}} \left[\prod_{j=0}^t f \left(\frac{H(z_j)}{E^*} \right)^{x_j} \right]. \quad (150)$$

Invoking the α -subdepolarizing property on the $\mathbb{E}_{z_t \sim z_{t-1}}$ expectation allows the simplification

$$2^{n/2} \langle + | \sigma | z \rangle \geq b^{\sum_{j=0}^t x_j} \mathbb{E}_{z_1 \sim z} \cdots \mathbb{E}_{z_{t-1} \sim z_{t-2}} \left[f \left(\frac{H(z_{t-1})}{E^*} (1 - \alpha) \right)^{x_t} \prod_{j=0}^{t-1} f \left(\frac{H(z_j)}{E^*} \right)^{x_j} \right]. \quad (151)$$

Invoking it again on the $\mathbb{E}_{z_{t-1} \sim z_{t-2}}$ allows the expectation values on the right-hand-side to be simplified to

$$\mathbb{E}_{z_1 \sim z} \cdots \mathbb{E}_{z_{t-2} \sim z_{t-3}} \left[f \left(\frac{H(z_{t-2})}{E^*} (1 - \alpha)^2 \right)^{x_t} f \left(\frac{H(z_{t-2})}{E^*} (1 - \alpha) \right)^{x_{t-1}} \prod_{j=0}^{t-2} f \left(\frac{H(z_j)}{E^*} \right)^{x_j} \right]. \quad (152)$$

Continuing this until all expectations are evaluated yields the expression $w(\sigma)$. $\square$

Proof of Prop. 15. Let $j_0 := \lceil [\ln(1 - \eta) - \ln(|\mathcal{E}|)] / \ln(1 - \alpha) \rceil \leq \lceil [-\ln(1 - \eta) + \ln(|\mathcal{E}|)] / \alpha \rceil$, so that $|\mathcal{E}|(1 - \alpha)^{j_0} \leq 1 - \eta$. Recall that $f(x) = 0$ when $x \leq 1 - \eta$ and note that $(1 - \alpha)^j \geq 1 - j\alpha$. Working from the definition of $w(\sigma)$, we have

$$\sum_{\sigma \in \Gamma} w(\sigma) = \prod_{j=0}^{\infty} \sum_{x_j=0}^{\infty} b^{x_j} f(|\mathcal{E}|(1 - \alpha)^j)^{x_j} = \prod_{j=0}^{\infty} \frac{1}{1 - bf(|\mathcal{E}|(1 - \alpha)^j)} \quad (153)$$

$$\geq \prod_{j=0}^{\infty} \exp(bf(|\mathcal{E}|(1 - \alpha)^j)) = \prod_{j=0}^{j_0-1} \exp(bf(|\mathcal{E}|(1 - \alpha)^j)) \quad (154)$$

$$= \exp \left(\frac{b}{\eta} \sum_{j=0}^{j_0-1} (|\mathcal{E}|(1 - \alpha)^j - (1 - \eta)) \right) \quad (155)$$

$$= \exp \left(\frac{b}{\eta} \left(\frac{|\mathcal{E}| - |\mathcal{E}|(1 - \alpha)^{j_0}}{\alpha} - (1 - \eta)j_0 \right) \right) \quad (156)$$

$$\geq \exp \left(\frac{b}{\eta} \left(\frac{|\mathcal{E}| - (1 - \eta)}{\alpha} - (1 - \eta)j_0 \right) \right) \quad (157)$$

$$\geq \exp \left(\frac{b}{\eta} \left(\frac{|\mathcal{E}| - (1 - \eta)}{\alpha} - \frac{(1 - \eta) \ln \left(\frac{|\mathcal{E}|}{1 - \eta} \right)}{\alpha} \right) \right) \quad (158)$$

$$= \exp \left(\frac{b|\mathcal{E}|}{\eta\alpha} \left(1 - \frac{1 - \eta}{|\mathcal{E}|} + \frac{1 - \eta}{|\mathcal{E}|} \ln \left(\frac{1 - \eta}{|\mathcal{E}|} \right) \right) \right) \quad (159)$$

which, recalling the definition of F in Eq. (15), proves the proposition. $\square$

Proof of Prop. 16. For any sequence $\sigma = (x_0, x_1, \dots)$ not in the set Λ_ℓ , there must exist some integer a such that $x_a > 0$ and $\sum_{j=0}^a x_j > \ell - a$ (otherwise it appears in the expansion of $(A + B)^\ell$). For each σ , let a_σ denote the minimum such integer. For each $a > 0$, let τ_a be the sequence with

$x_0 = \max(0, \ell - a)$ and $x_a = 1$, and $x_j = 0$ for all other j . Let τ_0 be the sequence with $x_0 = \ell + 1$ and $x_j = 0$ for all other j . Then we can let $\rho_\sigma = \sigma - \tau_{a_\sigma}$ (under entry-wise subtraction). Note that the x_0 entry for ρ_σ can be negative, but no smaller than $\min(0, -\ell + a_\sigma)$ and also no smaller than $-\sum_{j=1}^{\infty} x_j$. We have

$$w(\sigma) = w(\tau_{a_\sigma})w(\rho_\sigma) = b^{\max(1, \ell - a_\sigma + 1)} f(|\mathcal{E}|(1 - \alpha)^{a_\sigma}) w(\rho_\sigma) \quad (160)$$

We can break up the sum into two terms, associated with $a_\sigma \leq \ell$ or $a_\sigma > \ell$:

$$\sum_{\substack{\sigma \notin \Lambda_\ell \\ a_\sigma = a}} w(\sigma) = \sum_{a=0}^{\ell} b^{\ell - a + 1} f(|\mathcal{E}|(1 - \alpha)^a) \sum_{\substack{\sigma \notin \Lambda_\ell \\ a_\sigma = a}} w(\rho_\sigma) + \sum_{a=\ell+1}^{\infty} b f(|\mathcal{E}|(1 - \alpha)^a) \sum_{\substack{\sigma \notin \Lambda_\ell \\ a_\sigma = a}} w(\rho_\sigma) \quad (161)$$

Examine the second term. When $a_\sigma > \ell$, ρ_σ has no negative entries, meaning $\rho_\sigma \in \Gamma$. Moreover, for any fixed choice of a , the mapping $\sigma \mapsto \rho_\sigma$ restricted to σ for which $a_\sigma = a$ is an injective map. Thus, the sum $\sum_{\substack{\sigma \notin \Lambda_\ell \\ a_\sigma = a}} w(\rho_\sigma)$ is less than $\sum_{\sigma \in \Gamma} w(\sigma)$. Additionally, since $f(x) \leq x$, and $\sum_{a=\ell+1}^{\infty} (1 - \alpha)^a = \alpha^{-1}(1 - \alpha)^{\ell+1}$, the second term above is bounded by $b|\mathcal{E}|\alpha^{-1}(1 - \alpha)^{\ell+1} \sum_{\sigma \in \Gamma} w(\sigma)$.

Now examine the first term. When $a \leq \ell$, the map $\sigma \mapsto \rho_\sigma$ is still injective, but the x_0 entry of ρ_σ can be negative. If we sum over all possible sequences $(x_0, x_1, \dots)$ for which $x_j \geq 0$ when $j > 0$ and $x_0 \geq -\sum_{j=1}^{\infty} x_j$, every ρ_σ for which $a_\sigma = a$ will appear (exactly) once in the sum. Thus, we have

$$\sum_{\substack{\sigma \notin \Lambda_\ell \\ a_\sigma = a}} w(\rho_\sigma) \leq \sum_{x_1=0}^{\infty} \sum_{x_2=0}^{\infty} \cdots \sum_{x_0=-\sum_{j=1}^{\infty} x_j}^{\infty} b^{\sum_{j=0}^{\infty} x_j} \prod_{j=0}^{\infty} f(|\mathcal{E}|(1 - \alpha)^j)^{x_j} \quad (162)$$

$$= \sum_{x_1=0}^{\infty} \sum_{x_2=0}^{\infty} \cdots \sum_{x'_0=0}^{\infty} b^{x'_0} f(|\mathcal{E}|)^{x'_0 - \sum_{j=1}^{\infty} x_j} \prod_{j=1}^{\infty} f(|\mathcal{E}|(1 - \alpha)^j)^{x_j} \quad (163)$$

$$= \sum_{x_1=0}^{\infty} \sum_{x_2=0}^{\infty} \cdots \sum_{x'_0=0}^{\infty} (b f(|\mathcal{E}|))^{x'_0} \prod_{j=1}^{\infty} \left(\frac{f(|\mathcal{E}|(1 - \alpha)^j)}{f(|\mathcal{E}|)} \right)^{x_j} \quad (164)$$

$$= \frac{1}{1 - b f(|\mathcal{E}|)} \prod_{j=1}^{\infty} \frac{1}{1 - \frac{f(|\mathcal{E}|(1 - \alpha)^j)}{f(|\mathcal{E}|)}} \quad (165)$$

Since f is convex $f(|\mathcal{E}|(1 - \alpha)^j) \leq (1 - \alpha)^j f(|\mathcal{E}|)$ and thus the final expression is less than

$$\frac{1}{1 - b f(|\mathcal{E}|)} \prod_{j=1}^{\infty} \frac{1}{1 - (1 - \alpha)^j} . \quad (166)$$

We evaluate a bound on the logarithm as follows. Note that for a monotonically decreasing function g we have $\sum_{j=j_0}^{\infty} g(j) \leq \int_{j_0-1}^{\infty} g(u) du$.

$$\ln \left(\prod_{j=1}^{\infty} \frac{1}{1 - (1 - \alpha)^j} \right) = - \sum_{j=1}^{\infty} \ln(1 - (1 - \alpha)^j) = \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{(1 - \alpha)^{jk}}{k} \quad (167)$$

$$= \sum_{j=1}^{\infty} (1 - \alpha)^j + \sum_{k=2}^{\infty} \sum_{j=1}^{\infty} \frac{(1 - \alpha)^{jk}}{k} = \frac{1 - \alpha}{\alpha} + \sum_{k=2}^{\infty} \sum_{j=1}^{\infty} \frac{(1 - \alpha)^{jk}}{k} \quad (168)$$

$$\leq \frac{1 - \alpha}{\alpha} + \int_1^{\infty} dy \frac{1}{y} \int_0^{\infty} dx (1 - \alpha)^{xy} \quad (169)$$

$$= \frac{1 - \alpha}{\alpha} - \int_1^{\infty} dy \frac{1}{y} \frac{1}{y \ln(1 - \alpha)} \quad (170)$$

$$= \frac{1 - \alpha}{\alpha} - \frac{1}{\ln(1 - \alpha)} \leq \frac{2}{\alpha} \quad (171)$$

Combining these observations, we have

$$\sum_{\sigma \notin \Lambda_\ell} w(\sigma) \leq \left(\sum_{a=0}^{\ell} b^{\ell-a+1} f(|\mathcal{E}|(1 - \alpha)^a) \right) \frac{1}{1 - bf(|\mathcal{E}|)} e^{2/\alpha} + b|\mathcal{E}|\alpha^{-1}(1 - \alpha)^{\ell+1} \sum_{\sigma \in \Gamma} w(\sigma) \quad (172)$$

$$\leq \left(\sum_{a=0}^{\ell} b^{\ell-a+1} |\mathcal{E}|(1 - \alpha)^a \right) \frac{1}{1 - bf(|\mathcal{E}|)} e^{2/\alpha} + b|\mathcal{E}|\alpha^{-1}(1 - \alpha)^{\ell+1} \sum_{\sigma \in \Gamma} w(\sigma) \quad (173)$$

$$\leq b|\mathcal{E}|(1 - \alpha)^{\ell+1} \frac{1}{1 - \alpha - b} e^{2/\alpha} + b|\mathcal{E}|\alpha^{-1}(1 - \alpha)^{\ell+1} \sum_{\sigma \in \Gamma} w(\sigma) \quad (174)$$

which entails the proposition statement. $\square$

A.4 Tail bound for certain cost functions

Proposition 17. *Let $H = \sum_{j=1}^m \mathcal{C}_j$ be a MAX- k -CSP instance with m clauses, and let E^* be its optimal value. Let d_j be the total number of constraints involving the j th variable and let $C(E)$ be the number of assignments z such that $H(z) \leq E$. Define $D = nk^{-2}m^{-2} \sum_{j=1}^n d_j^2$. Then for any $0 \leq \eta \leq 1$*

$$C(E^*(1 - \eta)) \leq 2^{(1-\gamma)n} \quad (175)$$

where

$$\gamma = \left(\frac{|E^*|}{m} \right)^2 \frac{(1 - \eta)^2}{\ln(2) 2^{2k} k^2 D} \quad (176)$$

If $d_j \leq 2km/n$ for all j , then $D \leq 2$ and we can write

$$\gamma \geq \left(\frac{|E^*|}{m} \right)^2 \frac{(1 - \eta)^2}{2 \ln(2) 2^{2k} k^2}. \quad (177)$$

Furthermore, if each clause $\mathcal{C}_j$ involves exactly k distinct bits (with $k \geq 2$), which are chosen independently and uniformly at random for each j , then the expected value of D is upper bounded by $1 + n/(km)$, implying most random instances obey $D \leq 3$ so long as $m \geq n$.

Proof. We will use McDiarmid's inequality [64], which can be stated as follows. Suppose a real function H of n independent random variables $z_1, \dots, z_n$ has the property that changing the value of z_i while leaving the other $n - 1$ variables constant can change the value of H by at most an amount Δ_i . Then, when the z_i are chosen randomly, the probability that H deviates by more than δ from its mean is at most $e^{-2\delta^2/\sum_i \Delta_i^2}$.

Applying this to our problem, we may choose $z_i \in \{+1, -1\}$ uniformly at random for each $i = 1, \dots, n$, and note that since z_i participates in at most d_i clauses, changing z_i can change the value of H by at most $\Delta_i \leq 2^k d_i$ (recall that the value of a clause is restricted to the interval $[-1, 2^k - 1]$). The mean of H is precisely 0, and thus for $E < 0$, the quantity $C(E)/2^n$ is the probability that for a randomly chosen $z \in \{+1, -1\}^n$, $H(z)$ deviates beneath its mean by at least a value $|E|$. Thus, for $E < 0$, McDiarmid's inequality gives us the Gaussian tail bound

$$C(E) \leq 2^n \exp\left(-\frac{2E^2}{2^{2k} \sum_{j=1}^n d_j^2}\right) \quad (178)$$

$$\leq 2^n \exp\left(-\frac{2E^2 n}{2^{2k} k^2 m^2 D}\right) \quad (179)$$

Plugging in $E = E^*(1 - \eta)$ into Eq. (179) yields the quoted result. Next, recalling that $\sum_{j=1}^n d_j \leq km$ for MAX- k -CSP instances, and imposing $d_j \leq 2km/n$ for all j , implies that $D \leq 2$.

Now consider the case where each C_j is chosen randomly as described. Each d_j is given by the sum of m independent Bernoulli random variables that are equal to 1 with probability k/n and 0 otherwise (although note that the d_j are not independent for different values of j). The average value of d_j is given by $\mathbb{E}[d_j] = km/n$, and the average value of d_j^2 is given by $\mathbb{E}[d_j^2] = mk/n + m(m-1)k^2/n^2$. Thus, the expected value of D is given by $\mathbb{E}[D] = (m + n/k - 1)/m < 1.5$. A simple Markov inequality implies that the probability that D exceeds 3 is upper bounded by $\mathbb{E}[D]/3 < 1/2$, implying the quoted statement. $\square$

Proposition 18. *Let $\mathcal{J}_{n,k}$ denote the expected value of the optimal cost E^* when the cost function H is drawn from the k -spin model with n spins, as defined in Eq. (3). Then*

$$\mathcal{J}_{n,k} \leq -n \frac{\sqrt{2}}{\sqrt{\pi k}} \quad (180)$$

for any n, k for which n is a multiple of k . When n is not a multiple of k , we have $\mathcal{J}_{n,k} \leq -k \lfloor n/k \rfloor \frac{\sqrt{2}}{\sqrt{\pi k}}$. Note that as long as $n \geq k$, $k \lfloor n/k \rfloor \geq n/2$, and thus for all n, k , we can say that

$$\mathcal{J}_{n,k} \leq -n \frac{1}{\sqrt{2\pi k}} \quad (181)$$

Proof. First, let us fix k and consider varying n . Suppose $n_A < n_B$. We now construct a (correlated) joint ensemble over cost functions (H_A, H_B) where the marginal distribution on H_A is the k -spin model with $n = n_A$ and the marginal distribution on H_B is the k -spin model with $n = n_B$. First H_A is chosen randomly from the k -spin ensemble with $n = n_A$. Note that H_A has $\binom{n_A}{k}$ terms. Then H_B is generated by choosing the coefficients of these $\binom{n_A}{k}$ terms of H_B to be the same as those of H_A and choosing the other $\binom{n_B}{k} - \binom{n_A}{k}$ coefficients as i.i.d. standard Gaussian variables. Let E_A^* and E_B^* denote the optimal costs of H_A and H_B , and z_A^* and z_B^* the optimal assignments. We have $\mathbb{E}[E_A^*] = \mathcal{J}_{n_A,k}$ and $\mathbb{E}[E_B^*] = \mathcal{J}_{n_B,k}$, where $\mathbb{E}$ denotes expectation over the joint distribution defined above, since the marginal distributions over H_A and H_B are precisely the k -spin ensemble. However,

we also note that, due to the way the joint distribution was constructed, $\mathbb{E}[H_B(z_A^*)|H_A] = E_A^*$, where $\mathbb{E}[\cdot|H_A]$ denotes expectation conditioned on a fixed value of H_A . Moreover, $E_B^* \leq H_B(z_A^*)$ always holds. These observations imply that

$$\mathcal{J}_{n_A,k} \geq \mathcal{J}_{n_B,k} \quad \text{when } n_A \leq n_B \quad (182)$$

This statement means that $\mathcal{J}_{n,k} \geq \mathcal{J}_{\lfloor n/k \rfloor, k}$ and thus the proposition statement when n is not a multiple of k follows directly from the case when n is a multiple of k . Henceforth assume n is a multiple of k .

Now, we prove the formula for increasing k by induction. First, consider the base case. For $k = 1$, the expected optimal value $\mathcal{J}_{n,1} = -n\sqrt{2/\pi}$. This is because, for every instance, one can choose the value of $z_1, \dots, z_n$ such that each of the n degree-1 terms is negative (no frustration), and E^* is the sum of the absolute value of n independent random Gaussian variables, which can be readily calculated.

Now, suppose the proposition statement holds for $\mathcal{J}_{n,k-1}$ for all choices of n . We will show that for all choices of n , $\mathcal{J}_{n,k} \leq \mathcal{J}_{n',k-1}$ with $n' := (k-1)n/k$. Consider an instance H chosen from the k -spin model with n spins, where n is a multiple of k , defined by its coefficients $J_{i_1, \dots, i_k}$ as in Eq. (3). Let $t = n/k$. There are $\binom{n}{k}$ terms of H , and $t \binom{n-t}{k-1}$ of these terms will include exactly one of the variables $z_1, \dots, z_t$. Thus we can express $H = H_C + H_D$ where H_C includes these terms and H_D includes the rest of the terms. We can write H_C as

$$H_C(z_1, \dots, z_n) = \sqrt{\frac{k!}{n^{k-1}}} \sum_{i_1=1}^t z_{i_1} \left(\sum_{t < i_2 < \dots < i_k \leq n} J_{i_1, \dots, i_k} z_{i_2} \dots z_{i_k} \right) \quad (183)$$

Now consider the MAX-E($k-1$)-LIN2 instance on $n-t =: n'$ variables $z_{t+1}, \dots, z_n$

$$\bar{H}_C(z_{t+1}, \dots, z_n) = \sqrt{\frac{(k-1)!}{tn^{k-2}}} \sum_{t < i_2 < \dots < i_k \leq n} \left(\sum_{i_1=1}^t J_{i_1, \dots, i_k} \right) z_{i_2} \dots z_{i_k} \quad (184)$$

and note that

$$H_C(1, \dots, 1, z_{t+1}, \dots, z_n) = \sqrt{kt/n} \bar{H}_C(z_{t+1}, \dots, z_n). \quad (185)$$

The sum over $J_{i_1, \dots, i_k}$ from $i_1 = 1$ to t in Eq. (184) is a sum of t independent standard Gaussian-distributed random variables, which is itself Gaussian-distributed with mean zero and variance equal to t . This factor of t will precisely cancel the $1/\sqrt{t}$ in Eq. (184), allowing us to conclude that the ensemble over $\bar{H}_C$ is precisely the $(k-1)$ -spin ensemble with n' spins (note that n' is a multiple of $k-1$). Let $\bar{z}^*$ be the optimal assignment to $\bar{H}_C$, with energy $\bar{E}_C^*$ and note that $\mathbb{E}[\bar{E}_C^*] = \mathcal{J}_{n',k-1}$. We let z be equal to $\bar{z}^*$ with the additional assignments $z_1 = z_2 = \dots = z_t = 1$, so that $H_C(z) = \bar{E}_C^*$. Since the coefficients of H_D are chosen independently from H_C , we have $\mathbb{E}[H_D(z)] = 0$ when z is chosen as above. Thus $\mathbb{E}[H(z)] = \mathcal{J}_{n',k-1}$, and since the optimal value can only be smaller than $H(z)$, we have

$$\mathcal{J}_{n,k} \leq \mathcal{J}_{n',k-1}. \quad (186)$$

We now insert the bound on $\mathcal{J}_{n',k-1}$, finding that

$$\mathcal{J}_{n,k} \leq -\left(n - \frac{n}{k}\right) \frac{\sqrt{2}}{\sqrt{\pi}(k-1)} = -n \frac{\sqrt{2}}{k\sqrt{\pi}}. \quad (187)$$

which proves the proposition. $\square$

Proposition 19. When H is drawn randomly from the k -spin model with n spins, let $\mathcal{J}_{n,k}$ denote the expected value of E^* . For any δ

$$\Pr_H[E^* \geq \mathcal{J}_{n,k} + \delta n] \leq e^{-\delta^2 n/2}, \quad (188)$$

where $\Pr_H$ denotes probability over random choice of H from the k -spin ensemble.

Proof. We may think of the quantity $E^* \sqrt{n^{k-1}/k!}$ as a function of the $\binom{n}{k}$ coefficients $J_{i_1, \dots, i_k}$, as in Eq. (3). Note that this function is 1-Lipschitz as changing a single coefficient by some amount Δ can change $E^* \sqrt{n^{k-1}/k!}$ by at most the same amount Δ . Applying Lemma 1 of Ref. [15], we have for any $t > 0$

$$\Pr_H \left[E^* \sqrt{n^{k-1}/k!} \geq \mathbb{E}_H[E^*] \sqrt{n^{k-1}/k!} + t \right] \leq \exp \left(-\frac{t^2}{2 \binom{n}{k}} \right) \quad (189)$$

which, noting $\mathbb{E}_H[E^*] = \mathcal{J}_{n,k}$, and letting $\delta = t\sqrt{k!}/n\sqrt{n^{k-1}}$ can be rewritten as

$$\Pr_H [E^* \geq \mathcal{J}_{n,k} + \delta n] \leq \exp \left(-\frac{\delta^2 n^{k+1}}{2k! \binom{n}{k}} \right) \leq e^{-\delta^2 n/2}. \quad (190)$$

□

Proposition 20. Suppose H is drawn at random from the k -spin model. Let E^* denote its optimal value and let $C(E)$ be the number of assignments z such that $H(z) \leq E$. Then for any $0 \leq \eta \leq 1$, with probability at least $1 - 2^{-\gamma n+1}$ over choice of H ,

$$C(E^*(1 - \eta)) \leq 2^{(1-\gamma)n} \quad (191)$$

where

$$\gamma = \frac{(1 - \eta)^2}{32\pi \ln(2)k^2} \quad (192)$$

Proof. For a fixed bit string x and randomly chosen k -spin instance H , the values of each of the $m = \binom{n}{k}$ terms in the Hamiltonian are independently random since the coefficients are independent. Thus, $H(x)$ is distributed as the sum of m i.i.d. standard Gaussian variables with mean 0 and standard deviation $\sqrt{k!/n^{k-1}}$, which is equivalent to a Gaussian with mean 0 and standard deviation σ , with $\sigma^2 = mk!/n^{k-1} = n!/((n-k)!n^{k-1}) < n$. Thus for any t we have

$$\Pr_H [H(x) \leq -t] = \frac{1}{2} \text{Erfc} \left[\frac{t}{\sqrt{2\sigma^2}} \right] \leq \frac{1}{2} \text{Erfc} \left[\frac{t}{\sqrt{2n}} \right] \leq e^{-t^2/2n} = 2^{-t^2/(2n \ln(2))} \quad (193)$$

where $\Pr_H$ denotes probability over random choice of H from the k -spin ensemble. Since $C(E) = \sum_x \mathbb{1}(H(x) \leq E)$, where $\mathbb{1}$ is the indicator function, we can compute a bound on the expectation value of $C(E)$ as

$$\mathbb{E}_H[C(E)] = \sum_x \mathbb{E}_H[\mathbb{1}(H(x) \leq E)] = \sum_x \Pr_H [H(x) \leq E] \leq 2^n 2^{-E^2/(2n \ln(2))}. \quad (194)$$

Separately, let us apply Prop. 18 and Prop. 19 with $\delta = 1/k\sqrt{8\pi}$. We find

$$\Pr_H \left[E^* \geq -\frac{n}{2k\sqrt{2\pi}} \right] = \Pr_H \left[E^* \geq -n \frac{1}{k\sqrt{2\pi}} + n \frac{1}{2k\sqrt{2\pi}} \right] \leq \Pr_H [E^* \geq \mathcal{J}_{n,k} + \delta n] \quad (195)$$

$$\leq e^{-\delta^2 n/2} = \exp \left(-\frac{n}{16\pi k^2} \right) \quad (196)$$

Conditioned on $E^* \geq -n/(2k\sqrt{2\pi})$, which is true with high probability by Eq. (196), we can say from Eq. (194) that

$$\mathbb{E}_H[C(E^*(1-\eta))] \leq 2^{n\left(1 - \frac{(1-\eta)^2}{16k^2\pi \ln(2)}\right)} \quad (197)$$

and, by Markov's inequality

$$\Pr_H[C(E^*(1-\eta)) \geq 2^{(1-\gamma)n}] \leq 2^{-n\left(\frac{(1-\eta)^2}{16k^2\pi \ln(2)} - \gamma\right)} \quad (198)$$

Choosing $\gamma = (1-\eta)^2/(32k^2\pi \ln(2))$ yields the desired bound on $C(E^*(1-\eta))$. By the union bound, the chances this bound fails is at most the sum of the right-hand-sides of Eqs. (196) and (198), which gives

$$2^{-n\left(\frac{(1-\eta)^2}{32k^2\pi \ln(2)}\right)} + e^{-n\frac{1}{16\pi k^2}} \leq 2 \exp\left(-\frac{n(1-\eta)^2}{32k^2\pi}\right) = 2^{-\gamma n+1}, \quad (199)$$

which proves the proposition. $\square$

B Implementation of jump steps

We assume we have access to a $\text{poly}(n)$ size circuit that enacts a block-encoding of both the beginning and ending Hamiltonians, where (following standard convention) a (κ, a) -block encoding of an n -qubit Hamiltonian K is defined to be an $(n+a)$ -qubit unitary U for which

$$(|0\rangle^{\otimes a} \otimes I) U (|0\rangle^{\otimes a} \otimes I) = K/\kappa \quad (200)$$

Proposition 21 (Jump $K_1 \rightarrow K_2$ with parameters $(E_1, E_2, \Delta_1, \Delta_2, p, \delta)$). *Suppose K_1 and K_2 are n -qubit Hamiltonians and let $E_1, E_2, \Delta_1, \Delta_2, p, \delta$ be known positive parameters. Let U_i be a (κ_i, a) -block-encoding of K_i for $i \in 1, 2$. Suppose that the following properties are satisfied.*

- (A) K_1 has one eigenstate with energy at most E_1 , denoted by $|\psi_1\rangle$
- (B) $\|\Pi_2 |\psi_1\rangle\|^2 \geq p$, where Π_2 denotes the projector onto the subspace spanned by eigenvectors of K_2 with energy at most E_2
- (C) For $j \in \{1, 2\}$, at least one of the following holds:
 - (i) K_j has no eigenvalues in the interval $(E_j, E_j + \Delta_j)$
 - (ii) K_j can be represented as a diagonal matrix in either the computational basis or the Hadamard basis and each of its entries in that basis can be computed in $\text{poly}(n)$ classical time.

Then, there is a universal constant D such that we can construct a quantum circuit that implements a unitary U for which

$$\left\| U(|\psi_1\rangle \otimes |0\rangle^{\otimes a}) - \frac{\Pi_2 |\psi_1\rangle \otimes |0\rangle^{\otimes a}}{\|\Pi_2 |\psi_1\rangle\|} \right\| \leq \delta \quad (201)$$

and consists of $\text{poly}(n)$ gates along with

$$\frac{D\kappa_i}{\Delta_i\sqrt{p}} \log(\delta^{-1}) \log(p^{-1/2}\delta^{-1} \log(\delta^{-1})) \quad (202)$$

calls to U_j and controlled- U_j , for $j \in \{1, 2\}$. If K_j is diagonal in the computational basis and each of its entries can be computed in $\text{poly}(n)$ classical time, then no calls to U_j are necessary, and Δ_j is irrelevant.

When acting with U on $|\psi_1\rangle$, we say we are performing the jump $K_1 \rightarrow K_2$.

Proof. We can construct U through a combination of fixed-point amplitude amplification [44] and what essentially amounts to phase estimation, but enacted with the quantum singular value transformation (QSVT) framework [65]. First assume that conditions (A),(B), and (C)(i) are satisfied. For $j \in \{1, 2\}$, let $q_j : [-1, 1] \rightarrow [-1, 1]$ denote the step function

$$q_j(x) = \begin{cases} -1 & \text{if } x \leq 0 \\ 1 & \text{if } x \geq 0 \end{cases} \quad (203)$$

Let R_1 denote the unitary reflection operator which applies a -1 phase to the state $|\psi_1\rangle$ and acts as identity on the subspace orthogonal to $|\psi_1\rangle$. Similarly, let R_2 denote the unitary reflection that applies a -1 phase to the support of Π_2 . By conditions (A), (B) and (C)(i) we can say that

$$R_j = q_j \left(\frac{K_j - E_j - \frac{\Delta_j}{2}}{2\kappa_j} \right) \quad (204)$$

Let L be an integer to be specified later. Let $\delta' = \delta/4L$. Then, there is an odd polynomial $q'_j : [-1, 1] \rightarrow [-1, 1]$ with degree at most $d = O(\log(1/\delta')\kappa_j/\Delta_j)$ for which [65]

$$q'_j(x) \leq -1 + \delta' \quad \text{if } x \leq -\frac{\Delta_j}{4\kappa_j} \quad (205)$$

$$q'_j(x) \geq 1 - \delta' \quad \text{if } x \geq \frac{\Delta_j}{4\kappa_j} \quad (206)$$

which approximates the step function $q_j(x)$. Define the operator

$$R'_j = q'_j \left(\frac{K_j - E_j - \frac{\Delta_j}{2}}{2\kappa_j} \right) \quad (207)$$

Since q'_j is an odd polynomial of degree d whose range is in $[-1, 1]$ on the domain $[-1, 1]$, the quantum singular value transformation (QSVT) [65] allows a $(2\kappa_j, a + O(1))$ block-encoding of R'_j to be implemented using d calls to a block-encoding (and its inverse) of the operator $\frac{-E_j - \Delta_j/2}{2\kappa_j}I + \frac{K_j}{2\kappa_j}$, which itself can be constructed using a controlled- U_j gate via linear combination of block encoding with the identity.

Note that K_j , R_j and R'_j share a common set of eigenvectors. Using condition (C)(i), every eigenvalue of $\frac{-E_j - \Delta_j/2}{2\kappa_j}I + \frac{K_j}{2\kappa_j}$ satisfies either the condition in Eq. (205) or (206), and thus

$$\|R_j - R'_j\| \leq \delta' \quad (208)$$

Fixed-point amplitude amplification [44] describes how access to (controlled) reflection operators R_1 and R_2 allows one to construct a unitary V for which

$$\left\| V(|\psi_1\rangle \otimes |0\rangle^{\otimes a}) - \frac{\Pi_2 |\psi_1\rangle \otimes |0\rangle^{\otimes a}}{\|\Pi_2 |\psi_1\rangle\|} \right\| \leq \delta/2 \quad (209)$$

as long as $\|\Pi_2 |\psi_1\rangle\| \geq \sqrt{p}$. The construction uses $L = O(\log(1/\delta)/\sqrt{p})$ calls to controlled- R_1 and the same number of calls to controlled- R_2 . By replacing R_1 and R_2 by operators R'_1 and R'_2 to form U , at most $2L$ unitaries are modified, and each by at most an amount δ' . By the triangle inequality, errors accrue linearly and we have

$$\|U - V\| \leq 2L\delta' = \delta/2 \quad (210)$$

and this equation, together with Eq. (209), proves that U has the action claimed by the proposition. The number of calls U makes to the block encodings of K_1 and K_2 is precisely the L calls it makes to controlled- R'_1 and controlled- R'_2 times the d calls R'_1 and R'_2 make to controlled- K_1 and controlled- K_2 . Thus, the calls to K_1 and K_2 are doubly controlled.

If condition (C)(ii) is true for either K_1 , K_2 , or both, we modify the argument to construct the associated approximate reflection operator R'_j in a different way (actually, we will be able to construct an *exact* reflection operator). First assume that K_j is diagonal in the computational basis, and suppose that T bits are sufficient to represent the diagonal entries of K_j in binary. Let the operator R_j be implemented by four steps: first, using at most $A = \text{poly}(n)$ ancilla qubits initially in the state $|0\rangle$, we perform the operation

$$|i\rangle |0\rangle^{\otimes A} \mapsto |i\rangle |(K_j)_{ii}\rangle |0\rangle^{\otimes(A-T)} \quad (211)$$

using reversible classical arithmetic in superposition, where $(K_j)_{ii}$ denotes the i th diagonal entry of the diagonal operator K_j . Second, we compute whether $(K_j)_{ii} \leq E_j$ or $(K_j)_{ii} > E_j$ into another ancilla bit

$$|i\rangle |(K_j)_{ii}\rangle |0\rangle^{\otimes(A-T)} \mapsto |i\rangle |(K_j)_{ii}\rangle |(K_j)_{ii} \leq E_j\rangle |0\rangle^{\otimes(A-T-1)} \quad (212)$$

again using reversible classical arithmetic. Third, we apply a Pauli- Z gate to the $|(K_j)_{ii} \leq E_j\rangle$ bit. Fourth, we uncompute all the ancillas, resetting them to $|0\rangle$. This yields the overall operation

$$|i\rangle |0\rangle^{\otimes A} \mapsto (-1)^{((K_j)_{ii} \leq E_j)} |i\rangle |0\rangle^{\otimes A} \quad (213)$$

which is precisely the operation R_j . The complexity of implementing this operation is $\text{poly}(n)$ since we have assumed in condition (C)(ii) that all arithmetic is efficient. Note that this is possible regardless of how small Δ_j is due to the fact that we can use classical arithmetic to distinguish very near eigenvalues. If K_j is diagonal in the Hadamard basis, we may perform the exact same protocol with the modification that a layer of n single-qubit Hadamard gates is applied at the beginning and end of the circuit to rotate into the basis where K_j is diagonal. $\square$

Below, we show how to implement block-encodings of the Hamiltonian H_b needed to apply Prop. 21. Actually, we only show that a block-encoding can be implemented up to exponentially small precision; since the block-encoding is called at most an exponential number of times, this block-encoding error can be made smaller than the other errors that arise in Prop. 21. The construction outlined below is certainly not the most efficient method. For those interested in producing a block-encoding with optimized circuit complexity, we draw the reader's attention to explicit constructions of useful circuit primitives in Ref. [5].

Proposition 22 (Block-encoding H_b). *Suppose H is an efficiently computable cost function on n bits with known optimal value E^* . Suppose that $b < 1$ and $g : [-1, \infty) \rightarrow [-1, 0]$ is an efficiently computable function. Then for any $\delta \geq \exp(-\text{poly}(n))$, there is a unitary U that δ -approximates (in operator norm) a $(1 + b, \text{poly}(n))$ -block-encoding of the Hamiltonian H_b , defined as*

$$H_b = -\frac{X}{n} + b g\left(\frac{H}{|E^*|}\right), \quad (214)$$

and the circuit complexity of U is polynomial in n . Moreover, controlled- U and controlled-controlled- U can also be efficiently implemented.

Proof. The first term of H_b is diagonal in the Hadamard basis and the second term is diagonal in the computational basis. The diagonal elements in these bases are efficiently computable. Consider the second term. Let $h(z) = g(H(z)/|E^*|)$ and suppose the value of $h(z)$ can be exactly represented with $T = \text{poly}(n)$ bits. Note also that $\arcsin(h(x))$ can be represented to accuracy $\exp(-\Omega(P))$ with P bits. For any z , let θ_z denote the P -bit approximation to $\arcsin(h(z))$. We can form a $(1, \text{poly}(n))$ -block-encoding of $g(H/|E^*|)$ up to $\exp(-\text{poly}(n))$ error by constructing a unitary that for any n -qubit computational basis state $|z\rangle$, uses classical arithmetic in superposition to perform the following sequence of operations using $T + P + A + 1 = \text{poly}(n)$ ancilla qubits:

$$|0\rangle^{\otimes T} |0\rangle^{\otimes P} |0\rangle^{\otimes A} |z\rangle \mapsto |h(z)\rangle |0\rangle^{\otimes P} |0\rangle^{\otimes A} |z\rangle \quad (215)$$

$$\mapsto |h(z)\rangle |\theta_z\rangle |0\rangle^{\otimes A} |z\rangle \quad (216)$$

$$\mapsto |h(z)\rangle |\theta_z\rangle |0\rangle^{\otimes A} |z\rangle \quad (217)$$

$$\mapsto |h(z)\rangle |\theta_z\rangle (\sin(\theta_z) |0\rangle + \cos(\theta_z) |1\rangle) |0\rangle^{\otimes A} |z\rangle \quad (218)$$

$$\mapsto |h(z)\rangle |0\rangle^{\otimes P} (\sin(\theta_z) |0\rangle + \cos(\theta_z) |1\rangle) |0\rangle^{\otimes A} |z\rangle \quad (219)$$

$$\mapsto |0\rangle^{\otimes T} |0\rangle^{\otimes P} (\sin(\theta_z) |0\rangle + \cos(\theta_z) |1\rangle) |0\rangle^{\otimes A} |z\rangle \quad (220)$$

Noting that $\sin(\theta_z)$ approximates $h(z)$ up to error $\exp(-\Omega(P))$, we can see that this unitary $(1, T + P + A + 1)$ -block-encodes $g(H/|E^*|)$ up to the same overall error in operator norm. This block-encoding can be turned into a controlled-block-encoding with an arbitrary number of controls by controlling every operation in the above process.

A $(1, \text{poly}(n))$ -block-encoding of the first term of H_b can be constructed in a similar fashion by conjugating with Hadamard gates and noting it becomes diagonal in the computational basis. As H_b is a linear combination of the first and second terms with coefficients 1 and b , a $(1 + b, \text{poly}(n))$ -block-encoding of H_b can be given using the linear-combinations-of-block-encodings construction of Ref. [65]. This can be implemented as a controlled-block-encoding with an arbitrary number of controls by controlling all the operations involved in the circuit. $\square$

C The case where the optimal value is unknown

In our specification of the algorithm in Sec. 2, we assumed that the optimal value E^* of the cost function H is known ahead of time. In practice, this may often not be the case. In this section, we argue that if the algorithm described in the main text runs in time $O^*(2^{(0.5-c)n})$ when E^* is known, then there is always a slightly more sophisticated algorithm that also runs in time $O^*(2^{(0.5-c)n})$, perhaps with larger polynomial overheads.

This is easy to see in the case of MAX- k -CSP, since in this case there are m constraints, where each constraint takes either the value -1 or the value $s/(2^k - s)$ for some known quantity s equal to the number of assignments that satisfy the constraints. Thus, the cost of any input string can always be expressed as a rational number p/q with $q = (2^k - 1)!$, and $|p| \leq m(2^k - 1)!(2^k - 1)$. Thus we could always simply enumerate over all possible values of p , which, viewing k as a constant, adds overhead of only $\text{poly}(n)$. As we have a theoretical guarantee on the runtime when we choose E^* correctly, we can always terminate the algorithm at that runtime and never risk exceeding the $O^*(2^{(0.5-c)n})$ scaling.

However, if we allow the constraints to be weighted with arbitrary weights, as is the case for the k -spin model, this strategy may not work, as there may be an exponential number of possible values of E^* . To handle this more general situation, we come up with something more sophisticated.

Proposition 23. *Let H be a cost function for which there are known values q , Q , and ε such that $Q/q \leq \text{poly}(n)$, $q \leq |E^*| \leq Q$, and $\varepsilon \geq e^{-\Omega(n)}$ is a lower bound on the separation between the optimal value of H and the second-lowest non-optimal value. Suppose that, if E^* were known, there is a known choice for b and η for which Algorithm 1 runs in time $O^*(2^{(0.5-c)n})$. Then there is another quantum algorithm that runs in time $O^*(2^{(0.5-c)n})$ even when E^* is unknown.*

Proof. First examine step 2 of the algorithm as described in the pseudocode in Algorithm 1. Even when we do not know E^* , we may try to run this step with a guess W for E^* , in combination with choices b' and η' for the parameters b and η . These choices only impact the second term of H_b , as defined in Eq. (5). Looking at that term, note that the non-zero part of the function $b' g_{\eta'}(x/W)$ can be written as

$$b' g_{\eta'}(x/W) = b' \frac{\frac{x}{W} + 1 - \eta'}{\eta'} \quad (221)$$

$$= \frac{b'}{W\eta'}x + \frac{b'(1 - \eta')}{\eta'} \quad (222)$$

$$= Q^{-1}\phi x + \theta\phi, \quad (223)$$

where

$$\theta = \frac{W(1 - \eta')}{Q} \quad \phi = \frac{b'Q}{\eta'W} \quad (224)$$

Any choice of (W, η', b') that gives rise to the same (θ, ϕ) via Eq. (224) will result in an equivalent step 2 of the algorithm. In other words, specifying (W, η', b') is one more degree of freedom than is necessary.

For a given (θ, ϕ) choice, we let $b_{\theta, \phi}$ and $\eta_{\theta, \phi}$ denote the b' and η' values that would arise from choosing $W = |E^*|$ along with θ and ϕ . Turning around Eq. (224), we find

$$\eta_{\theta, \phi} = 1 - \frac{Q\theta}{|E^*|} \quad (225)$$

$$b_{\theta, \phi} = \frac{|E^*|}{Q}\phi - \phi\theta \quad (226)$$

Note that the region $(\theta, \phi) \in [0, 1] \times [0, Q/q]$ leads to a set of $(b_{\theta, \phi}, \eta_{\theta, \phi})$ that cover the region $[0, 1] \times [0, 1]$. Note also that the magnitude of the partial derivatives $\partial b_{\theta, \phi}/\partial \phi$, $\partial b_{\theta, \phi}/\partial \theta$, $\partial \eta_{\theta, \phi}/\partial \phi$, $\partial \eta_{\theta, \phi}/\partial \theta$ are each bounded above by Q/q on this region. Thus, if we cast a net of gridpoints over the region $[0, 1] \times [0, Q/q]$ for parameters (θ, ϕ) with spacing $\delta q/2Q$ in each dimension, we can say that for the “correct” choices (b, η) (known ahead of time), one of the grid points (θ, ϕ) gives rise to

$$b - \delta \leq b_{\theta, \phi} \leq b \quad \eta - \delta \leq \eta_{\theta, \phi} \leq \eta \quad (227)$$

Additionally, note that for any constants $b', \eta' < 1$, the partial derivatives of the function $c(b', \eta') = b'F(1 - \eta')/\eta'$ is C -Lipschitz over the input region $(b', \eta') \in [0, b] \times [0, \eta]$, for some constant C which depends on η and b , but not n . By choosing $\delta = 1/(\sqrt{2}nC)$, we can guarantee that the gridpoint (θ, ϕ) that leads Eq. (227) to hold will have

$$c(b_{\theta, \phi}, \eta_{\theta, \phi}) \geq c(b, \eta) - 1/n. \quad (228)$$

Thus, this choice of θ and ϕ leads to an algorithm with runtime $\text{poly}(n)2^{(0.5-c(b,\eta)+1/n)n} = \text{poly}(n)2^{(0.5-c(b,\eta))n}$, where the second equality follows by absorbing the constant into the $\text{poly}(n)$. Note that, since $\delta = 1/\text{poly}(n)$ and $Q/q = \text{poly}(n)$, there are at most $\text{poly}(n)$ gridpoints. We have established that one of these gridpoints will successfully prepare $|\psi_b\rangle$ in step 2 of the algorithm (although we cannot be sure which gridpoint it was).

Now we must implement step 3 without knowing E^* . This is tricky because we do not know which outcome to amplify. However, here we can do a binary search. Let us assume that we have chosen the “correct” gridpoint (we come back to this later). Under this assumption, we can assume that step 2 prepares $|\psi_b\rangle$. Now, we perform a computational basis measurement on $|\psi_b\rangle$ yielding outcome z , but we perform amplitude amplification on the outcome that $H(z) \leq U$, for some choice of U . If $E^* \leq U$, then the output will indeed be a string z for which $H(z) \leq U$, and we will learn that $E^* \leq U$. If the output is a string z for which $H(z) > U$, this gives us high confidence that $E^* > U$ since, if not, we would have found a string for which $H(z) \leq U$ with probability at least $1 - e^{-\Omega(n)}$. By choosing U repeatedly via binary search, we can find a $\varepsilon/2$ -close estimate for E^* in $O(\log(\varepsilon^{-1}))$ time steps. (We must choose the failure probability of the algorithm small enough that all of the binary search steps succeed with high probability.) This precision is sufficiently small to exactly resolve the optimal cost of H , and thus we can run the algorithm with this estimate for E^* and it will succeed in the stated runtime.

We do the above procedure on every gridpoint in succession. The correct gridpoint is guaranteed to output an optimal assignment with high probability. Every other gridpoint may do something different but will always output an assignment with greater cost, and thus, after we have run the algorithm for all the gridpoints, we will be able to output an optimal assignment with high probability simply by choosing the lowest-cost among all choices. $\square$

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Quantum Speedups for Group Relaxations of Integer Linear Programs

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Abstract

Integer Linear Programs (ILPs) are a flexible and ubiquitous model for discrete optimization problems. Solving ILPs is NP-Hard yet of great practical importance. Super-quadratic quantum speedups for ILPs have been difficult to obtain because classical algorithms for many-constraint ILPs are global and exhaustive, whereas quantum frameworks that offer super-quadratic speedup exploit local structure of the objective and feasible set. We address this via quantum algorithms for Gomory’s group relaxation. The group relaxation of an ILP is obtained by dropping nonnegativity on variables that are positive in the optimal solution of the linear programming (LP) relaxation, while retaining integrality of the decision variables. We present a competitive feasibility-preserving classical local-search algorithm for the group relaxation, and a corresponding quantum algorithm that, under reasonable technical conditions, achieves a super-quadratic speedup. When the group relaxation satisfies a nondegeneracy condition analogous to, but stronger than, LP nondegeneracy, our approach yields the optimal solution to the original ILP. Otherwise, the group relaxation tightens bounds on the optimal objective value of the ILP, and can improve downstream branch-and-cut by reducing the integrality gap; we numerically observe this on several practically relevant ILPs. To achieve these results, we derive efficiently constructible constraint-preserving mixers for the group relaxation with favorable spectral properties, which are of independent interest.

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1 Introduction

1.1 Motivation

Quantum algorithmic speedups for combinatorial optimization have been heavily studied for more than two decades [FGGS00, FGG14, Mon18, ABI⁺, Mon20], both due to the practical importance of these problems, as well as the theoretical optimism emerging from the derivation of many rigorous (usually quadratic) speedups based on amplitude amplification, amplitude estimation, quantum walks, or some combination thereof. However, motivated by the need for larger theoretical quantum speedups if a quantum advantage is to be realized on hardware [BMN⁺21, OHZ⁺25], recent years have seen a renewed interest in the development of frameworks for super-quadratic speedups [Has18, DPCB23, CHO⁺24] and the rigorous characterization of heuristics such as the Quantum Approximate Optimization Algorithm (QAOA) [FGG14] and the Quantum Adiabatic Algorithm (QAA) [FGGS00].

A unified framework for studying many discrete optimization problems is through the study of *integer linear programs* (ILPs). With $A \in \mathbb{Z}^{m \times n}$ and $b \in \mathbb{Z}^m$, define

$$\mathcal{P} := \{x \in \mathbb{Z}_{\geq 0}^n : Ax = b\}.$$

Given a vector $c \in \mathbb{R}^n$, the integer linear programming problem is to solve

$$\text{OPT} := \min\{c^\top x : x \in \mathcal{P}\}. \quad (\text{ILP})$$

ILP is a classic NP-complete problem underpinning countless applications in computer science, operations research, and decision sciences more generally. It naturally encodes many combinatorial optimization problems, as well as problems from finance, economics, industrial engineering, and the algorithmic geometry of numbers [Sch87, GLS88, WN99, CCZ14]. As a consequence of this flexibility, the study of algorithms for solving ILPs has gained significant interest, and there exist several popular commercial (Gurobi [Gur24] and CPLEX [CPL09]) and open-source solvers (SCIP [HBB⁺25]). From a practical point of view however, problems at interesting scales can often be solved, even when the asymptotic scaling of the algorithm is exponential. Polynomial speedups over this exponential scaling are significant as they directly expand the domain of problems that can be handled in this way.

Although special instances of ILP are polynomial-time solvable—such as when the feasible region is an integral¹ polyhedron—general instances remain computationally intractable. The difficulty of ILP stems from integrality: it destroys convexity, replacing a well-structured polyhedron with a discrete subset of the integer lattice. Unlike linear programs (LPs), ILPs lack local convexity properties that could inform efficient local search procedures. Even when the number of variables and constraints is small, or the feasible set appears relatively simple (for example, binary ILPs where the decision variables are $x \in \{0, 1\}^n$), determining feasibility or optimality can encode well-known NP-Complete problems such as subset-sum and 3-SAT. Assuming the exponential time hypothesis [IP01], these problems cannot be solved in polynomial time. From a practical point of view however, ILPs are solved regularly as part of established computational pipelines. Designing effective algorithms for solving general integer programming problems remains a central and enduring challenge in theoretical and applied computer science.

It is natural to ask what speedups quantum algorithms can provide for ILPs. In the black-box setting an ILP can be made to encode unstructured search, and the largest speedup possible is quadratic due to Grover’s search algorithm [Gro96]. For larger speedups (and practical relevance), we must focus on the white-box setting. In this setting, naïvely applying

¹A polyhedron is integral if it equals the convex hull of its integer points.

Grover’s algorithm to solve (ILP) would amount to brute-force search in a sufficiently large box $\{x \in \mathbb{Z}_{\geq 0}^n : \|x\|_\infty \leq M\}$ for suitable M , yielding a runtime $\mathcal{O}((M+1)^{n/2} \text{poly}(n, L))$, where L denotes the input length of the problem data. This scales with the size of the *ambient* space and ignores structure in the constraints. By contrast, the best (exact) classical ILP algorithms exploit geometry-of-numbers to decompose and enumerate the feasible region, and determine the optimal solution using $(\log(2n))^{\mathcal{O}(n)} \cdot \text{poly}(n, L)$ operations [Dad12, RR23]. A recent line of work [Mon18, AK17, Mon20, CMYP22] has improved upon direct Grover based methods by using quantum walks to accelerate the branch-and-cut algorithms that are used by many integer programming (IP) solvers, but there is no clear path to to obtain the super-quadratic speedups required to realize a quantum advantage in practice [BMN⁺21, OHZ⁺25].

A central obstacle to super-quadratic speedups for ILPs is that practically relevant instances typically carry many (often tight) constraints [AAA⁺24]. This is unsurprising: in general, the canonical formulation (ILP) does not yield a succinct, algorithmically useful *explicit* description of the feasible region $\mathcal{P}$, making direct constraint handling challenging even classically. Instead, general IP solvers use refined but ultimately exhaustive frameworks which proceed by incrementally tightening *relaxations* of (ILP) to implicitly describe $\mathcal{P}$ rather than maintaining feasibility, and thus cannot exploit the local geometry of the objective on the feasible set. By contrast, the quantum frameworks that promise super-quadratic gains—short-path and its generalizations [Has18, DPCB23, CHO⁺24], quantum dynamical algorithms [CHW⁺25, HOA⁺25], and heuristics such as QAOA [FGG14] and quantum annealing [FGGS00, KN98]—are explicitly local and benefit from mixers that preserve (or softly enforce) feasibility. Designing such local exploration for general ILPs is precisely the challenge: classical local methods (e.g., simulated annealing, parallel tempering, tabu search, memetic algorithms) succeed mainly on special cases with few constraints or efficiently enumerable feasible sets; the latter assumption is particularly strong, since even (approximately) uniform sampling over $\mathcal{P}$ is NP-Hard.

This mismatch motivates our central question:

Question 1. *Is there a class of algorithms for integer linear programming that*

1. *Is applicable to general ILPs with many constraints,*
2. *Is based on local exploration of a feasible space, and as a consequence,*
3. *Is a viable candidate for super-quadratic quantum speedups?*

In this paper, we answer Question 1 in the affirmative and give the first, to our knowledge, analysis of a framework that satisfies all the above conditions. Central to our results is a relaxation of (ILP) introduced by Gomory [Gom65, Gom67, Gom69] termed the *group relaxation* (also known as the *group problem*). The next sub-section introduces relaxations of ILPs in general and Gomory’s group relaxation more specifically, and discusses its significance and applicability to ILP solving. We then go on to outline our main contributions, and discuss their role in satisfying the criteria of Question 1.

1.2 Relaxations of ILPs

Modern integer-programming solvers—commercial (Gurobi [Gur24], CPLEX [CPL09]) and open-source (SCIP [HBB⁺25])—follow a relax–tighten–search paradigm. A relaxation of (ILP) is a problem obtained by dropping some of the constraints in the original formulation. Relaxations are thus problems of the form $\text{OPT}' := \min\{c^\top x : x \in \mathcal{P}'\}$, with $\mathcal{P} \subseteq \mathcal{P}'$ and $c^\top x \geq \text{OPT}'$ for any $x \in \mathcal{P}$. The challenge is to design relaxations that are both tractable and useful, since there is a natural tradeoff between the computational effort required to solve the relaxation, and the quality of the lower bound it provides for OPT.

The canonical choice is the LP relaxation, obtained by dropping integrality:

$$\text{OPT}_{\text{LP}} := \min \left\{ c^\top x : x \in \mathcal{P}_{\text{LP}} \right\}, \quad (\text{LP})$$

where

$$\mathcal{P}_{\text{LP}} := \{x \in \mathbb{R}_{\geq 0}^n : Ax = b\}.$$

The upshot of working with the LP relaxation is that it can be solved efficiently using *interior point methods* [Wri97], and clearly $\text{OPT}_{\text{LP}} \leq \text{OPT}$, since $\mathcal{P} := \mathcal{P}_{\text{LP}} \cap \mathbb{Z}^n$. If the optimal solution to (LP) is already integral, it automatically certifies optimality for the ILP. Unfortunately, aside from very restrictive classes of ILPs, such as those in which the constraint matrix A is *totally unimodular*², this is rarely the case. Moreover, a simple rounding of solutions to LP relaxations typically fails to yield high quality solutions to (ILP) without incurring exponential overhead.

Alternatively, the first nontrivial integer programming algorithms proposed combining information obtained from the solution to the LP relaxation with logical disjunction to describe and enumerate $\mathcal{P}$. Given an optimal solution x_{LP}^* to (LP), *cutting plane algorithms*, which were first introduced by Gomory [Gom58], generate an improved formulation of (ILP) whenever $x_{\text{LP}}^* \notin \mathcal{P}$. This tightened formulation is obtained by adding a linear constraint that is satisfied by every point in $\mathcal{P}$, but is not obeyed by x_{LP}^* . Such an inequality is called a *cut*. This process continues until enough cuts have been added so that the resulting LP relaxation has an integral optimal solution, thereby solving (ILP). In Land and Doig's algorithm [LD60], whenever $x_{\text{LP}}^* \notin \mathcal{P}$ the feasible region of the integer program is divided into subregions satisfying $x_{\text{LP}}^* \notin \mathcal{P}_1$ and $x_{\text{LP}}^* \notin \mathcal{P}_2$. A canonical example is to branch on a fractional coordinate $x_{\text{LP},j}^* \notin \mathbb{Z}$ by considering $\mathcal{P}_1 := \{x \in \mathcal{P} : x_j \leq \lfloor x_{\text{LP},j}^* \rfloor\}$ and $\mathcal{P}_2 := \{x \in \mathcal{P} : x_j \geq \lceil x_{\text{LP},j}^* \rceil\}$. This process continues until a global certificate of optimality (or infeasibility) is obtained. Over the years, many different strategies for decomposing $\mathcal{P}$ have been developed along these lines, and the resulting family of methods are termed as *branch-and-bound*; see [Ach07] for a modern survey. Using cutting planes within branch-and-bound gives rise to *branch-and-cut*. Variants of these algorithms underlie the software in [Gur24, CPL09, HBB⁺25].

Stronger relaxations are therefore central. Working with a relaxation whose feasible region more closely approximates the original feasible region $\mathcal{P}$ produces tighter bounds on OPT , and reduces the search footprint. Gomory's *group relaxation* [Gom65, Gom67, Gom69] is a principled strengthening derived from an optimal LP basis. Assume $\text{rank}(A) = m$ and let $\mathcal{B} \subset [n]$ denote an optimal basis of the LP relaxation. We can partition the variables and data according to $\mathcal{B}$ and its complement $\mathcal{N} := [n] \setminus \mathcal{B}$, writing

$$A = [A_{\mathcal{B}}, A_{\mathcal{N}}], \quad c = [c_{\mathcal{B}}, c_{\mathcal{N}}], \quad x = [x_{\mathcal{B}}, x_{\mathcal{N}}].$$

The group relaxation is formed by dropping the nonnegativity constraints on $x_{\mathcal{B}}$ (i.e., the entries that took positive values in the optimal solution of the LP relaxation) in (ILP), leading to:

$$\text{OPT}_{\mathcal{B}} := \min \left\{ c^\top x : x \in \mathcal{P}_{\mathcal{B}} \right\}, \quad (\text{ILP}_{\mathcal{B}})$$

where

$$\mathcal{P}_{\mathcal{B}} := \{(x_{\mathcal{B}}, x_{\mathcal{N}}) \in \mathbb{Z}^m \times \mathbb{Z}_{\geq 0}^{n-m} : Ax = b\}.$$

²A matrix A is totally unimodular (TU) if every square submatrix has determinant in $\{-1, 0, 1\}$. When A is TU and b is integral, the polyhedron $\{x \in \mathbb{R}_{\geq 0}^n : Ax = b\}$ is integral, so the LP relaxation attains an integer optimum. TU matrices arise in network and flow models, including bipartite matching/assignment, transportation, and min-cost flow.

It is immediate that $\text{OPT}_B \leq \text{OPT}$, since $\mathcal{P} = \mathcal{P}_B \cap \mathbb{Z}_{\geq 0}^n$. Note that $\mathcal{P}_B \not\subseteq \mathcal{P}_{\text{LP}}$ in general, because the group relaxation permits x_B to take negative values. However, $\text{OPT}_{\text{LP}} \leq \text{OPT}_B$. To see this, note that for $(x_B, x_N) \in \mathcal{P}_B$, we may use invertibility of A_B and write

$$c_B^\top x_B + c_N^\top x_N = c_B^\top A_B^{-1} b + (c_N - A_N^\top (A_B^{-1})^\top c_B)^\top x_N,$$

where $\bar{c}_N := c_N - A_N^\top (A_B^{-1})^\top c_B \geq 0$ are the usual LP *reduced costs* for the nonbasic indices. It is well-known [Sch98, WN99] that, upon eliminating x_B via $x_B = A_B^{-1}(b - A_N x_N)$, the group relaxation (ILP_B) can be equivalently expressed as

$$\text{OPT}_B = \min_{x_N \in \mathbb{Z}_{\geq 0}^{n-m}} \left\{ c_B^\top A_B^{-1} b + \bar{c}_N^\top x_N : A_B^{-1} A_N x_N \equiv A_B^{-1} b \pmod{\mathbb{Z}^m} \right\}. \quad (1)$$

Since B is optimal $c_B^\top A_B^{-1} b = \text{OPT}_{\text{LP}}$, which combined with nonnegativity of $\bar{c}_N$ asserts that feasible solutions to (1) satisfy

$$c_B^\top A_B^{-1} b + \bar{c}_N^\top x_N \geq \text{OPT}_{\text{LP}}.$$

As a consequence, $\text{OPT}_B \geq \text{OPT}_{\text{LP}}$ and together, we have the chain $\text{OPT}_{\text{LP}} \leq \text{OPT}_B \leq \text{OPT}$.

The group relaxation can, under suitable conditions, solve the original ILP to optimality. Gomory originally showed [Gom65] that (ILP_B) is an “asymptotic relaxation”: it recovers the ILP optimum whenever the right-hand side b is sufficiently large [WN99, Theorem 5.6]. More generally, Lasserre [Las04, Las09] identified a discrete nondegeneracy condition under which the group relaxation is exact: $\text{OPT}_B = \text{OPT}$ unless there exists another optimal LP basis $B' \neq B$ with $\text{OPT}_{B'} = \text{OPT}_B$. A certificate of optimality can also be obtained efficiently. If a group-optimal solution $x^* \in \mathcal{P}_B$ happens to be feasible for the original ILP, then x^* is optimal for the ILP because $\text{OPT}_B \leq \text{OPT}$.

Even when the group relaxation is degenerate (so $\text{OPT}_B < \text{OPT}$), it still provides strictly stronger information than the plain LP relaxation. As a bounding oracle, it yields tighter lower bounds, often allowing branch-and-bound to fathom nodes that would pass an LP bound test. As a tightening oracle, the *corner polyhedron* [Gom65, Gom67, Gom69]—the convex hull of group-feasible points—has been tremendously impactful in integer programming due to its connection to cutting planes [CCZ14, Chapter 6]. The geometry of the corner polyhedron exposes facet inequalities valid for the original ILP that typically dominate standard LP-derived cuts, further shrinking the integrality gap [JLN⁺09, Chapter 19].

Finally, while $\mathcal{P}_B$ (and hence the feasible set of (1)) can appear to be infinite as written, it is known that whenever the group relaxation is feasible there exists an optimal solution with nonbasic coordinates bounded by their orders, i.e., $x_j^* \in \{0, \dots, u_j - 1\}$ for $j \in N$, where u_j is determined by the lattice generated by A_B [Sha68, Las04]. Accordingly, one may work over an *optimum-preserving* finite feasible subset

$$\bar{\mathcal{P}}_B := \left\{ x_N \in \bigoplus_{j=1}^{n-m} \mathbb{Z}_{u_j} : A_B^{-1} A_N x_N \equiv A_B^{-1} b \pmod{\mathbb{Z}^m} \right\}.$$

In this bounded-residue view, feasibility is captured by a linear congruence over a finite abelian group and the feasible points form a coset. For more details, see Section 3. This structure is well suited to quantum primitives—finite abelian groups admit efficient quantum Fourier transforms [Kit95], and uniform superpositions over residue classes can be prepared coherently.

1.3 Contributions

The main contributions in this paper can be summarized as follows:

1. We give a competitive classical algorithm for solving Gomory’s group relaxation via a feasibility-preserving local search over a finite abelian group.
2. We design a quantum algorithm within the generalized short-path framework [CHO⁺24], provide efficient state-preparation and block-encodings, and identify conditions under which the speedup is *super-quadratic*.
3. We construct *constraint-preserving mixers* over the group relaxation’s feasible set—(i) a product-of-cycles with explicit log-Sobolev bounds and (ii) an expander with constant spectral gap. We also show that the ground states of the corresponding mixing Hamiltonians can be prepared efficiently. Our approach therefore provides feasibility-preserving, good-gap primitives for short-path/QAOA/QAA on group relaxations; and these are of independent interest.

We begin first by informally stating our main result as follows.

Theorem 1 (Main Result, informal statement of Theorems 9 & 11). *Let $\mathcal{B}$ be an optimal basis for the LP relaxation of (ILP). Define the set of optimal solutions to the (finite) group relaxation associated with this basis to be*

$$\overline{\mathcal{P}}_{\mathcal{B}}^* := \{x_{\mathcal{N}}^* \in \overline{\mathcal{P}}_{\mathcal{B}} : \text{OPT}_{\text{LP}} + \bar{c}_{\mathcal{N}}^\top x_{\mathcal{N}}^* = \text{OPT}_{\mathcal{B}}\}.$$

We construct a quantum algorithm that, with high probability, finds an optimal solution $x_{\mathcal{N}}^ \in \overline{\mathcal{P}}_{\mathcal{B}}^*$ with runtime*

$$\mathcal{O}\left(\left(|\overline{\mathcal{P}}_{\mathcal{B}}|/|\overline{\mathcal{P}}_{\mathcal{B}}^*|\right)^{\frac{1}{2}-\alpha(n)} \text{poly}(n, L)\right),$$

where $\alpha(n) > 0$ and satisfies $\alpha(n) = \mathcal{O}_n(1)$. Under mild technical assumptions, $\alpha(n) = \Theta_n(1)$, in which case the speedup is genuinely super-quadratic.

This theorem combines each of our main contributions. Firstly, as discussed, the classical algorithm that we introduce locally explores the feasible space of the group relaxation and hence incurs a cost of $\mathcal{O}\left((|\overline{\mathcal{P}}_{\mathcal{B}}|/|\overline{\mathcal{P}}_{\mathcal{B}}^*|) \text{poly}(n, L)\right)$. Our quantum algorithm is therefore an improvement over a direct Grover-like speedup of this classical algorithm. In fact, this improvement occurs under the technical conditions that make $\alpha(n) = \Theta_n(1)$. These conditions will be discussed in more detail in the technical sections, and we will give constructive examples where they are satisfied.

We also note that each of these developments are in fact *necessary* to obtain this improvement. The framework of [CHO⁺24] as well as several other quantum frameworks that may enable super-quadratic speedups require a “mixer” Hamiltonian that drives the exploration of the feasible space. In the following subsections, we explore the main technical ideas behind each contribution.

Solving the group relaxation with null space walks As discussed in our motivations, quantum algorithms for discrete optimization can struggle to handle constraints in a natural way. For example, a large class of methods are based on quantum Hamiltonians, which encode the cost function in a diagonal Hermitian operator and perform “mixing” operations, via a non-diagonal Hamiltonian, to transform from some initial configuration to the optimal solution, i.e. a minimal energy configuration of the diagonal Hamiltonian. To handle constraints, a penalty

term is often added to the objective, or, in situations where the constraint set is simple (e.g., a single Hamming weight constraint) a constraint-preserving mixer is used. This typically has detrimental effects on performance, however, since either approach often disrupts the spectral properties of the relevant Hamiltonians which determine the efficiency of such approaches. This is connected to a similar difficulty for *classical* algorithms, where most algorithms that handle constraints are global and exhaustive. Local algorithms that explore the feasible space directly, either resort to penalty terms or are limited to constrained spaces that can be efficiently “enumerated.” Since the enumeration of the feasible space of an ILP is NP-complete in general, this is only applicable to very specific ILPs.

The first ingredient of our quantum algorithm for the group relaxation is therefore a corresponding classical algorithm that is only based on a local search. Classical methods for solving the group relaxation are rooted in Gomory’s original work, and run in time $\mathcal{O}(n |\det A_B|)$, see, e.g., [AWW02, Proposition 4.4]. These involve searching for the shortest path between two elements (the identity and an element determined by the right-hand side vector b) in a directed Cayley graph associated with a particular finite abelian group G , where the presented runtime uses the upper bound $|G| \leq |\det A_B|$. The group G is related to the integer span of the constraint matrix $A_B^{-1} A_N$. This algorithm does not directly explore the feasible space locally; it actually always first encounters infeasible points before finding the solution. Nevertheless, the algorithm does use more local information than an exhaustive search method. There is also no current quantum algorithmic framework that facilitates a path to super-quadratic speedups for this shortest-path approach, and we leave this as an interesting potential direction for future work.

To ameliorate these difficulties, we move the search to the null space of the constraint matrix, $A_B^{-1} A_N$. Specifically, we reduce (ILP_B) to a problem with a domain Z that is a finite abelian group. The feasible set for this finite problem is a coset $\bar{\mathcal{P}}_B = \hat{x} + K \subseteq Z$, where $\hat{x} \in Z$ is any feasible residue vector and $K \leq Z$ is the kernel of the map $\mathbf{z} \mapsto A_B^{-1} A_N \mathbf{z} \pmod{\mathbb{Z}^m}$, $\mathbf{z} \in Z$. Note that $|\hat{x} + K| = |K|$. In Section 3, we construct a random walk on K , where each step of the walk remains within the feasible set by design. This approach ensures that feasibility is preserved at every step, and allows us to efficiently prepare a uniform superposition over feasible solutions to initialize the algorithm. We show in Lemma 1 that an arbitrary generating set for K can be found in $\mathcal{O}(\text{poly}(n, L))$ classical time.

It remains to argue that the local search algorithm is comparable to preexisting classical methods. Particularly, our algorithm is competitive with Gomory’s shortest-path approach when $|K|$ is comparable to $|G|$. In Section 3.5, we demonstrate a family of ILPs that satisfy this condition. However, one can determine this for any particular problem efficiently.

Equipped with our local search algorithm for the group relaxation, we are now ready to explore corresponding quantum speedups. While we rely on the framework of [CHO⁺24] to obtain rigorous runtimes, we point out that this walk is potentially useful to derive mixing Hamiltonians for a variety of other quantum algorithms. These include popular heuristics such as the QAOA and quantum annealing, as well as algorithms such as the quantum adiabatic algorithm whose runtime analysis diverges from that considered here.

Generalized Short Path algorithm Our work extends the generalized short path³ (SP) algorithm of [CHO⁺24], which itself builds on the SP framework of Hastings [Has18] and Dalzell et al. [DPCB23]. These algorithms refine the quantum adiabatic algorithm (QAA) [FGGS00] by introducing a piecewise-linear transformation of the cost Hamiltonian and replacing slow adiabatic evolution with two quantum “jumps” between ground states. Let $\mathcal{X}$ be a finite set,

³Note that here and throughout, “short path” has nothing to do with the shortest path problem.

and consider the optimization problem $\min_{x \in \mathcal{X}} f(x)$, encoded as a quantum Hamiltonian H with $H|x\rangle = f(x)|x\rangle$. Let $E^* < 0$ denote the ground state energy, i.e., the optimal objective value $f(x^*)$. The generalized SP Hamiltonian is

$$H_\mu = -D(P) + \mu \theta_\eta(H/|E^*|), \quad (2)$$

where $\theta_\eta(x) := \min\{0, (x + 1 - \eta)/\eta\}$ is a piecewise-linear function, $\mu > 0$ is a tunable parameter, and $D(P)$ is the discriminant matrix of the Markov chain over the feasible set (see Definition 3). The algorithm then starts by preparing the ground state of the discriminant matrix, and then prepares the ground state $|\psi_\mu\rangle$ of an intermediate Hamiltonian H_μ where the chosen μ depends on the properties of the random walk and the cost function. The key idea is that the probability measure $|\psi_\mu|^2$ is more concentrated on the optimal solutions than the stationary distribution of the classical random walk. By using quantum amplitude amplification, this algorithm overall uses super-quadratically fewer queries to the cost function H than existing classical algorithms.

Using the generators of the null space K , one can instantiate a feasibility-preserving product-of-cycles Cayley walk with explicit spectral gap bounds. One can also leverage the nice expansion properties of Cayley graphs, and apply the *Alon-Roichmain theorem* [AR94] to construct an expander walk with spectral gap lower bounded by a constant. Both walks yield runtime guarantees that scale with the number of feasible solutions, and achieve a constant exponent improvement over direct amplitude amplification on K under mild technical conditions. For each walk, we show how to prepare the corresponding ground states efficiently, and construct block-encodings for both $D(P)$ and H . The result is an end-to-end, feasibility-preserving pipeline for group relaxations that advances short path (and, by extension, QAOA/QAA) beyond prior unconstrained-centric applications.

Discussion It is clear that our result satisfies most of the criterion laid out in Question 1. Specifically, the group relaxation can be applied to general ILPs with many constraints, the new classical algorithm is based on local search, and a super-quadratic quantum speedup can be obtained.

The only remaining question is that of *usefulness*. After all, Theorem 1 only offers us a way to solve a relaxation of the ILP, and not the ILP itself, and it is not immediately clear whether there is an impact for the latter problem. Solving the group relaxation can be helpful for solving the ILP in two ways. Firstly, when the group relaxation is nondegenerate (see, Definition 7), the group relaxation solves the ILP directly. In this case, the new classical algorithm that we are accelerating is in fact solving the ILP itself. Whether the quantum algorithm achieves a quantum speedup over *state-of-the-art* classical algorithms for such ILPs depends not only on the size of the feasible set $|\overline{\mathcal{P}}_{\mathcal{B}}|$, but on the performance of state-of-art solvers for these problems (it could be the case, for example that ILPs with nondegenerate group relaxations are easier for these solvers).

A different setting is when the group relaxation is used as the bounding procedure for an outer Branch-and-Bound algorithm. It is known that the performance of Branch-and-Bound algorithm is heavily influenced by the strength of the bounding procedure that is used at each node, and a stronger bounding procedure can substantially reduce the number of nodes that must be explored by the branching procedure. A quantum speedup for the group relaxation can therefore indirectly accelerate a branch-and-bound solver if the resulting optimality gap is lesser than that obtained from the continuous LP relaxation. In Section 5, we test this hypothesis on a mixture of benchmark and synthetically generated integer problems. We find that the group relaxation indeed substantially reduces the gap compared to the LP relaxation.

Furthermore, we observe that for a substantial fraction of tested problems, solving the group relaxation produces a solution for the original ILP.

Beyond pure ILPs, the group relaxation machinery extends naturally to *mixed-integer linear programs* (MILPs), which are problems of the form:

$$\text{OPT}_{\text{MILP}} := \min \left\{ c^\top x : Ax = b, x \in \mathbb{Z}_{\geq 0}^p \times \mathbb{R}_{\geq 0}^{n-p} \right\}. \quad (\text{MILP})$$

MILPs blend integer and continuous decisions and thus enable one to model a wide range of challenging applications. Wolsey [Wol71] showed that, after clearing denominators in the continuous block by choosing

$$\mathbf{D} := \text{lcm} \{ |\det A_{\mathcal{B}}| : \mathcal{B} \in \mathcal{B} \cap \{p+1, \dots, n\} \}$$

and substituting $z_j = \mathbf{D} x_j$ for the continuous variables, every optimal solution to (MILP) is obtained by solving the resulting ILP. This reduction allows the group relaxation techniques developed here to apply verbatim. That said, computing a practically useful $\mathbf{D}$ (and the requisite bases) at scale is nontrivial; devising scalable strategies for this step is beyond the scope of this work and left to future investigation.

1.4 Related work

Integer linear programming We give a high-level overview of algorithmic development for ILP. For a thorough treatment of the field, we refer the reader to the classic texts [GLS88, WN99, Sch87, CCZ14]. Gomory’s cutting plane method was the first algorithm for solving (mixed) integer programming problems [Gom58]. Papadimitriou [Pap81] proved that ILP was in NP, and as a by product, gave a pseudopolynomial-time dynamic programming algorithm for ILPs with a fixed number of constraints.

A significant breakthrough was made by Lenstra [LJ83], who showed that ILPs are polynomial time solvable when the number of variables is fixed. A careful complexity analysis shows that Lenstra’s algorithm runs in time $2^{\mathcal{O}(n^2)}$ times a polynomial in the encoding size L of the problem data. Kannan improved the leading factor to $n^{\mathcal{O}(n)}$ shortly after [Kan83]. Both algorithms fundamentally rely on lattice point enumeration techniques. *Lenstra-type* algorithms utilize a thinnest direction to decompose the IP into lower dimensional subproblems which correspond to parallel lattice hyperplanes. *Kannan-type algorithms* generalize Lenstra-type algorithms by decomposing the feasible region along lattice shifts of a linear subspace.

Building on these ideas, Dadush [Dad12] developed an improved Kannan-type algorithm for the more general problem of *convex integer programming*, where the task is to decide whether a convex set $\mathcal{K} \subseteq \mathbb{R}^n$ contains an integer point. In his thesis, Dadush showed that his algorithm runs in time $2^{\mathcal{O}(n)} n^n \cdot \text{poly}(L)$. He further observed that, assuming a conjecture of Kannan and Lovász [KL88] on the best volume-based lower bound for the covering radius of a convex body with respect to a lattice⁴, the overall complexity could be tightened to $(\log(2n))^{\mathcal{O}(n)} \cdot \text{poly}(L)$. In a recent breakthrough, Reis and Rothvoss [RR23] gave a constructive resolution of the *subspace flatness conjecture*, thereby enabling Dadush’s algorithm to attain this improved bound.

The running time of the algorithm described in [RR23] cannot be reduced in general, and whether there exists a single exponential time algorithm for exact ILP remains open.⁵ Alternatively, Dadush, Eisenbrand and Rothvoss [DER25] give an algorithm based on reducing the

⁴Intuitively, the covering radius is the smallest scaling of a convex body that guarantees a lattice point in every translate.

⁵Kannan and Lovász [KL88] give an instance that saturates the $\log(n)$ lower bound for the covering radius.

exact ILP problem to a sequence of approximate ones. These intermediate problems are solved using an algorithm for approximate IP from Dadush [Dad14], which runs in time $2^{\mathcal{O}(n)}$. This approach leads to a framework for exactly solving general ILPs in time $2^{\mathcal{O}(n)} \cdot n^n$, but matches the state of the art $(\log(2n))^{\mathcal{O}(n)}$ complexity bound for problems with solutions whose entries are at most $\text{poly}(n)$.

Quantum algorithms for integer and combinatorial optimization By now quantum proposals for combinatorial optimization have been investigated for over two decades. The Quantum adiabatic algorithm [FGGS00] and Quantum Approximate Optimization algorithm [FGG14] are the most thoroughly studied frameworks.

There are also universal quantum speedups for backtracking [Mon18, AK17], which themselves furnish universal quantum speedups for branch-and-bound [Mon20, CMYP22]. However, the speedup here is quadratic. More recently, Decoded Quantum Interferometry [JSW⁺25] has been proposed for solving optimization problems over finite fields. This approach is based on Regev’s reduction [Reg09], and obtains exponential speedups for approximately solving the *optimal polynomial intersection* problem.

The group relaxation and the corner polyhedron Gomory introduced and studied the group relaxation and the corner polyhedron in [Gom65, Gom67, Gom69]. Wolsey [Wol71] generalized the group relaxation to *mixed integer programming* shortly after, and introduced *extended relations*. Extended relations are all of the possible group relaxations one can get upon dropping successively fewer nonnegativity constraints. One of the extended relaxations is guaranteed to solve the ILP to optimality, since the original ILP trivially corresponds to the extended relaxation obtained from dropping no non-negativity constraints.

There are several studies that investigate strategies for mitigating poor performance in the degenerate setting [BS77, GNS73, Wol73, WN99]. Empirical studies on the tightness of the group relaxation and the closely related *corner relaxation* are provided in [FM08, CMN12].

In addition to characterizing the nondegeneracy condition, Lasserre [Las04] showed that group relaxations are related to a generalized residue formula from Brion and Vergne [BV97], which serves as the foundation for efficient lattice point enumeration in polytopes when the dimension is a fixed constant [Bar94, BP99, DLHTY04].

Paper organization

The rest of this paper is organized as follows. Section 2 establishes notation and reviews the necessary background material, covering finite abelian groups, lattices, Markov chains, linear programming and the generalized short-path framework from [CHO⁺24]. In Section 3, we develop the null space formulation of the group relaxation, and present a classical Markov-chain local search over the kernel. In Section 4, we give the quantum algorithm: we specify state preparation and block-encodings, and analyze two mixers (product-of-cycles and expander walks) to obtain our main runtime guarantees. Section 5 reports empirical results on applying the group relaxation to synthetic cutting-stock instances and every pure ILP found in MIP LIB 2017 [GHG⁺21]. Appendix A details practical strategies for optimizing the null space computations, and Appendix B discusses an alternative speedup avenue (Gibbs sampling) and how this setting interacts with our assumptions.

2 Preliminaries

Let $[d] = \{1, \dots, d\}$, and $\mathbb{R}_{\geq 0}^n$ (resp. $\mathbb{R}_{> 0}^n$) denote the nonnegative (resp. positive) orthant. We represent the $n \times 1$ all-ones (resp. all-zeros) vector by $\mathbf{1}_n \in \mathbb{R}^n$ (resp. $\mathbf{0}_n \in \mathbb{R}^n$). Let I_m denote the $m \times m$ identity matrix. Given a matrix A , let A_j refer to the j th column of A . More generally, given a subset $S \subseteq [n]$ of indices, let A_S denote the submatrix whose columns are indexed in S . Similarly, a vector v has subvectors v_S obtained by collecting only those entries indexed in S . Given a vector over $\mathbb{R}$, or equivalently a real-valued function whose domain is finite, the support $\text{supp}(v)$ of v is the collection of indices such that $v_i \neq 0$. We let $\|\cdot\|_p$ denote the usual ℓ_p -norm on $\mathbb{R}^d$ for $p \in [1, \infty]$, and the ℓ_0 -norm counts the number of non-zero entries. We use $\mathbf{e}_j \in \mathbb{R}^n$ to denote the j -th, n -dimensional standard basis vector.

2.1 Abelian groups and lattices

For the basic definitions and properties of groups, we refer the reader to any abstract algebra textbook such as [Lan12]. Here we provide some key results and definitions pertinent to this work.

A group G is called *abelian* if its group binary operation is commutative. In the abelian setting, it is customary to represent the group operation using the plus symbol, its identity using 0, its inverse using a minus sign, etc. All groups considered in this work are abelian. For an integer k and $g \in G$, let kg denote repeated addition of g or its inverse k times, depending on the sign of k . Let $\cong$ denote group isomorphism. We will use the notation $S \leq G$ to denote subgroups $F = S$ of G , and $G/S = \{g + S : g \in G\}$ to denote the quotient group, whose elements are cosets $g + S = \{g + s : s \in S\}$. Given $a, b \in G$, we write $a = b \pmod{S}$ if $a \in b + S$. To maintain brevity, we frequently use a to refer to the coset $a + S$, and context should make clear when the quotient space is being discussed.

Given a subset $S \subseteq G$, let $\langle S \rangle$ denote the smallest subgroup of G containing S . We say that S *generates* G if $\langle S \rangle = G$. Equivalently, S generates G if every $g \in G$ can be obtained from finite products and inverses of elements in S . A group G is called *cyclic* if it has a single generator g , i.e., $G = \langle g \rangle$. The *order* of a group G is simply its cardinality as a set $|G|$. We also refer to the order of an element g to mean the order of $\langle g \rangle$.

Let $\mathbb{Z}_r$ denote the additive group of integers modulo r . The *fundamental theorem of finite abelian groups* states that every nontrivial finite abelian group G is isomorphic to a direct sum

$$G \cong \mathbb{Z}_{r_1} \oplus \mathbb{Z}_{r_2} \cdots \oplus \mathbb{Z}_{r_k},$$

for integers $r_1, \dots, r_k \geq 2$. There are several canonical choices for the integers r_j ; here we typically consider the *invariant factor* decomposition, whereby $r_j | r_{j+1}$ (i.e., r_j divides r_{j+1}), for all $j \in [k-1]$. In this case, the r_j are called invariant factors, and the decomposition is unique.

A *norm* $\|\cdot\| : G \rightarrow \mathbb{R}_{\geq 0}$ over an abelian group G is a real-valued function that is positive definite, subadditive, and inverse symmetric ($\| -g \| = \| g \|$). Of interest in this work is a group norm defined below.

Definition 1 (Cyclic metric). For $v \in \bigoplus_{i=1}^d \mathbb{Z}_{a_i}$ and nonzero $w \in \mathbb{R}^d$, define

$$\|v\|_{\text{cyc}, 1, w} := \sum_{i=1}^d |w_i| \max\{v_i, a_i - v_i\}.$$

For our purposes, a *lattice* is a discrete abelian subgroup of $\mathbb{R}^m$ given by

$$\Lambda = \left\{ \sum_{i=1}^n z_i B_i : z_i \in \mathbb{Z} \right\}$$

where $B_1, \dots, B_n \in \mathbb{R}^m$ are linearly independent. See [Cas96] for a comprehensive overview. If the B_i also form a basis ($n = m$), the lattice is *full rank*. We also denote Λ by $B\mathbb{Z}^n$, where B has columns B_i . When $\ker(B) \neq 0$, i.e., the columns are linearly dependent, $B\mathbb{Z}^n$ may not form a lattice but will in certain cases, such as if B has rational entries. A square integer matrix U is *unimodular* if $\det U = \pm 1$. Such a matrix always has unimodular inverse, and two full-rank lattices $A\mathbb{Z}^n, B\mathbb{Z}^n$ are equal iff $A = BU$ for some unimodular U . We can think of U as a change of basis matrix for a lattice, changing the fundamental cell while preserving the volume of that cell.

Any integer matrix $A \in \mathbb{Z}^{m \times n}$ can be expressed in *Smith normal form* (SNF) $A = URV$, where U, V are unimodular with dimension m, n respectively, and $R = \text{diag}(r_1, \dots, r_k)$ is a diagonal integer matrix with $r_i | r_{i+1}$ for all $i \in [k-1]$. The rightmost r_i may be zero. The SNF can be computed in time polynomial in the dimension and log of the matrix norm [Ili89], with improved performance in sparse instances. Given a full-rank lattice $A\mathbb{Z}^m$ with square matrix A , the quotient group $Q = \mathbb{Z}^m / A\mathbb{Z}^m$ is finite and

$$|Q| = |\det A|.$$

In fact, the SNF of A fully determines the group structure in the sense that

$$\mathbb{Z}^m / A\mathbb{Z}^m \cong \bigoplus_{j=1}^m \mathbb{Z}_{r_j}$$

is the invariant factor decomposition. Specifically, Q has independent generators which are the columns u_j of U (as cosets), where each u_j generates the cyclic group $\mathbb{Z}_{r_j}$.

2.2 Basics of finite Markov chains

Let $\mathcal{X}$ be a finite set of size N . A *Markov chain* on $\mathcal{X}$ is a random process that moves between elements of $\mathcal{X}$ according to an $N \times N$ transition matrix P , where $P(x, y)$ is the probability of moving from x to y . The matrix P is stochastic: $\sum_{y \in \mathcal{X}} P(x, y) = 1$ for all $x \in \mathcal{X}$. Probability distributions ρ on $\mathcal{X}$ evolve according to $\rho^{(t+1)} = \rho^{(t)}P$. After t steps the state is $\rho^{(t)} = \rho^{(0)}P^t$, $\rho^{(0)} = \rho$. A distribution π is *stationary* if $\pi P = \pi$. For any function $f : \mathcal{X} \rightarrow \mathbb{R}$,

$$(Pf)(x) = \sum_{y \in \mathcal{X}} P(x, y)f(y).$$

The *reversed Markov chain* $\mathcal{M}^* = (\mathcal{X}, P^*, \pi)$ shares the stationary distribution π and has transition matrix P^* defined by

$$\pi(x)P(x, y) = \pi(y)P^*(y, x).$$

The chain is *reversible* if $P^* = P$, and the chain is *symmetric* if $P = P^\top$.

For reversible chains, there exists an inner product space over which P is diagonalizable, with eigenvalues

$$1 = \lambda_1 > \lambda_2 \geq \dots \geq \lambda_n \geq -1,$$

where $\lambda_1 = 1$ and $\lambda_2 < 1$. The *spectral gap* is

$$\delta = 1 - \max\{|\lambda| : \lambda \in \{\lambda_2, \dots, \lambda_n\}, \lambda \neq \pm 1\}.$$

The chain is *aperiodic* if the set $\mathcal{T}(x) = \{t \geq 1 : P^t(x, x) > 0\}$ satisfies $\gcd(\mathcal{T}(x)) = 1, \forall x \in \mathcal{X}$. This can always be assured by making the chain *lazy*:

$$P_{\text{lazy}} = \frac{I + P}{2},$$

which only halves the spectral gap but preserves the stationary distribution.

The *mixing time* of a Markov chain is the amount of time it takes for the distance to stationarity to be small:

$$t_{\text{mix}}(\varepsilon) := \min \{t : d(t) \leq \varepsilon\},$$

where $d(t) := \sup_{\mu} \text{TV}(\mu P^t, \pi)$, and TV is the *total variation distance*

$$\text{TV}(\mu, \nu) := \frac{1}{2} \|\mu - \nu\|_1.$$

The relationship between mixing time and the spectral gap for a reversible Markov chain can be expressed as

$$t_{\text{mix}}(\varepsilon) = \mathcal{O} \left(\delta^{-1} \cdot \text{polylog} \frac{1}{\varepsilon} \right).$$

We also introduce the concept of log-Sobolev inequalities.

Definition 2 (Log-Sobolev inequality). *A Markov chain $\mathcal{M} = (\mathcal{X}, P, \pi)$ satisfies a log-Sobolev inequality if for all $f : \mathcal{X} \rightarrow \mathbb{R}$,*

$$\frac{1}{2} \mathbb{E}_{x \sim \pi} \mathbb{E}_{y \sim x} [(f(x) - f(y))^2] \geq \omega (\mathbb{E}_{x \sim \pi} [f(x) \ln f(x)] - \mathbb{E}_{x \sim \pi} [f(x)] \ln \mathbb{E}_{x \sim \pi} [f(x)]).$$

The largest such ω is called the *log-Sobolev constant*, and is related to the spectral gap via $\omega \leq \delta$.

The *discriminant matrix* is a useful operator for analyzing random walks:

Definition 3 (Discriminant matrix). *For $\mathcal{M} = (\mathcal{X}, P, \pi)$, the discriminant matrix is*

$$D(P)_{ji} = \sqrt{P_{ij} P_{ji}^*} = \left(\text{diag}(\pi)^{1/2} P \text{diag}(\pi)^{-1/2} \right)_{ij}.$$

Lastly, we also define the concept of pseudo-Lipschitzness.

Definition 4 (*P-pseudo Lipschitz norm*). *Let $\mathcal{M} = (\mathcal{X}, P, \pi)$ be a Markov chain. The P-pseudo Lipschitz norm of $f : \mathcal{X} \rightarrow \mathbb{R}$ is defined to be*

$$\|f\|_P := \max_{x \in \mathcal{X}} \mathbb{E}_{y \sim x} [(f(x) - f(y))^2].$$

2.3 Background on linear programming

The *linear optimization* problem and its dual can be written in standard form as

$$\min_{x \in \mathbb{R}^n} \left\{ c^\top x : Ax = b, x \geq 0 \right\}, \quad \max_{y \in \mathbb{R}^m} \left\{ b^\top y : A^\top y \leq c \right\},$$

where $A \in \mathbb{Z}^{m \times n}$ and $b \in \mathbb{Z}^m$ define a collection of m linear constraints. Here and throughout, we assume A is full rank. For our purposes, a *basis* $\mathcal{B} \subseteq [n]$ is a subset of m indices such that the vectors $\{A_j\}_{j \in \mathcal{B}}$ form a basis over $\mathbb{R}^m$ in the usual sense. A vector $x \in \mathbb{R}^n$ is called *feasible* if x satisfies the constraints, i.e., $Ax = b$ and $x \geq 0$. Moreover, a *basic feasible solution* $x = (x_{\mathcal{B}}, x_{\mathcal{N}})$ is a feasible solution such that its nonzero entries are indexed by $\mathcal{B}$. Note that the values of $x_{\mathcal{B}}$ are uniquely determined by the system $A_{\mathcal{B}} x_{\mathcal{B}} = b$. The associated basic dual solution is $y_{\mathcal{B}} = (A_{\mathcal{B}}^\top)^{-1} c_{\mathcal{B}}$.

Let $\mathcal{B}$ denote the collection of all feasible bases of the LP relaxation. For any basis $\mathcal{B} \in \mathcal{B}$, we define the complement $\mathcal{N} := [n] \setminus \mathcal{B}$, and the complementary parts $A_{\mathcal{N}}, c_{\mathcal{N}}$ of the constraints and the objective function, respectively. We will assume that $m = \Theta(n)$. Associated to

any basis $\mathcal{B}$, the *reduced costs* (or dual slacks) are given by $\bar{c} := c - A_{\mathcal{N}}^{\top} y_{\mathcal{B}}$. The reduced cost $\bar{c}_j$ for $j \in [n]$ quantifies the rate of improvement in the objective function per unit increase in x_j , subject to feasibility.

Let $(x^*, y^*) \in \mathbb{R}_{\geq 0}^n \times \mathbb{R}^m$ denote a primal-dual solution corresponding to basis $\mathcal{B} \in \mathcal{B}$. We say that $\mathcal{B}$ is an *optimal* basis if, in addition to primal-dual feasibility, *complementarity slackness* is satisfied:

$$x^* \odot (c - A^{\top} y^*) = \mathbf{0}_n$$

where $\odot$ denotes the element-wise product. When the LP is nondegenerate, $\mathcal{B} = \text{supp}(x^*)$ and $\mathcal{N} = \{i \in [n] : x_i^* = 0\}$.

When the problem data (A, b, c) is integer, we let L denote the binary input length:

$$L := \sum_{i=1}^m \sum_{j=1}^n [1 + \lfloor \log_2 |A_{ij}| \rfloor] + \sum_{i=1}^m [1 + \lfloor \log_2 |b_i| \rfloor] + \sum_{j=1}^n [1 + \lfloor \log_2 |c_j| \rfloor].$$

The optimal partition $(\mathcal{B}, \mathcal{N})$ can be found in time $\mathcal{O}(n^3 L)$ using interior point methods [Wri97].

2.4 The generalized short path algorithm

Short path (SP) algorithms, first proposed by Hastings [Has18] and later refined by Dalzell et al. [DPCB23], represent a methodological advancement over the QAA [FGGS00]. Below, we outline the core concepts.

While quantum computers cannot efficiently prepare the ground state for general Hamiltonians, certain Hamiltonians have ground states that are easy to construct. The QAA leverages this by starting with an initial Hamiltonian H_0 whose ground state is simple to prepare—commonly, the transverse-field operator $H_0 = -\sum_{i \in [n]} \sigma_i^x$, where σ_i^x is the Pauli X operator on qubit $i \in [n]$. The algorithm then defines a time-dependent Hamiltonian

$$H_{\text{QAA}}(t) := (1 - \tau(t))H_0 + \tau(t)H,$$

where $\tau : [0, T] \rightarrow [0, 1]$ is a smooth, monotonic function satisfying $\tau(0) = 0$ and $\tau(T) = 1$ so that $H_{\text{QAA}}(0) = H_0$ and $H_{\text{QAA}}(T) = H$. According to the *adiabatic theorem* [Mes58], simulating the Schrödinger equation

$$i \frac{d}{dt} |x(t)\rangle = H_{\text{QAA}}(t) |x(t)\rangle$$

for sufficiently large T prepares the ground state of H , provided that the minimum gap $\delta_{\min}$ between the lowest two eigenvalues is nonzero. Here

$$\delta_{\min} := \min_{t \in [0, T]} \{\lambda_2(H_{\text{QAA}}(t)) - \lambda_1(H_{\text{QAA}}(t))\},$$

where λ_1 and λ_2 are the smallest and second smallest eigenvalues, respectively.

The challenge is that $\delta_{\min}$ tends to shrink rapidly with system size, necessitating prohibitive simulation times T . Roughly, T scales polynomially with the inverse gap. Obtaining analytical bounds on the spectral gap is notoriously difficult [HS12], but more importantly, the QAA is hampered by “localization” phenomena—quantum analogues of a classical particle getting trapped in a local minimum. Localization can induce (super) exponentially small gaps, resulting in run times that may be exponentially slower than classical brute-force search [AKR10].

SP algorithms avoid first-order phase transitions via two key modifications:

1. Instead of using the standard adiabatic Hamiltonian H_{QAA} , the algorithm uses a slightly different interpolation between the mixer Hamiltonian H_0 and the final Hamiltonian H which encodes a truncated and normalized cost function.

2. The algorithm does not follow the adiabatic path, but performs two successive jumps called short jump and long jump to avoid the points where the spectral gap of the Hamiltonian decays (super) exponentially.

The *generalized* quantum SP algorithm, as developed in [CHO⁺24], further provides a principled framework for achieving super-quadratic quantum speedups over not just over unstructured search, but also over classical Markov Chain search in combinatorial optimization. The key ingredient is a generalized mixer Hamiltonian over the Markov Chain graph of interest. Below, we review the algorithm and conditions for super-quadratic speedups.

2.4.1 Short path Hamiltonian and algorithmic steps

Let $\mathcal{M} = (\mathcal{X}, P, \pi)$ be a reversible, aperiodic Markov chain over a finite set $\mathcal{X}$, where π is the stationary distribution of P . The generalized SP Hamiltonian H_μ is defined as in (2). This construction generalizes the original SP framework to arbitrary reversible Markov chains and cost functions, as formalized in [CHO⁺24]. We define $E^* := \min_{z \in \mathcal{X}} H(z)$, and subtract a scalar from the cost function to ensure $E^* < 0$.

We outline the generalized SP algorithm in Algorithm 1. The algorithm begins by preparing the ground state $|\sqrt{\pi}\rangle = \sum_{z \in \mathcal{X}} \sqrt{\pi(z)}|z\rangle$ of $-D(P)$, a weighted superposition over the feasible set $\mathcal{X}$. Next, quantum phase estimation and amplitude amplification are employed to carry out the short jump. This enables us to prepare the ground state $|\psi_\mu\rangle$ of H_μ for a suitably chosen value of μ . Following this, amplitude amplification is applied to project the state onto the optimal solution subspace, taking advantage of the overlap created by the short jump, leading to the long jump. Finally, a measurement is performed in the computational basis to obtain a feasible solution, which is optimal with high probability. Each step is implemented using block-encoded quantum circuits [CGJ19] and fixed-point amplitude amplification [YLC14], ensuring polynomial overhead in the relevant parameters.

Algorithm 1 GENERALIZEDSHORTPATHALGORITHM

Input: Algorithmic parameters μ, η . Problem parameters H, P, π, E^* .

Output: A global minimizer of H .

- 1: Prepare $|\sqrt{\pi}\rangle$, the ground state of $-D(P)$.
 - 2: **Short Jump:** Given block-encodings of $-D(P)$ and H_μ , prepare $|\psi_\mu\rangle$ (the ground state of H_μ) up to exponentially small error using [DPCB23, Proposition 1].
 - 3: **Long Jump:** Given block-encodings of H_μ and H , prepare $\frac{\Pi^*|\psi_\mu\rangle}{\|\Pi^*|\psi_\mu\rangle\|}$ up to exponentially small error using procedure outlined in [DPCB23, Proposition 1], where Π^* orthogonally projects onto the space of global minimizers in $\mathcal{X}$.
 - 4: Measure the resulting state in the computational basis.
-

2.4.2 Runtime of the short path algorithm

The generalized short path algorithm makes use of the following input assumptions:

Assumption 1. *The following subroutines are used as primitives in the Generalized Quantum SP framework.*

1. **Initial State Preparation:** We assume the existence of a unitary U_π implementable using $\text{poly}(n)$ gates, such that $U_\pi|0\rangle = |\sqrt{\pi}\rangle := \sum_{z \in \mathcal{X}} \sqrt{\pi(z)}|z\rangle$.

2. **Block-encoding of Markov Chain:** We assume the existence of a unitary $U_{D(P)}$, implementable with $\text{poly}(n)$ gates, that is a (κ_1, a) -block-encoding of $D(P)$ for $\kappa_1 = \mathcal{O}(\text{poly}(n))$.
3. **Block-encoding of Cost Function:** We assume the existence of a unitary U_H , implementable with $\text{poly}(n)$ gates, that is a (κ_2, a) -block-encoding of H for $\kappa_2 = \mathcal{O}(\text{poly}(n))$.

Each of the input assumptions are satisfied in the settings we consider in this paper. We provide explicit procedures for initial state preparation and constructing the necessary block-encodings in Section 4.1.

Our analysis of the short path framework requires two additional technical assumptions called γ -spectral density and Δ_P -stability. The first of these concerns properties of the spectral density near the ground state energy.

Definition 5 (γ -Spectral Density – Definition 3.2 [CHO⁺24]). *The cost Hamiltonian H , with ground state energy $E^* < 0$, is said to satisfy the γ -spectral density condition with respect to the stationary distribution π if:*

$$\pi(E \leq (1 - \eta)E^*) \leq \pi(E^*)^\gamma,$$

for some $\eta \in (0, 1)$ and $\gamma > 0$.

To show a true super-quadratic speedup, we require $\gamma = \Theta(1)$. In this regime, the *near-optimal* sublevel set $\{x \in \mathcal{X} : \langle x | H | x \rangle \leq (1 - \eta)E^*\}$ does not carry measure that grows super-polynomially relative to the set of exact optima. If, for every $\eta \in (0, 1)$, no constant γ exists (i.e., the near-optimal sublevel set is disproportionately large), then simple uniform sampling or rapidly mixing chains would find $(1 - \eta)$ -optimal solutions in subexponential time, yielding a subexponential approximation scheme. Thus, for “hard” instances where no such scheme is expected, modeling with a *constant* γ is reasonable: the near-optimal sublevel set is not exponentially larger (in measure) than the optimal set.

Meanwhile, Δ_P -stability is a certain “smoothness” condition that we now define.

Definition 6 (Δ_P -stability – Definition 3.1 [CHO⁺24]). *Let $\mathcal{M} = (\mathcal{X}, P, \pi)$ be a Markov chain and define $\tilde{\theta}_\eta(x) := \theta_\eta\left(\frac{x}{|E^*|}\right)$. We say that H is $\Delta_P(\eta)$ -stable under $\mathcal{M}$ if*

$$\mathbb{E}_{y \sim x}[\tilde{\theta}_\eta(H(y))] \leq \tilde{\theta}_\eta(H(x) + \Delta_P(\eta)), \quad \forall x \in \mathcal{X}.$$

From [CHO⁺24, Lemma 4.26], we have $\sqrt{\|H\|_P} \geq \Delta_P(\eta), \forall \eta \in (0, 1)$. When the value of η is not important, we simply write Δ_P . The Δ_P -stability condition reflects the underlying structure of the problem, that is when Δ_P is large, the problem effectively becomes an unstructured search, and the achievable speedup is typically that of Grover’s algorithm. In fact, the advantage over Grover’s algorithm $\alpha(n)$ in Theorem 9 is inversely proportional to Δ_P .

In Section 3, we will show that the null space random walk we design satisfies all of the aforementioned assumptions for the family of group relaxations considered in this work. When the Markov chain P satisfies a Log-Sobolev inequality (Definition 2), we have the following bound on the runtime of the generalized SP algorithm.

Theorem 2 (Log-Sobolev Version [CHO⁺24, Theorem 4.2]). *Let $\mathcal{M} = (\mathcal{X}, P, \pi)$ be a reversible, aperiodic Markov chain, and let $H : \mathcal{X} \rightarrow \mathbb{R}$ be a diagonal, Δ_P -stable Hamiltonian (Definition 6) with ground state energy $E^* < 0$. Suppose $\mathcal{M}$ has a log-Sobolev constant lower bounded by ω . Then γ -spectral density (Definition 5) is satisfied for*

$$\gamma = \frac{\omega(-(1 - \eta)E^* - \mathbb{E}_\pi[H])^2}{\|H\|_P \ln(1/\pi(E^*))}.$$

Also if μ satisfies

$$\mu < \mu^* := \frac{2}{3}\gamma\omega \ln\left(\frac{1}{\pi(E^*)}\right)$$

and

$$\frac{1}{3\gamma \ln(1/\pi(E^*))\sqrt{\pi(E \leq (1-\eta)E^*)}} - \max\left(\frac{4\ln(|\mathcal{X}|)}{\omega}, \frac{3|E^*|^2(1-\eta)^2}{\Delta_P^2}\right) \geq 2, \quad (3)$$

then there exists a short path algorithm that determines the ground state of H over $\mathcal{X}$ with running time

$$\mathcal{O}\left(\text{poly}(n)\omega^{-1}[\pi(E^*)^{-1}]^{\left(\frac{1}{2} - \frac{\eta(1-\eta)|E^*|\mu}{2\ln(1/\pi(E^*))\Delta_P}\right)}\right).$$

As remarked in [CHO⁺24, Remark 4.4], the additional conditions in the above theorem—on top of spectral density, stability and the maximum-allowed μ —are considerably mild since the denominator in the first term of (3) is exponentially small.

There is also a version of the generalized SP runtime that only depends on the spectral gap of the walk which is slightly weaker, limiting its applicability.

Theorem 3 (Spectral Gap Version [CHO⁺24, Theorem 4.4]). *Let $\mathcal{M} = (\mathcal{X}, P, \pi)$ be a reversible, aperiodic Markov chain, and let $H : \mathcal{X} \rightarrow \mathbb{R}$ be a diagonal, Δ_P -stable Hamiltonian with ground state energy $E^* < 0$, P -pseudo Lipschitz norm $\|H\|_P$. In addition, suppose $\mathcal{M}$ has a spectral gap lower bounded by δ . Then γ -spectral density is satisfied for*

$$\gamma = \frac{\sqrt{\delta}(-(1-\eta)E^* - \mathbb{E}_\pi[\tilde{\theta}_\eta(H)])}{\sqrt{\|H\|_P} \ln(1/\pi(E^*))},$$

and suppose that $\frac{\delta}{2} > \pi(E^*)^\gamma$. Also if μ satisfies

$$\mu < \mu^* := \frac{\delta}{4}$$

and

$$\frac{1}{2\sqrt{\pi(E \leq (1-\eta)E^*)}} - \max\left(\frac{4\ln(|\mathcal{X}|)}{\delta}, \frac{3|E^*|^2(1-\eta)^2}{\Delta_P^2}\right) \geq 2, \quad (4)$$

then there exists a short path algorithm that determines the ground state of H over $\mathcal{X}$ with running time

$$\mathcal{O}\left(\text{poly}(n)\delta^{-1}[\pi(E^*)^{-1}]^{\left(\frac{1}{2} - \frac{\eta(1-\eta)|E^*|\mu}{2\ln(1/\pi(E^*))\Delta_P}\right)}\right).$$

The following provides sufficient conditions under which the SP algorithm provides a super-quadratic speedup over classically sampling from π .

Corollary 1 (Super-Quadratic Condition [CHO⁺24, Corollary 4.5]). *Let $\mathcal{M} = (\mathcal{X}, P, \pi)$ be a reversible, aperiodic Markov chain, and let $H : \mathcal{X} \rightarrow \mathbb{R}$ be a diagonal Hamiltonian with ground state energy $E^* < 0$. Suppose that the chain $\mathcal{M}$ and cost-function H are such that the conditions of Theorem 2 or 3 are satisfied for some $\mu = \Theta(1)$ and*

$$\frac{|E^*|}{\Delta_P} = \Theta(\log(1/\pi(E^*))).$$

Then under Assumption 1, the short path algorithm finds the minimizer of H over $\mathcal{X}$ with running-time bounded by

$$\mathcal{O}\left([\pi(E^*)^{-1}]^{\frac{1}{2}-\alpha} \text{poly}(n)\right),$$

where

$$\alpha = \frac{\eta(1-\eta)|E^*|\mu}{2\Delta_P \ln(1/\pi(E^*))} = \Theta(1).$$

We note that even though the conditions in Corollary 1 seem complicated and somewhat contrived, the speedup factor (say in the setting of Theorem 2) simplifies to $\alpha(n) = \Omega\left(\frac{\omega n}{\Delta_P}\right)$ for typical hard combinatorial optimization problems. This because typically $|E^*| = \Theta(n)$ and $\ln(1/\pi(E^*)) = \mathcal{O}(n)$ since the spectral density condition holds (otherwise we can find approximate solution in sub-exponential time). As an example, consider the hypercube walk P , where $D(P) = \frac{1}{n} \sum_{i \in [n]} \sigma_i^x$. Since $\omega = \Omega(1/n)$ for this walk, $\alpha = \Omega(1)$ as long as the change in the objective function after a bit-flip, bounded by Δ_P is $\mathcal{O}(1)$. For general problems, the challenge is to construct a random walk that has large Log-Sobolev constant and small pseudo-Lipschitz norm or Δ_P . In the next sections, we describe the Markov chain construction to satisfy these conditions for a certain class of integer programming problems.

3 Solving group relaxations through local search

In this section, we first review important details regarding the group relaxation in order to derive a formulation of this problem amenable to local search. We then show how to solve the reformulated problem using *Markov chain search*, and analyze the complexity of the resulting scheme.

3.1 The group relaxation

Let $\mathcal{B}$ denote an optimal basis for the LP relaxation of (ILP). Recall the definition of the group relaxation induced by this basis

$$\text{OPT}_{\mathcal{B}} := \min \left\{ c^\top x : x \in \mathcal{P}_{\mathcal{B}} \right\}, \quad (\text{ILP}_{\mathcal{B}})$$

where $\mathcal{P}_{\mathcal{B}} := \{(x_{\mathcal{B}}, x_{\mathcal{N}}) \in \mathbb{Z}^m \times \mathbb{Z}_{\geq 0}^{n-m} : Ax = b\}$.

Since any $x = (x_{\mathcal{B}}, x_{\mathcal{N}}) \in \mathcal{P}_{\mathcal{B}}$ satisfies $Ax = b$, we may write

$$c^\top x = c_{\mathcal{B}}^\top x_{\mathcal{B}} + c_{\mathcal{N}}^\top x_{\mathcal{N}} = c_{\mathcal{B}}^\top A_{\mathcal{B}}^{-1}(b - A_{\mathcal{N}}x_{\mathcal{N}}) + c_{\mathcal{N}}^\top x_{\mathcal{N}} = c_{\mathcal{B}}^\top A_{\mathcal{B}}^{-1}b + \bar{c}_{\mathcal{N}}^\top x_{\mathcal{N}}.$$

Given that $A_{\mathcal{B}}$ is invertible, we can also reformulate the system $A_{\mathcal{B}}x_{\mathcal{B}} + A_{\mathcal{N}}x_{\mathcal{N}} = b$ as

$$A_{\mathcal{B}}^{-1}A_{\mathcal{N}}x_{\mathcal{N}} + x_{\mathcal{B}} = A_{\mathcal{B}}^{-1}b \implies x_{\mathcal{B}} = A_{\mathcal{B}}^{-1}(b - A_{\mathcal{N}}x_{\mathcal{N}}).$$

Thus, once $x_{\mathcal{N}} \in \mathbb{Z}_{\geq 0}^{n-m}$ is fixed, $x_{\mathcal{B}} \in \mathbb{Z}^m$ is uniquely determined. Accordingly, (ILP _{$\mathcal{B}$}) is equivalent to

$$\text{OPT}_{\mathcal{B}} = \min_{x_{\mathcal{N}} \in \mathbb{Z}_{\geq 0}^{n-m}} \left\{ c_{\mathcal{B}}^\top A_{\mathcal{B}}^{-1}b + \bar{c}_{\mathcal{N}}^\top x_{\mathcal{N}} : A_{\mathcal{B}}^{-1}A_{\mathcal{N}}x_{\mathcal{N}} = A_{\mathcal{B}}^{-1}b \pmod{\mathbb{Z}^m} \right\}. \quad (5)$$

Since $\mathcal{B}$ is optimal for (LP), the nonbasic part of the reduced cost is nonnegative,

$$\bar{c}_{\mathcal{N}}^\top := c_{\mathcal{N}}^\top - c_{\mathcal{B}}^\top A_{\mathcal{B}}^{-1}A_{\mathcal{N}} \geq 0,$$

and $c_B^\top A_B^{-1} b = \text{OPT}_{\text{LP}}$, so we have $\text{OPT}_B \geq \text{OPT}_{\text{LP}}$.

To further refine this formulation, consider the Smith normal form $A_B = URV$, where U and V are unimodular matrices, and $R = \text{diag}(r_1, \dots, r_m)$ with

$$|\det A_B| = \prod_{j=1}^m r_j, \quad r_1 \mid r_2 \mid \dots \mid r_m.$$

Each $r_j > 0$, as A_B is full-rank. Multiplying both sides of the constraint in (5) by RV , we obtain

$$\text{OPT}_B = \text{OPT}_{\text{LP}} + \min_{x_N \in \mathbb{Z}_{\geq 0}^{n-m}} \left\{ \bar{c}_N^\top x_N : U^{-1} A_N x_N = U^{-1} b \pmod{R\mathbb{Z}^m} \right\}.$$

Labeling the data $\tilde{\mathbf{A}} = U^{-1} A_N \in \mathbb{Z}^{m \times (n-m)}$, $\mathbf{b} = U^{-1} b \in \mathbb{Z}^m$, and $\mathbf{c} = \bar{c}_N$, we arrive at

$$\text{OPT}_B = \text{OPT}_{\text{LP}} + \min_{\mathbf{x} \in \mathbb{Z}_{\geq 0}^{n-m}} \left\{ \mathbf{c}^\top \mathbf{x} : \tilde{\mathbf{A}} \mathbf{x} = \mathbf{b} \pmod{R\mathbb{Z}^m} \right\}. \quad (6)$$

Note it may be the case that some columns of $\tilde{\mathbf{A}}$ are already in $R\mathbb{Z}^m$. For those columns, we can simply set $x_j = 0$, as this process never increases the objective value, nor changes the LHS of the constraints modulo $R\mathbb{Z}^m$. Denoting by $\mathbf{A} \in \mathbb{Z}^{m \times d}$, with $d \leq n - m$, the matrix obtained upon dropping these columns, problem (6) simplifies to

$$\text{OPT}_B = \text{OPT}_{\text{LP}} + \min_{\mathbf{x} \in \mathbb{Z}_{\geq 0}^d} \left\{ \mathbf{c}^\top \mathbf{x} : \mathbf{A} \mathbf{x} = \mathbf{b} \pmod{R\mathbb{Z}^m} \right\}. \quad (\widetilde{\text{ILP}}_B)$$

We thus describe an instance of a group relaxation via the tuple $(\mathbf{A}, \mathbf{b}, \mathbf{c}, \{r_j\}_{j=1}^m)$.

Gomory originally showed that, under certain conditions, the group relaxation is *asymptotically tight*—that is, it solves the original ILP exactly when the right-hand side is sufficiently large.

Theorem 4 (Theorem 5.6 in [WN99]). *Let $\mathcal{B}$ be any feasible basis for the LP relaxation of the ILP problem. The group relaxation defined by $\mathcal{B}$ solves the ILP problem to optimality if*

$$A_B^{-1} b \geq \max_{i,j} \left\{ |(A_B^{-1} A_N)_{ij}| \right\} \cdot |\det A_B| \mathbf{1}_{n-m}.$$

Lasserre later established a *nondegeneracy condition*:

Definition 7 (Nondegeneracy condition for group relaxations [Las04]). *Denote the set of all optimal bases to the LP relaxation of (ILP) by $\mathcal{B}$. Let $\mathcal{B} \in \mathcal{B}$ be an optimal LP basis. If there exists another basis $\mathcal{B}' \in \mathcal{B}$ such that $\text{OPT}_B = \text{OPT}_{B'}$ then the problem is degenerate. Otherwise, the group relaxation defined by $\mathcal{B}$ is nondegenerate, and solves (ILP) to optimality.*

This nondegeneracy condition can be viewed as a similar statement to the usual notion of nondegeneracy in linear programming, in the following sense: the situation in which the group relaxation only provides a lower bound on OPT arises, necessarily, when there is another LP basis $\mathcal{B}'$ whose group relaxation yields the same lower bound. We emphasize that group nondegeneracy is strictly stronger than LP nondegeneracy. Moreover, while LP nondegeneracy can often be certified a posteriori (e.g., via reduced costs and basic variable signs), no obvious efficient a posteriori certificate is available for the group case. That said, there is a simple *sufficient* a posteriori check for exactness: if a group-optimal solution is ILP-feasible (i.e., satisfies the dropped nonnegativity), then $\text{OPT}_B = \text{OPT}$; conversely, failure of this check does not certify degeneracy, since there may be another basis which induces a group relaxation that solves the ILP.

3.2 Null space formulation

Recall from $(\text{ILP}_{\mathcal{B}})$ of the prior section that the feasible set to the group relaxation is

$$\{\mathbf{x} \in \mathbb{Z}_{\geq 0}^d : \mathbf{A}\mathbf{x} = \mathbf{b} \pmod{R\mathbb{Z}^m}\}.$$

This set is infinite, provided it is nonempty. However, note that the constraints are linear equations over the finite group $\mathcal{R} := \mathbb{Z}^m / R\mathbb{Z}^m \cong \bigoplus_{i=1}^m \mathbb{Z}_{r_i}$. Thus, the order of $\mathbf{A}_j$ is finite, and necessarily divides r_m . Reducing each component modulo r_m never increases the objective value $\mathbf{c}^\top \mathbf{x}$, since $\mathbf{c} \geq 0$. We can thus take the feasible set to be a subset of $Z := \mathbb{Z}_{r_m}^d$ while preserving optimality and feasibility. We note that the ambient space of $\mathbf{x}$ can be even further reduced with additional refinements, and explore this approach in Appendix A.

Suppose $\mathbf{A}\mathbf{x} = \mathbf{b} \pmod{R\mathbb{Z}^m}$ is solvable, with some solution $\hat{\mathbf{x}} \in Z$. Then, the set of solutions can be written as the coset $\hat{\mathbf{x}} + K$. Here, K is the kernel of $\mathbf{A}$ as a homomorphism from Z to $\mathcal{R}$

$$K = \{\mathbf{x} \in Z : \mathbf{A}\mathbf{x} = \mathbf{0} \pmod{R\mathbb{Z}^m}\}. \quad (7)$$

Thus, as typical with more familiar linear systems over vector spaces, one can parameterize the feasible set of the group relaxation using any feasible solution together with the kernel of some group homomorphism. We shall see, however, that characterizing the kernel is somewhat more challenging than the vector space setting.

Supposing the group problem is feasible, we can formulate the objective function in terms of the kernel subgroup as follows

$$\min_{\mathbf{x} \in \hat{\mathbf{x}} + K} \mathbf{c}^\top \mathbf{x} + c_B^\top A_B^{-1} b = \min_{\mathbf{x} \in K} \mathbf{c}^\top (\hat{\mathbf{x}} + \mathbf{x} \pmod{r_m \mathbb{Z}^d}) + c_B^\top A_B^{-1} b =: \min_{\mathbf{x} \in K} f(\mathbf{x}). \quad (8)$$

The constant shift $c_B^\top A_B^{-1} b$ ensures that $\min_{\mathbf{x} \in K} f(\mathbf{x}) = \text{OPT}_{\mathcal{B}}$.

3.3 Computing the null space

Our formulation (8) is only useful algorithmically provided there are efficient methods for

- (a) producing a feasible solution $\hat{\mathbf{x}} \in Z$ to $\mathbf{A}\mathbf{x} = \mathbf{b} \pmod{R\mathbb{Z}^m}$; and
- (b) accessing elements $\mathbf{x} \in K$.

We shall show how to accomplish both of these tasks in classical polynomial time using tools from computational number theory [Coh00]. We remark that one can formulate (8) as an instance of the abelian Hidden Subgroup Problem by considering the task of finding the kernel of K as a hidden subgroup of Z . This is a problem for which quantum computers are known to be effective. However, because $\mathbf{A}$ is stored in classical memory rather than provided via an oracle, the setting is somewhat distinct, and in any case, an efficient classical algorithm which produces a set of generators is preferable.

Solving $\mathbf{A}\mathbf{x} = \mathbf{b} \pmod{R\mathbb{Z}^m}$ is complicated by the fact that $\mathcal{R}$ is rectangular, as opposed to square: we would rather work in a space of the form $a\mathbb{Z}^m$ for some positive integer a because it remains invariant under unimodular transformations. This can be accomplished with some preconditioning. Note that $\mathbf{x} \in Z$ satisfies $\mathbf{A}\mathbf{x} = \mathbf{b} \pmod{R\mathbb{Z}^m}$, if and only if

$$B\mathbf{A}\mathbf{x} = B\mathbf{b} \pmod{r_m \mathbb{Z}^m},$$

where $B = \text{diag}(r_m/r_1, \dots, r_m/r_{m-1}, 1)$. We shall presently see how this aides us.

Given $B\mathbf{A} \in \mathbb{Z}^{m \times d}$, we first compute the Smith normal form $B\mathbf{A} = UTV$ for $U \in \mathbb{Z}^{m \times m}$, $V \in \mathbb{Z}^{d \times d}$ unimodular and $T = \text{diag}(t_1, t_2, \dots, t_k)$ consisting of invariant factors, with $k = \min(m, d)$. Multiplying by U^{-1} , the system reduces to

$$TV\mathbf{x} = \mathbf{b}' \pmod{r_m \mathbb{Z}^m}, \quad (9)$$

where $\mathbf{b}' := U^{-1}B\mathbf{b}$. To solve this system, observe that by unimodularity, the operation $\mathbf{y} \mapsto V^{-1}\mathbf{y} \pmod{r_m \mathbb{Z}^d}$ defines an automorphism on $\mathbb{Z}$. Thus, any solution $\mathbf{x} \in \mathbb{Z}$ to (9) can be obtained uniquely from some $\mathbf{y} \in \mathbb{Z}$ that solves

$$T\mathbf{y} = \mathbf{b}' \pmod{r_m \mathbb{Z}^m}. \quad (10)$$

The problem has thus been reduced to m independent linear equations modulo r_m

$$t_i y_i = \mathbf{b}'_i \pmod{r_m}$$

whose solutions are well known from number theory. A solution exists if and only if

$$\gcd(r_m, t_i) \mid \mathbf{b}'_i \quad (11)$$

for each $i \in [k]$, and in this case a particular solution $\hat{\mathbf{y}}_i$ can be found through the extended Euclidean algorithm. Thus, $V^{-1}\hat{\mathbf{y}} \pmod{r_m \mathbb{Z}^d}$ is a feasible solution in $\mathbb{Z}_{r_m}^d$.

It remains to determine the kernel K , and this can be done by setting $\mathbf{b}' = 0$. In this case, the i th equation in (11) has $\gcd(r_m, t_i)$ solutions: the multiples of $r_m / \gcd(r_m, t_i)$. The number of solutions is thus $\prod_{i \in [k]} \gcd(r_m, t_i)$. The solutions $\mathbf{x} \in \mathbb{Z}_{r_m}^d$ to (9) are obtained as

$$\mathbf{x} = \sum_{i=1}^k n_i h_i, \quad n_i \in \{0, 1, \dots, \gcd(r_m, t_i) - 1\}. \quad (12)$$

where $h_i = V_i^{-1} \frac{r_m}{\gcd(r_m, t_i)} \pmod{r_m \mathbb{Z}^d}$ and all addition is entrywise modulo r_m . To summarize,

$$K \cong \bigoplus_{i=1}^k \mathbb{Z}_{\gcd(r_m, t_i)} \quad (13)$$

with corresponding cyclic generators $\mathcal{S} = (h_i)_{i=1}^k$.

The procedure outlined above computes a generating set for K , and shows that transforming the problem in $(\widetilde{\text{ILP}}_{\mathcal{B}})$ to (8) is classically efficient, enabling us to efficiently compute a generating set for K . We summarize the overall procedure in Algorithm 2.

Algorithm 2 NULLGENFINDING

Input: Constraints from Group relaxation $(\mathbf{A}, \{r_j\}_{j=1}^m)$

Output: An independent generating set $\mathcal{S}$ for $K \leq \mathbb{Z}_{r_m}^d$, the solution space to $\mathbf{A}\mathbf{x} = \mathbf{0} \pmod{R\mathbb{Z}^m}$.

- 1: Compute the SNF of $B\mathbf{A} \in \mathbb{Z}^{m \times d}$: $B\mathbf{A} = UTV$, $T = \text{diag}(t_1, \dots, t_k)$, $B = \text{diag}(r_m/r_j)_{j=1}^m$.
 - 2: For each $i \in [k]$ compute $\gcd(r_m, t_i)$ and set $z_i = r_m / \gcd(r_m, t_i)$.
 - 3: Set $h_i = z_i V_i^{-1} \pmod{r_m \mathbb{Z}^d}$ for $i \in [k]$
 - 4: Output $\mathcal{S} = (h_1, \dots, h_k)$.
-

Proof of correctness is given by the above discussion. The runtime of Algorithm 2 can be bounded as follows.

Lemma 1 (Runtime of NULLGENFINDING). *Suppose one has data $(\mathbf{A}, \mathbf{b}, \mathbf{c}, \{r_j\}_{j=1}^m)$ corresponding to the group relaxation of an ILP stored in classical memory. There is a classical algorithm that outputs cyclic generators $\mathcal{S} = (h_i)_{i=1}^k$ for the kernel K of $\mathbf{A}$ in $\mathbb{Z}$ using time $\tilde{\mathcal{O}}(\text{poly}(\log\|\mathbf{A}\|_\infty, d))$. In particular, $K \cong \mathbb{Z}_{|\langle h_1 \rangle|} \oplus \cdots \oplus \mathbb{Z}_{|\langle h_k \rangle|}$.*

Proof. For dense $\mathbf{A}$, the Smith normal form can be computed using

$$\mathcal{O}(\text{poly}(\log\|\mathbf{A}\|_\infty, d))$$

operations, where $\|\mathbf{A}\|_\infty := \max_{i \in [d], j \in [d]} |\mathbf{A}_{ij}|$, see [Sto96]. The only other operations involved are multiplication by V^{-1} and modular arithmetic (V^{-1} can be obtained along with V “on the fly” during the SNF calculation, by composing the intermediate row and column operations). These costs, are subdominant, hence the runtime follows simply. The last statement from theorem follows from V^{-1} being automorphism for $\mathbb{Z}_{r_m}^d$. $\square$

3.4 Markov-chain search algorithms for the group relaxation

In this section, we design a random-walk-based approach for sampling optimal solutions of the group relaxation. We will show how to construct a random walk over a coset corresponding to the set of feasible solutions, enabling one to run Markov chain search to solve the group relaxation. The motivation for considering Markov-chain search is primarily for analysis: it is easier to demonstrate quantum speedup in this framework compared with a classical algorithm that directly samples uniformly over the feasible coset.

As done in the previous section, by exploiting the algebraic structure of the group relaxation, one can efficiently generate a basis for the kernel of the linear map defined by the coefficient matrix with the range mapped to $\mathcal{R}$. Hence, provided any feasible solution to the group relaxation, we can determine the optimal solution by solving a problem over the null space.

An efficient procedure for obtaining a generating set for K buys us nothing on its own. It does however enable one to construct a Markov Chain that explores the feasible region of the group relaxation, and mixes to a stationary distribution π in $\text{poly}(n)$ steps. Letting π^* denote the total probability that a sample from π is a global minimizer of f , drawing $\mathcal{O}((\pi^*)^{-1} \log(\delta^{-1}))$ samples from π suffices to ensure that we encounter the global minimizer with probability at least $1 - \delta$. This algorithm, following [CHO⁺24], is called the *Markov Chain Search* (MCS). We outline this framework in Algorithm 3. Under the stated prerequisites, MCS finds the global minimizer of f in time $\mathcal{O}((\pi^*)^{-1} \cdot \text{poly}(n, L))$. The major component is the construction of the random walk over the feasible set, using the generator-finding procedure from the previous section.

A natural choice for our setting is a *Cayley walk* generated by the set $\mathcal{S} = \{h_1, \dots, h_k\}$ returned by Algorithm 2, which leads to a walk over an undirected graph. In particular, a step of the walk corresponds to uniformly at random picking a generator $h_j \in \mathcal{S}$, then uniformly at random adding h_j or $-h_j$ to the current point modulo $r_m \mathbb{Z}^d$ producing another element of $K \subset \mathbb{Z}$. Let $u_j = |\langle h_j \rangle|$ for $j \in [k]$. Via the isomorphism

$$K \cong \bigoplus_{i \in [k]} \mathbb{Z}_{u_i}, \tag{14}$$

one can see that this Cayley walk corresponds to a product of cycle walks over the cyclic groups $\mathbb{Z}_{u_1}, \dots, \mathbb{Z}_{u_k}$. Thus, the stationary distribution π is the uniform superposition over K , and satisfies $\pi^* = |K^*|/|K|$. The log-Sobolev (LS) constant of the cycle walk on $\mathbb{Z}_{u_i}$ is $\Theta(1/u_i^2)$ [DSC96, Example 4.2], and the product of k cycles has a LS constant lower bounded

Algorithm 3 GROUPMARKOVCHAINSEARCH

Prerequisites: Group relaxation data $(\mathbf{A}, \mathbf{b}, \mathbf{c}, \{r_j\}_{j=1}^m)$, distribution π such that $\pi(\text{OPT}_B)$ is the probability that a sample from π is a global minimizer, a Markov Chain with transition matrix P that preserves $\mathbf{A}\mathbf{x} = \mathbf{b} \pmod{R\mathbb{Z}^m}$, mixing time $t_{\text{mix}}(\pi^*/2) = \mathcal{O}(\text{poly}(n))$.

Input: Description of $\text{poly}(n)$ time procedures to evaluate f and perform a step of the Markov Chain described by P , failure probability ϵ . Known feasible point $\hat{\mathbf{x}} \in \mathcal{P}_B$.

Output: A global minimizer $\mathbf{x}^*$ of the group relaxation.

- 1: Set $i = 0$, set $z^{(0)} = \hat{\mathbf{x}}$, and construct $f(\mathbf{x})$ via Equation (8).
 - 2: **while** $i \leq \frac{2}{\pi(\text{OPT}_B)} \log(1/\epsilon)$ **do**
 - 3: Simulate t_{mix} steps of P to obtain sample $\tilde{z}$.
 - 4: **if** $f(\tilde{z}) \leq f(z^{(i)})$ **then**
 - 5: Set $z^{(i+1)} = \tilde{z}$
 - 6: **else**
 - 7: Set $z^{(i+1)} = z^{(i)}$
 - 8: **end if**
 - 9: $i \leftarrow i + 1$
 - 10: **end while**
 - 11: Output $z^{(i)}$.
-

by $\Omega(1/(k \min_{i \in [k]} u_i^2))$ [DSC96, Lemma 3.2]. This also happens to be the same order as the spectral gap.

Given the above facts, the next result follows immediately.

Theorem 5 (Markov chain search for group relaxations). *Algorithm 3, with a Cayley walk over K as the input Markov chain, finds an optimal solution to the group relaxation with high probability, and runtime at most $\mathcal{O}(|K|/|K^*| \cdot \text{poly}(n, L))$.*

3.5 Comparison with the shortest path on the Cayley graph

The classical approach for solving $(\widetilde{\text{ILP}}_B)$ leverages the nonnegativity of $\mathbf{c}$ to reduce the group relaxation to finding the shortest path in a Cayley directed graph (digraph) [Gom65, Sha68]. Unlike MCS, this algorithm does not explicitly depend on $|K|$. It instead depends on $|G|$, where

$$G := (\mathbf{A}\mathbb{Z}^d + R\mathbb{Z}^m)/R\mathbb{Z}^m \leq \mathbb{Z}^m/R\mathbb{Z}^m. \quad (15)$$

In words, the group G is the range of the map $\mathbf{x} \in \mathbb{Z}_{\geq 0}^d \mapsto \mathbf{A}\mathbf{x} \pmod{R\mathbb{Z}^m}$. In this section, we recall the shortest-path approach of Gomory and present a setting under which $|K| = |G|$, making the shortest-path and MCS approaches comparable. Hence, we want to identify settings in which our quantum algorithm has a super-quadratic speedup over Gomory's shortest-path approach.

Let $\mathcal{G}$ be the corresponding Cayley digraph over G generated by

$$\mathcal{A} = \{\mathbf{A}_1 \pmod{R\mathbb{Z}^m}, \dots, \mathbf{A}_d \pmod{R\mathbb{Z}^m}\},$$

where an edge incident on $\mathbf{y} \in G$ corresponding to $\mathbf{y} \mapsto [\mathbf{y} + \mathbf{A}_j] \pmod{R\mathbb{Z}^m}$ has a weight of $\mathbf{c}_j \geq 0$. Define

$$\mathbf{b}' = (\mathbf{b} \pmod{R\mathbb{Z}^m}) \in G.$$

Feasible solutions correspond to directed paths from $\mathbf{0}$ to $\mathbf{b}'$. Since $\mathbf{c}$ is nonnegative, the optimal solution is the feasible solution with smallest ℓ_1 -norm, i.e., the shortest path from $\mathbf{0}$ to $\mathbf{b}'$ in $\mathcal{G}$.

This approach has complexity dependent on $|G|$, as there are $\mathcal{O}(|\mathcal{A}||G|)$ edges in $\mathcal{G}$. Thus, shortest-path algorithms can solve the problem with $\mathcal{O}(n|G|)$ time. In contrast, the null space approach depends on $|K|$. In general, either $|K|$ or $|G|$ may be larger. More precisely, (7) and the first isomorphism theorem says that $G \cong \mathbb{Z}_{r_m}^d / K$, and in particular $r_m^d = |K| |G|$. As discussed in Appendix A, choosing $\bigoplus_{i=1}^d \mathbb{Z}_{s_i}$, where $s_i := |\langle \mathbf{A}_i \rangle|$, as the group relaxation domain can result in a smaller kernel K' . Then the relation is $\prod_{i=1}^d s_i = |K'| |G|$. As expected, the range and kernel sizes are inversely related for fixed domain, but in general either could be larger than the other.

In fact, it appears one can construct families of instances where either the range space or kernel space algorithms can be asymptotically faster than the other. The goal of the following result is to show there at least exists a seemingly generic family of ILPs, where $|G| = |K|$.

Theorem 6. *There is a constructable family of integer linear programming problems, parameterized by $m \in \mathbb{Z}_+$, whose group relaxations satisfy $G \cong K$ (in particular, $|G| = |K|$).*

Proof. Consider arbitrary positive integers t, m . Let $n = 2m$, $r = t^2$, $R = rI_m$, and $T = tI_m$. Pick unimodular matrices $U, V, U', V' \in \mathbb{Z}^{m \times m}$. We then let the ILP constraint matrix be $A = [URV | U'TV'] = [A_B | A_N] \in \mathbb{Z}^{m \times n}$. Additionally, we require U, V are chosen such that A_B^{-1} has all non-negative entries.

We want the right-hand side vector $b \in \mathbb{Z}_{\geq 0}^m$ to be such that the original ILP is feasible. To do this, we first look at the group relaxation constraint of

$$U^{-1}A_N \mathbf{x} = U^{-1}b \pmod{r\mathbb{Z}^m}.$$

The feasibility of the group relaxation hinges on

$$T\mathbf{z} = b \pmod{r\mathbb{Z}^m}$$

having a solution. By a discussion in Section 3.3, this is true if and only if $t \mid b_j, \forall j \in [m]$. Choosing b as such and keeping all entries non-negative, one solution is then $\hat{\mathbf{x}} = (V')^{-1} \begin{pmatrix} b \\ t \end{pmatrix} \pmod{r\mathbb{Z}^n}$. From Section 3.1, ILP feasibility then hinges on

$$A_B^{-1}(b - A_N \hat{\mathbf{x}}) \geq 0,$$

which holds by our assumptions on A_B^{-1} .

We also want to be able to plant our choice of basis $\mathcal{B}$ as an optimal one. To perform the planting, we only need to ensure primal and dual feasibility of the corresponding LP basic feasible solution (BFS). The primal BFS is $x_{\text{LP}} = (x_B, x_N) = (A_B^{-1}b, \mathbf{0}_{n-m})$ and the associated dual solution is $y_{\text{LP}} = (A_B^\top)^{-1}c_B$. By the earlier paragraph, we already have $x_B \geq 0$, ensuring primal feasibility. Dual feasibility requires the reduced costs to be nonnegative:

$$c_N^\top - c_B^\top A_B^{-1} A_N \geq 0,$$

which we can ensure by choosing c .

From equation (13), the null space K has cardinality

$$|K| = \prod_{j=1}^m \gcd(r, t) = t^m.$$

Meanwhile, the cardinality of G satisfies

$$|G| = \prod_{i=1}^m \frac{r_m}{\gcd(r_m, t_i)} = \prod_{i=1}^m \frac{t^2}{t} = |K|$$

Hence for each m we have a collection of ILPs corresponding to the set of all possible choices for t, U, V, U', V' , where $|K| = |G|$ for all of them. $\square$

To summarize, the only difference between the construction in Theorem 6 and the set of all ILPs with $n = 2m$ is the requirement on the invariant factors of A_B and A_N , i.e. the invariant factors of A_B are all equal and each factor is exactly the square of a factor of A_N .

Note from (15), G satisfies $|G| \leq |\det A_B|$ and $|A| \leq n$, which is why the shortest-path approach for the group relaxation is usually presented with an $\mathcal{O}(n |\det A_B|)$ runtime. Here we present a set of sufficient conditions on an ILP under which $|K|$ in the null space formulation is not significantly larger, i.e. in $\mathcal{O}(\text{poly}(n, m) |\det A_B|)$. Unlike the previous theorem, we relax some of the constraints on n and the invariant factors of A_B . There are no conditions on A_N .

Theorem 7. *Let $c \in \mathbb{R}^n, A \in \mathbb{Z}^{m \times n}, b \in \mathbb{Z}^m$ define an integer linear program, and $\mathcal{B}$ be an optimal LP basis. Let $R = \text{diag}(r_j)_j^m = UA_BV \in \mathbb{Z}^{m \times m}$ be the SNF of A_B . Suppose $n \leq 2m + \mathcal{O}(\log(m))$ and that there exists an integer $j = \mathcal{O}(\log(m))$ such that for all integers k satisfying $j \leq k \leq m$, we have $r_j = r_m$. Then, we can construct a null space formulation of the ILP group relaxation where $|K| = \mathcal{O}(\text{poly}(m, n) |\det A_B|)$.*

Proof. We follow the procedure outlined in Section 3.2 with the domain $\mathbb{Z}_{r_m}^d$. Without loss of generality, we take $d = n - m$ here. By construction, $n - m \leq m + \mathcal{O}(\log(m))$. Note trivially $|K| \leq |\mathbb{Z}_{r_m}^d|$. Also $|\det A_B| = \prod_{i=1}^m r_i = \mathcal{O}(\text{poly}(m) r_m^{m-j})$, which one can easily see is the same order of $|\mathbb{Z}_{r_m}^d| = r_m^d$. $\square$

The main point is that $|K|$ tends to not be much larger than $|\det A_B|$ for problems with around twice as many variables as constraints and with an A_B that has the majority of its invariant factors being equal.

4 Quantum short path algorithm for the group relaxation

In this section we apply the quantum short path algorithm to the group relaxation. We provide two distinct instantiations of the framework. The first relies on the product-of-cycles walk (defined in Section 3.4) and uses Theorem 2, while the second constructs an expander from the product-of-cycles walk and applies Theorem 3. Before devling into these algorithms, we show that the quantum input assumption (Assumption 1) is satisfied for both walks.

4.1 Quantum state preparation and block encoding

We recall below the input requirements of the SP algorithm.

Initial State Preparation A critical prerequisite for the SP framework is the efficient preparation of $|\sqrt{\pi}\rangle$. For loading the initial state for either the product of cycles walk or the expander walk, we can utilize the following procedure outline in Algorithm 4. Step 3 can be carried out using $\mathcal{O}\left(k \max_{u_j \in \{u_i; h_i \in \mathcal{S}\}} u_j\right)$ gates via [SV23, GR02]. Steps 4 and 5 can be performed by reversibly encoding $\mathcal{O}(\text{poly}(n))$ -sized classical circuits for applying $\mathbf{B}$ and $\mathbf{B}^{-1}$. The rest of the operations are either classical polynomial time or require $\mathcal{O}(\text{poly}(n))$ quantum gates.

Algorithm 4 STATE PREPARATION FOR UNIFORM DISTRIBUTION OVER K

Input: Generating set $\mathcal{S}$

Output: A quantum state $|\sqrt{\pi}\rangle$ encoding a uniform distribution over K

- 1: Construct a matrix $\mathbf{B} \in \mathbb{Z}^{d \times k}$ using elements of $\mathcal{S}$ as columns.
 - 2: Determine $u_j = |\langle h_j \rangle|$ for $h_j \in \mathcal{S}$
 - 3: Load superposition over $\bigoplus_i^k \mathbb{Z}_{u_i}$
 - 4: Apply a unitary $\mathcal{U}_{\mathbf{B}}$ that performs $|\alpha\rangle|0\rangle \mapsto |\alpha\rangle|\mathbf{B}\alpha\rangle$
 - 5: Apply unitary $\mathcal{U}_{\mathbf{B}_{\text{inv}}}$ that performs $|\alpha\rangle|\gamma\rangle|0\rangle \mapsto |\alpha\rangle|\gamma\rangle|\mathbf{B}^{-1}\gamma\rangle$
 - 6: CNOT third (control) and first (target) registers, apply $\mathcal{U}_{\mathbf{B}_{\text{inv}}}^\dagger$
 - 7: Trace out first and third registers, i.e. the first and third registers will be definitely all zeroes (unentangled), hence we can ignore them.
-

Block-encodings of Markov Chain and Cost Function Furthermore, we require efficient block-encodings of both the Markov chain discriminant matrix $D(P)$ and the cost Hamiltonian H . We say that unitary $\mathcal{U}(M)$ is (κ, a) -block encoding of an n -qubit operator M of

$$(\langle 0|^{\otimes a} \otimes I)\mathcal{U}(M)(|0\rangle^{\otimes a} \otimes I) = M/\kappa.$$

Specifically, for SP in general, we want to block-encode H defined as

$$H|\mathbf{x}\rangle = f(\mathbf{x})|\mathbf{x}\rangle,$$

for the problem cost function f (8), and $D(P)$ defined by

$$D(P)_{\mathbf{x}, \mathbf{x}'} = \sqrt{P(\mathbf{x}, \mathbf{x}')P(\mathbf{x}', \mathbf{x})},$$

when the chain P is reversible.

Since Cayley walks form regular graphs, P is symmetric and $D(P) = P$. Hence, we only need to block-encode P (more specifically, its lazy variant). Note that there is an efficient classical circuit that computes $P(\mathbf{x}, \mathbf{x}')$ for any $\mathbf{x}, \mathbf{x}' \in K$, and hence we can efficiently construct unitary operations that perform:

$$\begin{aligned} |\mathbf{x}\rangle|j\rangle|0\rangle &\mapsto |\mathbf{x}\rangle|j\rangle|r_{\mathbf{x}j}\rangle \\ |\mathbf{x}\rangle|j\rangle|0\rangle &\mapsto |\mathbf{x}\rangle|j\rangle|c_{\mathbf{x}j}\rangle \\ |\mathbf{x}\rangle|\mathbf{x}'\rangle|0\rangle &\mapsto |\mathbf{x}\rangle|\mathbf{x}'\rangle|P(\mathbf{x}, \mathbf{x}')\rangle, \end{aligned}$$

where $r_{\mathbf{x}j} \in K$ is such that $P(\mathbf{x}, r_{\mathbf{x}j})$ is the j -th nonzero entry of the row $P(\mathbf{x}, \cdot)$, and $c_{\mathbf{x}j} \in K$ is such that $P(c_{\mathbf{x}j}, \mathbf{x})$ is the j -th nonzero entry of the column $P(\cdot, \mathbf{x})$. Additionally, P is $|\mathcal{S}| = \Theta(\log|K|) = \mathcal{O}(d \log r_m)$ row and column-sparse matrix. Hence, by [GSLW19, Lemma 48] we can prepare a $(\Theta(d \log r_m, \mathcal{O}(\text{poly}(n))))$ -block encoding to P .

The cost function in this case is f from Equation (8), which we can evaluate reversibly in quantum polynomial time. Specifically, the following is a $(\kappa, 1)$ -block-encoding for H :

$$\mathcal{U}(H) : |\mathbf{x}\rangle|0\rangle \mapsto \frac{f(\mathbf{x})}{\kappa}|\mathbf{x}\rangle|0\rangle + \sqrt{1 - \frac{f(\mathbf{x})^2}{\kappa^2}}|\mathbf{x}\rangle|1\rangle,$$

where κ ensures $|f(\mathbf{x})/\kappa| \leq 1$ and $\mathbf{x} \in K$ (for example, $\kappa = \|\mathbf{c}_1\| \cdot r_m$ suffices.) This operation involves reversibly implementing f and $\mathcal{O}(\log \kappa)$ additional gates to rotate $f(\mathbf{x})$ onto an amplitude.

We already have the ingredients for a quadratic quantum speedup. As discussed in Theorem 5, classical Markov Chain search, with a fast mixing chain, requires time

$$\mathcal{O}(|K|/|K^*|) \cdot \text{poly}(n, L)$$

to find an optimal solution. Using Algorithm 4, we can efficiently prepare a quantum state encoding $\sqrt{\pi}$ for the Cayley walk. As a consequence, we trivially obtain a quadratic speedup for over MCS for this problem, via generalized quantum minimum finding [vAGGdW20, Theorem 49]. We highlight this fact in a theorem:

Theorem 8 (Quadratic Speedup for group relaxations). *There is a quantum algorithm that, with high probability, finds an optimal solution to the group relaxation with runtime at most*

$$\mathcal{O}\left(\sqrt{|K|/|K^*|} \cdot \text{poly}(n, L)\right).$$

Utilizing the generalized SP algorithm discussed in Section 2.4.2, we can construct a quantum algorithm that improves upon the one above in the following sense. Our quantum algorithm achieves a runtime of $\mathcal{O}(|K|/|K^*|)^{\frac{1}{2}-\alpha} \cdot \text{poly}(n, L)$ for some $\alpha > 0$ determined by spectral properties of the walk over K , stability of the cost function (Definition 6), and spectral density (Definition 5) parameters. While α may not be a dimension-independent constant in general, we do provide sufficient conditions for ensuring it is.

4.2 Product of cycles walk approach

In this section we analyze the cost of the short path algorithm for solving the group relaxation when the underlying walk is the product-of-cycles walk described in Section 3.4. The runtime can be bounded as follows.

Theorem 9 (Runtime for solving the group relaxation). *With high probability, the generalized quantum SP algorithm finds the optimal solution to the group relaxation ($\widetilde{\text{ILP}}_{\mathcal{B}}$) with runtime at most*

$$\mathcal{O}\left(\left(\frac{|K|}{|K^*|}\right)^{\frac{1}{2}-\alpha} \cdot \text{poly}(n, L)\right).$$

If the set $\mathcal{S} = \{h_1, \dots, h_k\}$ outputted by NULLGENFINDING (Algorithm 2) satisfies:

$$\max_{j \in [k]} \|h_j\|_{\text{cyc}, 1, \mathbf{c}} = \Theta\left(\frac{|\text{OPT}_{\mathcal{B}}|}{\log(|K|/|K^*|)}\right), \quad (16a)$$

$$\max_{j \in [k]} |\langle h_j \rangle|^2 = \Theta\left(\frac{\log(|K|/|K^*|)}{k}\right), \quad (16b)$$

then α is a nonzero constant.

Proof. The quantum input assumption (see Assumption 1) holds by results from Section 4.1. Recall from Eq. (8) that $f : K \rightarrow \mathbb{R}_{\geq 0}$ is given by

$$f(\mathbf{x}) = \mathbf{c}^\top (\hat{\mathbf{x}} + \mathbf{x} \pmod{r_m \mathbb{Z}^d}) + \mathbf{c}_B^\top A_B^{-1} \mathbf{b}.$$

Define a shifted cost $\tilde{f}(\mathbf{x}) := f(\mathbf{x}) - C$ for some $C = \Theta(|\text{OPT}_{\mathcal{B}}|)$ so that $E^* = \min_{\mathbf{x} \in K} \tilde{f}(\mathbf{x}) < 0$.

By the definition of the P -pseudo-Lipschitz norm and P being a Cayley walk:

$$\begin{aligned}
\|\tilde{f}\|_P &= \|f\|_P \\
&= \frac{1}{3|\mathcal{S}|} \sum_{h_j \in \mathcal{S}} \sum_{a \in \{\pm 1, 0\}} (\langle \mathbf{c}, [\hat{\mathbf{x}} + \mathbf{x} + a \cdot h_j \pmod{r_m \mathbb{Z}^d}] - [\hat{\mathbf{x}} + \mathbf{x} \pmod{r_m \mathbb{Z}^d}] \rangle)^2 \\
&= \frac{1}{3|\mathcal{S}|} \sum_{h_j \in \mathcal{S}} \sum_{a \in \{\pm 1, 0\}} \left(\sum_{i=1}^d c_i ([\hat{x}_i + x_i + a \cdot h_{ji} \pmod{r_m}] - [\hat{x}_i + x_i \pmod{r_m}]) \right)^2 \\
&\leq \frac{1}{3|\mathcal{S}|} \sum_{h_j \in \mathcal{S}} \sum_{a \in \{\pm 1, 0\}} \left(\sum_{i=1}^d |c_i| \max\{r_m - h_{ji}, h_{ji}\} \right)^2 \\
&\leq \max_{h_j \in \mathcal{S}} \|h_j\|_{\text{cyc}, 1, \mathbf{c}}^2.
\end{aligned}$$

Also, as mentioned in Section 2.4.2, $\Delta_P \leq \sqrt{\|\tilde{f}\|_P}$.

From Theorem 2 and our assumptions, we take γ for the spectral density condition to be

$$\gamma = \frac{\omega(-(1-\eta)(\text{OPT}_B - C) - \mathbb{E}_\pi[\tilde{\theta}_\eta(\tilde{f})])^2}{(\max_{j \in [k]} \|h_j\|_{\text{cyc}, 1, \mathbf{c}})^2 \ln(|K|/|K^*|)} = \Theta(\omega \log(|K|/|K^*|)). \quad (17)$$

Note that in Theorem 2, the expression for γ contains $\mathbb{E}_\pi[H]$ in the numerator, where here $H = \tilde{f}$. However, it is implicit in the analysis of [CHO⁺24] that we can replace $\mathbb{E}_\pi[\tilde{f}]$ with $\mathbb{E}_\pi[\tilde{\theta}_\eta(\tilde{f})]$. Note that by construction

$$\text{OPT}_B - C \leq \mathbb{E}_\pi[\tilde{\theta}_\eta(\tilde{f})] \leq (1-\eta)(\text{OPT}_B - C).$$

Since $\pi(E^*) \neq 0$, we have that $|-(1-\eta)(\text{OPT}_B - C) - \mathbb{E}_\pi[\tilde{\theta}_\eta(\tilde{f})]|$ is nonzero and $\in \Theta(|\text{OPT}_B|)$. Hence the expression in (17) follows.

We will want $\gamma = \Theta(1)$. However, note that we will also want

$$\mu^* = \gamma \omega \ln(|K|/|K^*|) = \Theta(\omega^2 \log^2(|K|/|K^*|))$$

from Theorem 2 to be $\Theta(1)$. Hence both conditions require $\omega^{-1} = \Theta(\log(|K|/|K^*|))$. By the discussion in Section 3.4,

$$\omega = \Omega\left(\frac{1}{k \min_{j \in [k]} u_j^2}\right)$$

for the product-of-cycles walk, where $u_i := |\langle h_i \rangle|$. Hence our assumption implies both γ and μ^* are $\Theta(1)$.

We also need to check that condition (3) is satisfied. By the above, it suffices for

$$\begin{aligned}
&\frac{1}{3\gamma \ln(1/\pi(E^*)) \sqrt{\pi(E \leq (1-\eta)E^*)}} - \max \left(\frac{4 \ln(|K|)}{\omega}, \frac{3|\text{OPT}_B - C|^2(1-\eta)^2}{\left(\max_{j \in [k]} \|h_j\|_{\text{cyc}, 1, \mathbf{c}}\right)^2} \right) \\
&= \frac{\xi_1}{\ln(|K|/|K^*|)(|K^*|/|K|)^{\xi}} - \max \left(\xi_3, \xi_2 \log^2(|K|/|K^*|) \right) \geq 2,
\end{aligned}$$

for some constants $\xi, \xi_1, \xi_2, \xi_3 > 0$. This holds for sufficiently large problem size by our assumptions.

The runtime of the SP algorithm is

$$\mathcal{O} \left(\text{poly}(n) \omega^{-1} [\pi(E^*)^{-1}]^{\left(\frac{1}{2} - \frac{\eta(1-\eta)|E^*| \mu}{2 \ln(1/\pi(E^*)) \Delta_P}\right)} \right)$$

with $\mu < \mu^*$ then follows from Theorem 2.

Fixing $\eta \in (0, 1)$ and using that $\mu^* = \Theta(1)$, the super-quadratic condition becomes

$$\frac{|E^*|}{\Delta_P} \geq \frac{|E^*|}{\max_{j \in [k]} \|h_j\|_{\text{cyc}, 1, \mathbf{c}}} = \Theta \left(\frac{|\text{OPT}_{\mathcal{B}}|}{\max_{j \in [k]} \|h_j\|_{\text{cyc}, 1, \mathbf{c}}} \right) = \Theta(\log(|K|/|K^*|)),$$

which holds by our assumptions. Note that $|\text{OPT}_{\mathcal{B}} - C| = \Theta(|\text{OPT}_{\mathcal{B}}|)$. Thus we have that $\alpha > 0$ (cf. Corollary 1) is dimension-independent constant. This completes the proof. $\square$

Here, we seek to make the super-quadratic condition ($\alpha > 0$ constant) in Theorem 9 more transparent. For the condition in (16a), we have the following bound on generator-weight norm:

$$\max_{j \in [k]} \|h_j\|_{\text{cyc}, 1, \mathbf{c}} \leq r_m \|\mathbf{c}\|_{\infty} \max_{j \in [k]} \|h_j\|_{\ell_0}.$$

Suppose that the generators are sparse, in the sense that $\max_{j \in [k]} \|h_j\|_{\ell_0} = \mathcal{O}(1)$. This reduces the first condition to

$$\frac{\text{OPT}_{\mathcal{B}}}{\|\mathbf{c}\|_{\infty}} = \Theta(r_m \log(|K|/|K^*|)).$$

Suppose $|Z| = r_m^d = \Theta(\exp(n))$. This condition boils down to assuming the problem is hard, taking exponential time, but not super exponential time. For hardness, we are assuming $m = \Theta(n)$. Thus these assumptions yield $\max_{j \in [k]} |\langle h_j \rangle|^2 = \mathcal{O}(1)$. To see this, note by definition of Z , $|Z| = \Theta(\exp(n))$, and $n = \Theta(m)$ imply that $r_m = \|R\|_{\infty} = \mathcal{O}(1)$. However, $|\langle h_j \rangle| \leq r_m$.

Suppose also that reducing the problem to a search over $|K|$ does not result in subexponential runtime, i.e. $k = |\mathcal{S}| = \Theta(n)$. Thus, $|K| = \prod_{h_j \in \mathcal{S}} |\langle h_j \rangle| = \Theta(\exp(n))$. Thus, the second condition (16b) is satisfied.

Thus we end up with the following more intuitive conditions for hard, yet not super-exponentially hard, problems. To summarize, $\alpha > 0$ is a nonzero constant if

1. $n = \Theta(m)$;
2. the domain of the finite group relaxation Z , satisfies $|Z| = \Theta(\exp(n))$;
3. $\frac{\text{OPT}_{\mathcal{B}}}{\|\mathbf{c}\|_{\infty}} = \Theta(n)$;
4. restricting to the feasible set $|K|$ does not make the runtime subexponential;
5. and the generating set $\mathcal{S}$ obtained from Algorithm 2 is sparse : $\forall h_j \in \mathcal{S}, \|h_j\|_{\ell_0} = \mathcal{O}(1)$.

One will note that most of these conditions are light, except for the condition on the sparsity of the generators.

The following result shows that there does exist a family of ILPs whose group relaxations satisfy the conditions for super-quadratic speedups using the product-cycle walk. The following ILPs turn out to have a specific structure that enables them to be solved easily. However, the goal is simply to use this construction to show that our conditions are not vacuous.

Theorem 10. *There exists a family of integer linear programming problems whose group relaxations satisfy conditions (16a) and (16b) in Theorem 9.*

Proof. The family is constructed using the procedure outlined in the proof of Theorem 6, with some additional assumptions on the unimodular matrix V' chosen in the construction therein. To recall for the reader, the construction considers arbitrary positive integers t, m , with $n = 2m$ and $r = t^2$. Define $R = rI_m$ and $T = tI_m$. We pick unimodular matrices $U, V, U', V' \in \mathbb{Z}^{m \times m}$. The ILP constraint matrix is then $A = [URV|U'TV'] = [A_B|A_N] \in \mathbb{Z}^{m \times n}$, with cost c chosen so that $\mathcal{B}$ is an optimal basis. We now additionally specify $V' = I_m$.

Recall that our constraint on the cost function vector c is from the need for the reduced costs to be nonnegative:

$$c_N^\top - c_B^\top A_B^{-1} A_N \geq 0.$$

This is satisfied by the choice $c_B = \mathbf{0}_m$ and $c_N = \mathbf{1}_m$. The conditions for super-quadratic speedup are

$$\max_{j \in [k]} \|h_j\|_{\text{cyc}, 1, c} = \Theta\left(\frac{|\text{OPT}_{\mathcal{B}}|}{\log(|K|/|K^*|)}\right), \quad \max_{j \in [k]} |\langle h_j \rangle|^2 = \Theta\left(\frac{\log(|K|/|K^*|)}{k}\right).$$

The group relaxation constraint is

$$U^{-1} A_N \mathbf{x} = U^{-1} b \pmod{r\mathbb{Z}^m}.$$

Then let us follow the steps of Algorithm 2. Note that by construction, the SNF of $U^{-1} A_N$ is $U^{-1} U' T V'$. We first solve $T\mathbf{z} = 0 \pmod{r\mathbb{Z}^m}$, which from (12) has solution set

$$\{\ell t : \ell \in [t]\}^{\times m}.$$

We then transform the generating set $\{t\mathbf{e}_j : j \in [m]\}$ via $(V')^{-1}$ and mod by $r\mathbb{Z}^m$, which is trivial here as V' is the identity. Thus, $h_j = t\mathbf{e}_j$, for each $j \in [m]$, and there are $k = m$ generators of K . These generators each have order t , thus $\max_{j \in [k]} |\langle h_j \rangle|^2 = t^2 = \Theta_m(1)$. Additionally, a direct calculation shows that for any $t \geq 2$,

$$\|h_j\|_{\text{cyc}, 1, c} = c_j(t^2 - t) = \Theta_m(1).$$

Hence, $\max_{j \in [k]} \|h_j\|_{\text{cyc}, 1, c} = \Theta_m(1)$.

Thus, the two conditions for super-quadratic runtime become

$$\log(|K|/|K^*|) = \Theta(m), \quad |\text{OPT}_{\mathcal{B}}| = \Theta(m).$$

From the proof of Theorem 6, we can take $\hat{\mathbf{x}} = (V')^{-1} \begin{pmatrix} b \\ t \end{pmatrix} \pmod{r\mathbb{Z}^m} = \begin{pmatrix} b \\ t \end{pmatrix} \pmod{r\mathbb{Z}^m}$. Take $b = \ell t \mathbf{1}_m$, $\ell < r$. Thus our feasible coset has elements of the form

$$\ell + \sum_j t n_j \mathbf{e}_j \pmod{t^2 \mathbb{Z}^n}, \quad n_j \in \{0, 1, \dots, t-1\}.$$

This construction ensures that there is only one optimal solution, i.e. the element of the coset with smallest entries. We can pick ℓ to ensure that all the entries of the minimizer are nonzero. Hence we have $|K^*| = 1$ and $|\text{OPT}_{\mathcal{B}}| = \Theta(m)$ by the chosen cost function. Because $\log |K| = m \log t = \Theta_m(m)$, we see that for each $t \geq 2$ there is a family of problems satisfies both conditions of Theorem 9. $\square$

4.3 Expander walk approach

We can convert the product-of-cycles walk into an expander graph, with high probability, by randomly sampling additional elements from K . Specifically, if we use the minimal generating set $\mathcal{S}$ found using Algorithm 2 to sample $C \log|K|$ (for some constant $C > 0$) elements $\tilde{\mathcal{S}}$ uniformly at random from K , then with high-probability both $\tilde{\mathcal{S}}$ will generate K and the Cayley graph spanned by $\tilde{\mathcal{S}}$ is an expander (Algorithm 5). The latter part follows from the *Alon–Roichman theorem* [AR94], which asserts that the spectral gap is $\delta \geq \epsilon$ for a constant $\epsilon > 0$. Since the Cayley walk that we generate is with high probability a spectral expander, we refer to this approach as the *Expander Walk*.

Algorithm 5 EXPANDERGENERATION

Input: Constraints from Group relaxation $(\mathbf{A}, \{r_j\}_{j=1}^m)$

Output: A Cayley graph generated by a set $\tilde{\mathcal{S}} \subset K$ s.t. with high probability the outputted set both generates K and forms a Cayley graph that is a spectral expander.

- 1: Let $\mathcal{S} = \text{NullGenFinding}(\mathbf{A}, \{r_j\}_{j=1}^m)$, compute $|K|$.
 - 2: Use $\mathcal{S}$ to uniformly sample $C \log|K|$ elements from K and obtain $\tilde{\mathcal{S}} \subset K$.
 - 3: Output $\tilde{\mathcal{S}}$.
-

Since the expander has a constant spectral gap, we obtain an alternative runtime bound for solving the group relaxation. Compared with Theorem 9, the condition (16b) for α being dimension-independent is no longer required.

Theorem 11 (Runtime for solving the group relaxation – Expander Version). *With high probability, the generalized quantum SP algorithm finds the optimal solution to the group relaxation ($\widetilde{\text{ILP}}_{\mathcal{B}}$) with runtime at most*

$$\mathcal{O}\left((|K|/|K^*|)^{\frac{1}{2}-\alpha} \cdot \text{poly}(n, L)\right).$$

If the set $\mathcal{S} = \{h_1, \dots, h_k\} \subset Z$ output by EXPANDERGENERATION (Algorithm 5) is a generating set for K satisfying

$$\max_{j \in [k]} \|h_j\|_{\text{cyc}, 1, \mathbf{c}} = \Theta\left(\frac{|\text{OPT}_{\mathcal{B}}|}{\log(|K|/|K^*|)}\right),$$

then α is a positive constant independent of the problem dimension.

Proof. The input assumptions in Assumption 1 hold by construction (see, Section 4.1). The Cayley walk induced by $\tilde{\mathcal{S}}$ returned by EXPANDERGENERATION has a constant spectral gap $\delta = \Omega(1)$ with high probability. Hence, we have that $\mu^* = \Theta(\delta) = \Omega(1)$. Let $\tilde{f}$ be as in the proof of Theorem 9.

By Theorem 3, for any $\mu < \mu^*$, the short path algorithm runs in time

$$\mathcal{O}\left(\text{poly}(n) \delta^{-1} [\pi(E^*)^{-1}]^{\left(\frac{1}{2} - \frac{\eta(1-\eta)|E^*|\mu}{2 \ln(1/\pi(E^*)) \sqrt{\|\tilde{f}\|_P}}\right)}\right), \quad (18)$$

if $\frac{\delta}{2} > \pi(E^*)^\gamma$ and

$$\frac{1}{2\sqrt{\pi(E \leq (1-\eta)E^*)}} - \max\left(\frac{4 \ln(|K|)}{\delta}, \frac{3|E^*|^2(1-\eta)^2}{\Delta_P^2}\right) \geq 2.$$

Like in the proof of Theorem 9:

$$\Delta_P \leq \|\tilde{f}\|_P = \|f\|_P \leq \left(\max_{j \in [k]} \|h_j\|_{\text{cyc}, 1, c} \right)^2,$$

which holds because the expander walk is still a Cayley walk.

Also similar to the proof of Theorem 9, but using the expression for γ from Theorem 11, we have

$$\gamma = \frac{\sqrt{\delta}(-(1-\eta)(\text{OPT}_B - C) - \mathbb{E}_{\pi}[\tilde{\theta}_{\eta}(\tilde{f})])}{\max_{j \in [k]} \|h_j\|_{\text{cyc}, 1, c} \ln(|K|/|K^*|)} = \Theta(1),$$

with $C = \Theta(|\text{OPT}_B|)$ and by our assumptions. Hence, since $\delta = \Omega(1)$, we have $\frac{\delta}{2} > \pi(E^*)^{\Theta(1)}$ for sufficiently large problem size, implying the first condition is satisfied. The second condition reduces to

$$\begin{aligned} & \frac{1}{2\sqrt{\pi(E^*)^{\Theta(1)}}} - \max \left(\frac{4\ln(|K|)}{\delta}, \frac{3|\text{OPT}_B - C|^2(1-\eta)^2}{\left(\max_{j \in [k]} \|h_j\|_{\text{cyc}, 1, c} \right)^2} \right) \\ &= \frac{1}{2\sqrt{(|K^*|/|K|)^{\xi}}} - \max \left(\xi_2 \ln(|K|), \xi_3 \log^2(|K|/|K^*|) \right) \geq 2, \end{aligned}$$

by assumption and for some constants $\xi, \xi_2, \xi_3 > 0$. The second condition is satisfied for large enough problem size. Thus we have the runtime stated in Theorem 3.

Using that $\mu < \mu^* = \Theta(1)$, the super-quadratic condition becomes

$$\frac{|E^*|}{\Delta_P} \geq \frac{|E^*|}{\max_{j \in [k]} \|h_j\|_{\text{cyc}, 1, c}} = \Theta \left(\frac{|\text{OPT}_B|}{\max_{j \in [k]} \|h_j\|_{\text{cyc}, 1, c}} \right) = \Theta \left(\log \frac{|K|}{|K^*|} \right),$$

which is satisfied by our assumptions. Thus we have that $\alpha > 0$ (cf. Corollary 1) is a dimension-independent constant. This completes the proof. $\square$

5 Numerical results

We evaluate our algorithm on two complementary problem classes: a synthetic suite of CUT-GEN1 one-dimensional cutting-stock instances, which lets us generate many reproducible problems via parametric sweeps [GW95]; and (ii) a curated set of pure-integer ILPs from MIPLIB 2017, providing diverse, real-world benchmarks [GHG⁺21]. This pairing allows us to test behavior both in controlled settings (generation at scale) and across accepted benchmark models. Because the group relaxation is defined for any ILP and yields bounds stronger than the plain LP relaxation, with the minimization chain $\text{OPT}_{\text{LP}} \leq \text{OPT}_B \leq \text{OPT}$, these two families are natural testbeds.

We use the standard pattern-based ILP for the 1D-CSP: given stock length L , item widths w_i and demands d_i for $i \in [m]$, let P be the set of cutting patterns with counts $a_{ip} \in \mathbb{Z}_{\geq 0}$ satisfying $\sum_{i=1}^m a_{ip} w_i \leq L$. The problem at hand is thus

$$\min_{x \in \mathbb{Z}_{\geq 0}^{|P|}} \left\{ \sum_{p \in P} x_p : \sum_{p \in P} a_{ip} x_p \geq d_i \quad \forall i \in [m] \right\},$$

i.e., minimize the number of stock rolls subject to satisfying all demands.

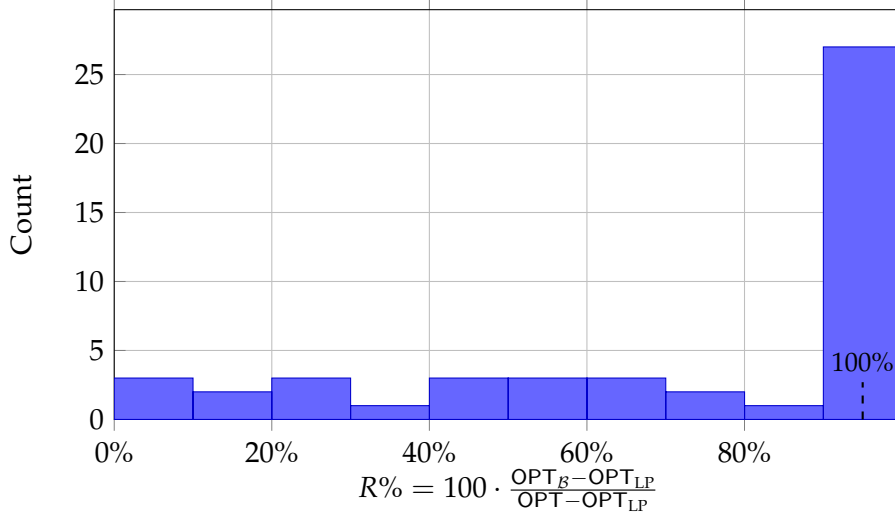


Figure 1: Histogram of $R\% = 100 \cdot (\text{OPT}_{\mathcal{B}} - \text{OPT}_{\text{LP}}) / (\text{OPT} - \text{OPT}_{\text{LP}})$ over the cutting-stock benchmark. Dashed line at 100% marks full gap closure by the group relaxation.

Instance generation follows the CUTGEN1 scheme of Gau and Wäscher [GW95]. We instantiate a grid over three knobs

$$m \in \{25, 50, 75\}, \quad v_2 \in \{0.25, 0.50, 0.75, 1.00\}, \quad \bar{d} \in \{5, 10, 20\}.$$

Here: m is the number of distinct item types; v_2 is the upper width ratio relative to the stock length L , and widths are sampled as integers in an interval proportional to L , typically

$$w_i \in [\lceil v_1 L \rceil, \lfloor v_2 L \rfloor],$$

where v_1 is a fixed lower ratio that is held constant across our runs.

Instance	OPT_{LP}	$\text{OPT}_{\mathcal{B}}$	OPT_{ILP}	$\Delta_{\text{LP} \rightarrow \text{ILP}}$	$\Delta_{\mathcal{B}}$	R_{abs}	$R\%$
qap10	332.57	336.00	340.00	7.43	4.00	3.43	46.2%
gen02	-4840.54	-4817.32	-4783.73	56.81	33.59	23.22	40.9%
ex10	100.00	100.00	100.00	0.00	0.00	0.00	N/A
supp19	12677206.00	12677206.00	12677552.00	346.00	346.00	0.00	0.0%

Table 1: Minimization results on pure ILP instances extracted from MIPLIB 2017 [GHG⁺21]. Column definitions: $\Delta_{\text{LP} \rightarrow \text{ILP}} := \text{OPT}_{\text{ILP}} - \text{OPT}_{\text{LP}}$, $\Delta_{\mathcal{B}} := \text{OPT}_{\text{ILP}} - \text{OPT}_{\mathcal{B}}$, $R_{\text{abs}} := \text{OPT}_{\mathcal{B}} - \text{OPT}_{\text{LP}}$, and $R\% := 100 \cdot R_{\text{abs}} / \Delta_{\text{LP} \rightarrow \text{ILP}}$.

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A Compressing the search space

This appendix discusses an approach to shrink the kernel group K by compressing the ambient space. Recall from equation (7) that K is a subgroup of $Z = \mathbb{Z}_{r_m}^d$. Though this set is finite, it is clearly not the minimal choice. To see this, let $s_j = |\langle \mathbf{A}_j \rangle|$ be the order of $\mathbf{A}_j$. Then, for any $\mathbf{x} \in K$ the coordinate $\mathbf{x}_j$ can always be reduced modulo $s_j := |\langle \mathbf{A}_j \rangle|$ while remaining in constraint. Explicitly, the orders are given by

$$s_j = \text{lcm} \left(\frac{r_1}{\gcd(r_1, \mathbf{A}_{1j})}, \dots, \frac{r_m}{\gcd(r_m, \mathbf{A}_{mj})} \right).$$

To see this, observe that any solution to $s_i \mathbf{A}_{ij} = 0 \pmod{r_i}$ is a multiple of $\gcd(r_i, \mathbf{A}_{ij})$. Any solution to these equations simultaneously must therefore be a common multiple. By definition of order, the smallest s_j satisfying these equations is therefore the *least* common multiple.

Note that r_m is a common multiple of the $\gcd(r_i, \mathbf{A}_{ij})$, hence $s_j | r_m$. This implies a natural homomorphism

$$Z \rightarrow Z' := \bigoplus_{i=1}^d \mathbb{Z}_{s_i}$$

obtained by reducing each component modulo s_i . The kernel K can be reduced under this map to some $K' \leq Z'$. In this way, we obtain a new, generally smaller kernel space containing the original optimal solutions. This procedure provides a benefit when

$$\prod_{j=1}^d s_j \ll r_m^d.$$

A set of generators for K' can be obtained from the generators (h_i) of K by $h'_i = h_i \pmod{Z'}$. However these generators do not necessarily provide a direct sum decomposition of K . Such a decomposition is desirable for several reasons. At a high level, it allows for efficient search over the group elements without repetitions. More specifically to our purposes, we will find the cyclic decomposition useful for considering Markov chain approaches: the Cayley graph becomes a product of cycles, which has provably nice mixing properties.

To obtain cyclic generators for K' , we utilize Algorithm 4.1.10 of Cohen [Coh00] for computing K' . The set up is as follows: let $\varphi(x) = x \pmod{Z'}$ be the homomorphism mapping Z to Z' (and thus K to K'). We note that φ is “effective,” meaning images and pre-image elements are trivial to compute. The domain space Z is already in SNF (as a direct sum of cyclic groups, each order dividing the previous), while the codomain Z' can be put in SNF efficiently (especially given that the corresponding group relation matrix $\text{diag}(s_j)_j$ is diagonal). Let $(\alpha_1, \dots, \alpha_q)$ be the cyclic generators of Z with invariant factors $z_1, \dots, z_q$, $q \leq d$. We note that φ is still effective with respect to the new generators $(\alpha_j)_{j=1}^q$.

The last ingredient is an effective description of K as a subgroup of Z . In particular, by Prop. 4.1.6 of [Coh00], any subgroup A of B is in 1-1 correspondence with integral matrices H_A in *Hermite Normal Form* (HNF). Such a matrix can be found in time polynomial in d , and typically faster than SNF. For our present case, H_K is the matrix obtained from an HNF algorithm on the block matrix

$$(h_1 h_2 \dots h_\ell | r_m I_{d \times d}). \quad (19)$$

We thus assume access to the appropriate descriptor H_K of K in Z .

With H_K in hand, Algorithm 4.1.10 outputs another HNF integer matrix L such that

$$(\alpha_1 \dots \alpha_q) L = (h'_1 \dots h'_p) \quad (20)$$

where $(h'_i)_{i=1}^p$ form the cyclic generators of K' and $p \leq \ell$. The runtime for this Algorithm is dominated by HNF calculations, which are again polynomial in d and the size of the invariant factors.

Here we also present an alternative and self-contained approach (Algorithm A) for compressing the kernel. This again produces a set of cyclic generators for K' .

Algorithm 6 NULLGENFINDINGCOMPRESS

Input: Constraints from Group relaxation $(\mathbf{A}, \{r_j\}_{j=1}^m)$

Output: An independent generating set $\mathcal{S}$ for $K' \leq \bigoplus_{j=1}^d \mathbb{Z}_{s_j}$, the solution space to $\mathbf{A}\mathbf{x} = \mathbf{0} \pmod{R\mathbb{Z}^m}$, where $s_j = |\langle \mathbf{A}_{:,j} \rangle|$.

- 1: Let $\{h_1, \dots, h_k\} = \text{NullGenFinding}(\mathbf{A}, \{r_j\}_{j=1}^m)$
 - 2: Compute $s_j = |\langle \mathbf{A}_{:,j} \rangle|$, the orders of the columns of $\mathbf{A}$ as elements of $\mathbb{Z}_{r_m}^m$.
 - 3: Construct $B = \text{diag}(r_m/s_1, \dots, r_m/s_k)$, $D = [h_1, \dots, h_k]$.
 - 4: Let $\{\tilde{m}_1, \dots, \tilde{m}_t\} = \text{NullGenFinding}(BD, \{r_m\}_{j=1}^m)$.
 - 5: Let $m_t = D\tilde{m}_t \pmod{r_m\mathbb{Z}^d}$.
 - 6: Compute SNF of $\mathbf{C} = [m_1, \dots, m_t | r_me_1, \dots, r_me_d] = U\mathbf{M}\mathbf{V}$.
 - 7: Let $\mathcal{S}'$ include columns of U such that $M_{jj} \notin \{0, 1\}$.
 - 8: Let $\mathcal{S} = \{u \pmod{SZ^d} \mid u \in \mathcal{S}'\}$.
 - 9: Output $\mathcal{S}$.
-

Lemma 2 (Runtime of NULLGENFINDINGCOMPRESS). *Suppose one has data $(\mathbf{A}, \mathbf{b}, \mathbf{c}, \{r_j\}_{j=1}^m)$ corresponding to the group relaxation of an ILP stored in classical memory. Let $S = \text{diag}(s_1, \dots, s_d)$, where s_j is the order of the j -th column of $\mathbf{A}$ taken $\pmod{R\mathbb{Z}^m}$. There is a classical algorithm that outputs cyclic generators $\mathcal{S} = (\tilde{h}_i)_{i=1}^v$ for $\tilde{K}$, the kernel of $\mathbf{A}$ in $\bigoplus_{j=1}^d \mathbb{Z}_{s_j}$. In particular, $K' \cong \mathbb{Z}_{|\langle \tilde{h}_1 \rangle|} \oplus \dots \oplus \mathbb{Z}_{|\langle \tilde{h}_v \rangle|}$. The algorithm uses time $\tilde{O}(\text{poly}(\log\|\mathbf{A}\|_\infty, d))$.*

Proof. We start by proving the validity of the above procedure. We again consider the generators output by Algorithm 2, $(h_j)_{j=1}^k$, which are all in $Z = \mathbb{Z}_{r_m}^d$.

Recall S is a diagonal matrix with the orders of the columns of $\mathbf{A}$ modulo $R\mathbb{Z}^m$. Note that reducing the element modulo $S\mathbb{Z}^d$ still produces a solution to $\mathbf{A}\mathbf{x} = \mathbf{0} \pmod{R\mathbb{Z}^m}$.

Let $D = [h_1, \dots, h_k]$. We then want to solve

$$D\mathbf{x} = \mathbf{0} \pmod{SZ^d},$$

which is to find the kernel of the map $\mathbf{y} \mapsto \mathbf{y} \pmod{SZ^d}$ in $\langle h_1, \dots, h_k \rangle \leq Z$. To do this, we solve a scaled system. We know that for each integer $j, 1 \leq j \leq d, s_j \mid r_m$. Let $B = \text{diag}(r_m/s_1, \dots, r_m/s_k)$. We then compute a generating set for the solution set to

$$BD\mathbf{x} = \mathbf{0} \pmod{r_m\mathbb{Z}^d},$$

i.e. run algorithm 2 getting generators $\tilde{m}_1, \dots, \tilde{m}_t \in Z$. We then know that the set of $m_t = D\tilde{m}_t \pmod{r_m\mathbb{Z}^d}$ span the kernel of the map $\mathbf{y} \mapsto \mathbf{y} \pmod{SZ^d}$ in $\langle h_1, \dots, h_k \rangle \leq Z$

Let us now work in the coset space $\mathbb{Z}^d/r_m\mathbb{Z}^d \cong Z$: we have

$$\tilde{Z} := \langle h_1 + r_m\mathbb{Z}^d \rangle \oplus \dots \oplus \langle h_k + r_m\mathbb{Z}^d \rangle \leq \langle \mathbf{e}_1 + r_m\mathbb{Z}^d \rangle \oplus \dots \oplus \langle \mathbf{e}_d + r_m\mathbb{Z}^d \rangle = \mathbb{Z}^d/r_m\mathbb{Z}^d.$$

We also have that

$$\tilde{K} := \langle m_1 + r_m\mathbb{Z}^d \rangle \oplus \dots \oplus \langle m_t + r_m\mathbb{Z}^d \rangle$$

forms the kernel of $\mathbf{y} \mapsto \mathbf{y} \pmod{SZ^d}$ in $\langle h_1 + r_m \mathbb{Z}^d \rangle \oplus \cdots \oplus \langle h_k + r_m \mathbb{Z}^d \rangle$.

Following [Chi17, Section 6.2], let

$$\mathbf{C} = [m_1, \dots, m_t | r_m e_1, \dots, r_m e_d].$$

We then compute the SNF of $\mathbf{C}$ to get $\mathbf{C} = U\mathbf{M}\mathbf{V}$. By unimodularity of $U \in \mathbb{Z}^{d \times d}$ we have

$$\tilde{Z} \leq \langle u_1 + r_m \mathbb{Z}^d \rangle \oplus \cdots \oplus \langle u_d + r_m \mathbb{Z}^d \rangle = \mathbb{Z}^d / r_m \mathbb{Z}^d,$$

where $u_j = U_{\cdot j}$. By construction $M_{jj}u_j \in \tilde{K}$. Since our goal is to find a generating set for the range of $\mathbf{y} \mapsto \mathbf{y} \pmod{SZ^d}$, we can ignore u_j where $M_{jj} = 1$.

We can also drop u_j such that $M_{jj} = 0$. This is because for such u_j , $\langle u_j + r_m \mathbb{Z}^d \rangle$ is not in $\tilde{Z}$. To see this: first, note that $M_{jj} = 0$ and the columns of U are linearly independent, so we have that for any integer $k, 0 < k < r_m, ku_j \notin \tilde{K}$. The only option then is either $u_j \notin \tilde{K}$ or $\langle u_j + r_m \mathbb{Z}^d \rangle \in \tilde{Z}$, i.e. the entire cyclic subgroup of u_j is in $\tilde{Z}$, no part is mapped to the kernel.

Suppose the later. Then we know that if we delete the all-zero columns from M to get M' then $r_mu_j \in \text{colspace}_{\mathbb{Z}}(UM')$. However, the columns of U are linearly independent and form a basis over $\mathbb{R}^d$, a contradiction. Hence $u_j \notin \tilde{K}$.

This implies

$$\tilde{Z}/\tilde{K} = \langle (u_1 + r_m \mathbb{Z}^d) + \tilde{K} \rangle \oplus \cdots \oplus \langle (u_v + r_m \mathbb{Z}^d) + \tilde{K} \rangle.$$

One will also note that we then immediately have

$$\tilde{Z}/\tilde{K} \cong \langle u_1 \pmod{SZ^d} \rangle \oplus \cdots \oplus \langle u_v \pmod{SZ^d} \rangle = K' \leq \bigoplus_{j=1}^d \mathbb{Z}_{s_j},$$

as desired.

The runtime follows from the runtime presented for SNF in Section 3.2. □

B A Gibbs sampling approach for quantum speedup

In this section, we present a Gibbs sampler that comes from applying a Metropolis filter (i.e. Metropolis-Hastings algorithm) to either of the Cayley walks described in Sections 4.2 and 4.3.

While Gibbs sampling for inverse-temperature $\beta > 0$ will outperform the uniform sampling approaches, the complexity of classical Gibbs sampling from the null space K for $\beta > 0$ is unclear. The reason we still mention the Gibbs sampler, is that if one shows that classical Gibbs sampling at some $\beta > 0$ can be done classically in $\mathcal{O}(\text{poly}(n))$ time and the conditions we present are met, then there is a super-quadratic speedup over the Gibbs sampler.

Beyond uniform sampling, one can also introduce a Metropolis filter (run the Metropolis-Hastings algorithm [LP17]) to perform Gibbs sampling over the feasible coset $\hat{\mathbf{x}} + K$ using a walk with the corresponding Gibbs measure as its stationary distribution. The Gibbs distribution over $\hat{\mathbf{x}} + K$ at inverse-temperature β is

$$\pi_{\beta}(x) \propto \exp(\beta \langle \mathbf{c}, x \rangle), x \in \hat{\mathbf{x}} + K,$$

for $\beta > 0$. The Metropolis filter performs an accept-reject step on top of another walk, referred to as the *base chain*. Specifically, we can use either the expander or product cycle walk. One

step of the Metropolis-adjusted chain is as follows: starting from x , accept a move to $y \neq x$ proposed by the base chain with probability $\min(1, \exp(\beta \langle \mathbf{c}, y - x \rangle))$.

For $\beta > 0$, the Gibbs sampler will provide a strict improvement over uniform sampling. Unfortunately, it appears there is not much known about the complexity of sampling for $\beta > 0$, implying that the current best theoretical result only applies to $\beta = 0$.

The below theorem shows that if there is a $\beta > 0$ such that the classical Metropolis chain satisfies a LSI with sufficiently large constant, then SP can provide a super-quadratic speedup over the Gibbs sampler. The next result follows from Theorem 2 and Corollary 1.

Theorem 12 (Conditional Super-Quadratic Speedup Over Gibbs Sampling). *Suppose there exists a $\beta > 0$ such that a Metropolis-adjusted Cayley Walk with stationary distribution π_β has LS constant lower bounded by $\omega = \Omega([\log(1/\pi_\beta(\text{OPT}_B))]^{-1})$. Then, under condition (16a) from Theorem 9, there exists a generalized quantum SP algorithm that runs in time*

$$\mathcal{O}\left([\pi_\beta(\text{OPT}_B)]^{-(1/2-\alpha)} \text{poly}(n, L)\right), \quad (21)$$

for $\alpha > 0$.

Note that the proof is effectively the same as Theorem 9, as adding the Metropolis filter can only decrease Δ_P , so the choices made for Δ_P in the proof of Theorem 9 remain valid. In addition, the Metropolis-adjusted Cayley walk can be performed with a constant number of queries to the base chain and cost function operators using the approach of [LHP⁺20] combined with the Cayley-walk implementation previously mentioned and Section 4.1. The above theorem implies that if there exists a $\beta > 0$ that satisfies the specified conditions, the runtime of both classical and quantum algorithms proposed in the main text can be improved as the distribution is tilted towards the optimal solutions. Unfortunately, for most general Gibbs measures, the distribution starts forming clusters and there is no local Markov chain that can efficiently sample from such a measure due to long range correlations.

C Notations

Symbol	Meaning
Core problem data and sets	
$A \in \mathbb{Z}^{m \times n}, b \in \mathbb{Z}^m, c \in \mathbb{R}^n$	Problem data (constraint matrix, RHS, cost)
$\mathbb{Z}_{\geq 0}^n, \mathbb{R}_{\geq 0}^n$	Nonnegative orthants over $\mathbb{Z}^n$ and $\mathbb{R}^n$
$\mathcal{P} := \{x \in \mathbb{Z}_{\geq 0}^n : Ax = b\}$	ILP feasible set
$\mathcal{P}_{\text{LP}} := \{x \in \mathbb{R}_{\geq 0}^n : Ax = b\}$	LP relaxation feasible set
$\text{OPT} := \min\{c^\top x : x \in \mathcal{P}\}$	ILP optimal value
$\text{OPT}_{\text{LP}} := \min\{c^\top x : x \in \mathcal{P}_{\text{LP}}\}$	LP optimal value
LP bases, partitions, and reduced costs	
$\mathcal{B} \subset [n]$	LP basis (with $ \mathcal{B} = m$ if $\text{rank}(A) = m$)
$\mathcal{N} := [n] \setminus \mathcal{B}$	Nonbasic index set
$A = [A_{\mathcal{B}}, A_{\mathcal{N}}]$	Partition of A by $(\mathcal{B}, \mathcal{N})$
$c = [c_{\mathcal{B}}, c_{\mathcal{N}}], x = [x_{\mathcal{B}}, x_{\mathcal{N}}]$	Matching partitions of c and x
$\bar{c}_{\mathcal{N}} := c_{\mathcal{N}} - A_{\mathcal{N}}^\top (A_{\mathcal{B}}^{-1})^\top c_{\mathcal{B}}$	Reduced costs for nonbasic indices
Group relaxation (primal formulations)	
$\mathcal{P}_{\mathcal{B}} := \{(x_{\mathcal{B}}, x_{\mathcal{N}}) \in \mathbb{Z}^m \times \mathbb{Z}_{\geq 0}^{n-m} : Ax = b\}$	Group relaxation feasible set
$\text{OPT}_{\mathcal{B}} := \min\{c^\top x : x \in \mathcal{P}_{\mathcal{B}}\}$	Group relaxation optimal value
$\text{OPT}_{\mathcal{B}} = \min_{x_{\mathcal{N}} \in \mathbb{Z}_{\geq 0}^{n-m}} \{c_{\mathcal{B}}^\top A_{\mathcal{B}}^{-1} b + \bar{c}_{\mathcal{N}}^\top x_{\mathcal{N}}\}$	Objective in nonbasic variables
Smith normal form and bounded formulation	
$A_{\mathcal{B}} = URV$	Smith normal form
$R = \text{diag}(r_1, \dots, r_m)$	Invariant factors of $A_{\mathcal{B}}$
$U^{-1} A_{\mathcal{N}} x_{\mathcal{N}} = U^{-1} b \pmod{R\mathbb{Z}^m}$	Feasibility in SNF coordinates
$\mathbf{A} := U^{-1} A_{\mathcal{N}}, \mathbf{b} := U^{-1} b, \mathbf{c} := \bar{c}_{\mathcal{N}}$	Data for group formulation
Groups, lattices, kernel and cosets	
$\Lambda := A\mathbb{Z}, \Lambda_{\mathcal{B}} := A_{\mathcal{B}}\mathbb{Z}$	Lattices generated by A and $A_{\mathcal{B}}$
$G := (\mathbf{A}\mathbb{Z}^d + R\mathbb{Z}^m) / R\mathbb{Z}^m$	Range-space quotient (finite abelian group)
$\mathcal{R} := \mathbb{Z}^m / R\mathbb{Z}^m$	Ambient range space
$Z := \mathbb{Z}_{r_m}^d$	Ambient space for finite group relaxation
$K := \{\mathbf{h} \in Z : \mathbf{A}\mathbf{h} = \mathbf{0} \pmod{R\mathbb{Z}^m}\}$	Null space subgroup
$\hat{\mathbf{x}}$	Vector in Z with $\mathbf{A}\hat{\mathbf{x}} = \mathbf{b} \pmod{R\mathbb{Z}^m}$
$\bar{\mathcal{P}}_{\mathcal{B}} = \hat{\mathbf{x}} + K$	Feasible set (coset)
$\mathcal{S} = \{h_1, \dots, h_k\}, u_j := \langle h_j \rangle $	Generators of K and their orders
$K \cong \bigoplus_{j=1}^k \mathbb{Z}_{u_j}$	Cyclic decomposition of K
$\mathcal{P}_{\mathcal{B}}^*, K^*$	Group-relaxation optima; optimal subset in K
Continued on next page	

Symbol	Meaning
Markov chains and generalized short path (SP) constructs	
P	Transition matrix
π	Stationary distribution of P
$D(P) := \text{diag}(\pi)^{1/2} P \text{diag}(\pi)^{-1/2}$	Discriminant matrix
δ	Spectral gap of P
$t_{\text{mix}}(\varepsilon), \text{TV}(\mu, \nu)$	Mixing time; total variation distance
H, E^*	Cost Hamiltonian; ground state energy
$H_\mu := -D(P) + \mu \theta_\eta(H E^*)$	Short path Hamiltonian
$\theta_\eta(x) := \min\{0, (x + 1 - \eta)/\eta\}$	Thresholding function (SP)
$ \psi_\mu\rangle, \Pi^*$	Ground state of H_μ ; projector onto optima
$\ f\ _P := \max_x \mathbb{E}_{y \sim P(x, \cdot)}[(f(x) - f(y))^2]$	P -pseudo Lipschitz norm
Δ_P, γ, ω	Stability; spectral density; log-Sobolev constant